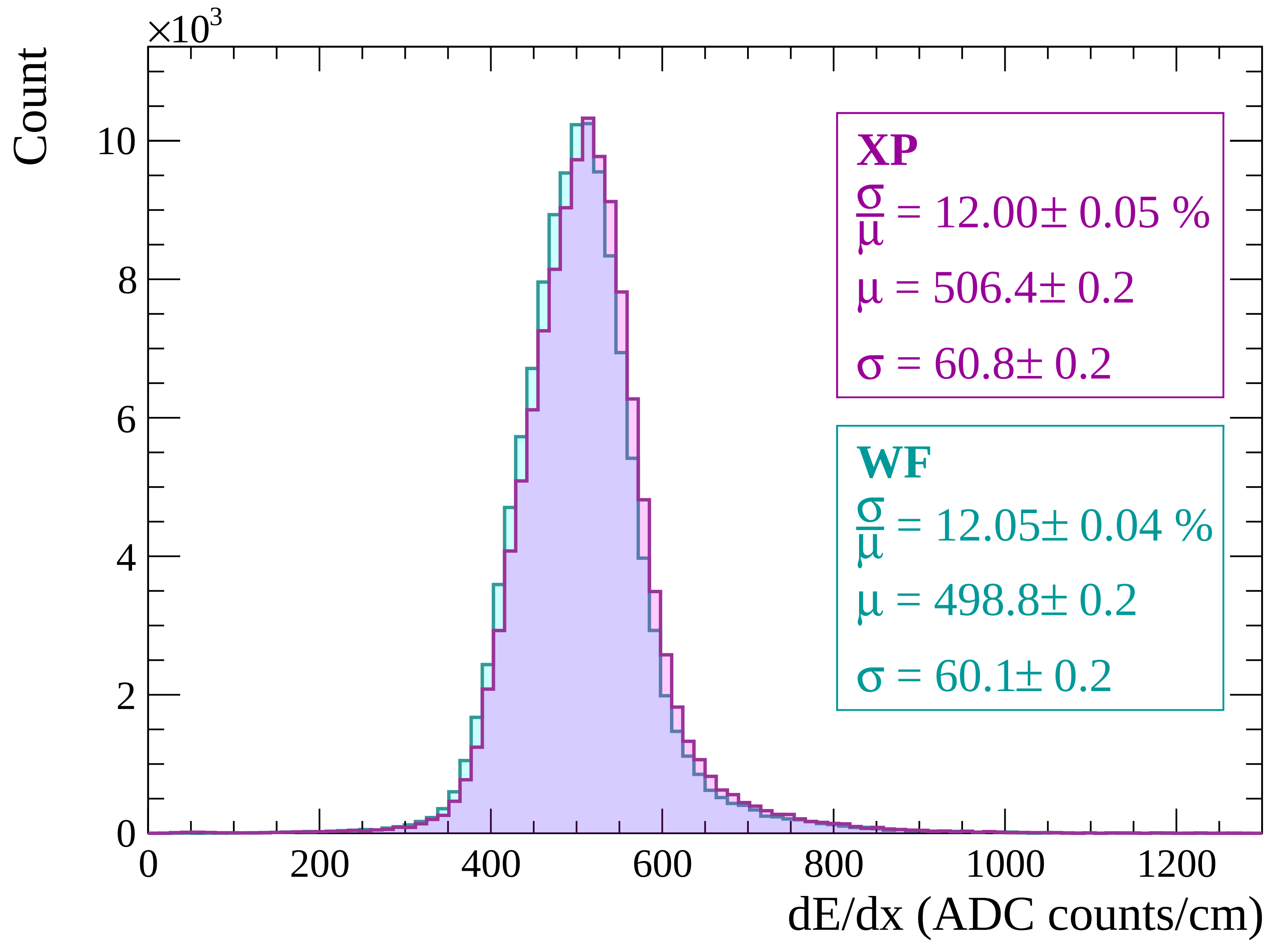
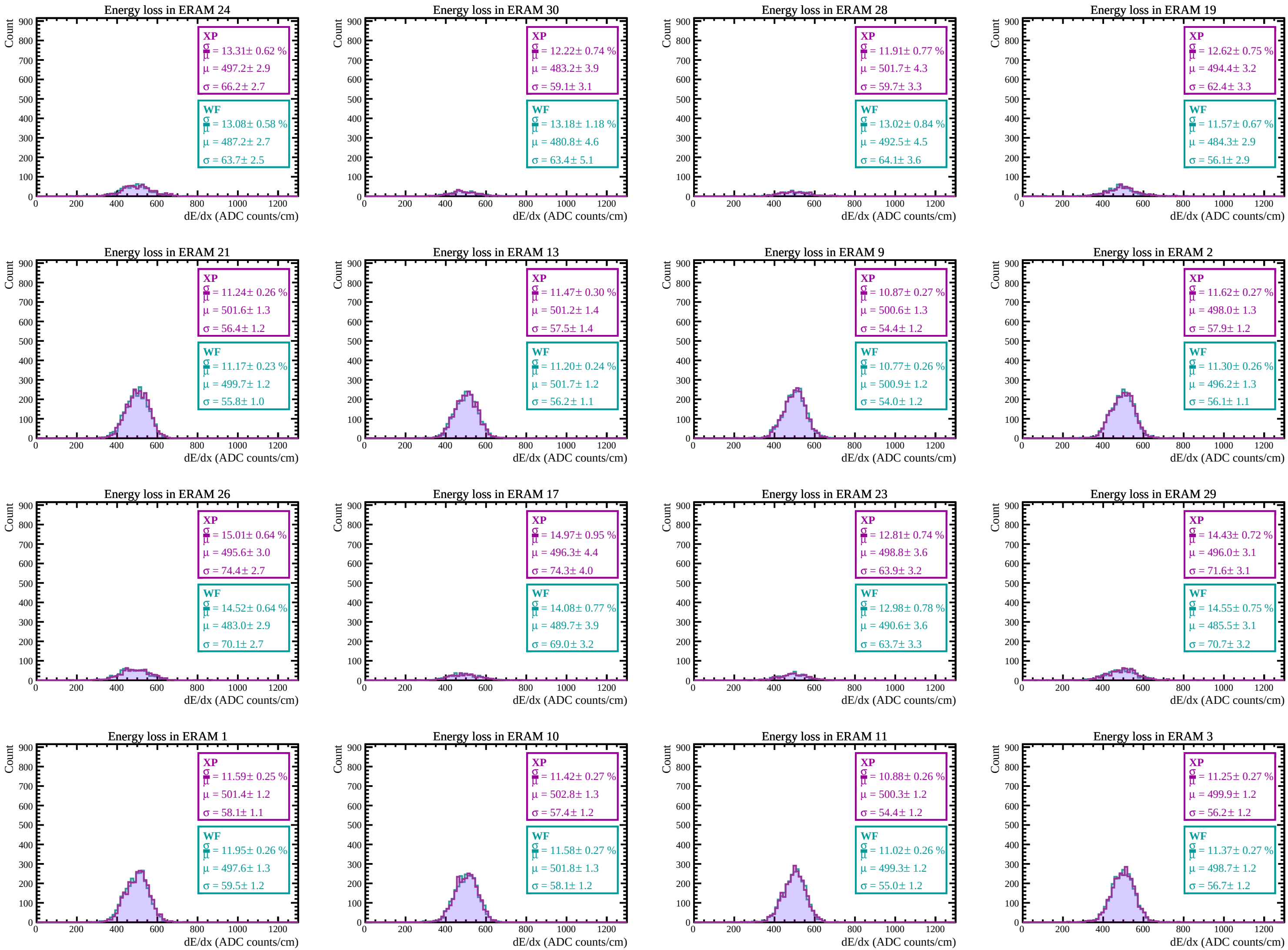
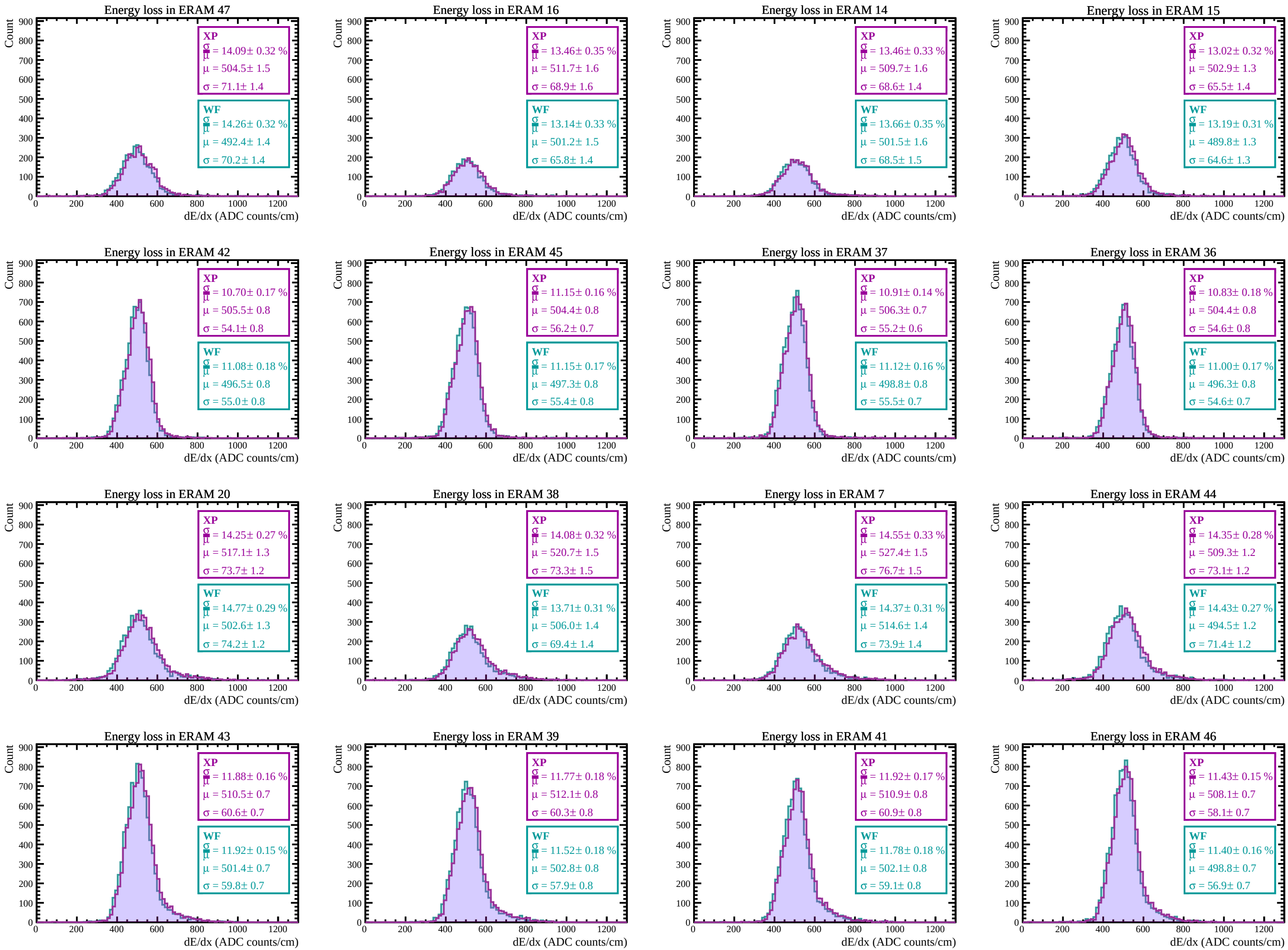
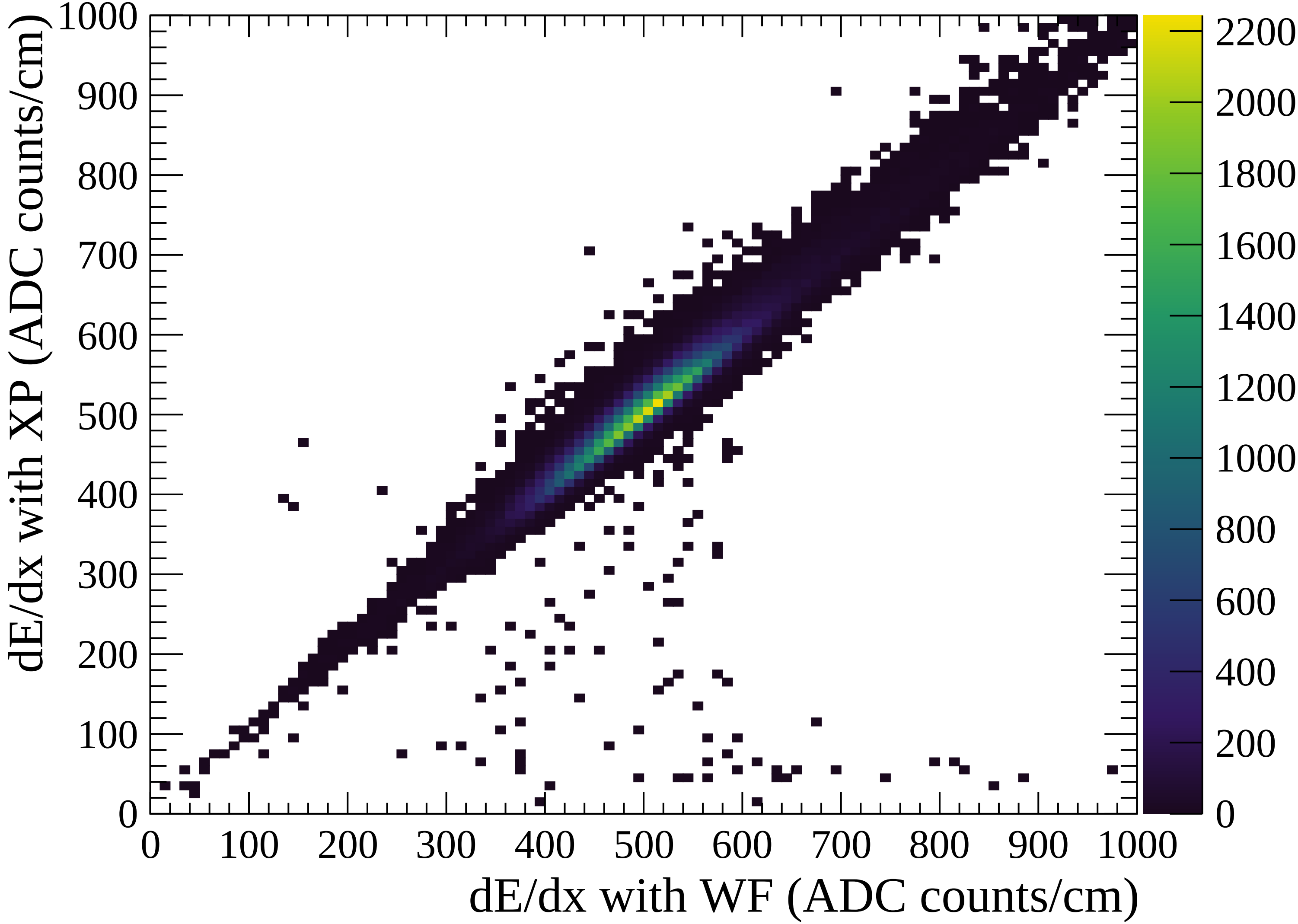
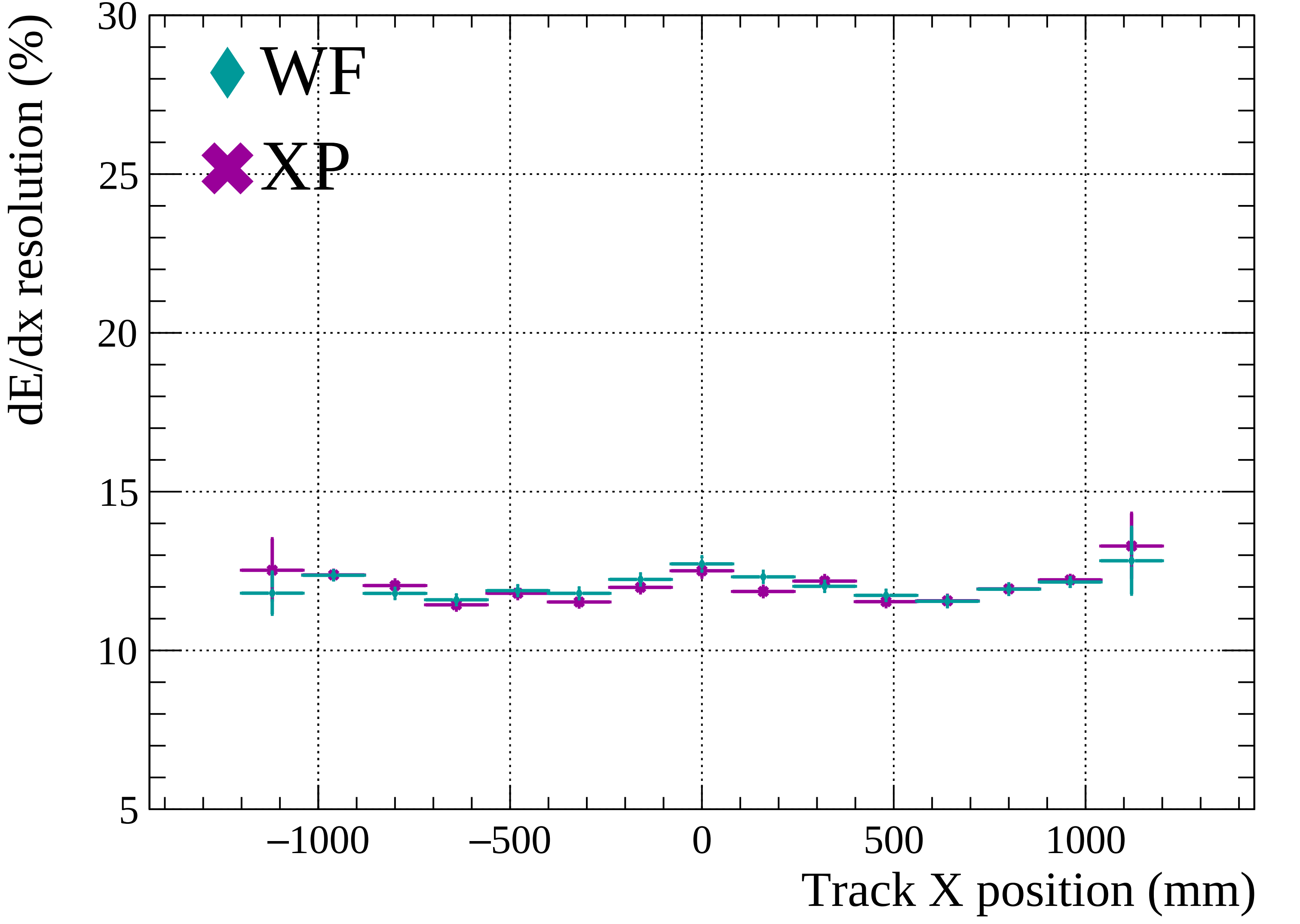


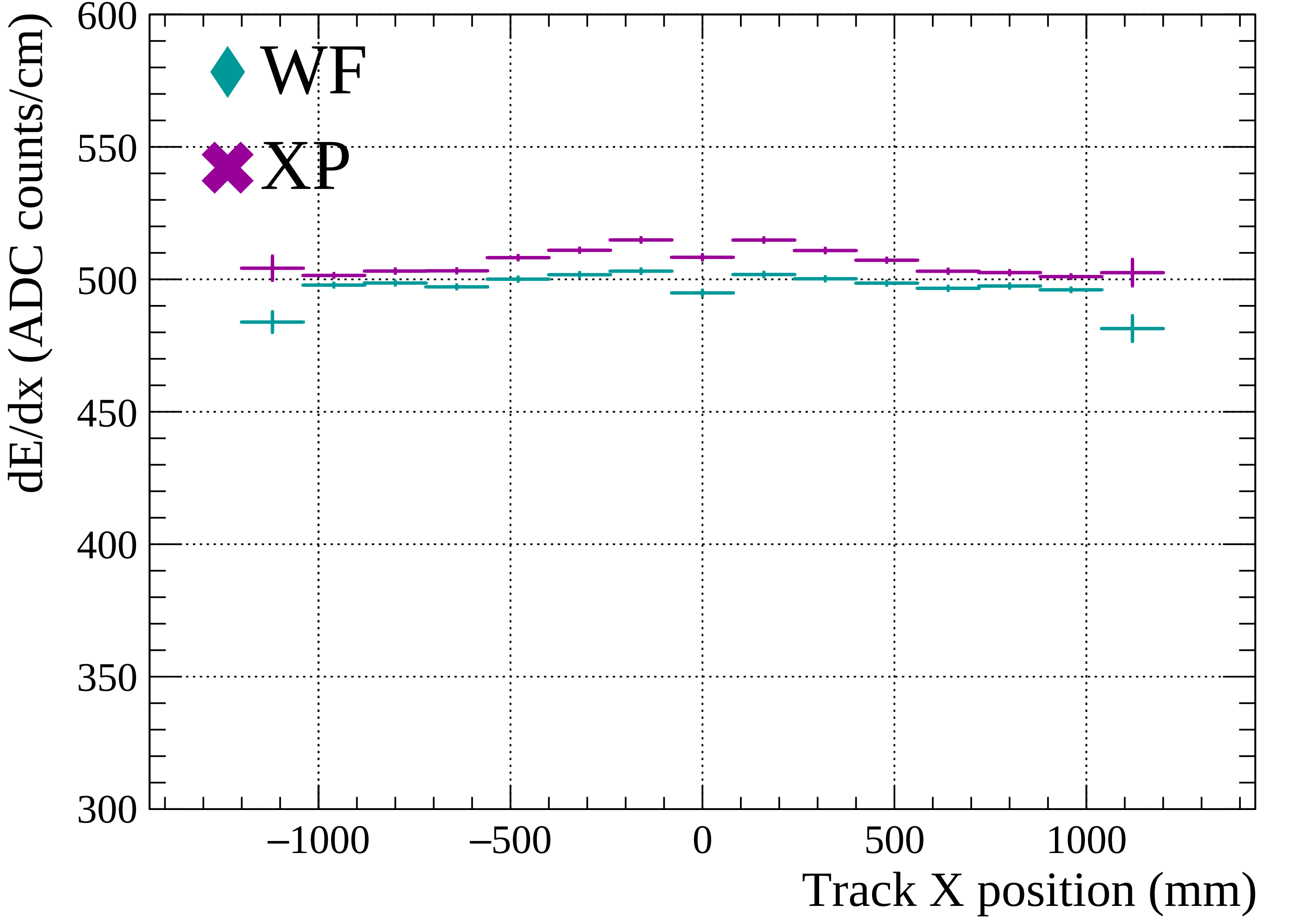
$\times 10^3$
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.00± 0.05 %
μ = 506.4± 0.2
σ = 60.8± 0.2
WF
$\frac{\sigma}{\mu}$ = 12.05± 0.04 %
μ = 498.8± 0.2
σ = 60.1± 0.2

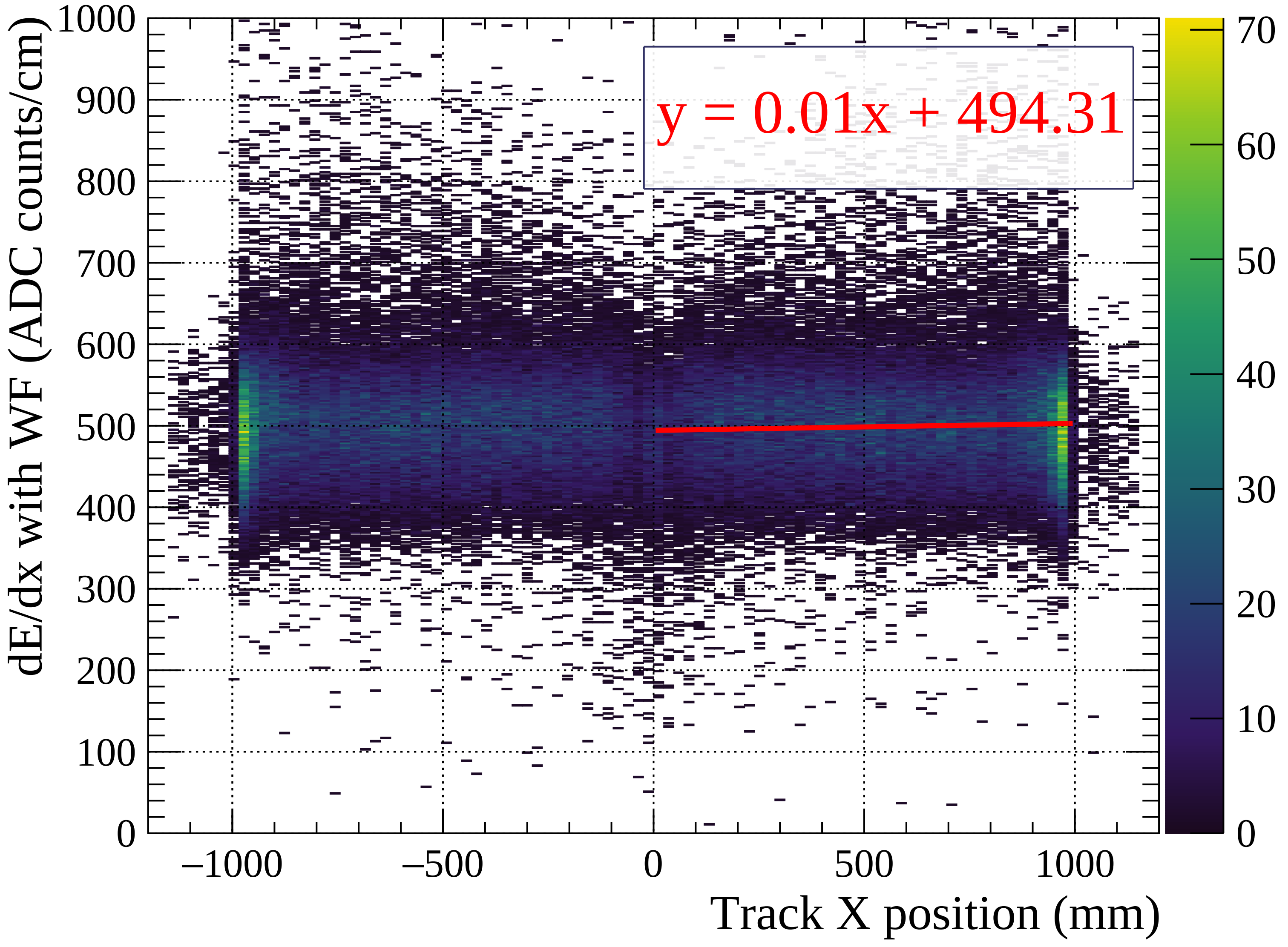

Energy loss in ERAM 24
XP
σ/μ = 13.31± 0.62 %
μ = 497.2± 2.9
σ = 66.2± 2.7
WF
σ/μ = 13.08± 0.58 %
μ = 487.2± 2.7
σ = 63.7± 2.5
Energy loss in ERAM 30
XP
σ/μ = 12.22± 0.74 %
μ = 483.2± 3.9
σ = 59.1± 3.1
WF
σ/μ = 13.18± 1.18 %
μ = 480.8± 4.6
σ = 63.4± 5.1
Energy loss in ERAM 28
XP
σ/μ = 11.91± 0.77 %
μ = 501.7± 4.3
σ = 59.7± 3.3
WF
σ/μ = 13.02± 0.84 %
μ = 492.5± 4.5
σ = 64.1± 3.6
Energy loss in ERAM 19
XP
σ/μ = 12.62± 0.75 %
μ = 494.4± 3.2
σ = 62.4± 3.3
WF
σ/μ = 11.57± 0.67 %
μ = 484.3± 2.9
σ = 56.1± 2.9
Energy loss in ERAM 21
XP
σ/μ = 11.24± 0.26 %
μ = 501.6± 1.3
σ = 56.4± 1.2
WF
σ/μ = 11.17± 0.23 %
μ = 499.7± 1.2
σ = 55.8± 1.0
Energy loss in ERAM 13
XP
σ/μ = 11.47± 0.30 %
μ = 501.2± 1.4
σ = 57.5± 1.4
WF
σ/μ = 11.20± 0.24 %
μ = 501.7± 1.2
σ = 56.2± 1.1
Energy loss in ERAM 9
XP
σ/μ = 10.87± 0.27 %
μ = 500.6± 1.3
σ = 54.4± 1.2
WF
σ/μ = 10.77± 0.26 %
μ = 500.9± 1.2
σ = 54.0± 1.2
Energy loss in ERAM 2
XP
σ/μ = 11.62± 0.27 %
μ = 498.0± 1.3
σ = 57.9± 1.2
WF
σ/μ = 11.30± 0.26 %
μ = 496.2± 1.3
σ = 56.1± 1.1
Energy loss in ERAM 26
XP
σ/μ = 15.01± 0.64 %
μ = 495.6± 3.0
σ = 74.4± 2.7
WF
σ/μ = 14.52± 0.64 %
μ = 483.0± 2.9
σ = 70.1± 2.7
Energy loss in ERAM 17
XP
σ/μ = 14.97± 0.95 %
μ = 496.3± 4.4
σ = 74.3± 4.0
WF
σ/μ = 14.08± 0.77 %
μ = 489.7± 3.9
σ = 69.0± 3.2
Energy loss in ERAM 23
XP
σ/μ = 12.81± 0.74 %
μ = 498.8± 3.6
σ = 63.9± 3.2
WF
σ/μ = 12.98± 0.78 %
μ = 490.6± 3.6
σ = 63.7± 3.3
Energy loss in ERAM 29
XP
σ/μ = 14.43± 0.72 %
μ = 496.0± 3.1
σ = 71.6± 3.1
WF
σ/μ = 14.55± 0.75 %
μ = 485.5± 3.1
σ = 70.7± 3.2
Energy loss in ERAM 1
XP
σ/μ = 11.59± 0.25 %
μ = 501.4± 1.2
σ = 58.1± 1.1
WF
σ/μ = 11.95± 0.26 %
μ = 497.6± 1.3
σ = 59.5± 1.2
Energy loss in ERAM 10
XP
σ/μ = 11.42± 0.27 %
μ = 502.8± 1.3
σ = 57.4± 1.2
WF
σ/μ = 11.58± 0.27 %
μ = 501.8± 1.3
σ = 58.1± 1.2
Energy loss in ERAM 11
XP
σ/μ = 10.88± 0.26 %
μ = 500.3± 1.2
σ = 54.4± 1.2
WF
σ/μ = 11.02± 0.26 %
μ = 499.3± 1.2
σ = 55.0± 1.2
Energy loss in ERAM 3
XP
σ/μ = 11.25± 0.27 %
μ = 499.9± 1.2
σ = 56.2± 1.2
WF
σ/μ = 11.37± 0.27 %
μ = 498.7± 1.2
σ = 56.7± 1.2
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
XP
σ/μ = 14.09± 0.32 %
μ = 504.5± 1.5
σ = 71.1± 1.4
WF
σ/μ = 14.26± 0.32 %
μ = 492.4± 1.4
σ = 70.2± 1.4
Energy loss in ERAM 16
XP
σ/μ = 13.46± 0.35 %
μ = 511.7± 1.6
σ = 68.9± 1.6
WF
σ/μ = 13.14± 0.33 %
μ = 501.2± 1.5
σ = 65.8± 1.4
Energy loss in ERAM 14
XP
σ/μ = 13.46± 0.33 %
μ = 509.7± 1.6
σ = 68.6± 1.4
WF
σ/μ = 13.66± 0.35 %
μ = 501.5± 1.6
σ = 68.5± 1.5
Energy loss in ERAM 15
XP
σ/μ = 13.02± 0.32 %
μ = 502.9± 1.3
σ = 65.5± 1.4
WF
σ/μ = 13.19± 0.31 %
μ = 489.8± 1.3
σ = 64.6± 1.3
Energy loss in ERAM 42
XP
σ/μ = 10.70± 0.17 %
μ = 505.5± 0.8
σ = 54.1± 0.8
WF
σ/μ = 11.08± 0.18 %
μ = 496.5± 0.8
σ = 55.0± 0.8
Energy loss in ERAM 45
XP
σ/μ = 11.15± 0.16 %
μ = 504.4± 0.8
σ = 56.2± 0.7
WF
σ/μ = 11.15± 0.17 %
μ = 497.3± 0.8
σ = 55.4± 0.8
Energy loss in ERAM 37
XP
σ/μ = 10.91± 0.14 %
μ = 506.3± 0.7
σ = 55.2± 0.6
WF
σ/μ = 11.12± 0.16 %
μ = 498.8± 0.8
σ = 55.5± 0.7
Energy loss in ERAM 36
XP
σ/μ = 10.83± 0.18 %
μ = 504.4± 0.8
σ = 54.6± 0.8
WF
σ/μ = 11.00± 0.17 %
μ = 496.3± 0.8
σ = 54.6± 0.7
Energy loss in ERAM 20
XP
σ/μ = 14.25± 0.27 %
μ = 517.1± 1.3
σ = 73.7± 1.2
WF
σ/μ = 14.77± 0.29 %
μ = 502.6± 1.3
σ = 74.2± 1.2
Energy loss in ERAM 38
XP
σ/μ = 14.08± 0.32 %
μ = 520.7± 1.5
σ = 73.3± 1.5
WF
σ/μ = 13.71± 0.31 %
μ = 506.0± 1.4
σ = 69.4± 1.4
Energy loss in ERAM 7
XP
σ/μ = 14.55± 0.33 %
μ = 527.4± 1.5
σ = 76.7± 1.5
WF
σ/μ = 14.37± 0.31 %
μ = 514.6± 1.4
σ = 73.9± 1.4
Energy loss in ERAM 44
XP
σ/μ = 14.35± 0.28 %
μ = 509.3± 1.2
σ = 73.1± 1.2
WF
σ/μ = 14.43± 0.27 %
μ = 494.5± 1.2
σ = 71.4± 1.2
Energy loss in ERAM 43
XP
σ/μ = 11.88± 0.16 %
μ = 510.5± 0.7
σ = 60.6± 0.7
WF
σ/μ = 11.92± 0.15 %
μ = 501.4± 0.7
σ = 59.8± 0.7
Energy loss in ERAM 39
XP
σ/μ = 11.77± 0.18 %
μ = 512.1± 0.8
σ = 60.3± 0.8
WF
σ/μ = 11.52± 0.18 %
μ = 502.8± 0.8
σ = 57.9± 0.8
Energy loss in ERAM 41
XP
σ/μ = 11.92± 0.17 %
μ = 510.9± 0.8
σ = 60.9± 0.8
WF
σ/μ = 11.78± 0.18 %
μ = 502.1± 0.8
σ = 59.1± 0.8
Energy loss in ERAM 46
XP
σ/μ = 11.43± 0.15 %
μ = 508.1± 0.7
σ = 58.1± 0.7
WF
σ/μ = 11.40± 0.16 %
μ = 498.8± 0.7
σ = 56.9± 0.7
Count
dE/dx (ADC counts/cm)

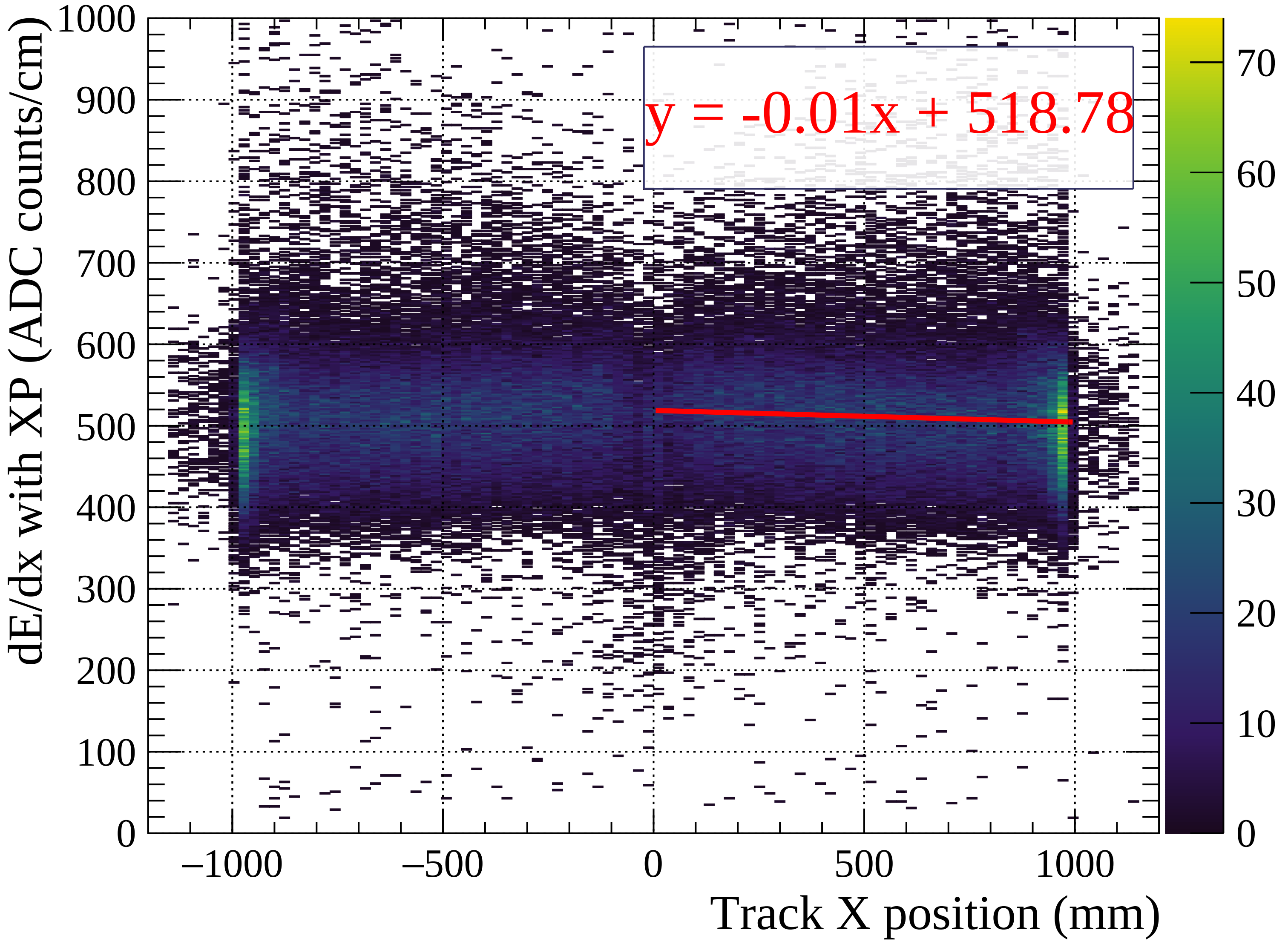

dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0


WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000

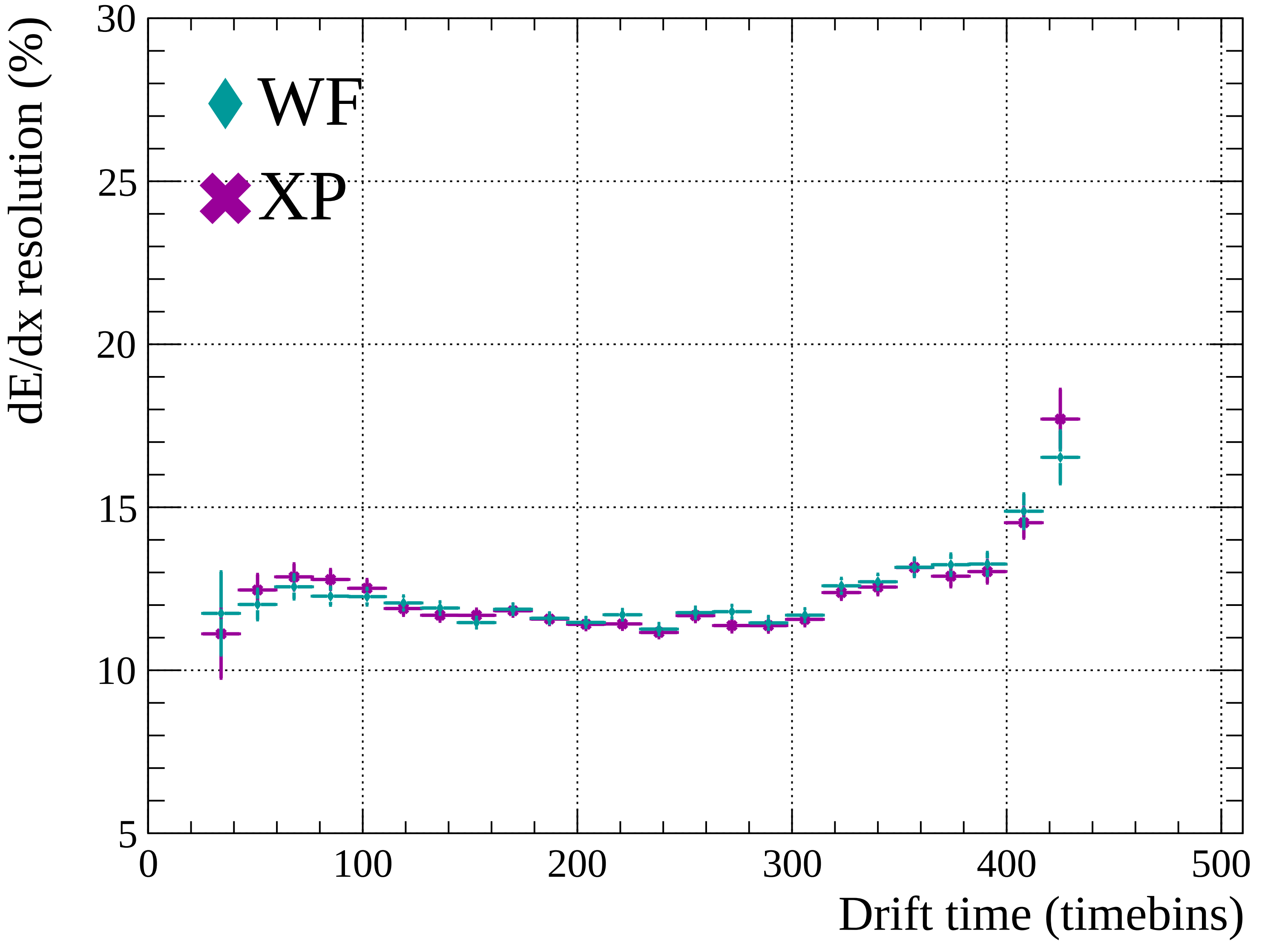

WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)



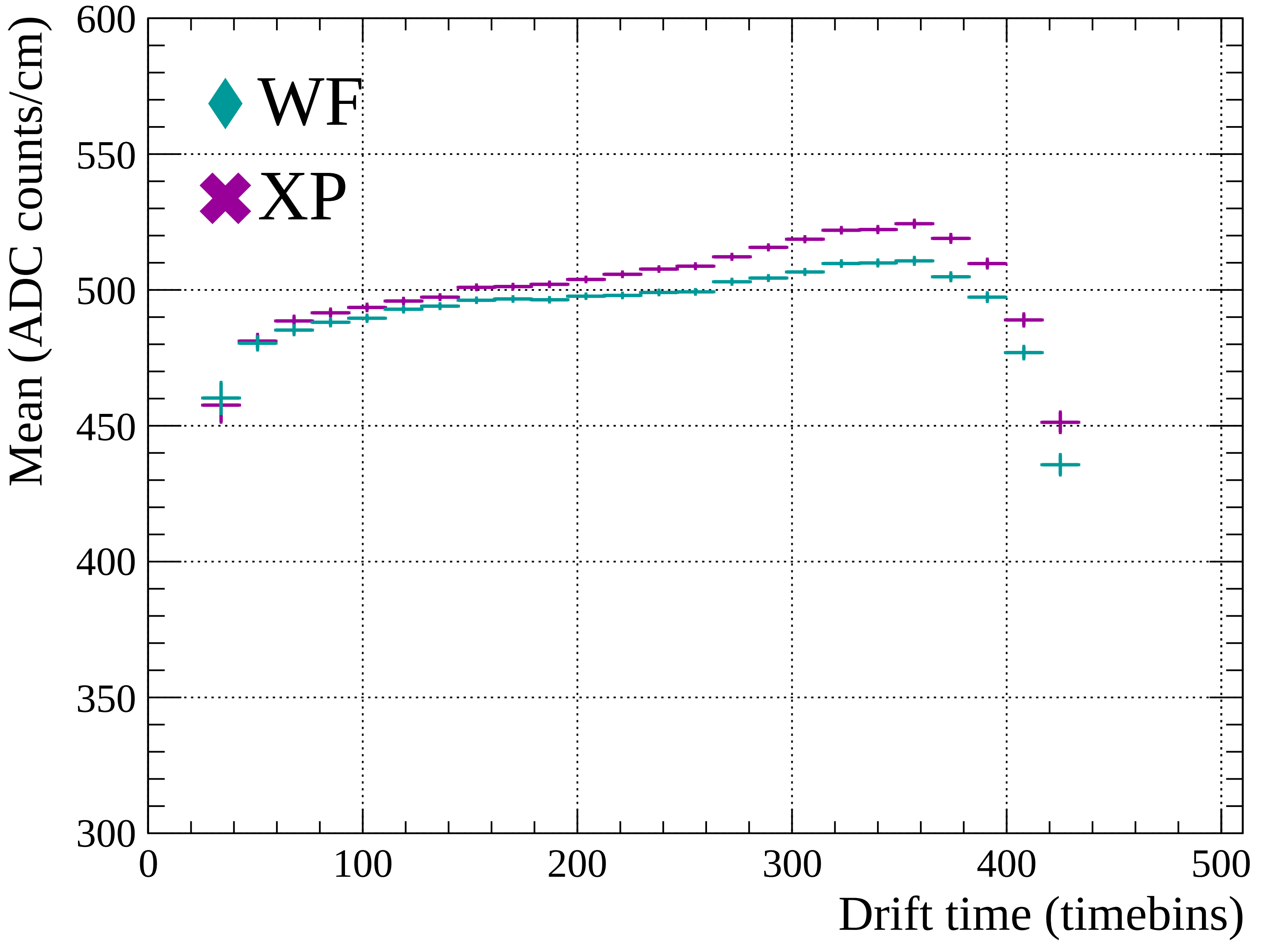
y = 0.01x + 494.31
dE/dx with WF (ADC counts/cm)
Track X position (mm)



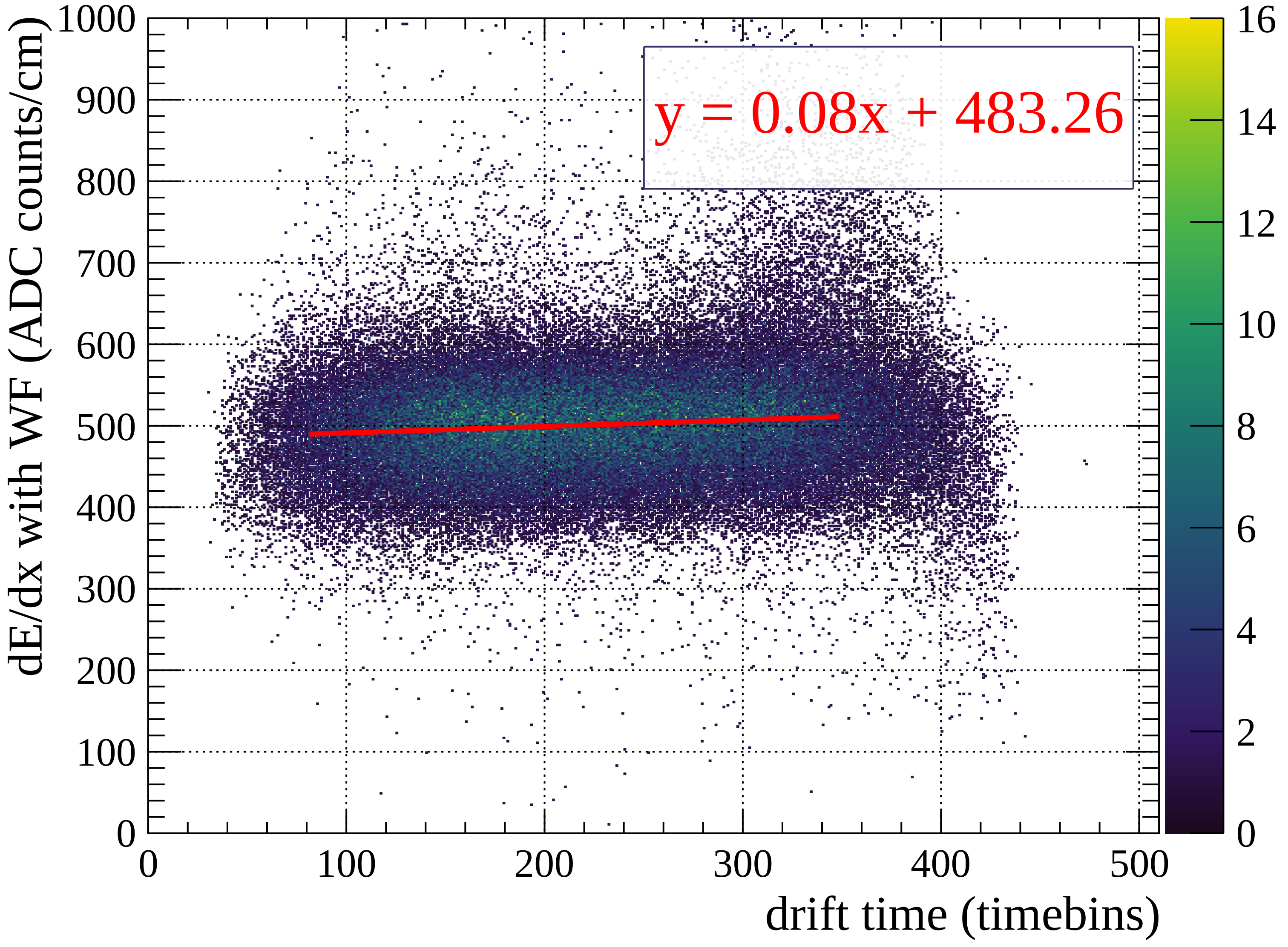
y = -0.01x + 518.78
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70

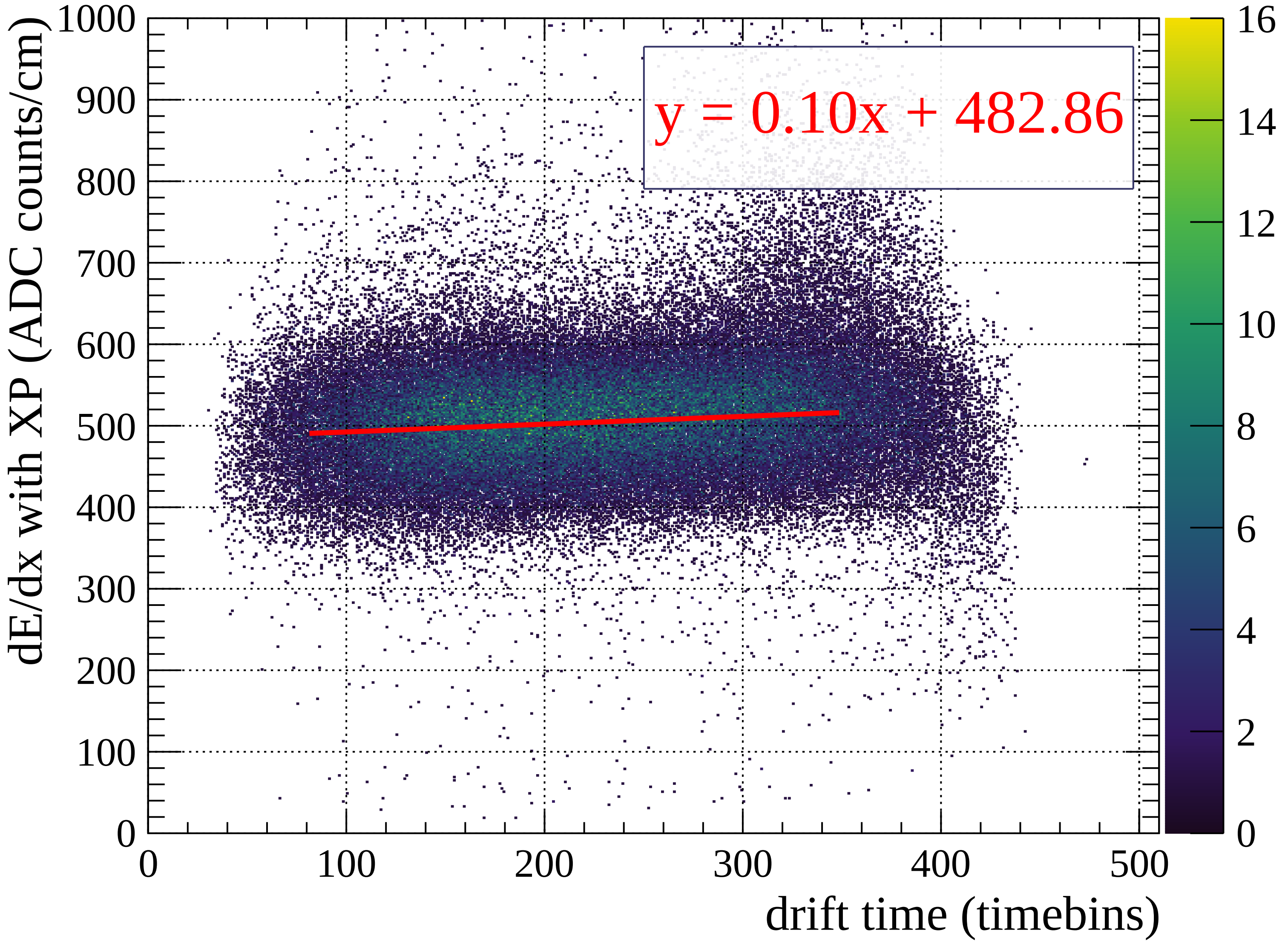


dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)
WF
XP

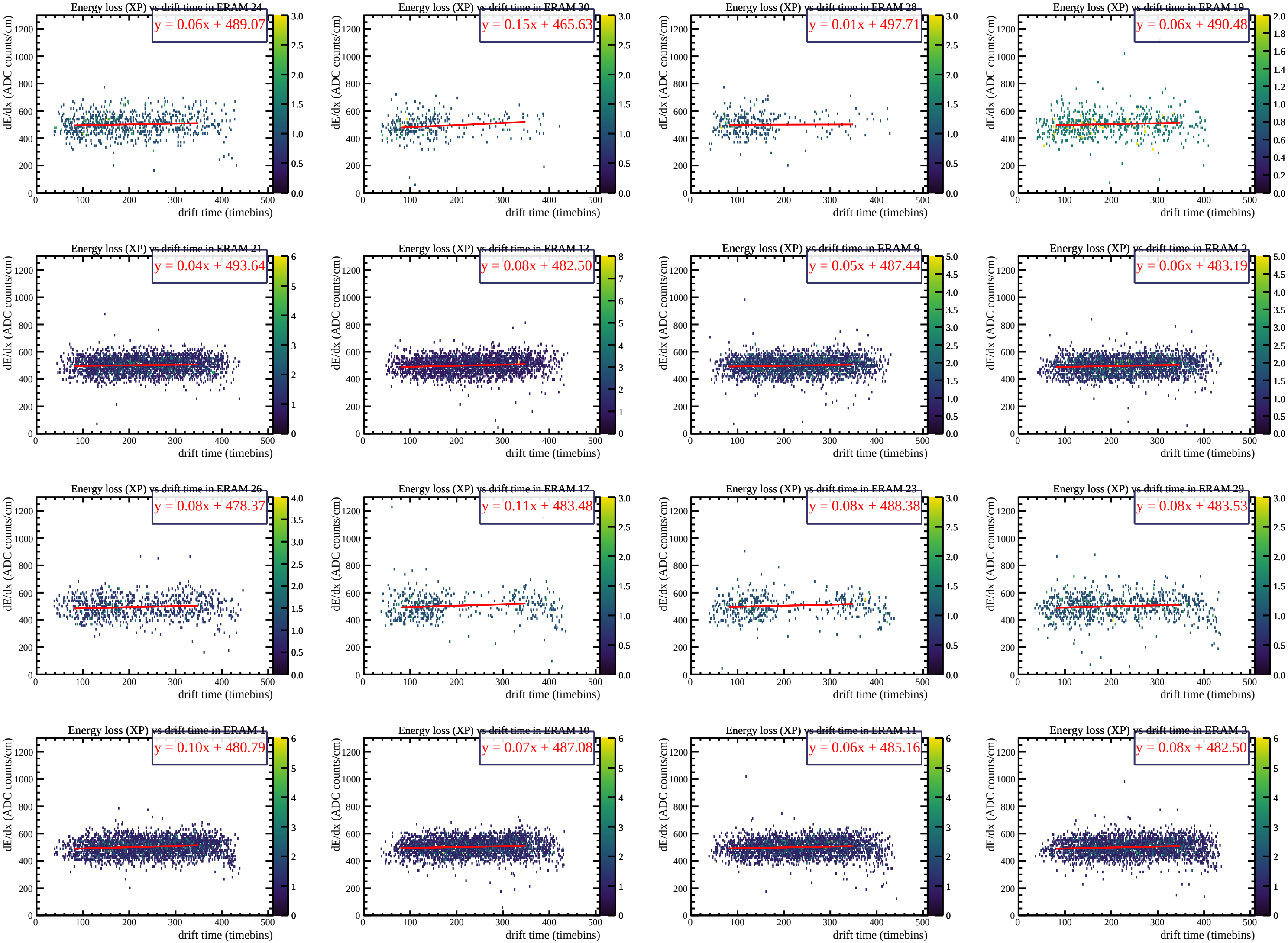


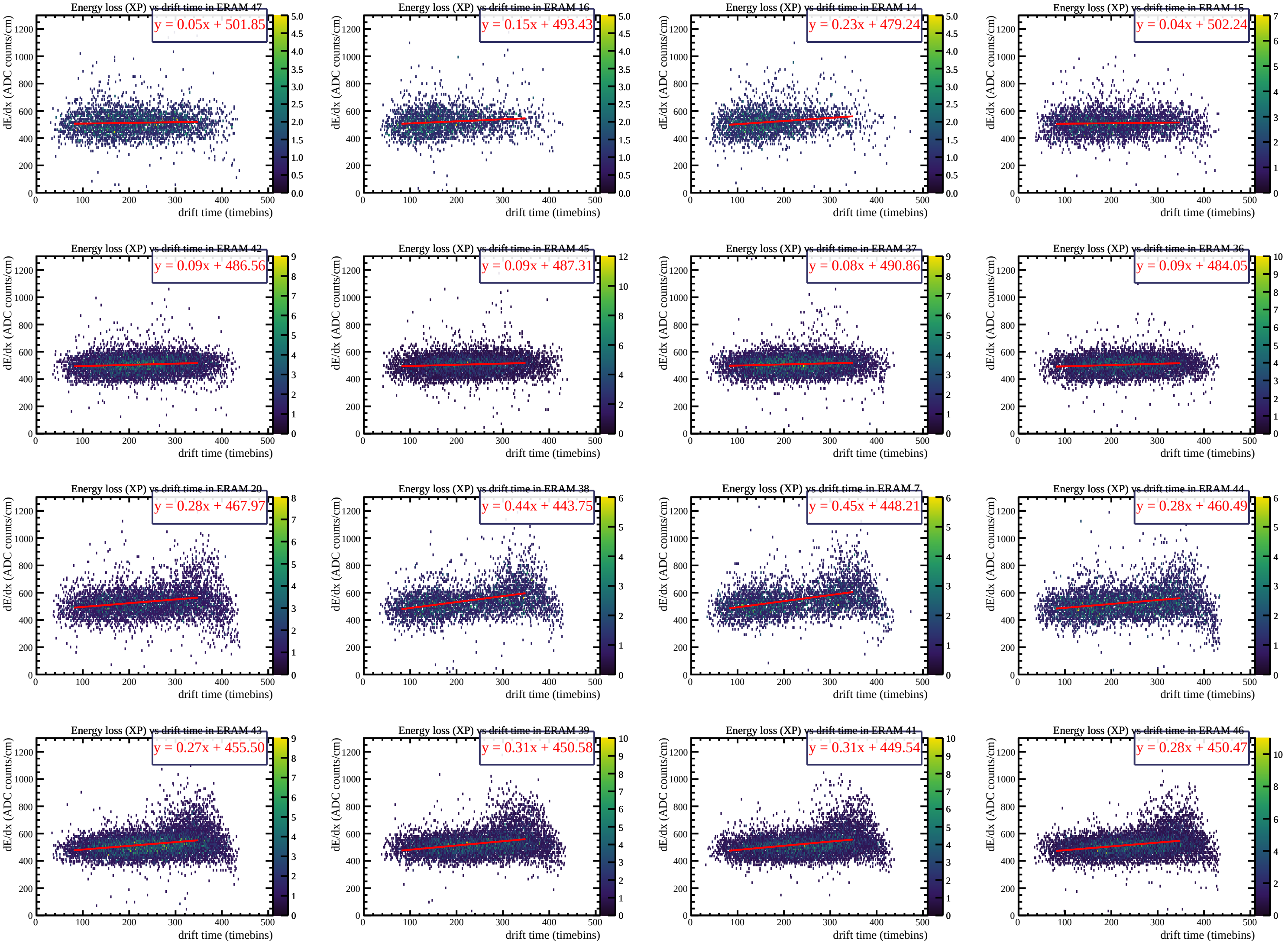
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

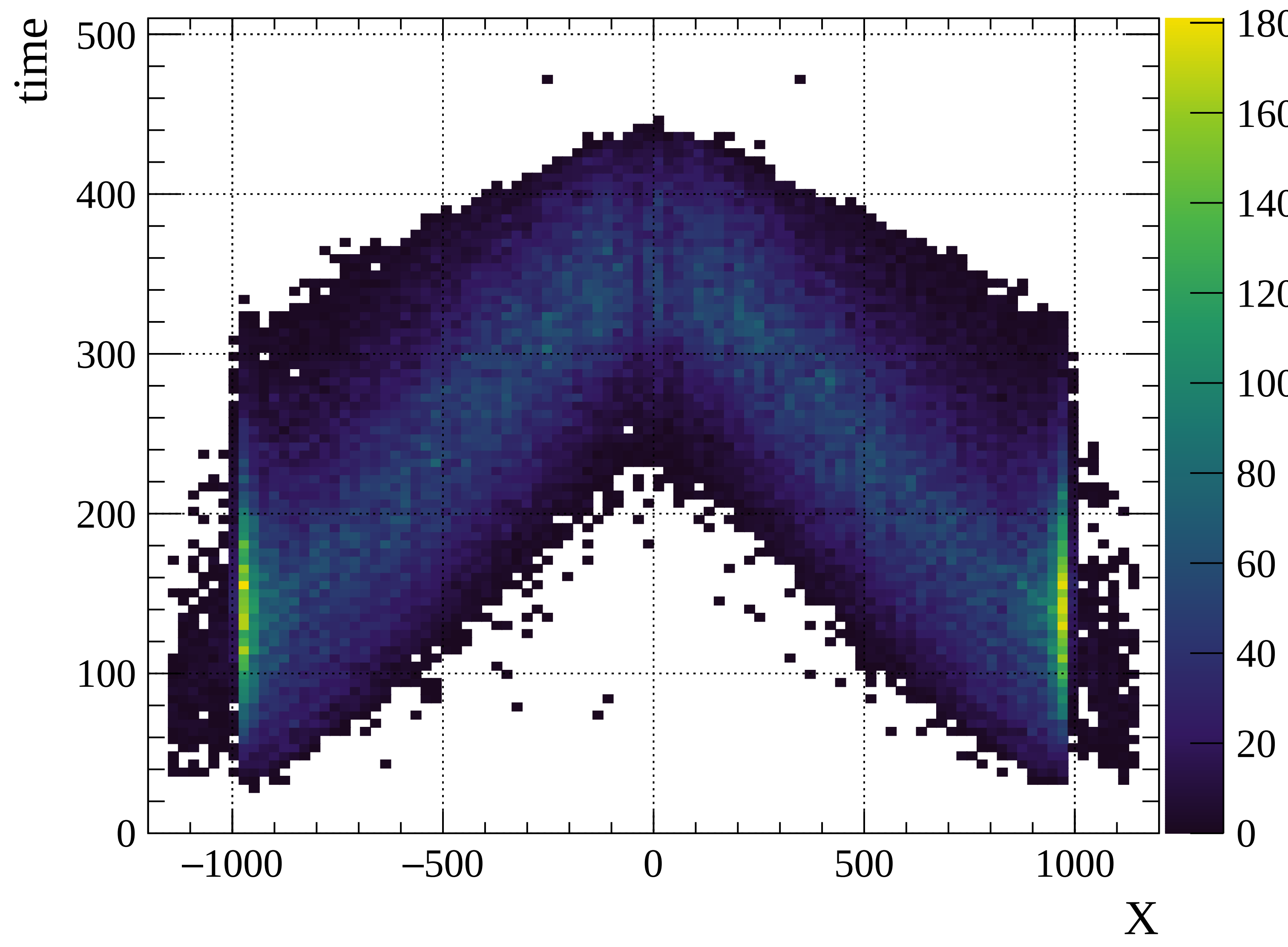

y = 0.08x + 483.26
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16

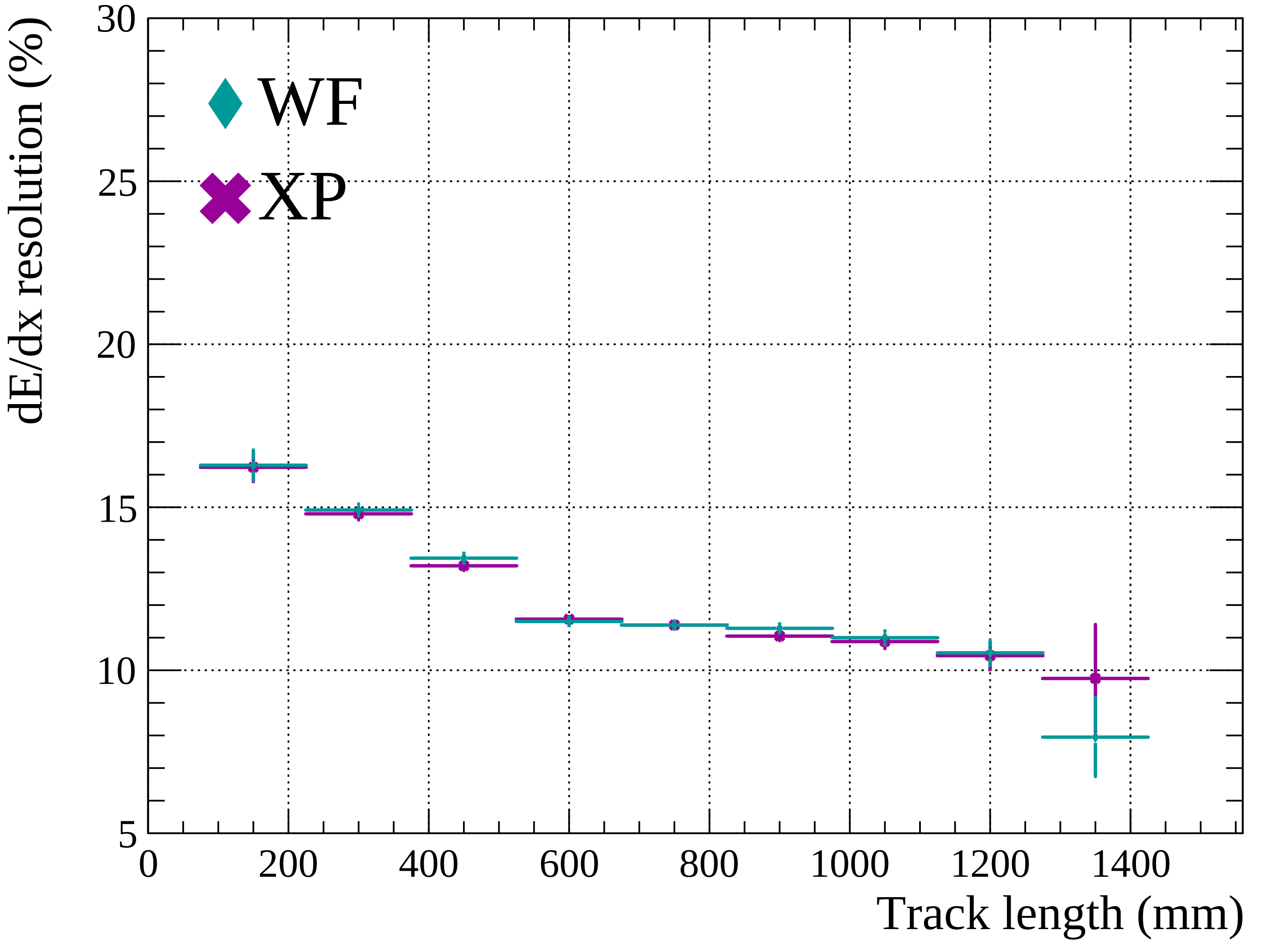


y = 0.10x + 482.86
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16

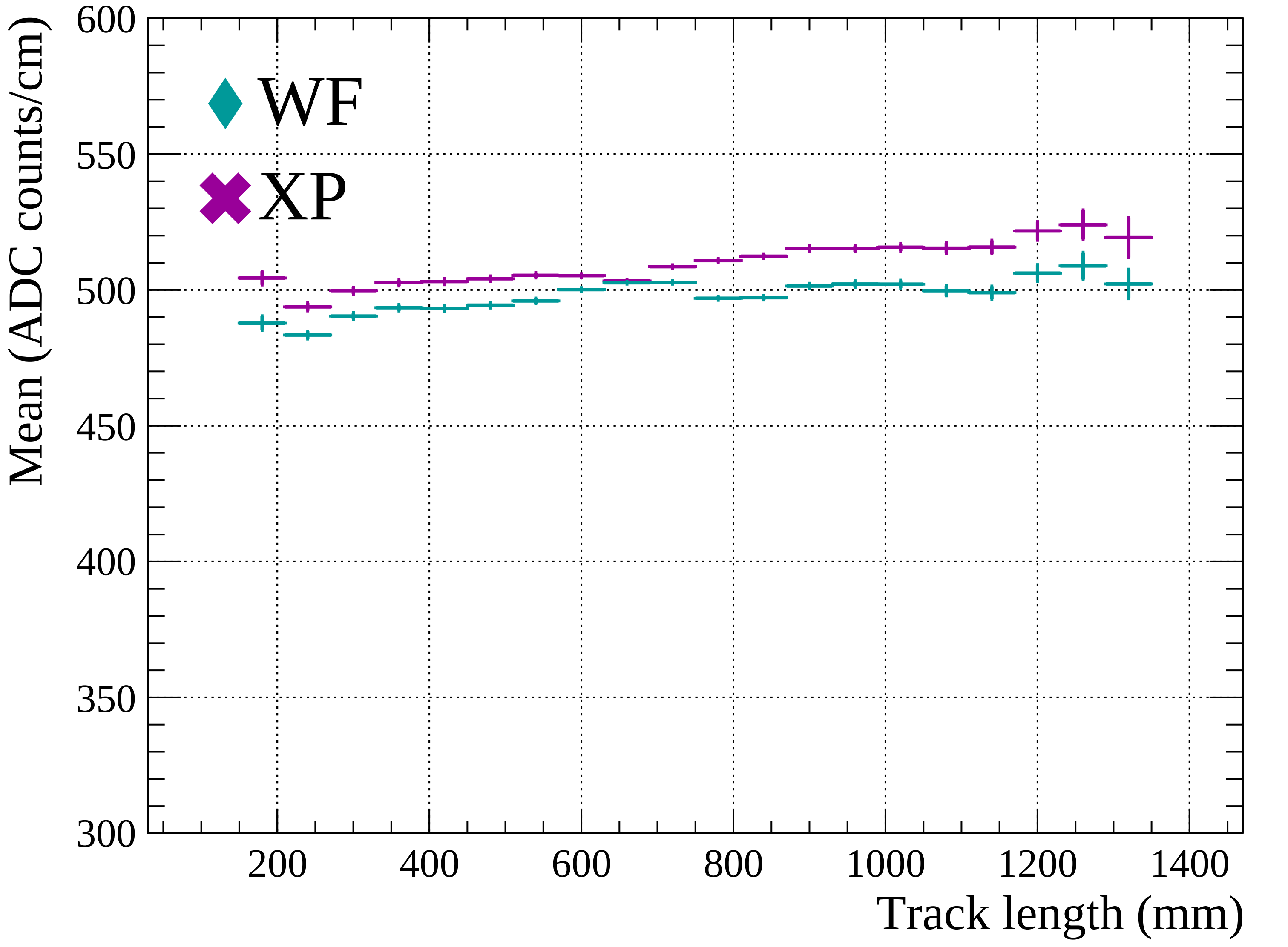

Energy loss (XP) vs drift time in ERAM 24
y = 0.06x + 489.07
Energy loss (XP) vs drift time in ERAM 30
y = 0.15x + 465.63
Energy loss (XP) vs drift time in ERAM 28
y = 0.01x + 497.71
Energy loss (XP) vs drift time in ERAM 19
y = 0.06x + 490.48
Energy loss (XP) vs drift time in ERAM 21
y = 0.04x + 493.64
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 482.50
Energy loss (XP) vs drift time in ERAM 9
y = 0.05x + 487.44
Energy loss (XP) vs drift time in ERAM 2
y = 0.06x + 483.19
Energy loss (XP) vs drift time in ERAM 26
y = 0.08x + 478.37
Energy loss (XP) vs drift time in ERAM 17
y = 0.11x + 483.48
Energy loss (XP) vs drift time in ERAM 23
y = 0.08x + 488.38
Energy loss (XP) vs drift time in ERAM 29
y = 0.08x + 483.53
Energy loss (XP) vs drift time in ERAM 1
y = 0.10x + 480.79
Energy loss (XP) vs drift time in ERAM 10
y = 0.07x + 487.08
Energy loss (XP) vs drift time in ERAM 11
y = 0.06x + 485.16
Energy loss (XP) vs drift time in ERAM 3
y = 0.08x + 482.50
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (XP) vs drift time in ERAM 47
y = 0.05x + 501.85
Energy loss (XP) vs drift time in ERAM 16
y = 0.15x + 493.43
Energy loss (XP) vs drift time in ERAM 14
y = 0.23x + 479.24
Energy loss (XP) vs drift time in ERAM 15
y = 0.04x + 502.24
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 486.56
Energy loss (XP) vs drift time in ERAM 45
y = 0.09x + 487.31
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 490.86
Energy loss (XP) vs drift time in ERAM 36
y = 0.09x + 484.05
Energy loss (XP) vs drift time in ERAM 20
y = 0.28x + 467.97
Energy loss (XP) vs drift time in ERAM 38
y = 0.44x + 443.75
Energy loss (XP) vs drift time in ERAM 7
y = 0.45x + 448.21
Energy loss (XP) vs drift time in ERAM 44
y = 0.28x + 460.49
Energy loss (XP) vs drift time in ERAM 43
y = 0.27x + 455.50
Energy loss (XP) vs drift time in ERAM 39
y = 0.31x + 450.58
Energy loss (XP) vs drift time in ERAM 41
y = 0.31x + 449.54
Energy loss (XP) vs drift time in ERAM 46
y = 0.28x + 450.47
dE/dx (ADC counts/cm)
drift time (timebins)

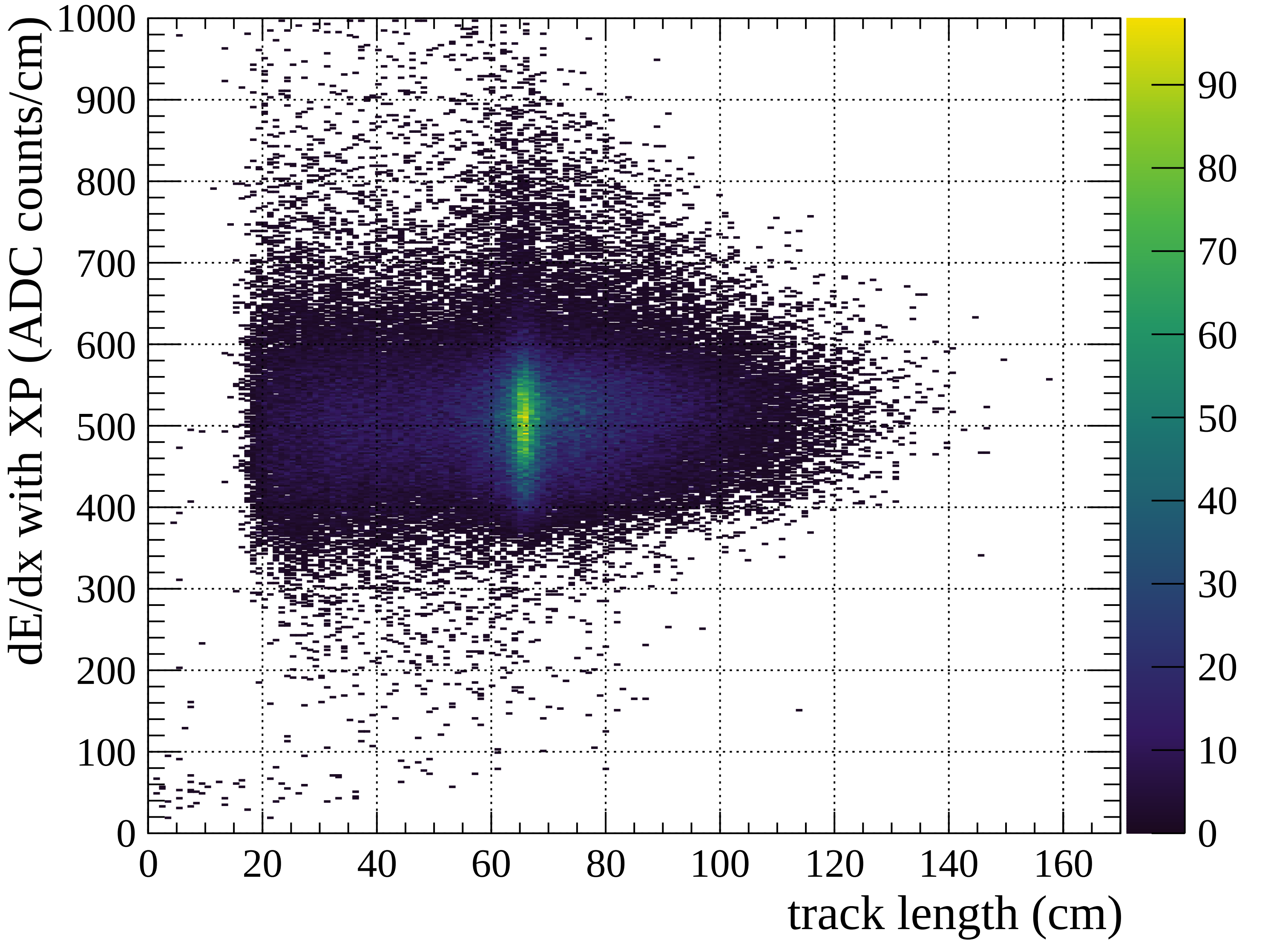

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
0
20
40
60
80
100
120
140
160
180

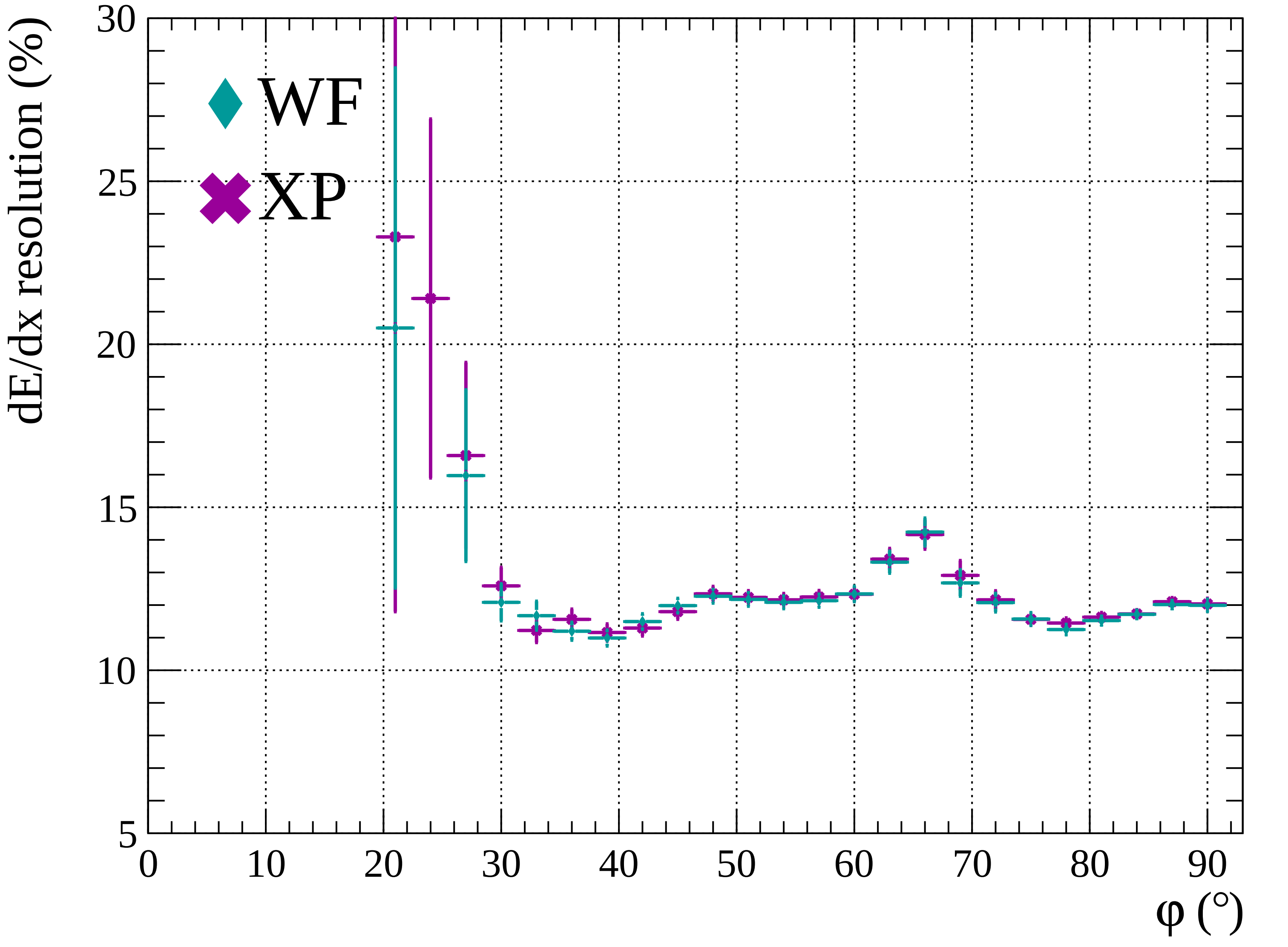


dE/dx resolution (%)
Track length (mm)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400

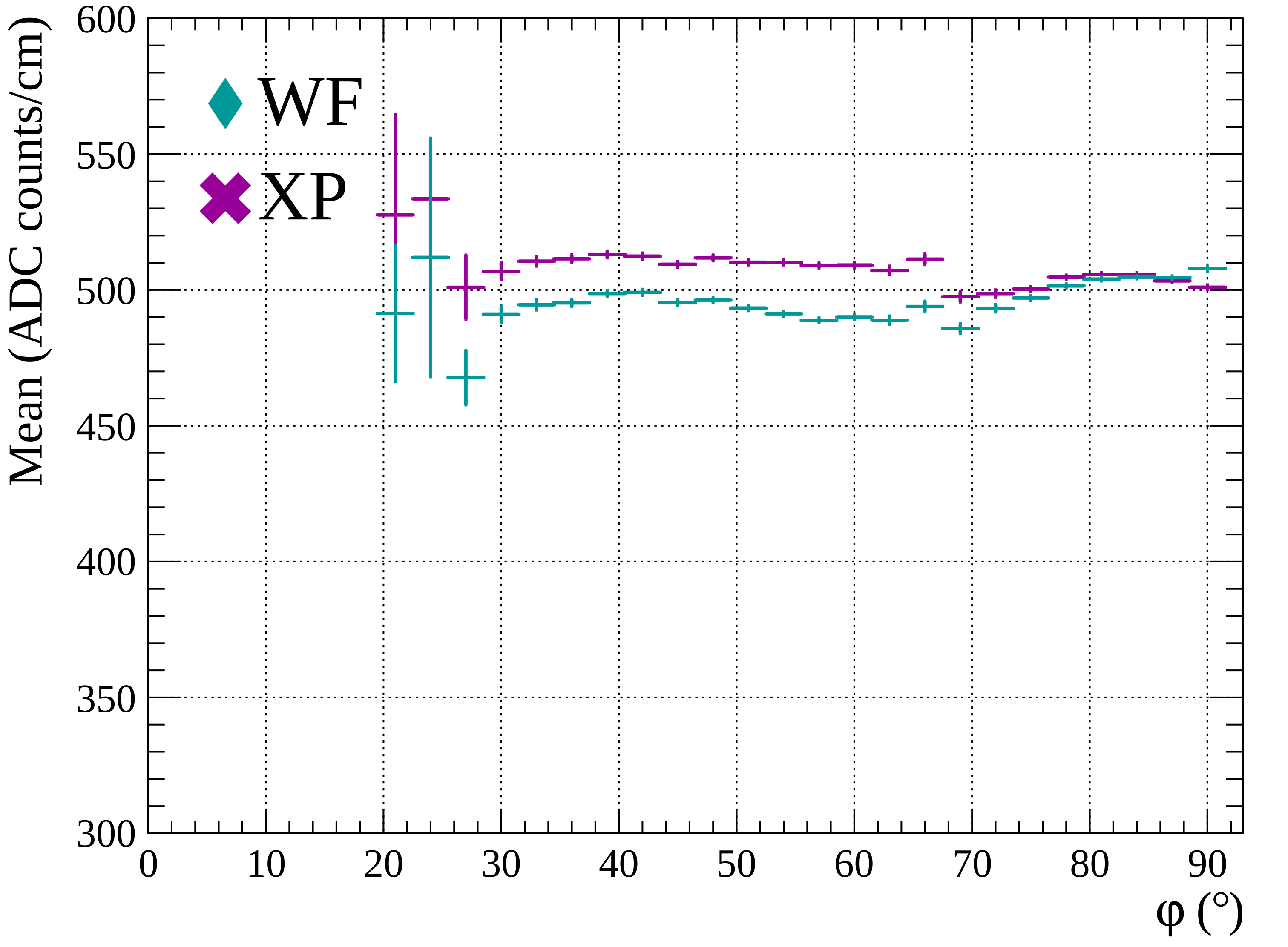


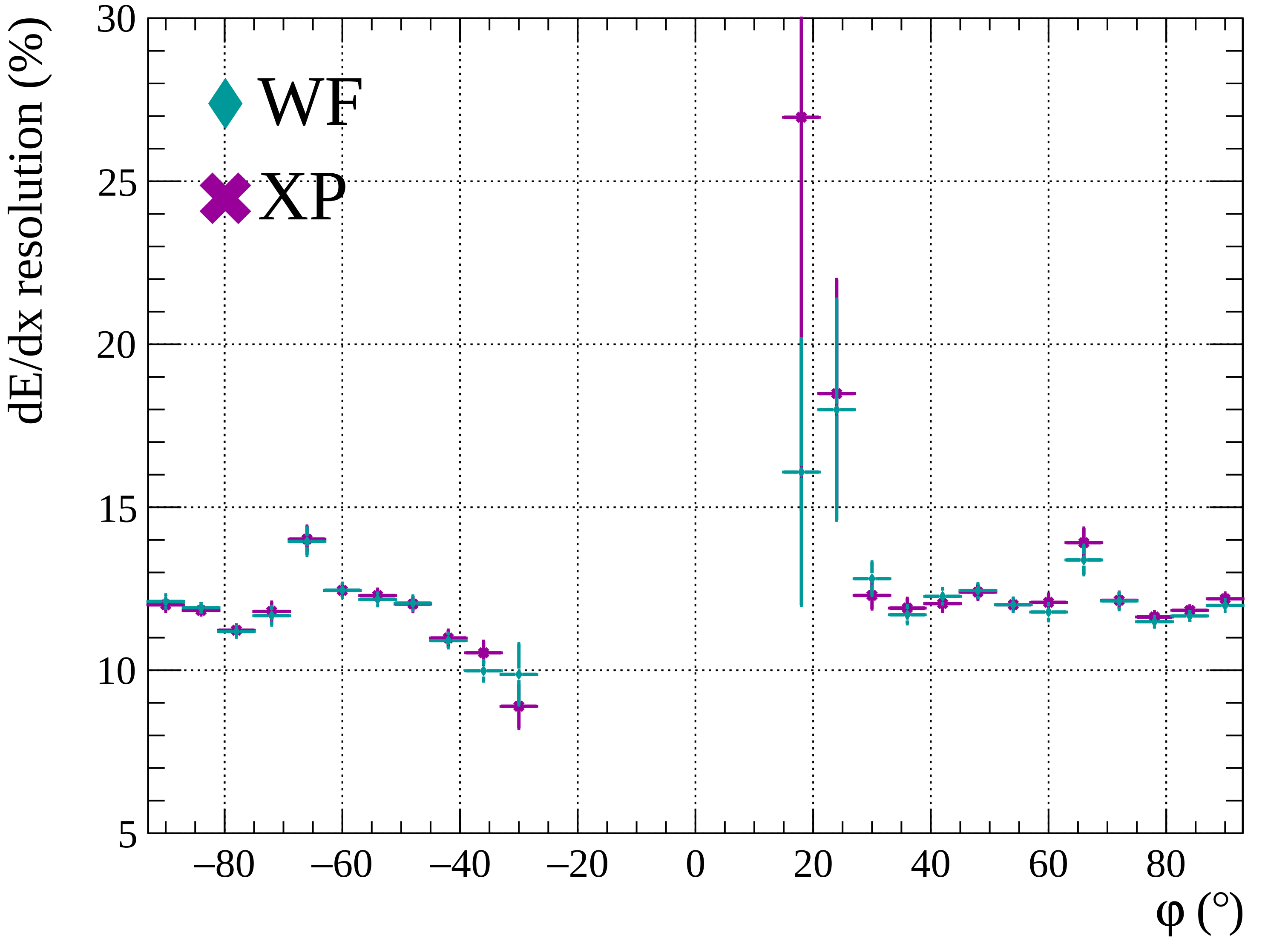
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
200
400
600
800
1000
1200
1400
Track length (mm)


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
90
80
70
60
50
40
30
20
10
0

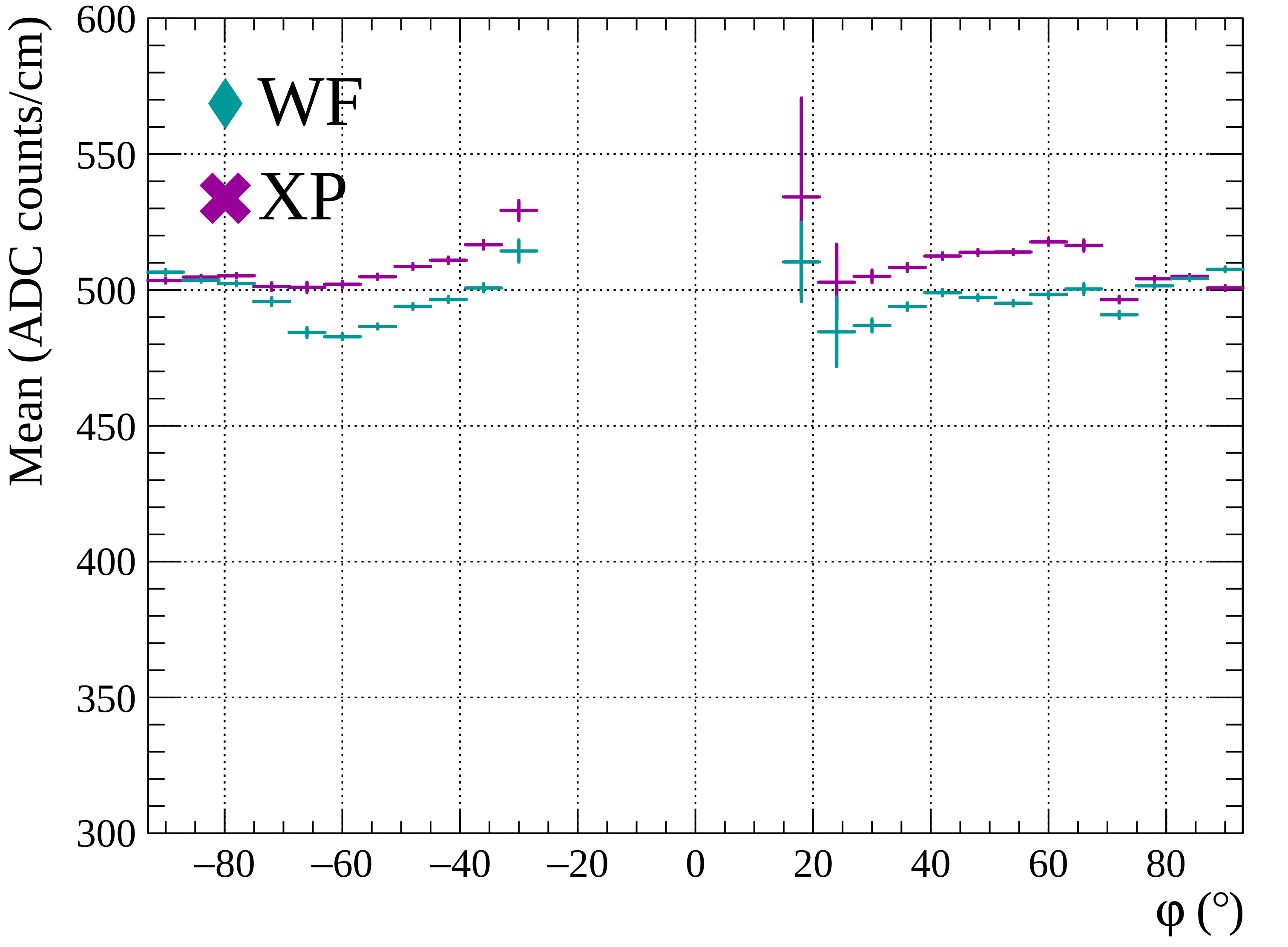


WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90

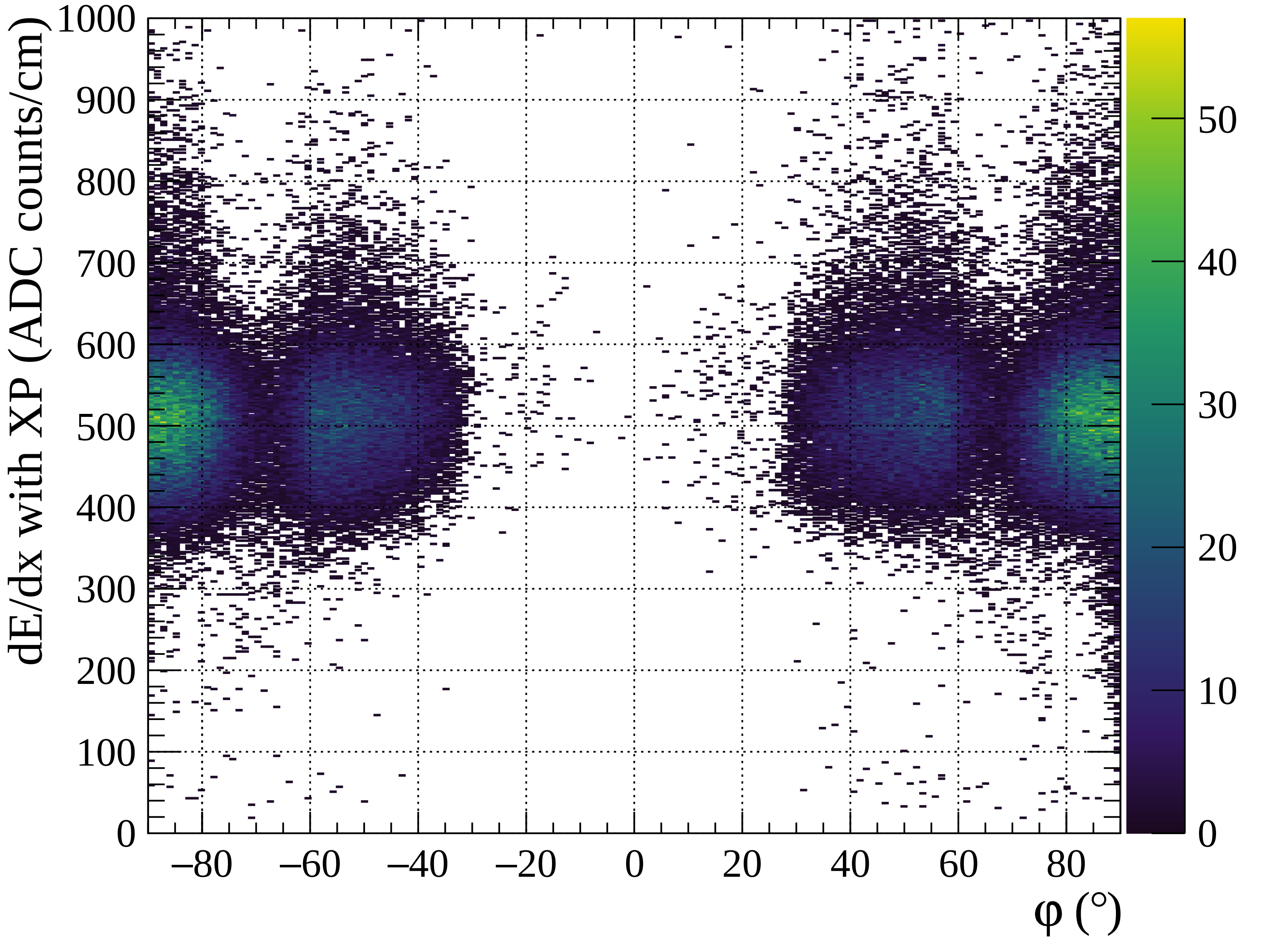

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
φ (°)



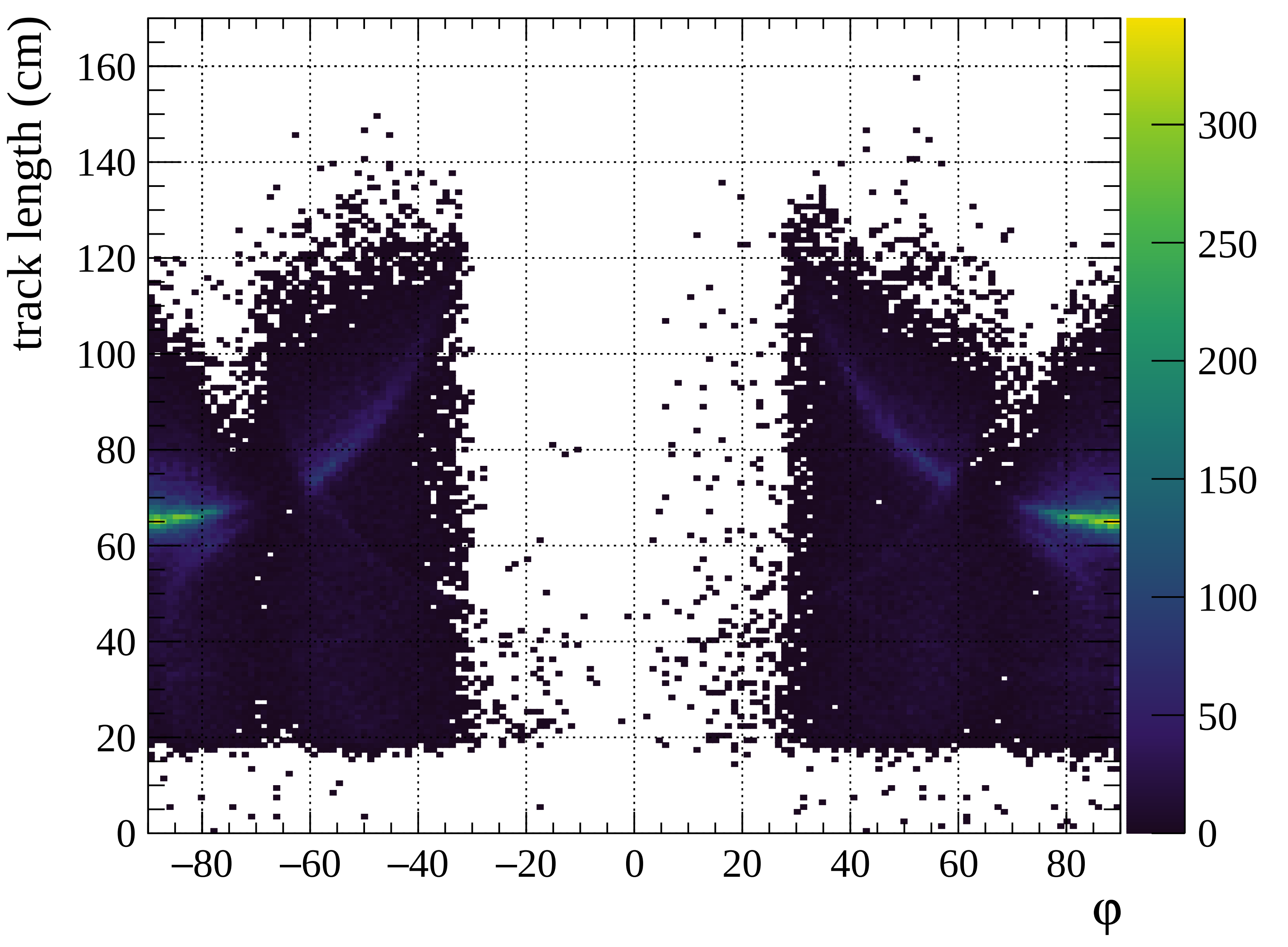
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

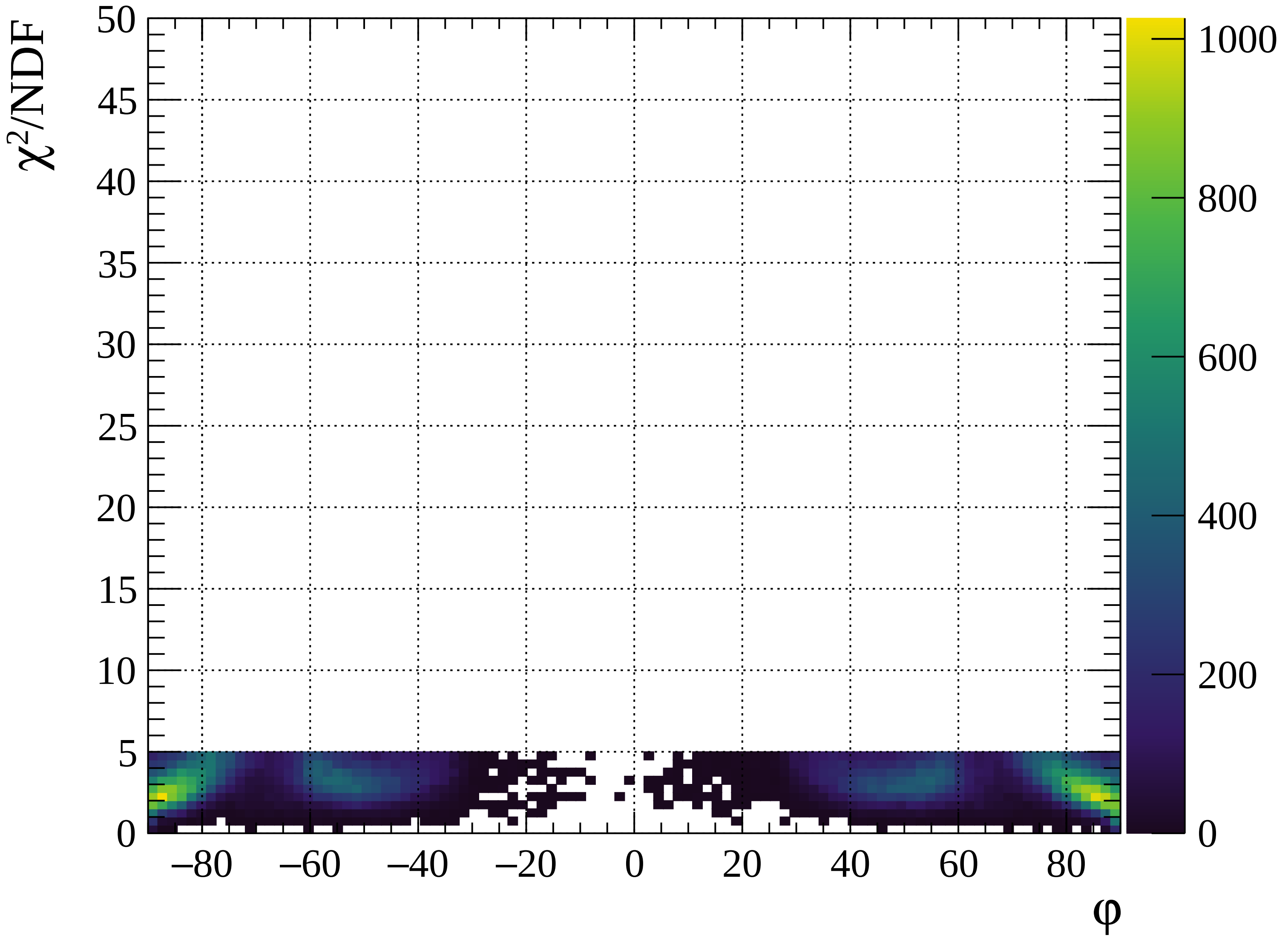


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

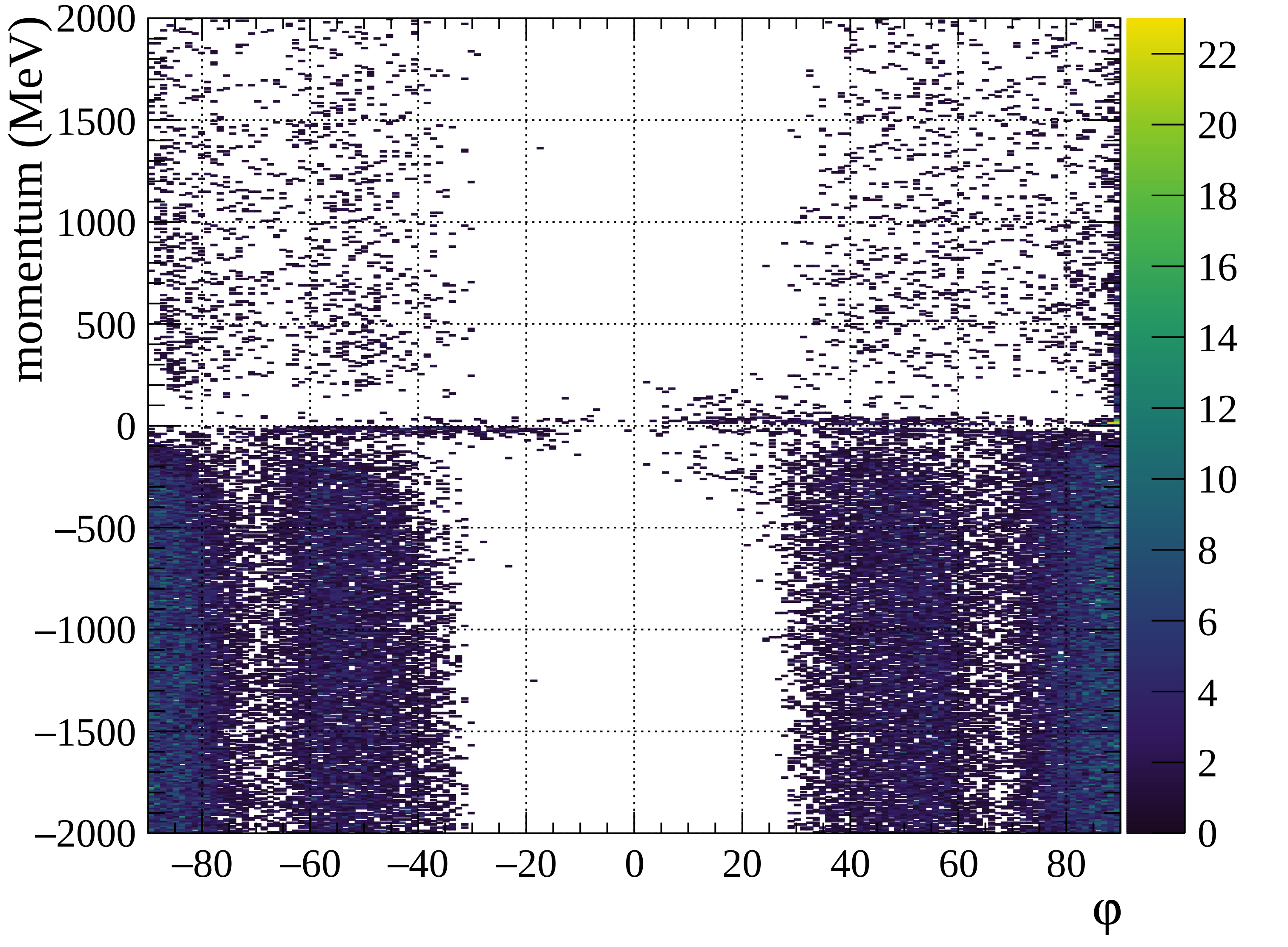


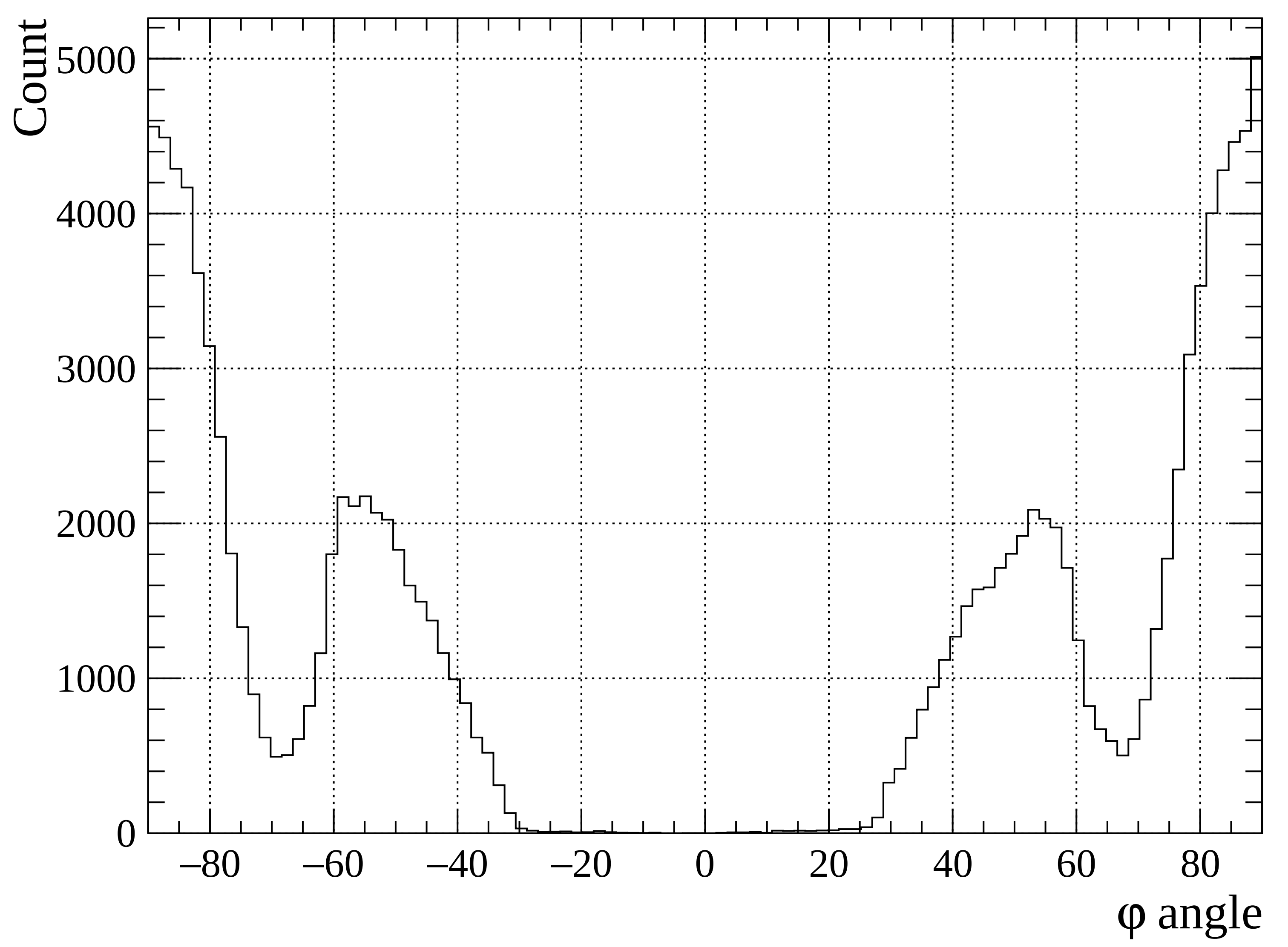
dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
10
20
30
40
50


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0

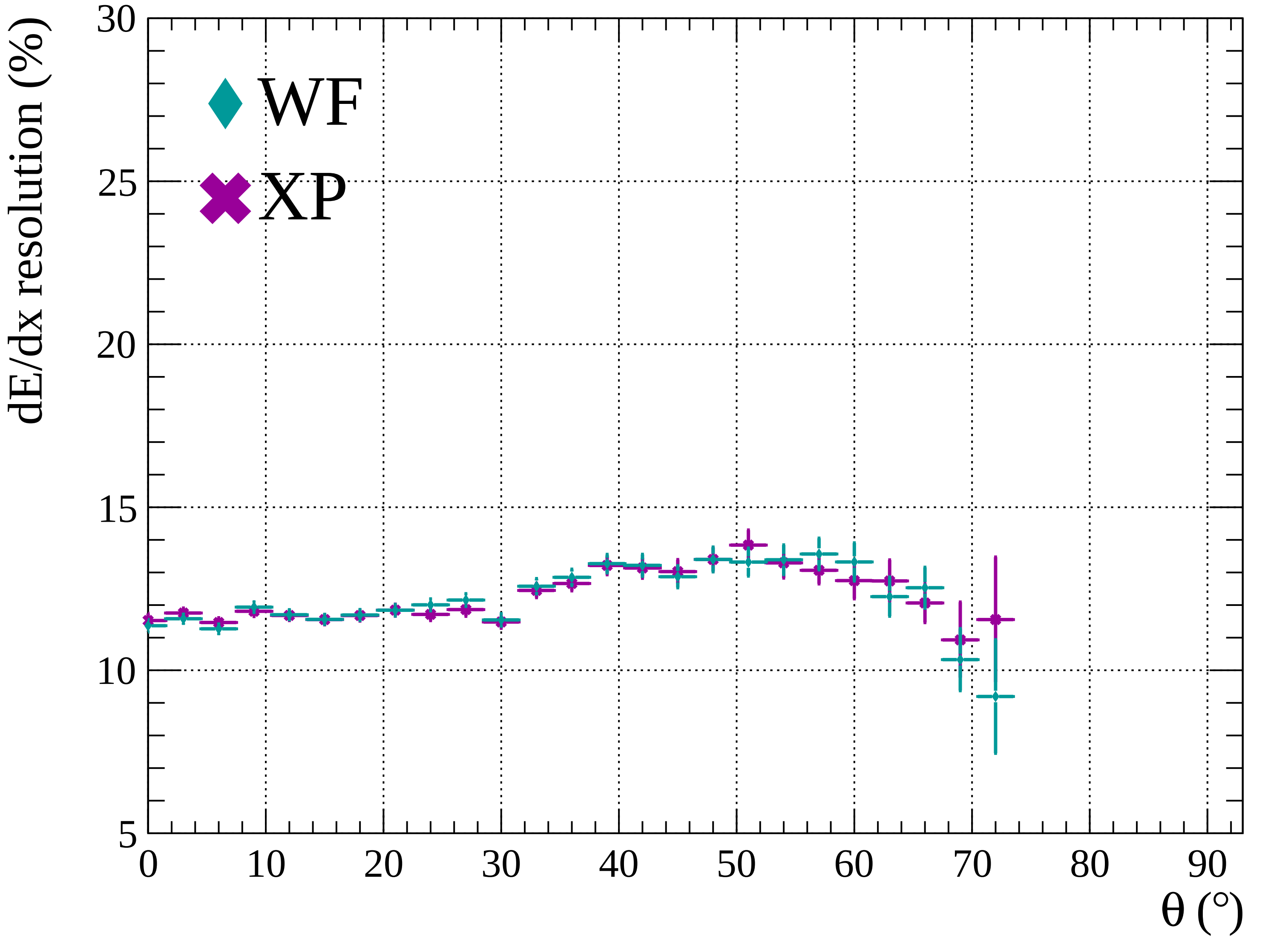


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
1000
800
600
400
200
0

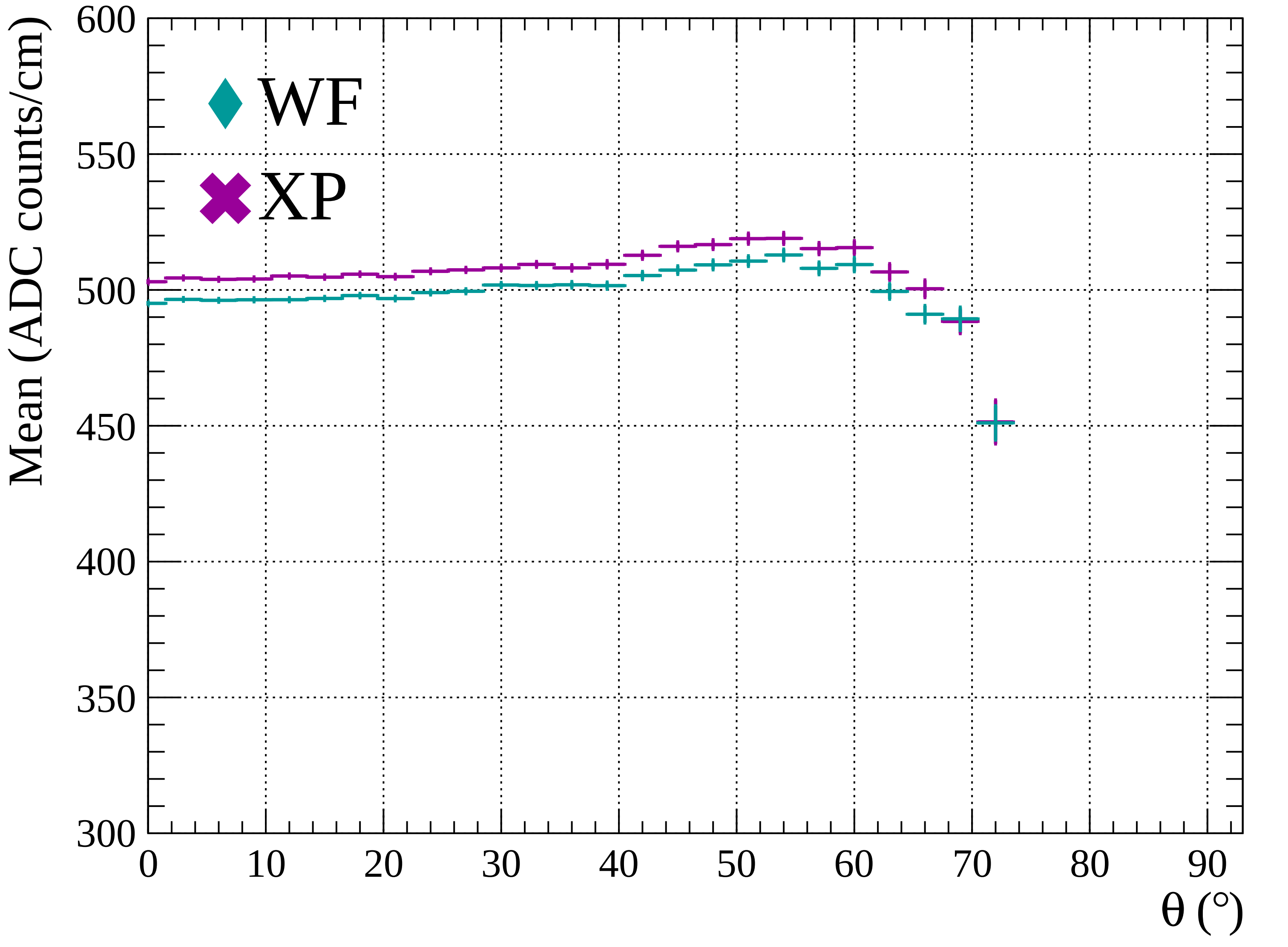

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
22
20
18
16
14
12
10
8
6
4
2
0



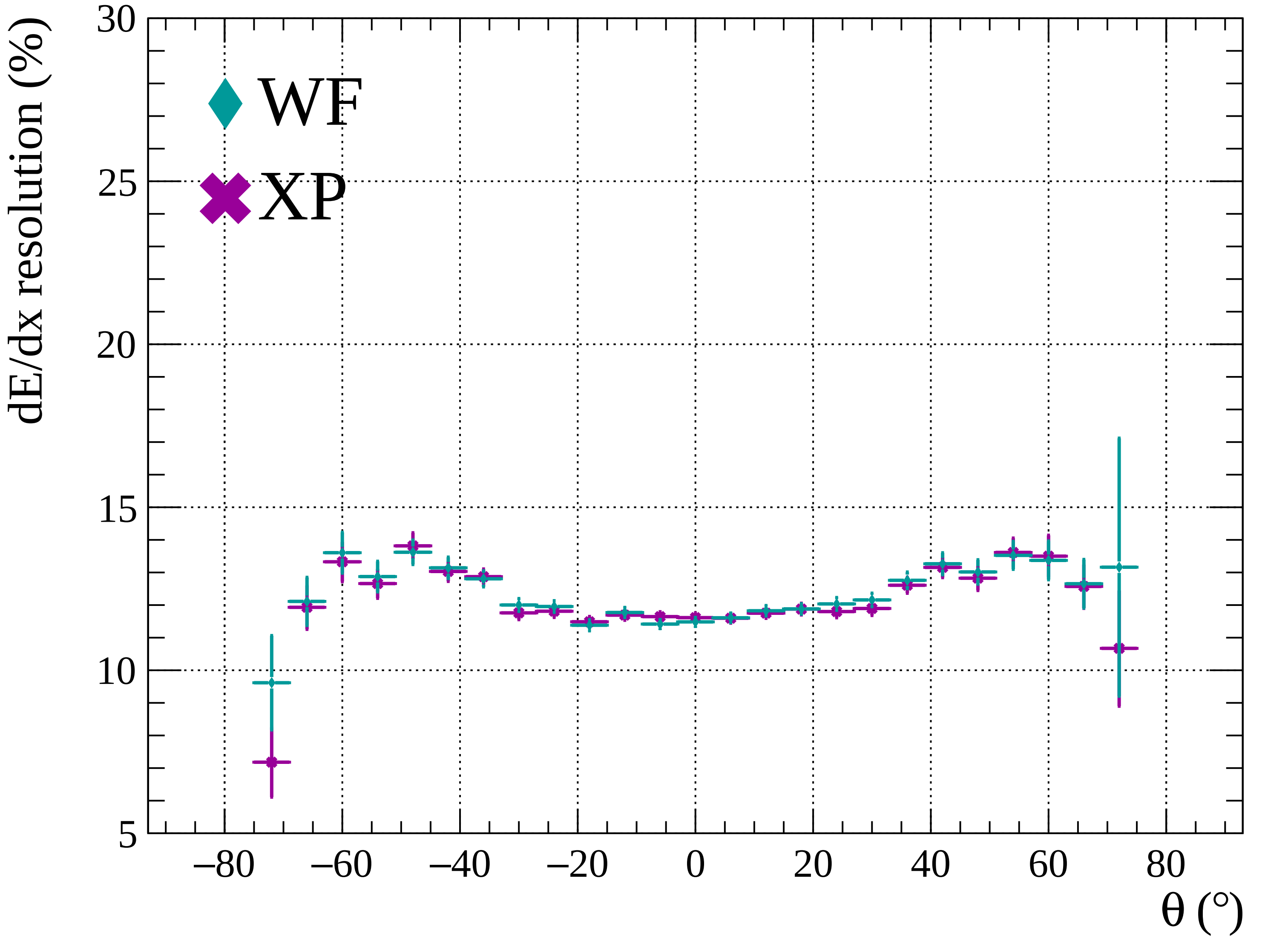
Count
5000
4000
3000
2000
1000
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



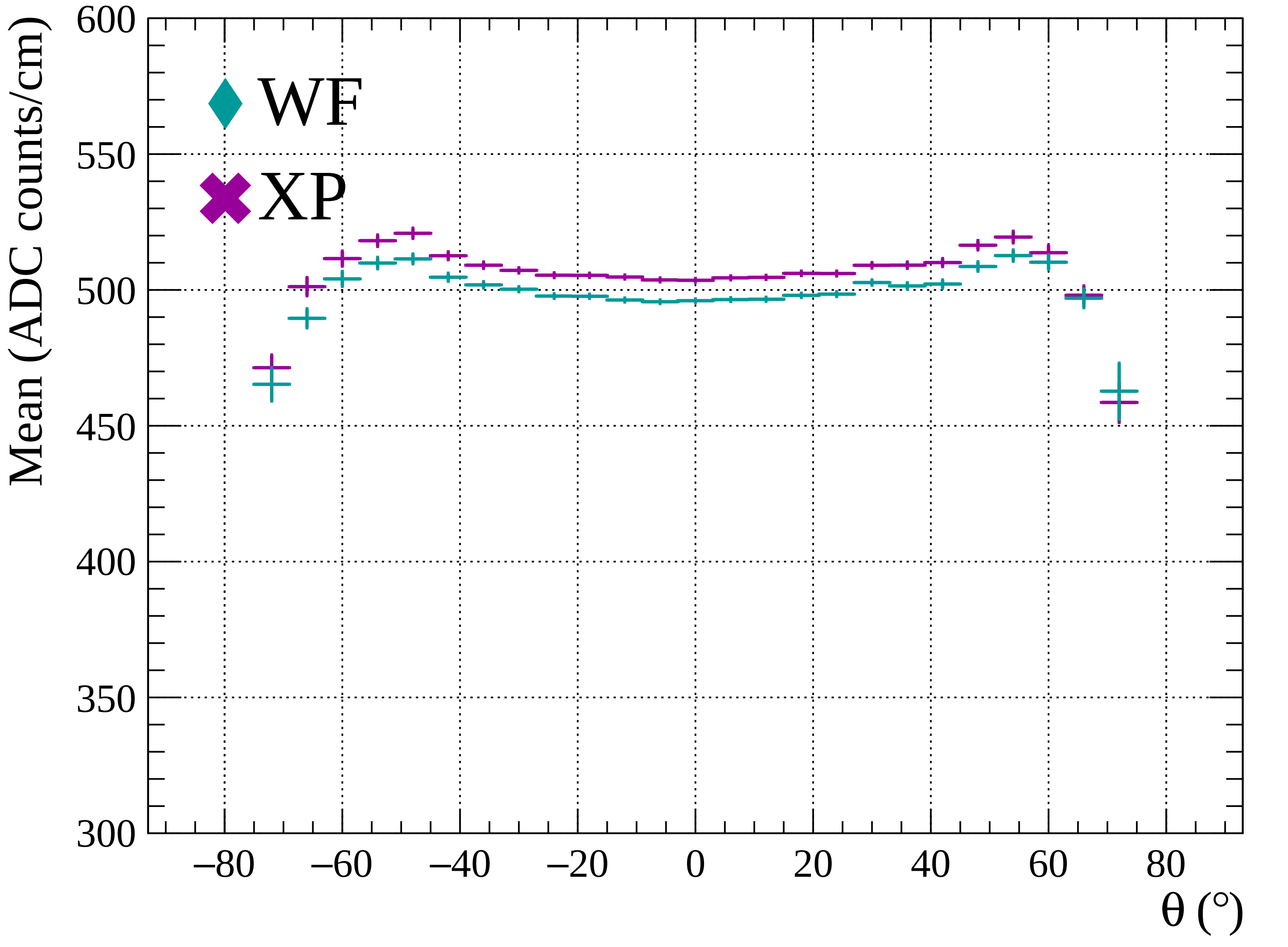
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



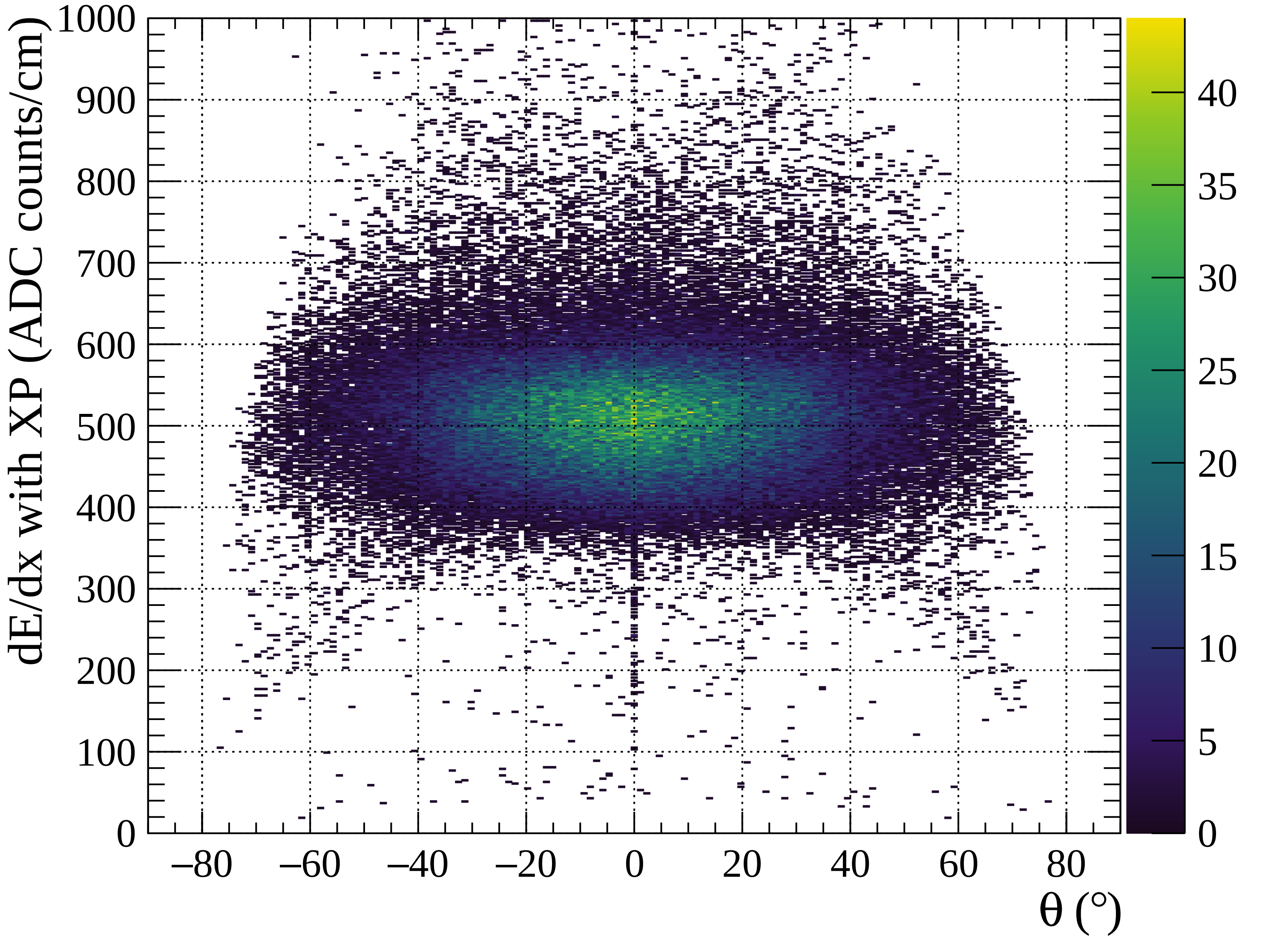
Mean (ADC counts/cm)
θ (°)
WF
XP
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



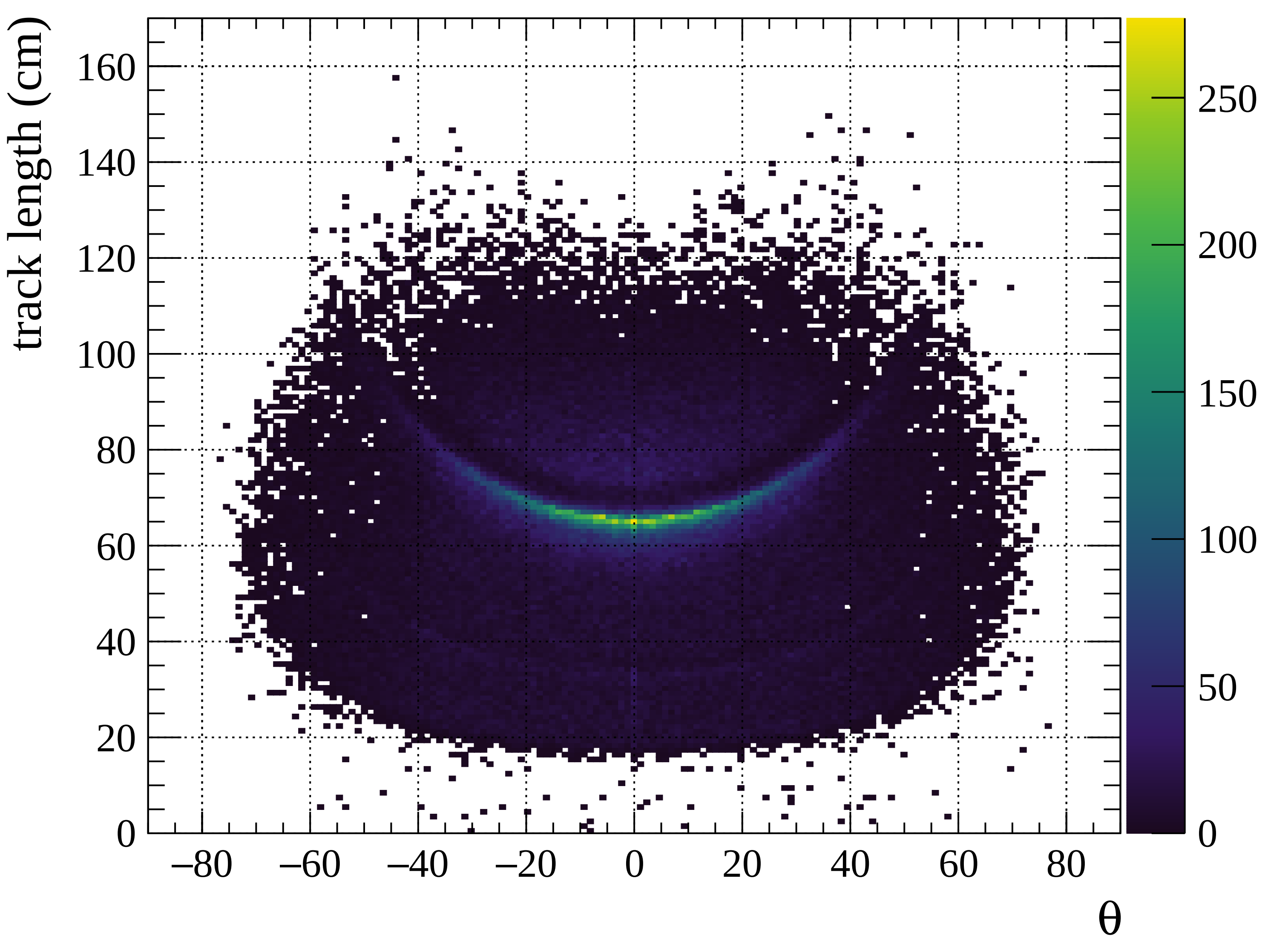
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



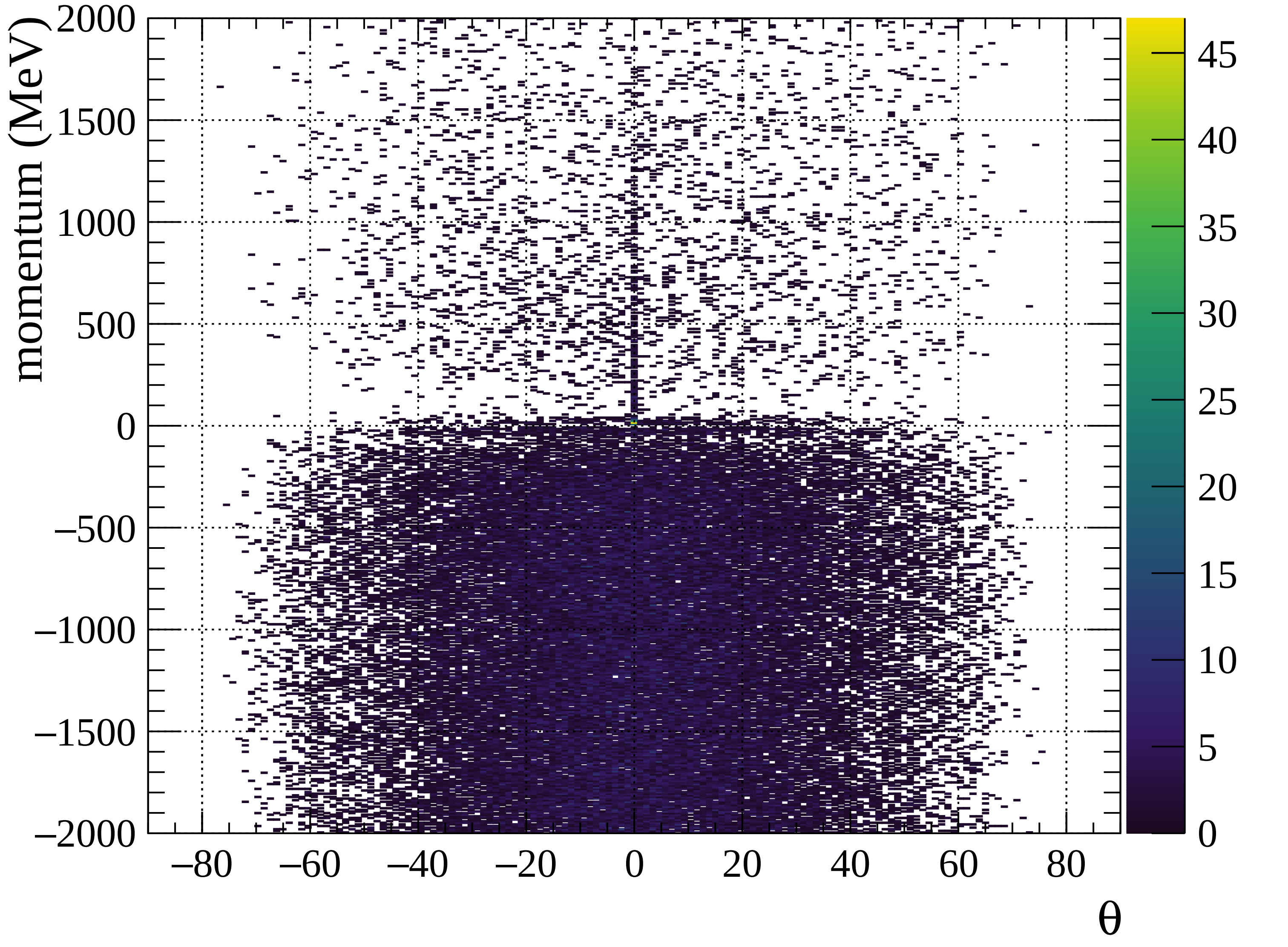
Mean (ADC counts/cm)
θ (°)
WF
XP
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

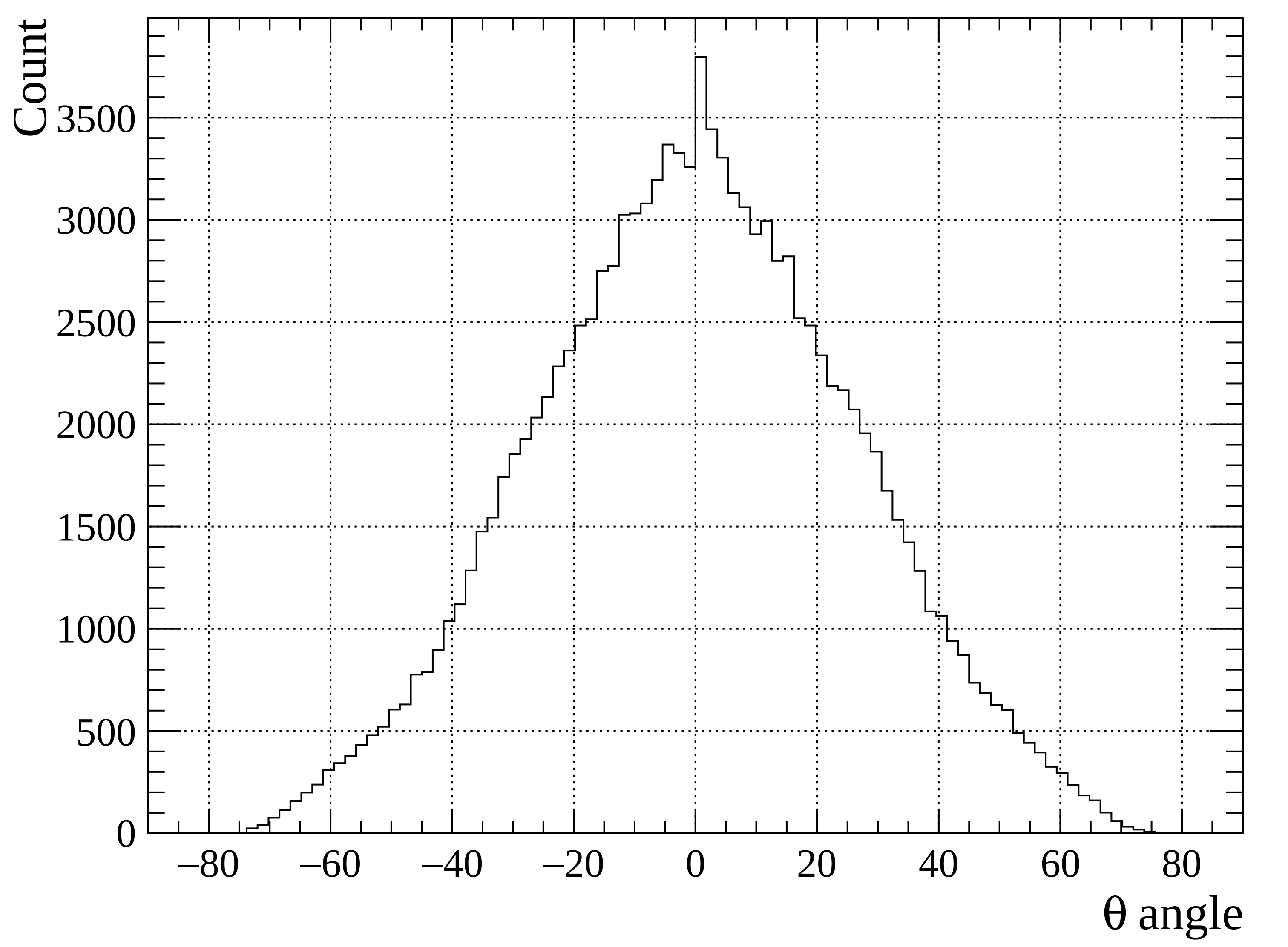


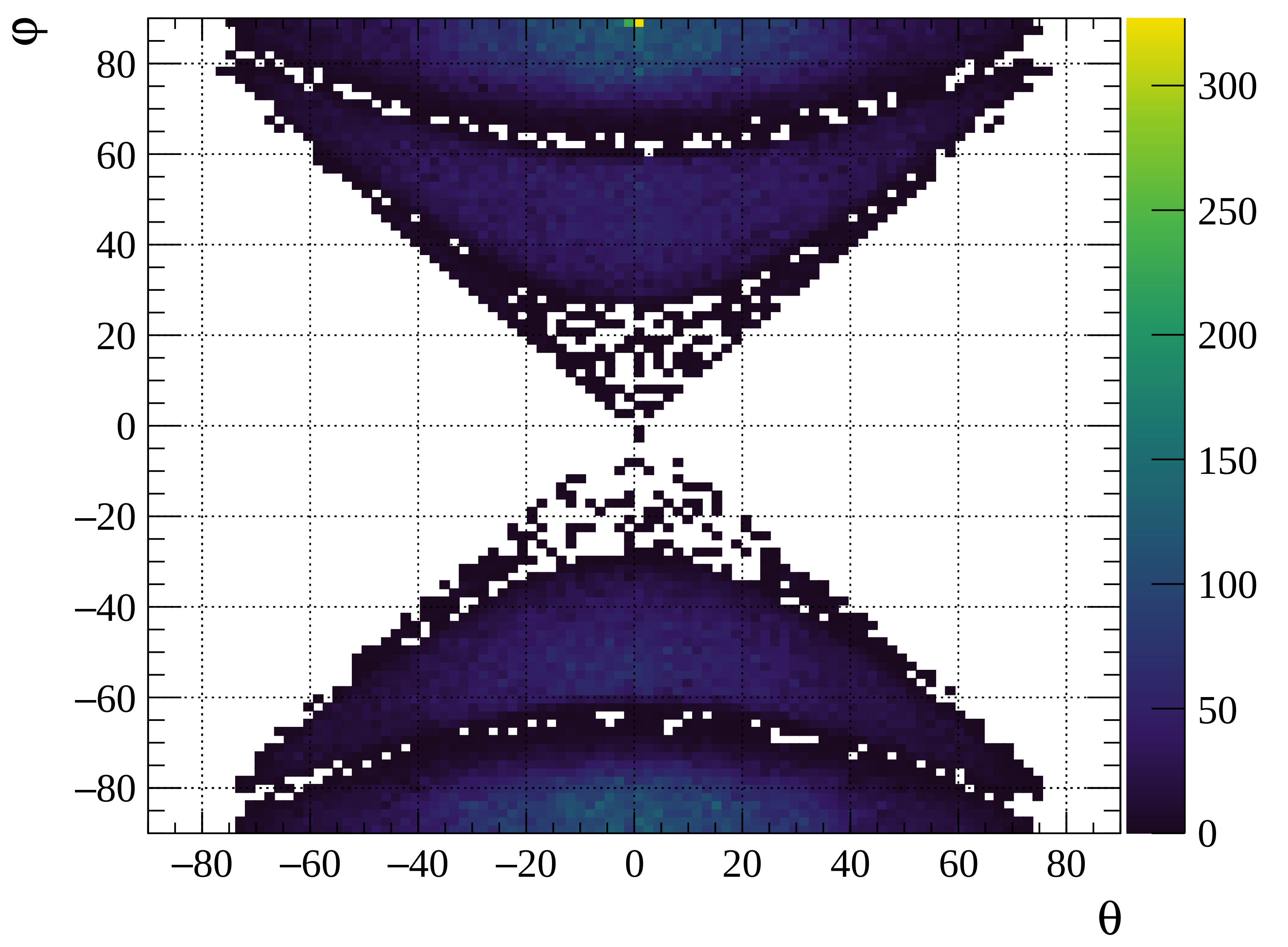
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
40
35
30
25
20
15
10
5
0

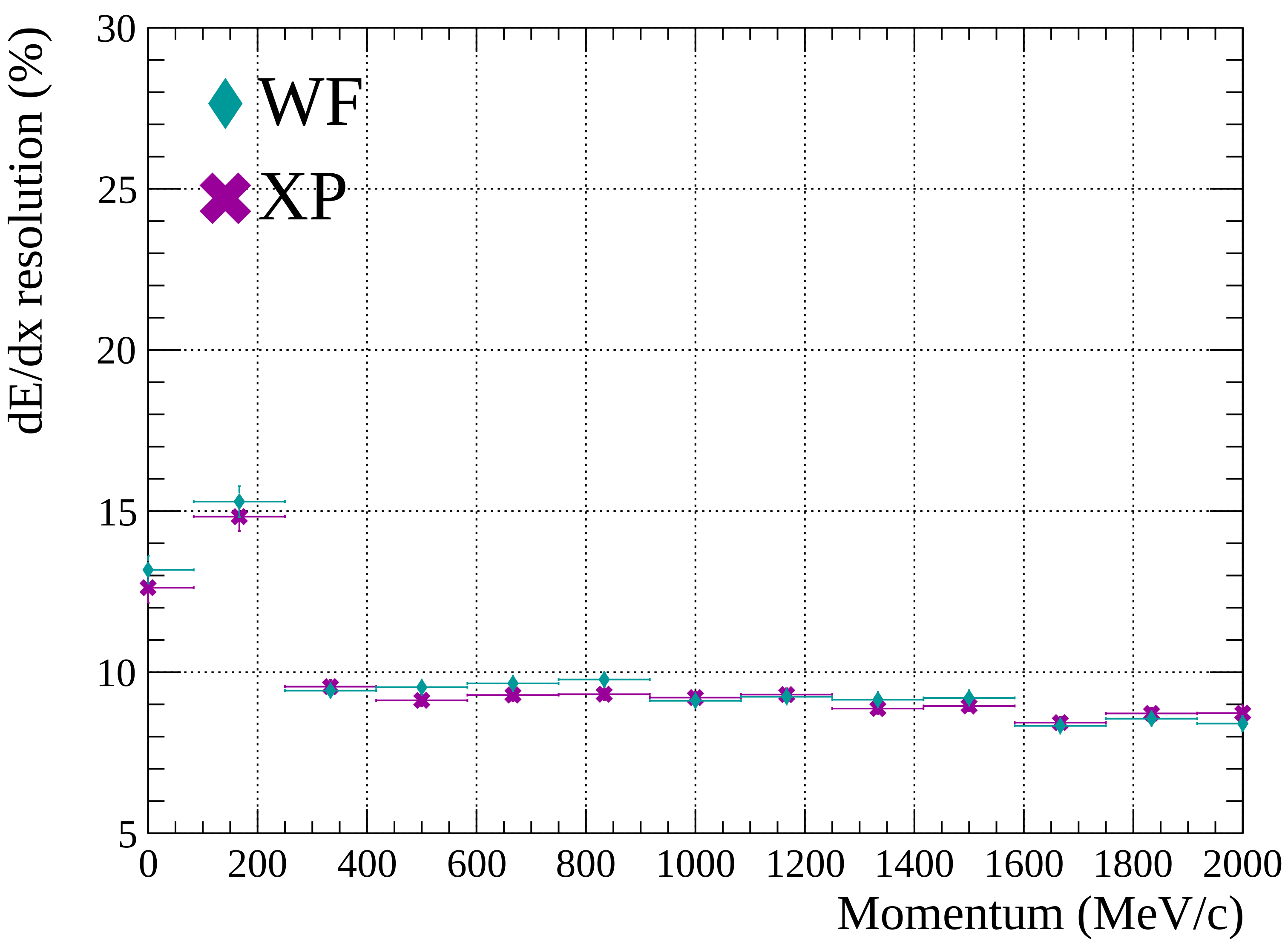


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
250
200
150
100
50
0

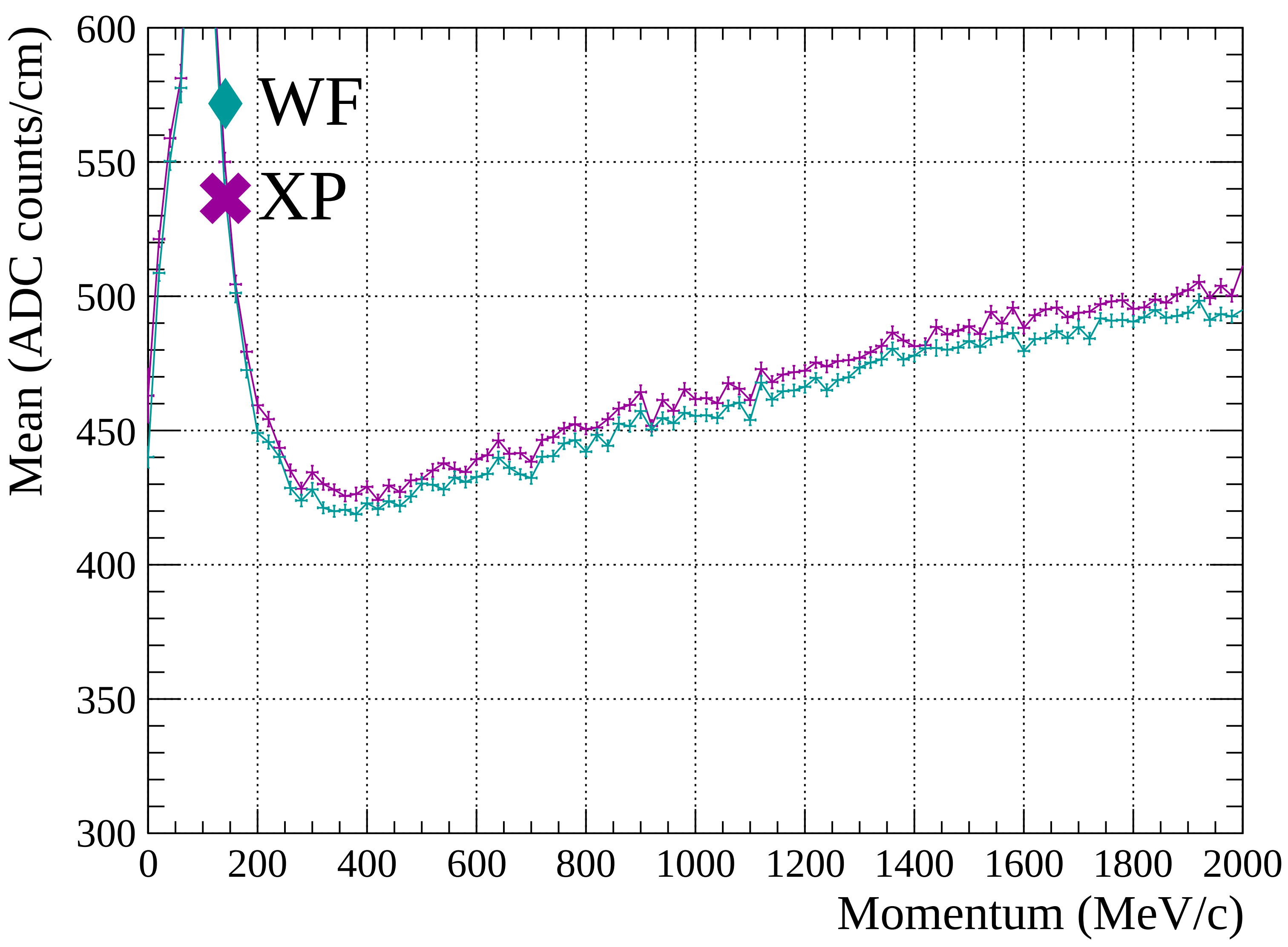

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
45
40
35
30
25
20
15
10
5
0

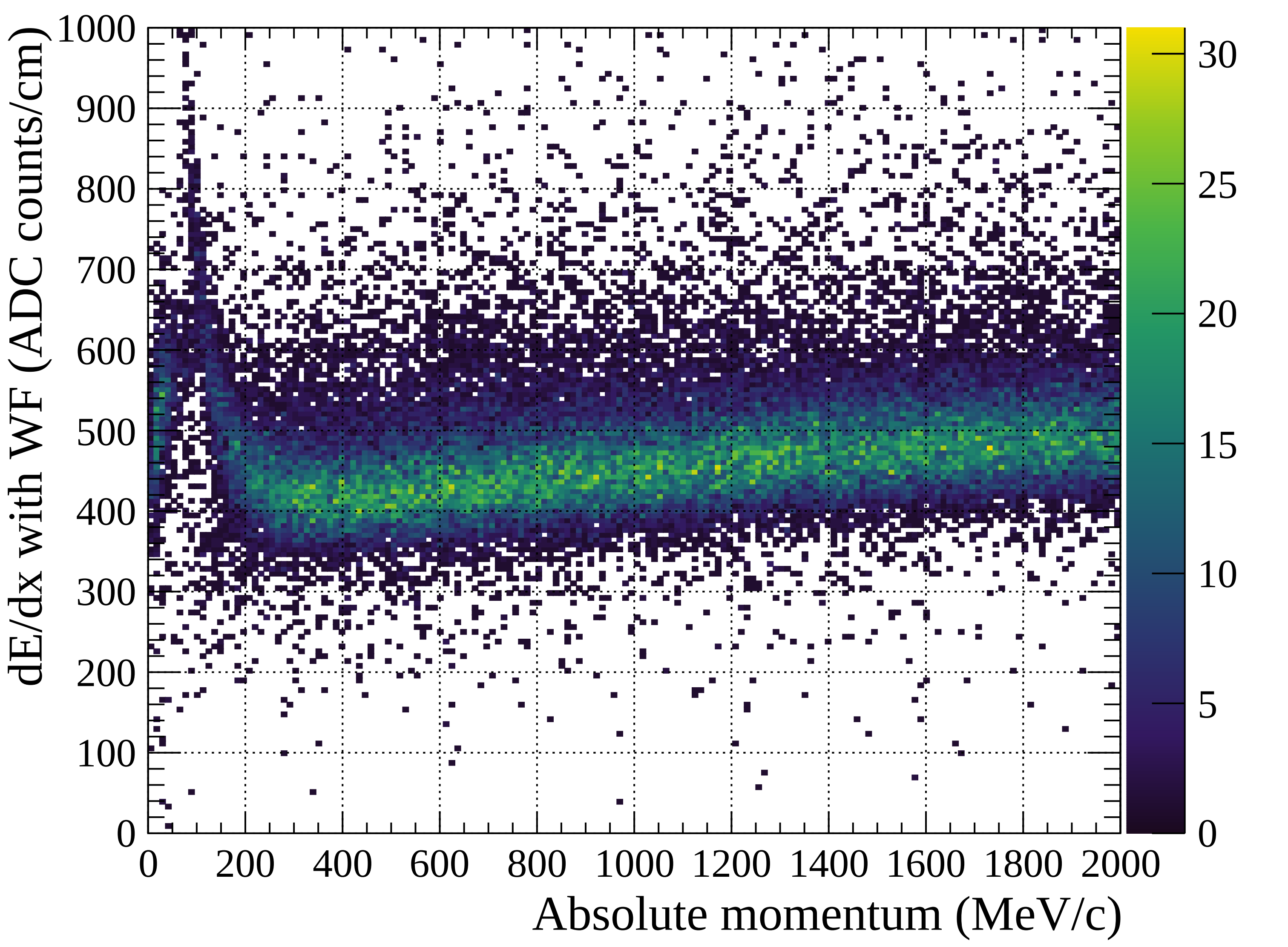

Count
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

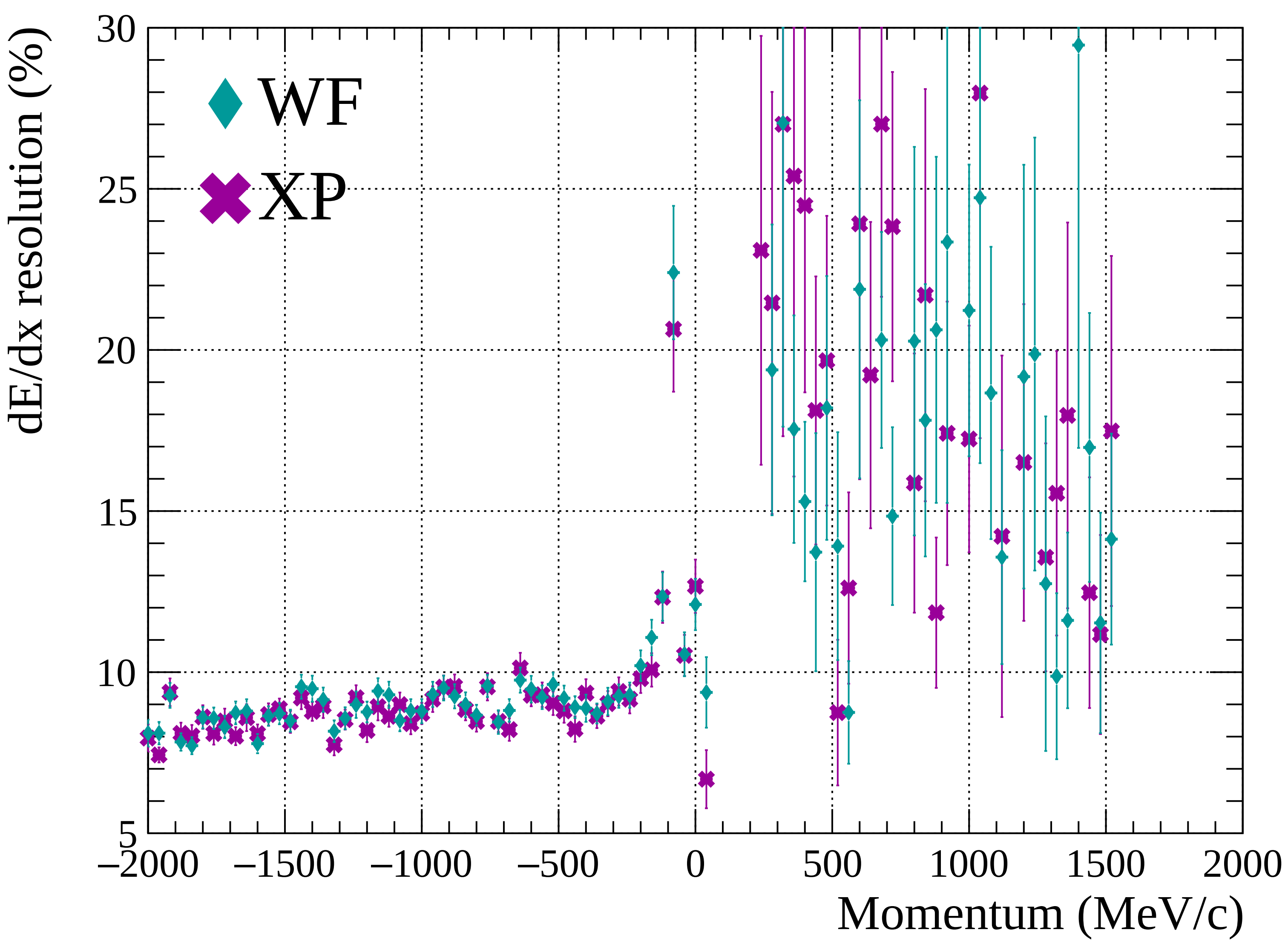

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
300
250
200
150
100
50
0

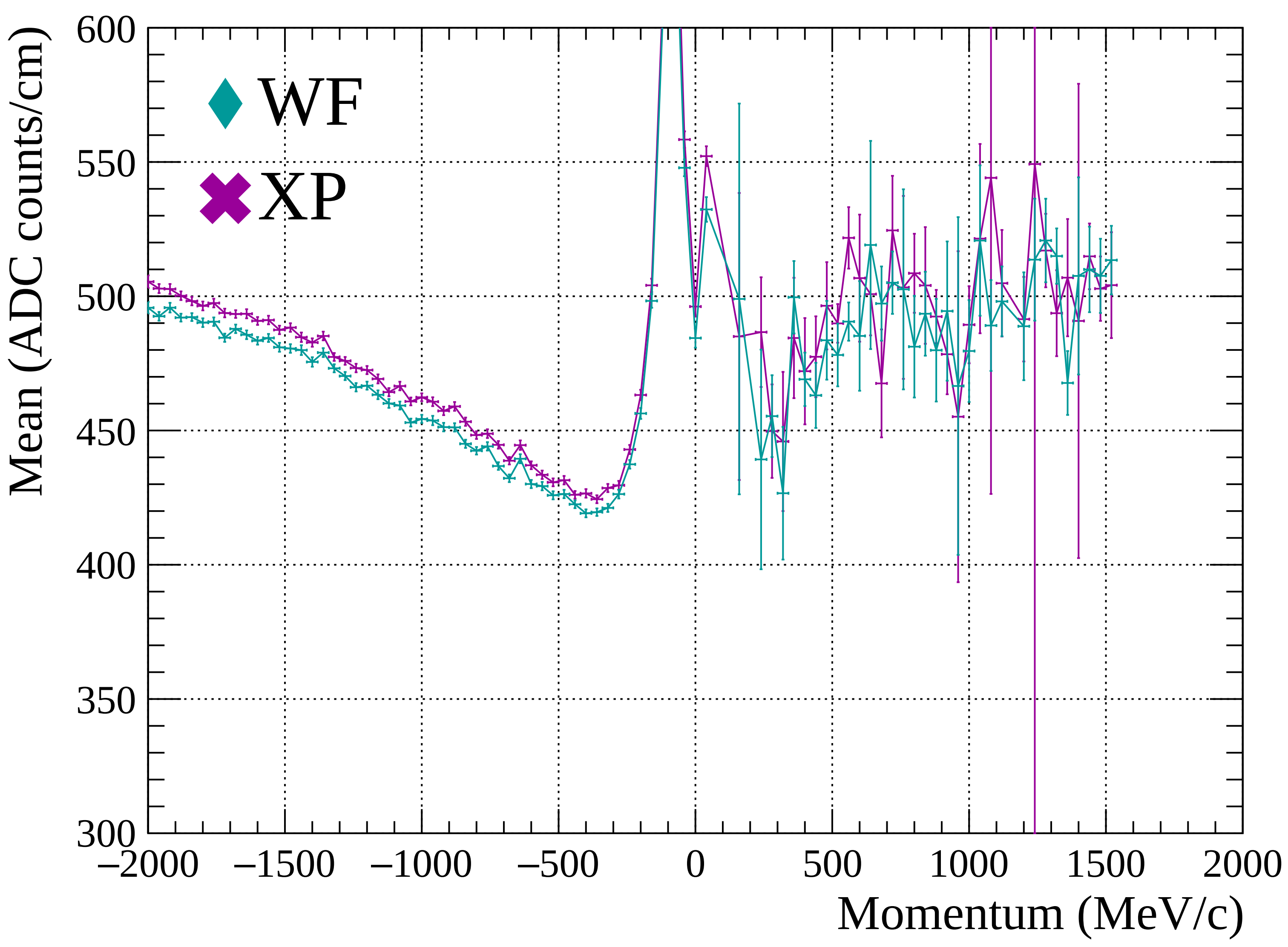


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

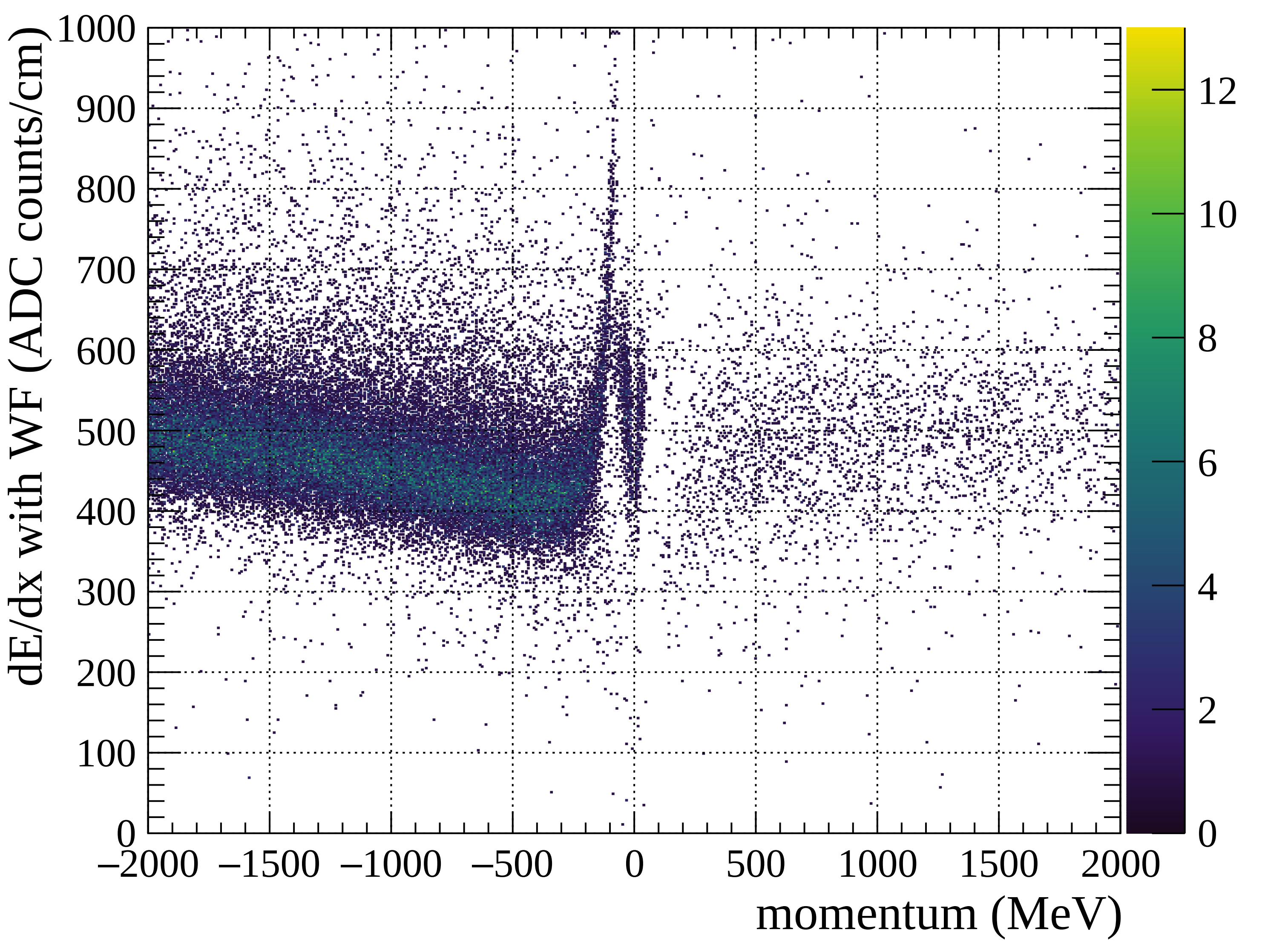

Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)
WF
XP

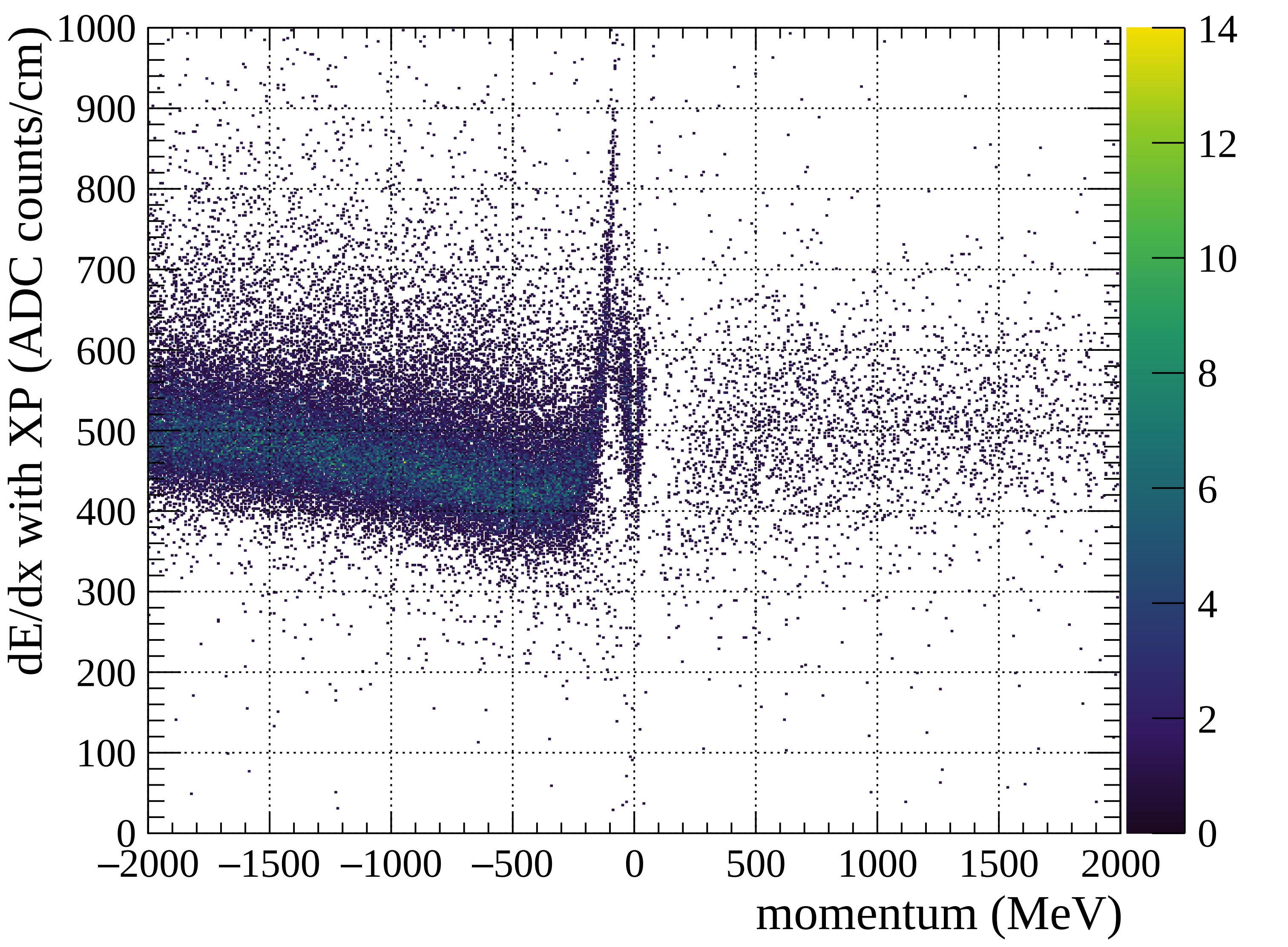

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30

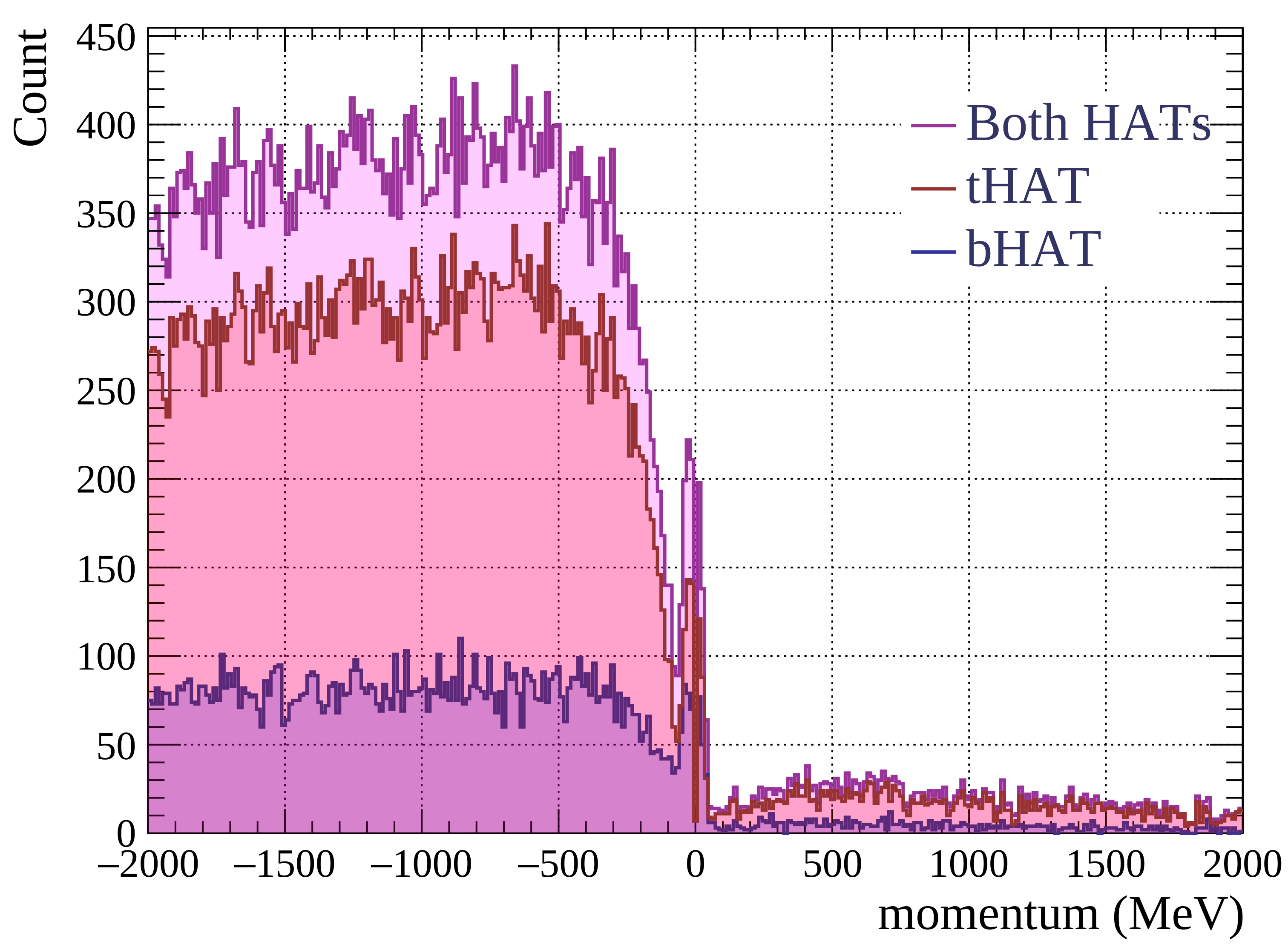

WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



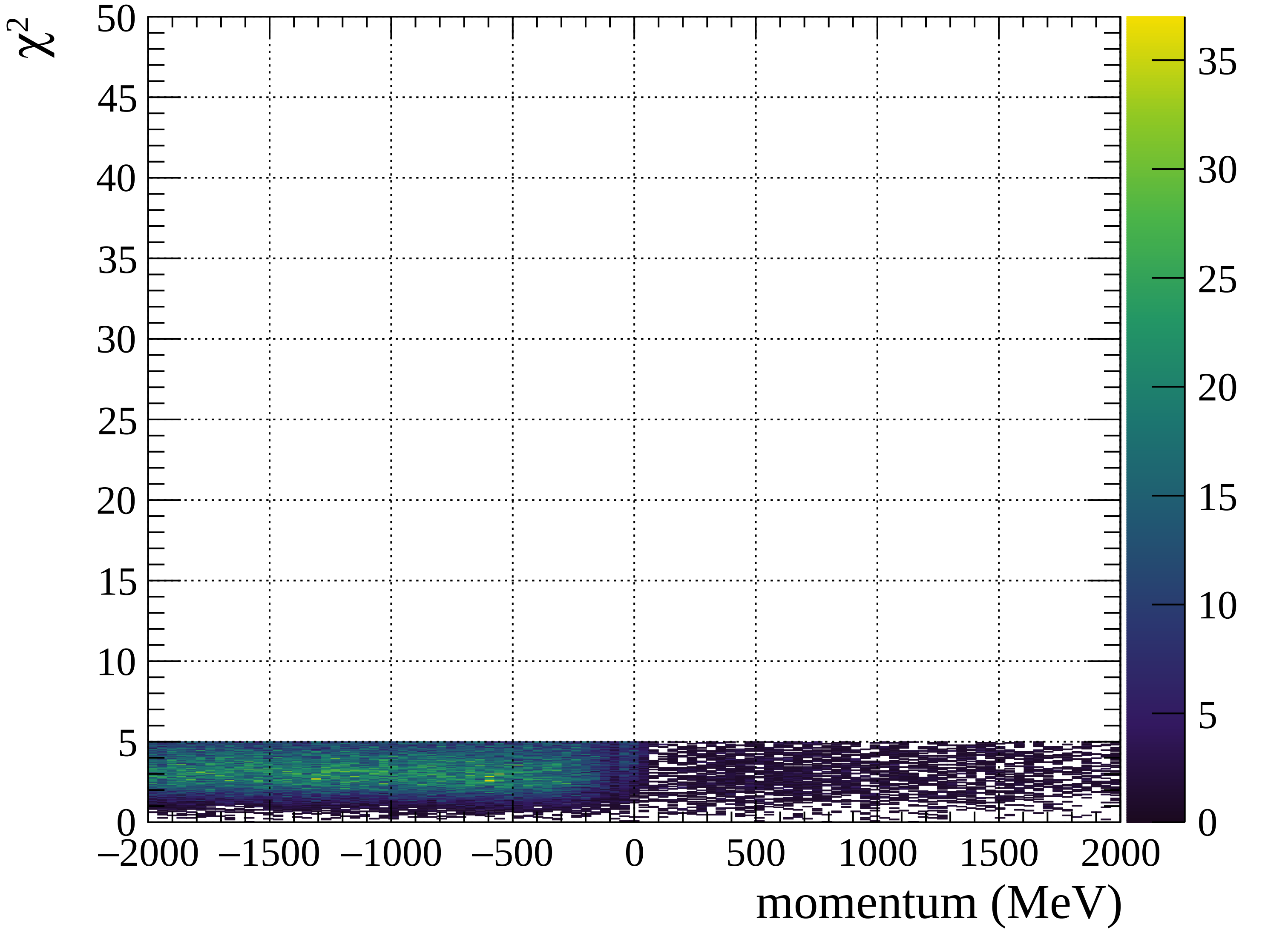
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
12
10
8
6
4
2
0

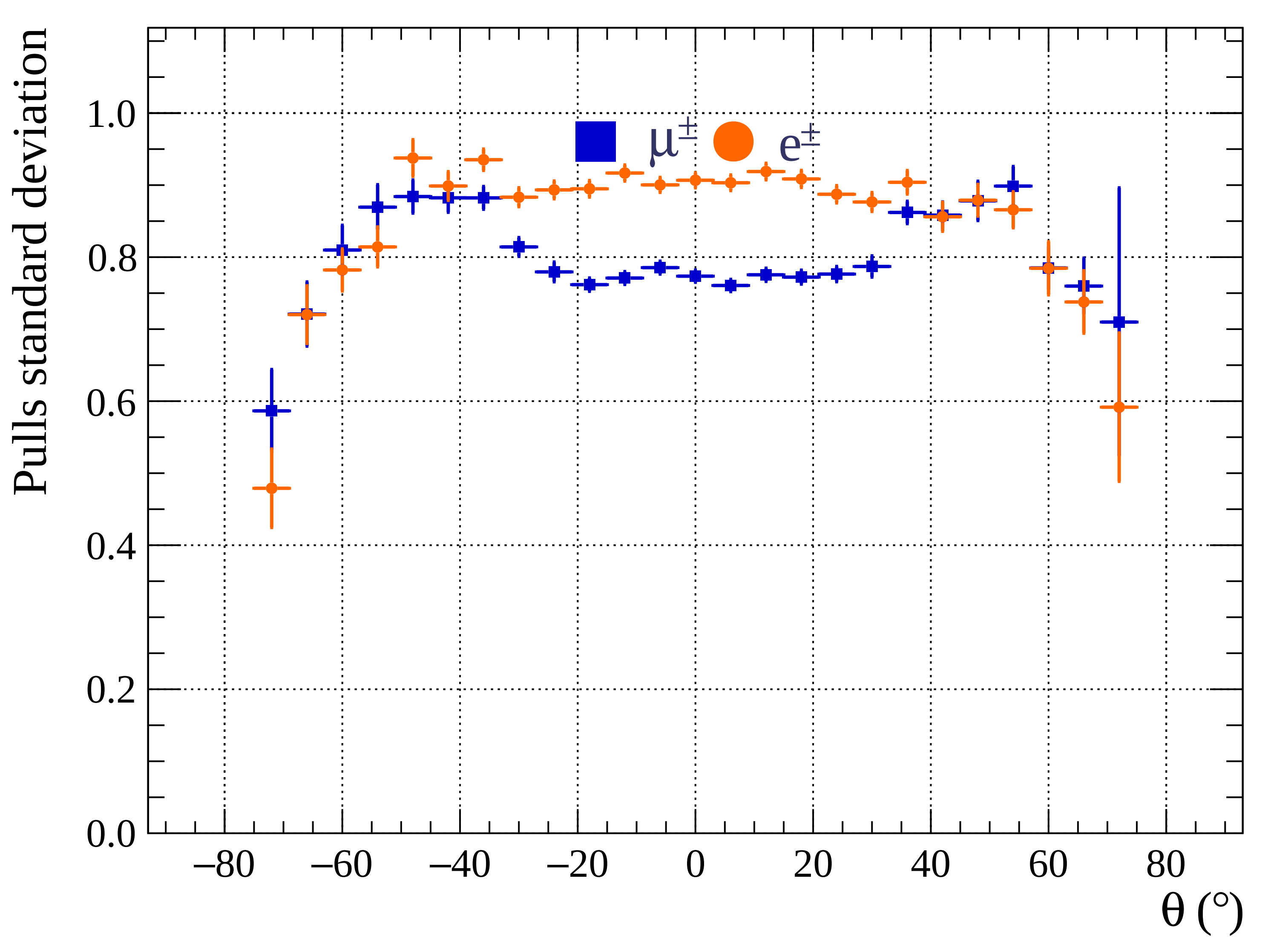

dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
14
12
10
8
6
4
2
0



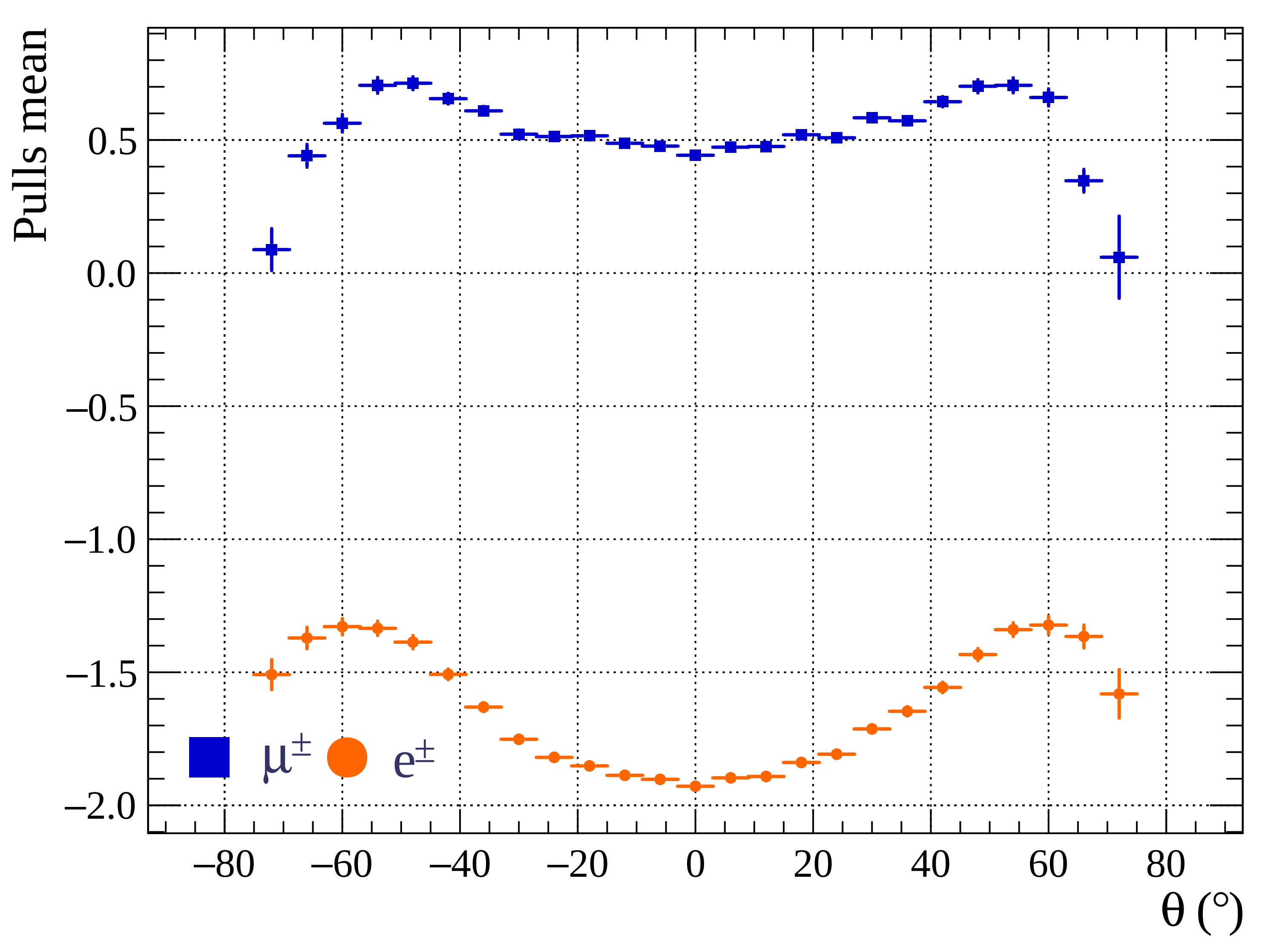
Count
450
400
350
300
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



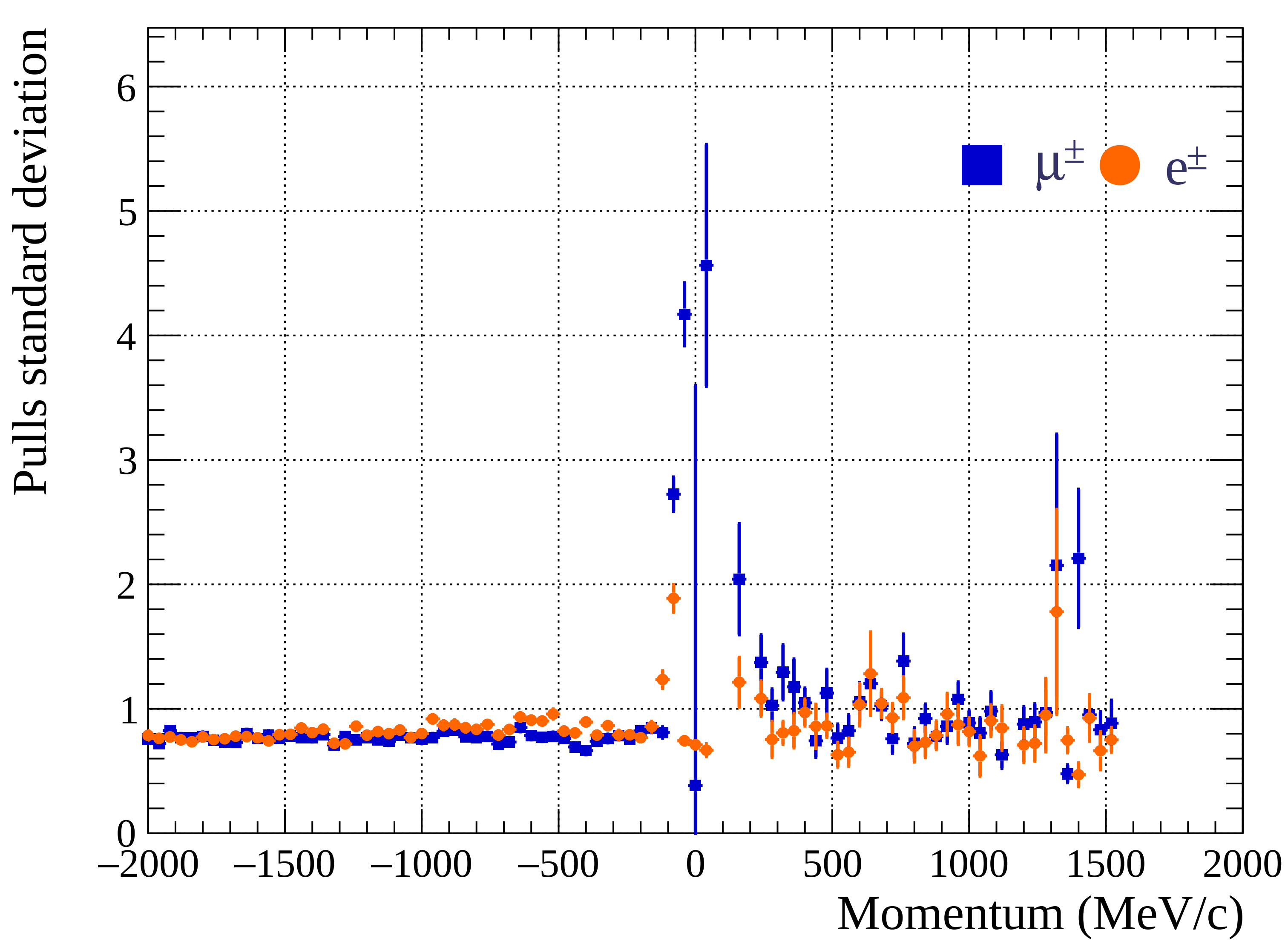
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
35
30
25
20
15
10
5
0

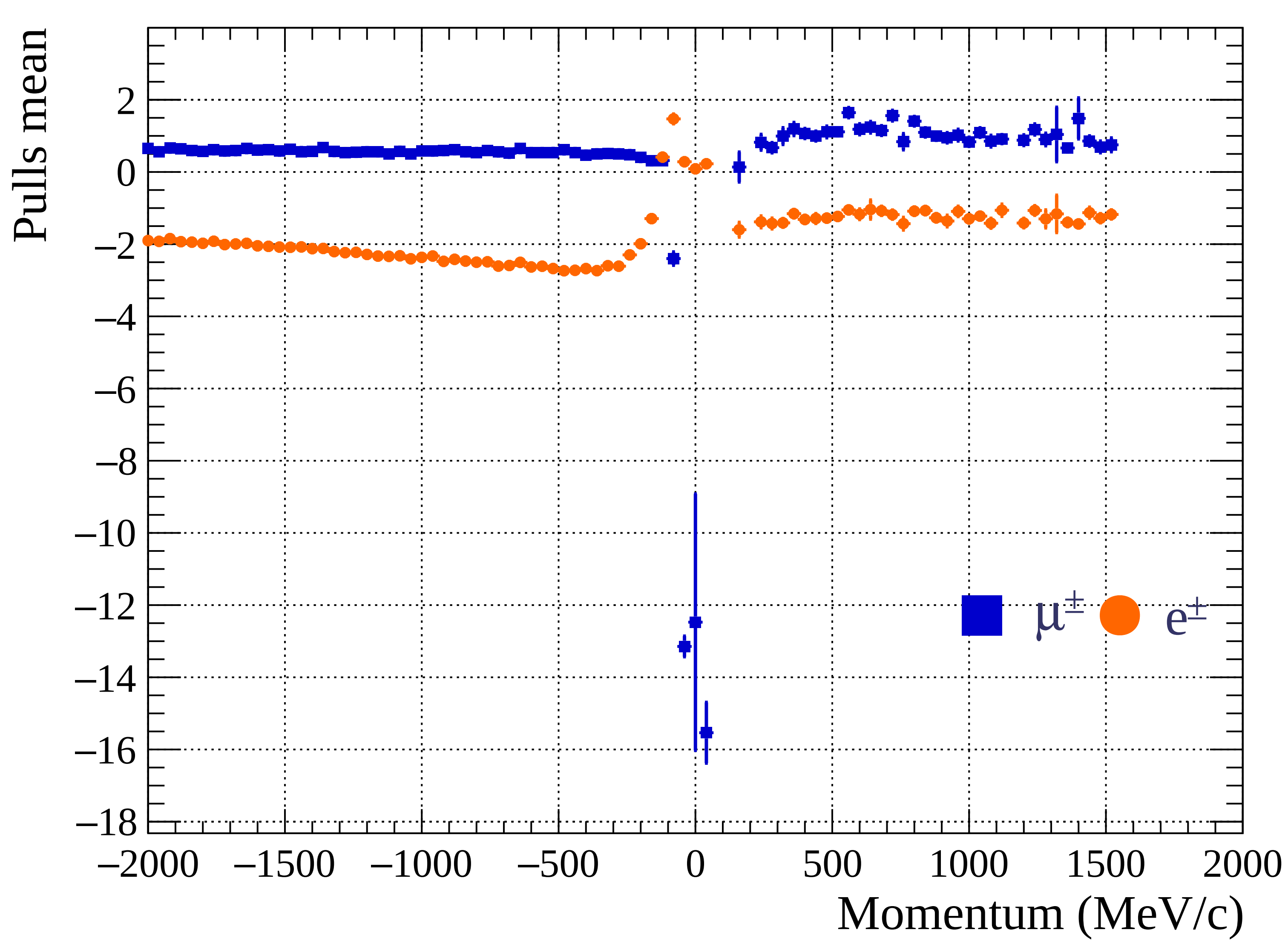


Pulls standard deviation
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

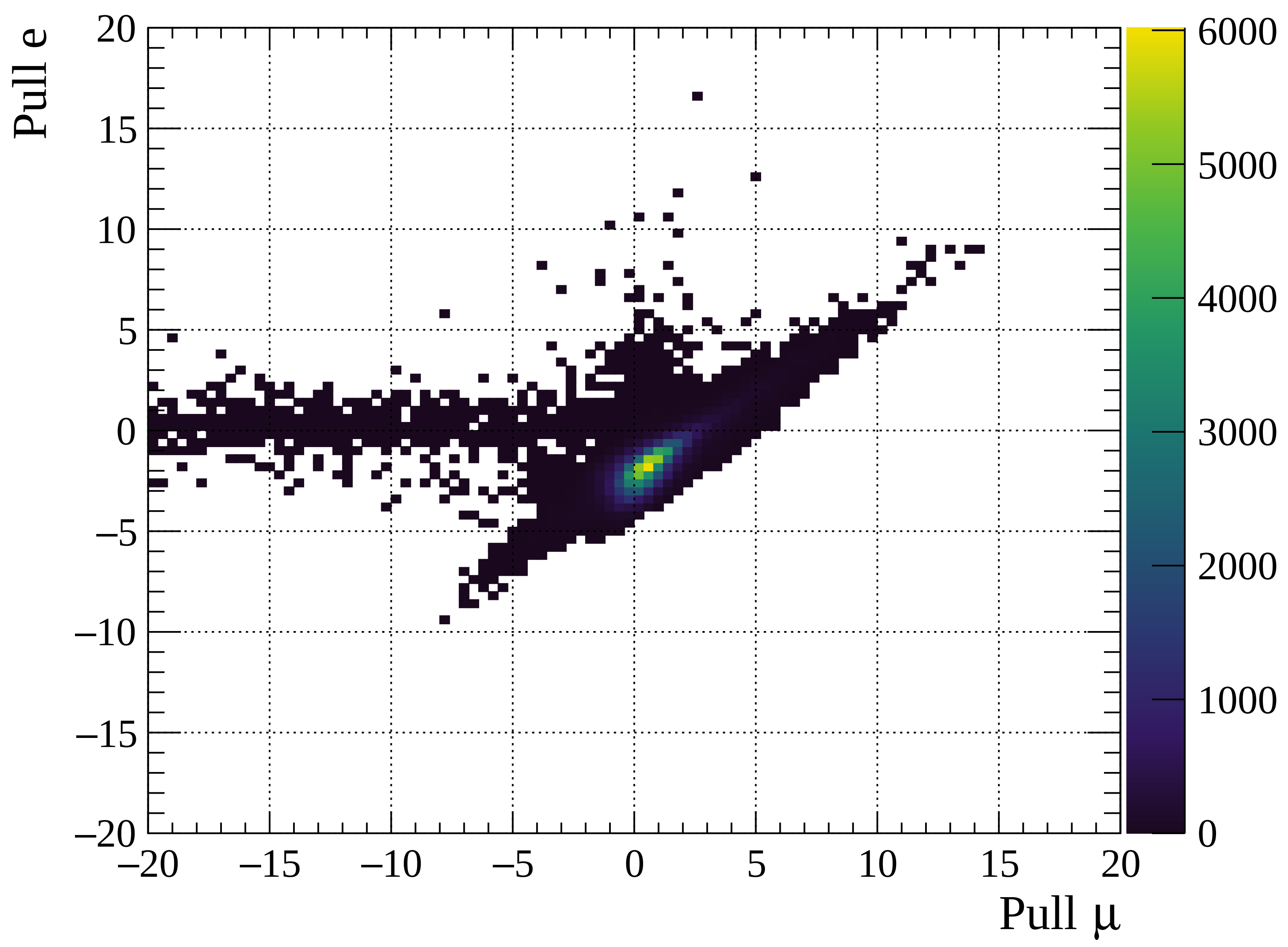


Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

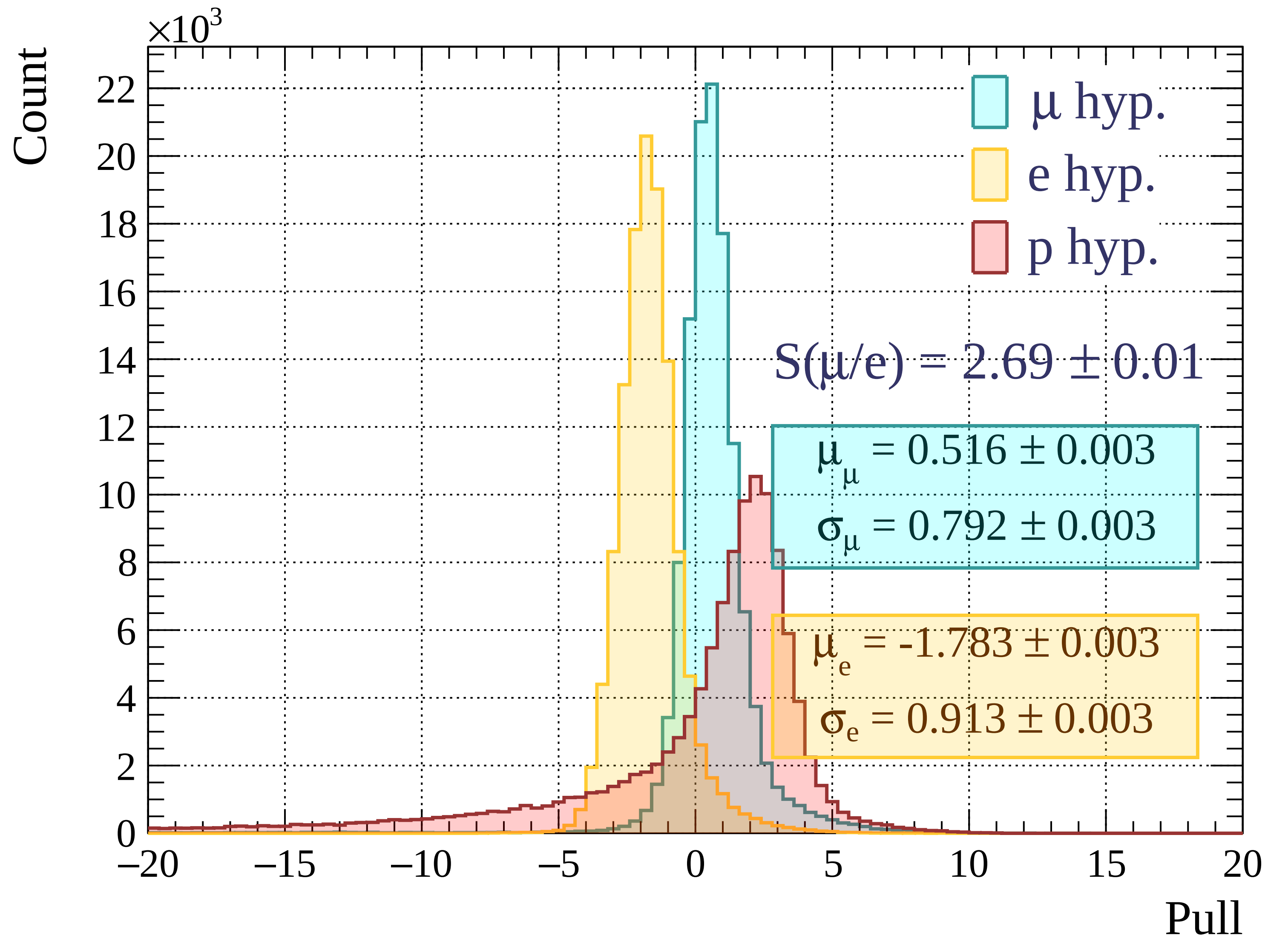

Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
1
2
3
4
5
6
−2000
−1500
−1000
−500
0
500
1000
1500
2000

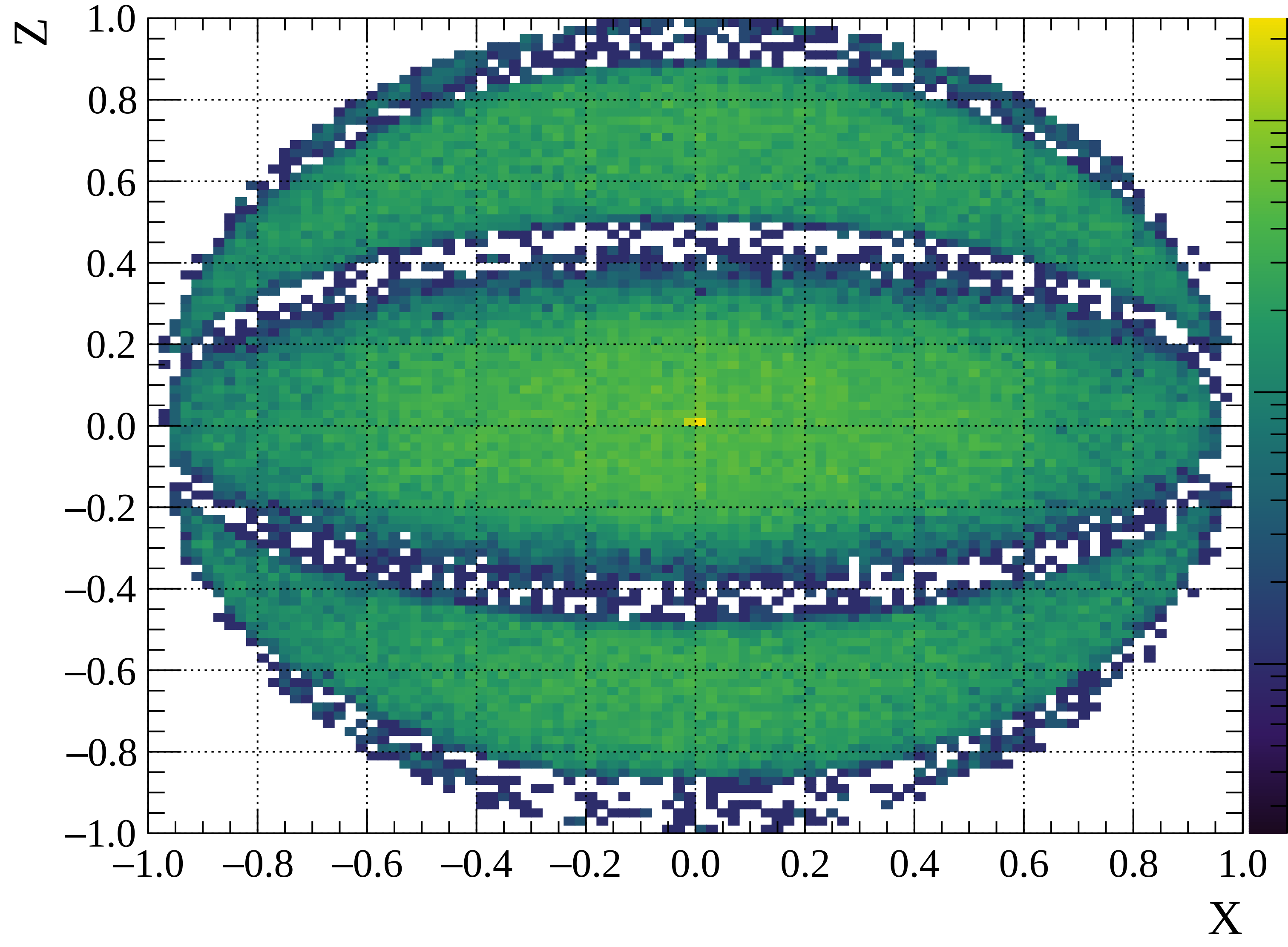


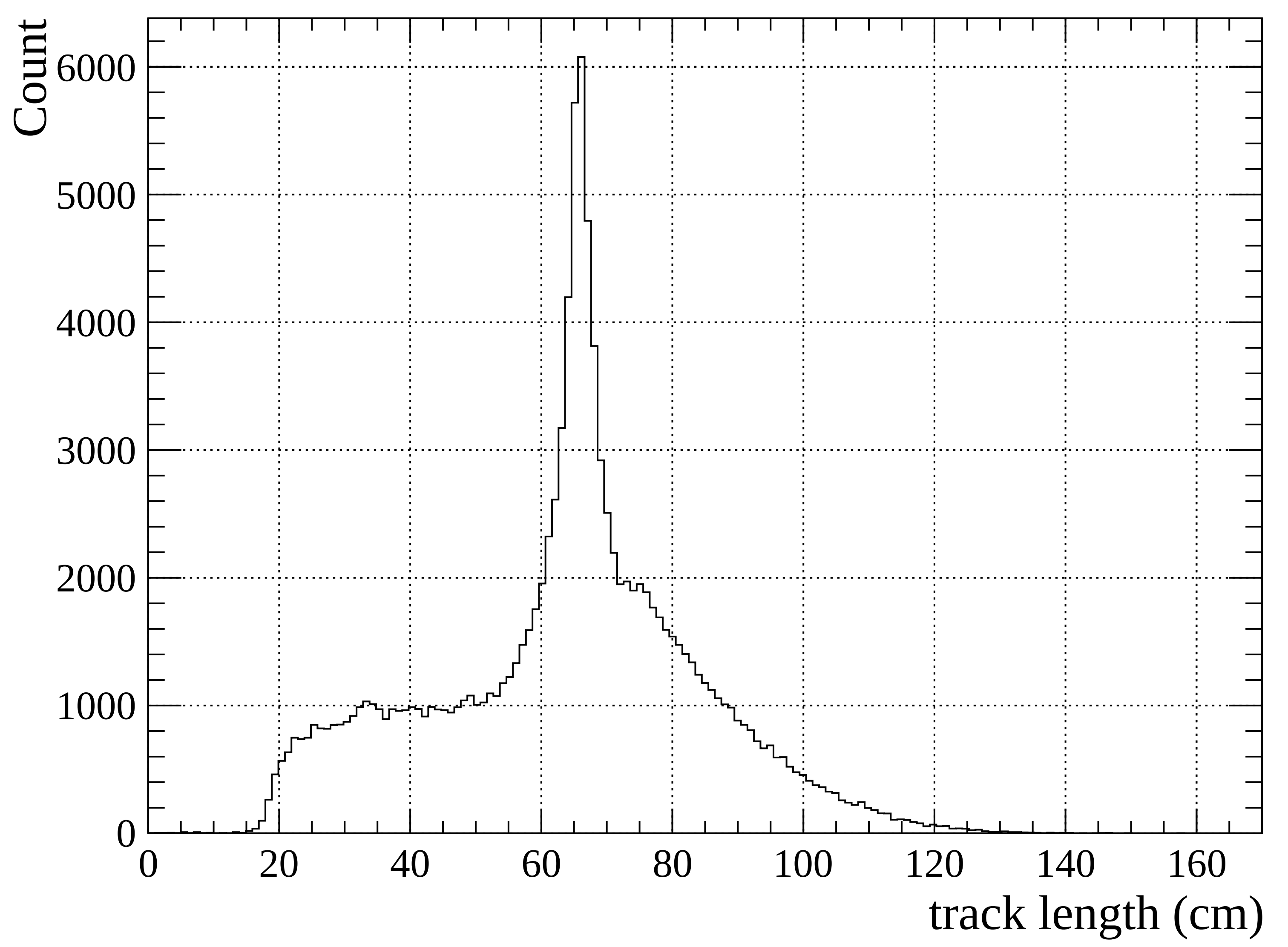
Pulls mean
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−18
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

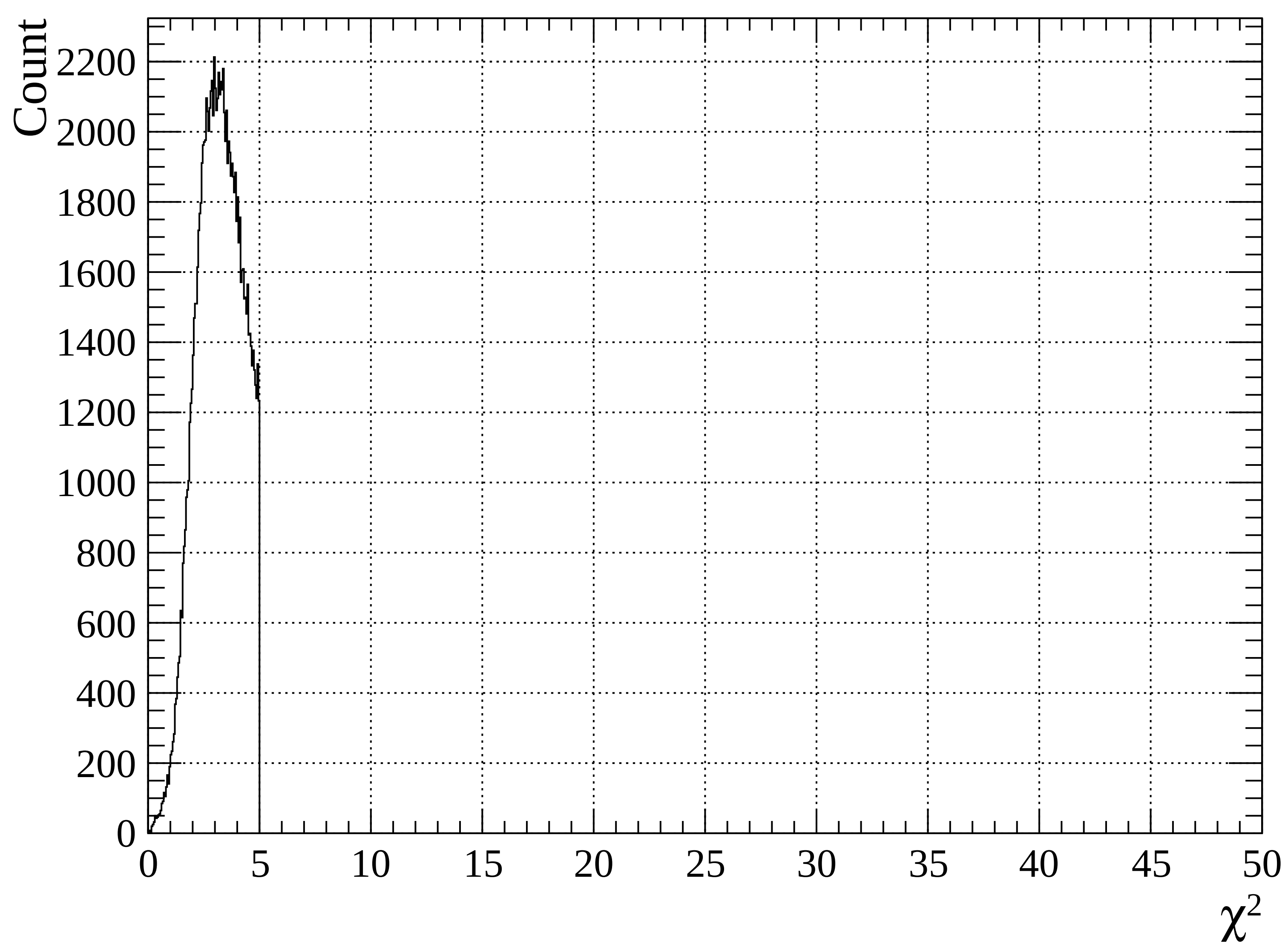


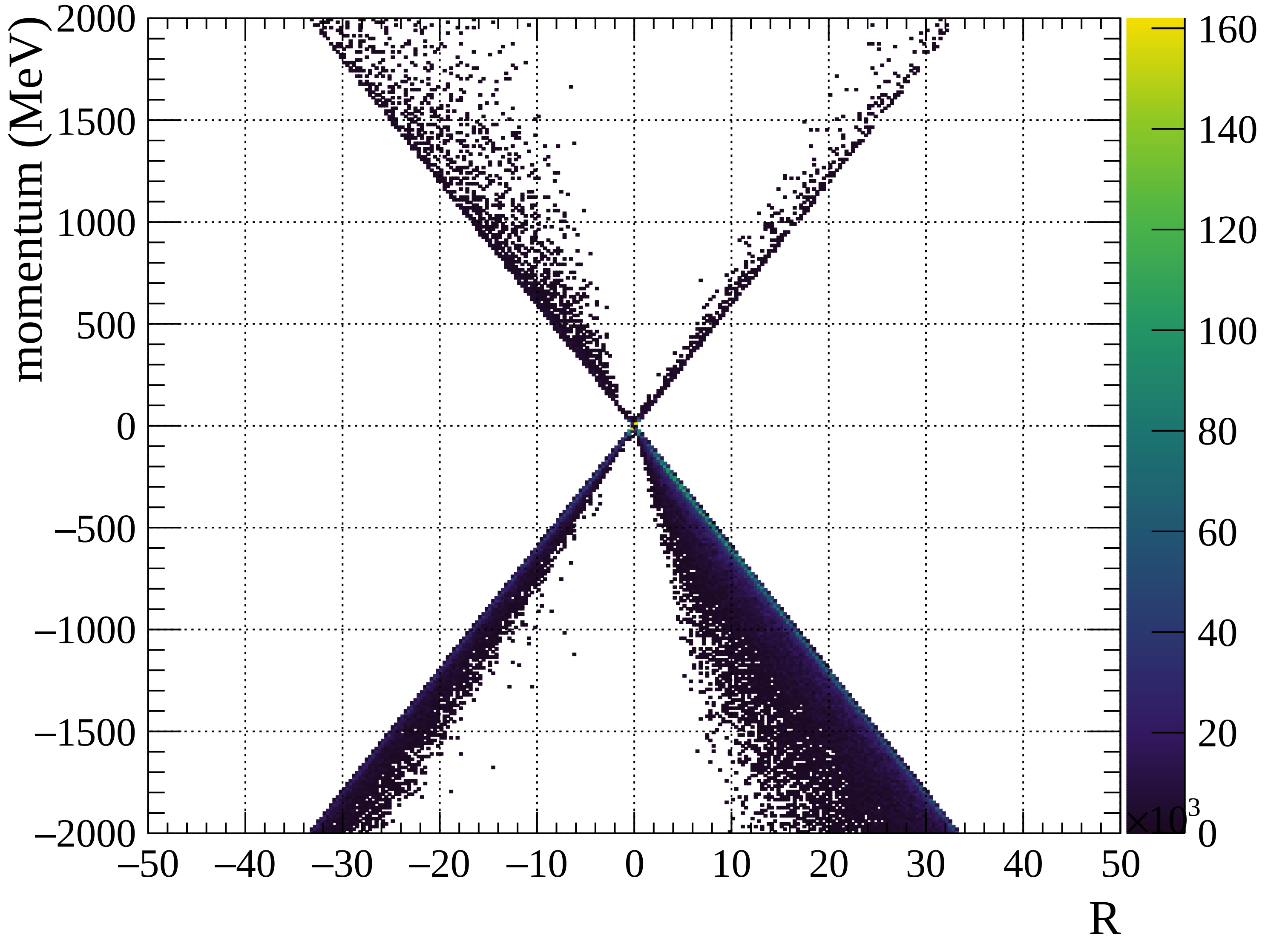
Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
6000
5000
4000
3000
2000
1000
0

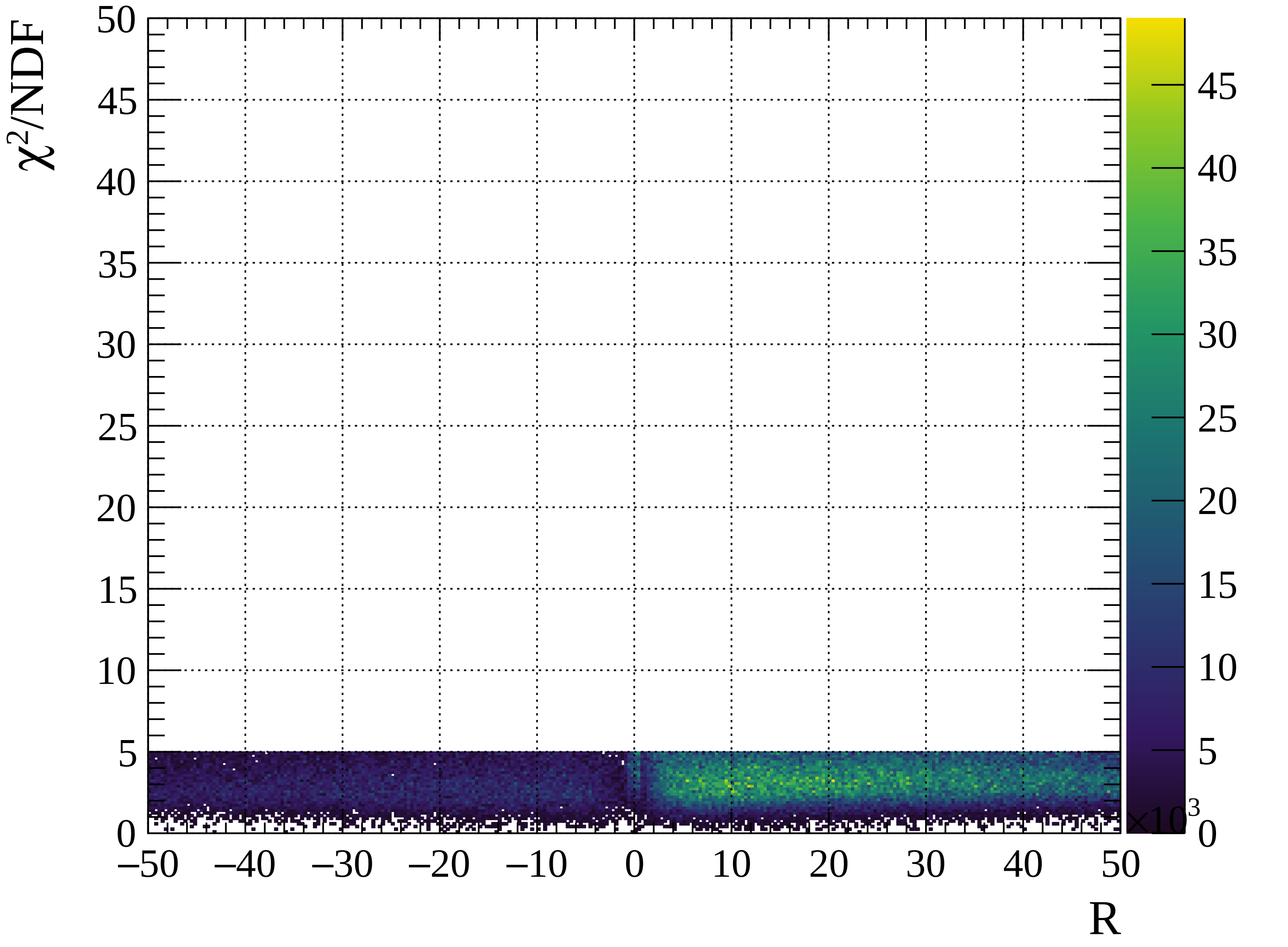

×10^3
Count
22
20
18
16
14
12
10
8
6
4
2
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.69 ± 0.01
μ_μ = 0.516 ± 0.003
σ_μ = 0.792 ± 0.003
μ_e = -1.783 ± 0.003
σ_e = 0.913 ± 0.003


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

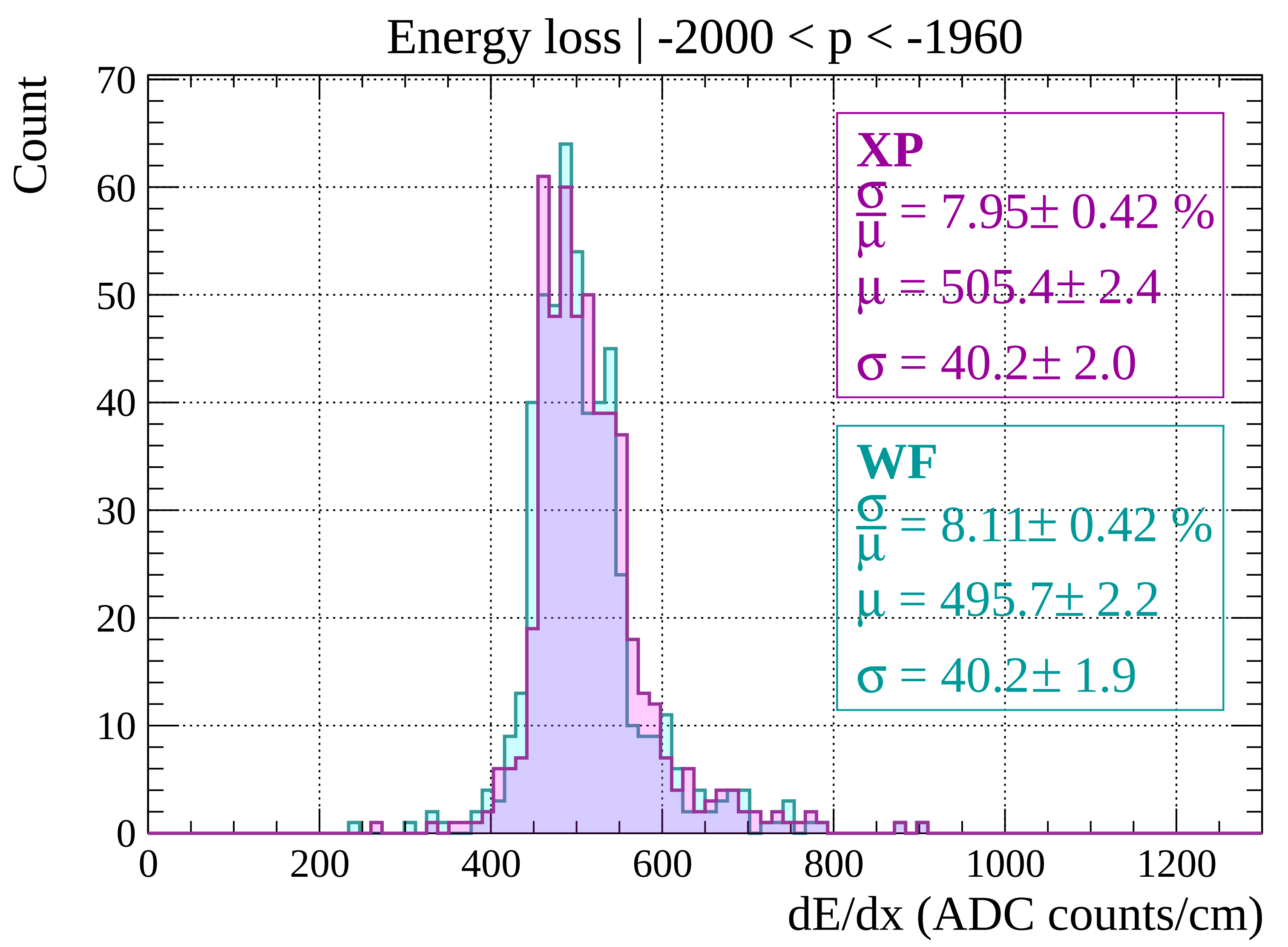

Count
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

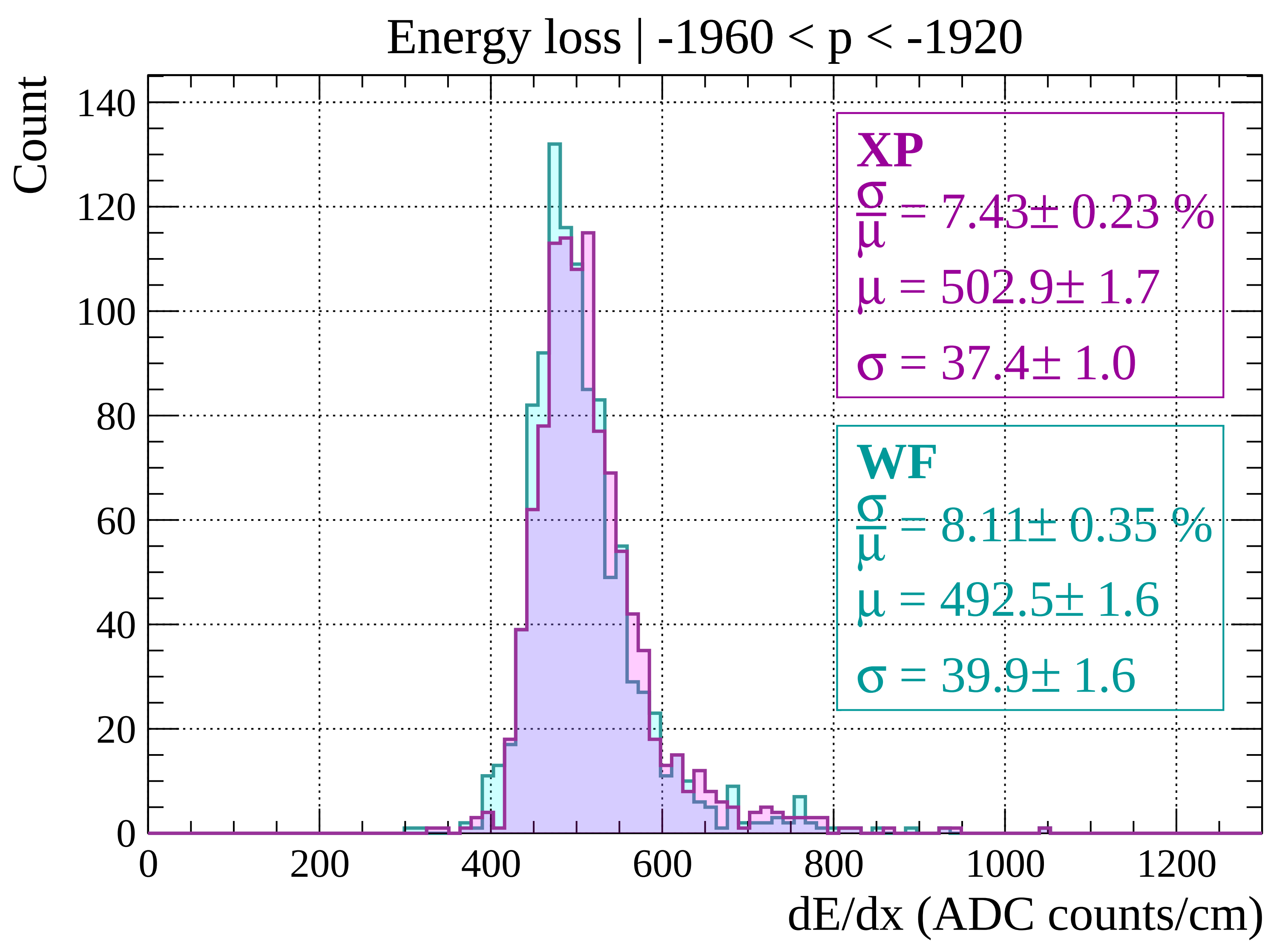

Count
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
5
10
15
20
25
30
35
40
45
50
χ^2


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
160
140
120
100
80
60
40
20
0
×10^3

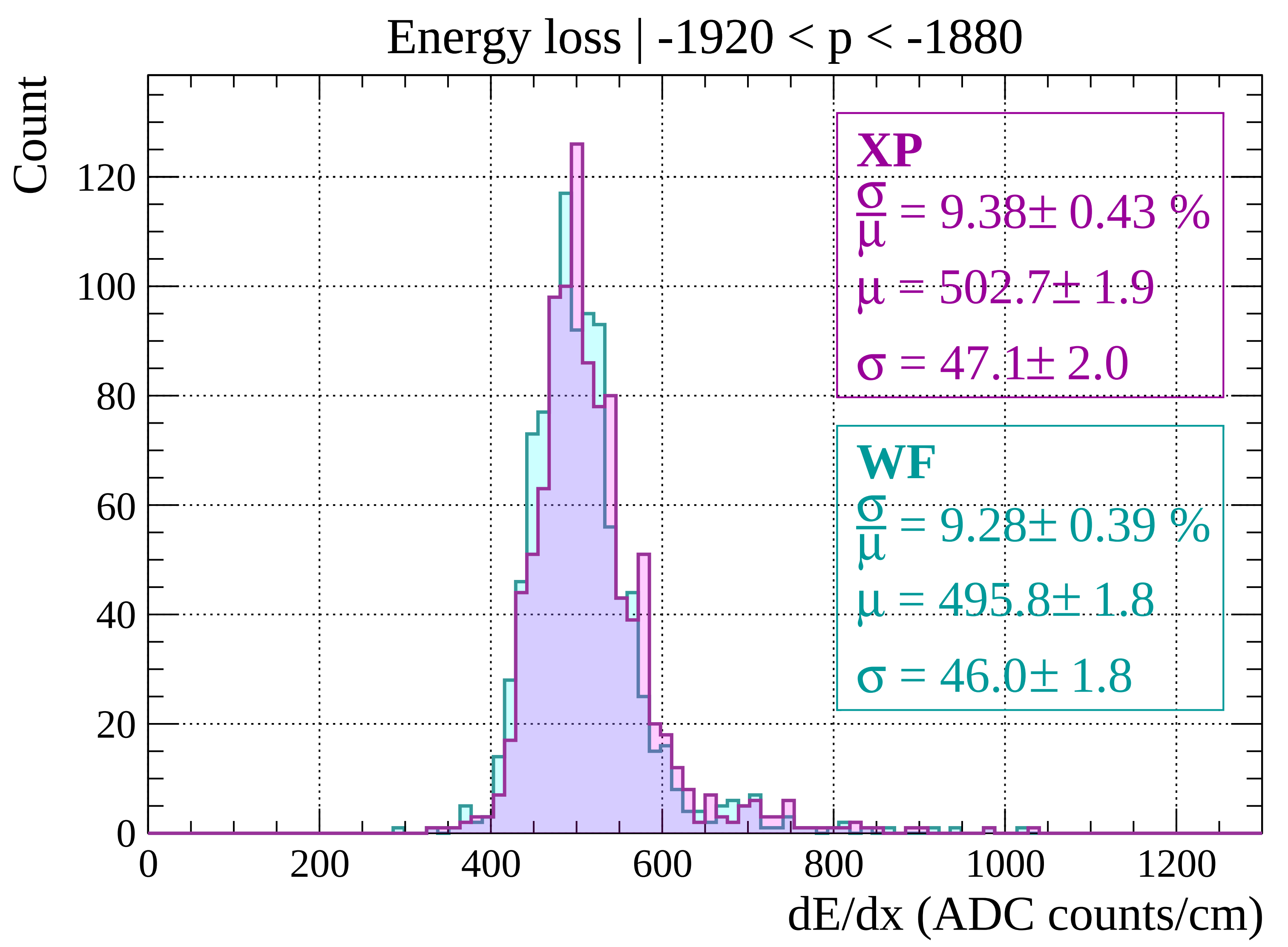


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
$\times 10^3$

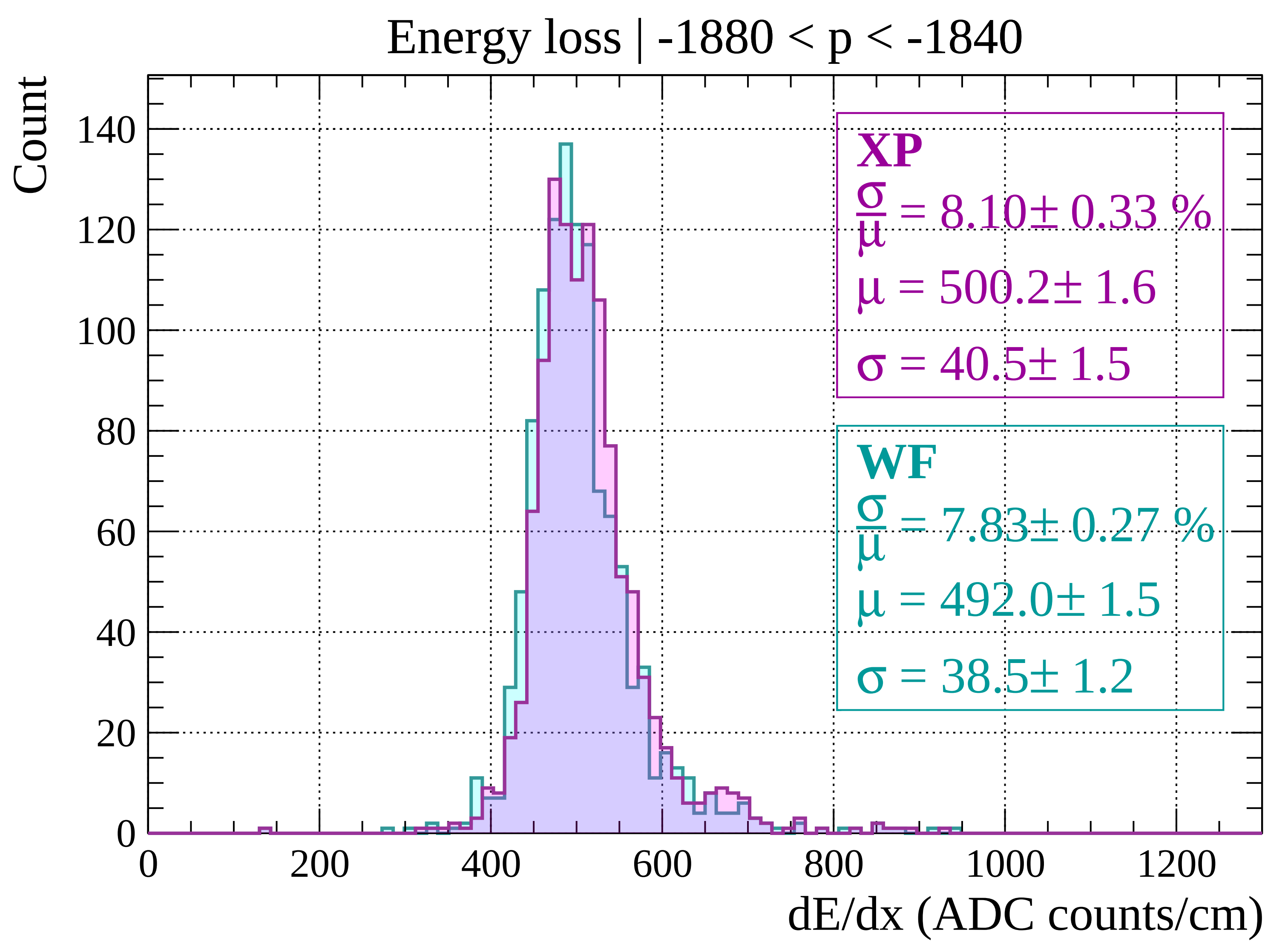

Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 7.95± 0.42 %
μ = 505.4± 2.4
σ = 40.2± 2.0
WF
σ/μ = 8.11± 0.42 %
μ = 495.7± 2.2
σ = 40.2± 1.9

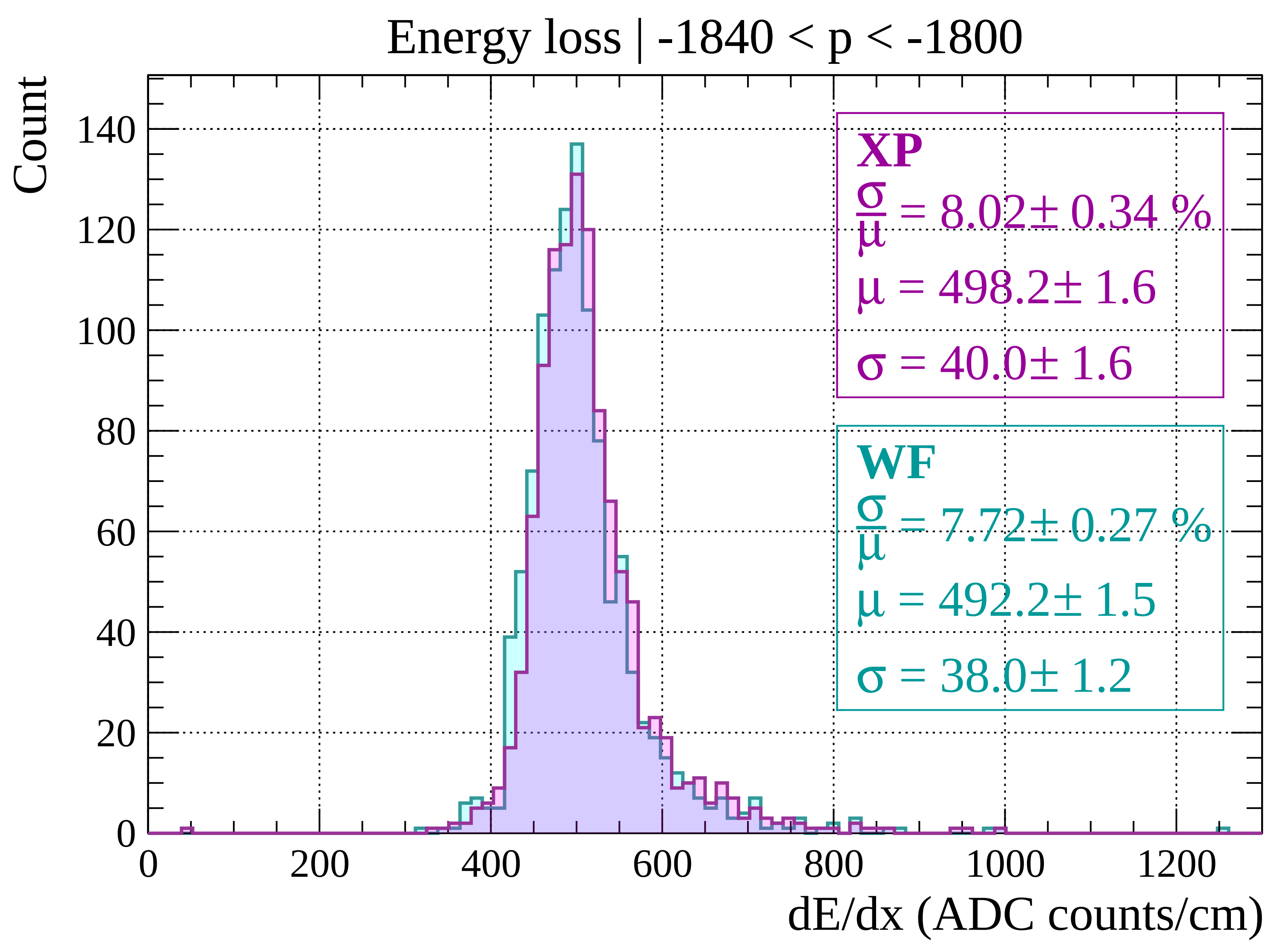


Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 7.43± 0.23 %
μ = 502.9± 1.7
σ = 37.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.11± 0.35 %
μ = 492.5± 1.6
σ = 39.9± 1.6

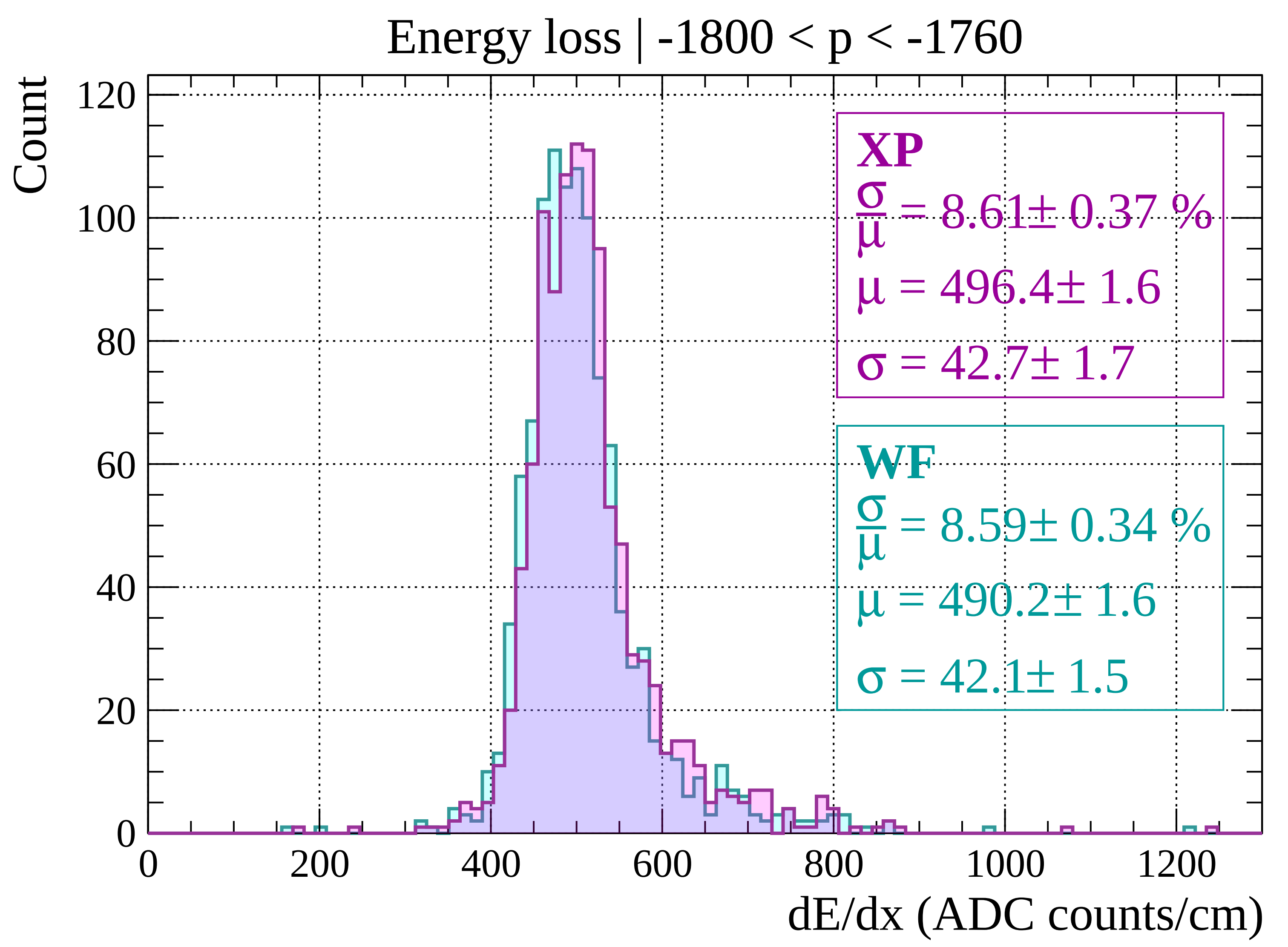


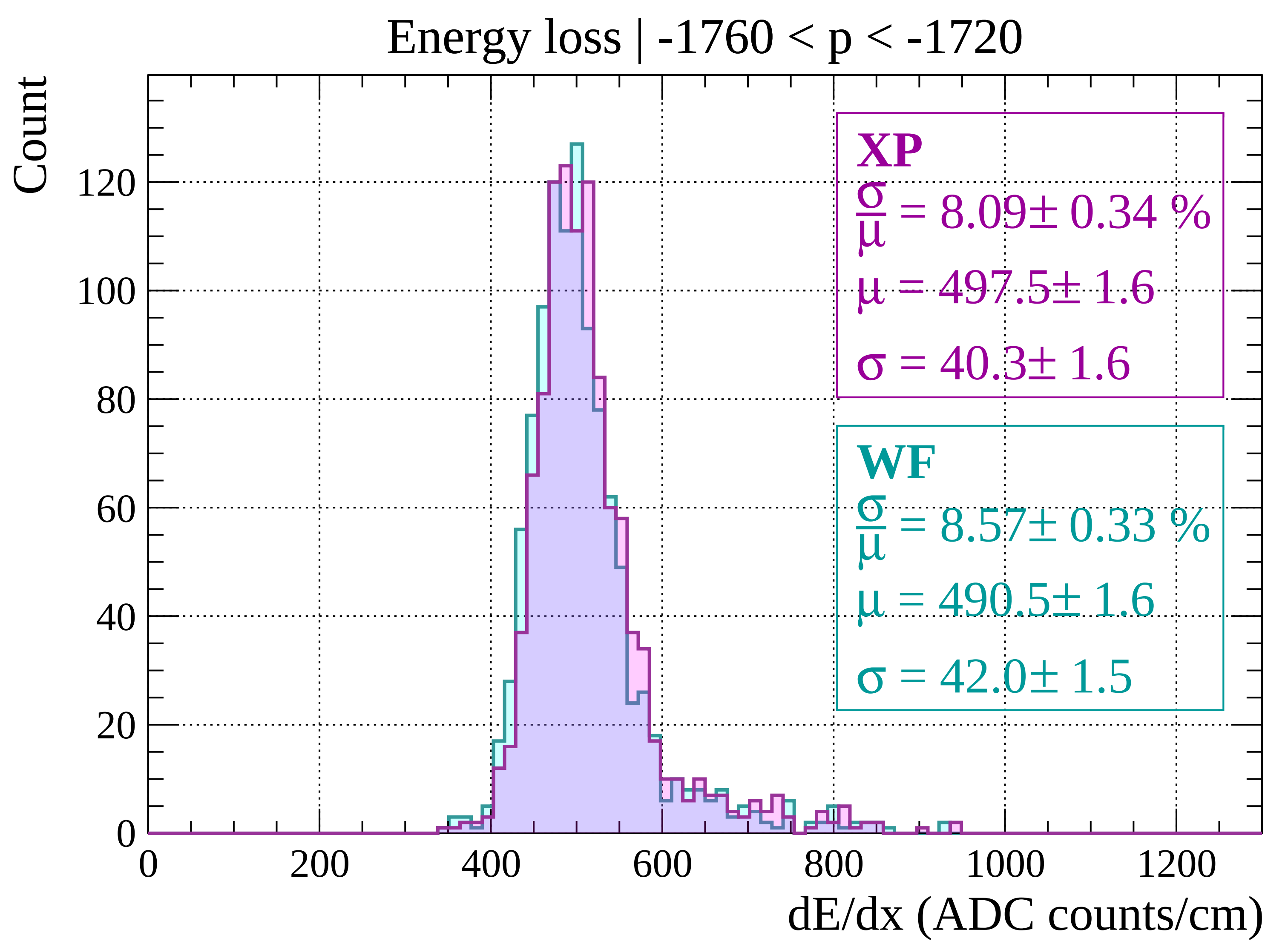
Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 9.38± 0.43 %
μ = 502.7± 1.9
σ = 47.1± 2.0
WF
σ/μ = 9.28± 0.39 %
μ = 495.8± 1.8
σ = 46.0± 1.8


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.10± 0.33 %
μ = 500.2± 1.6
σ = 40.5± 1.5
WF
$\frac{\sigma}{\mu}$ = 7.83± 0.27 %
μ = 492.0± 1.5
σ = 38.5± 1.2

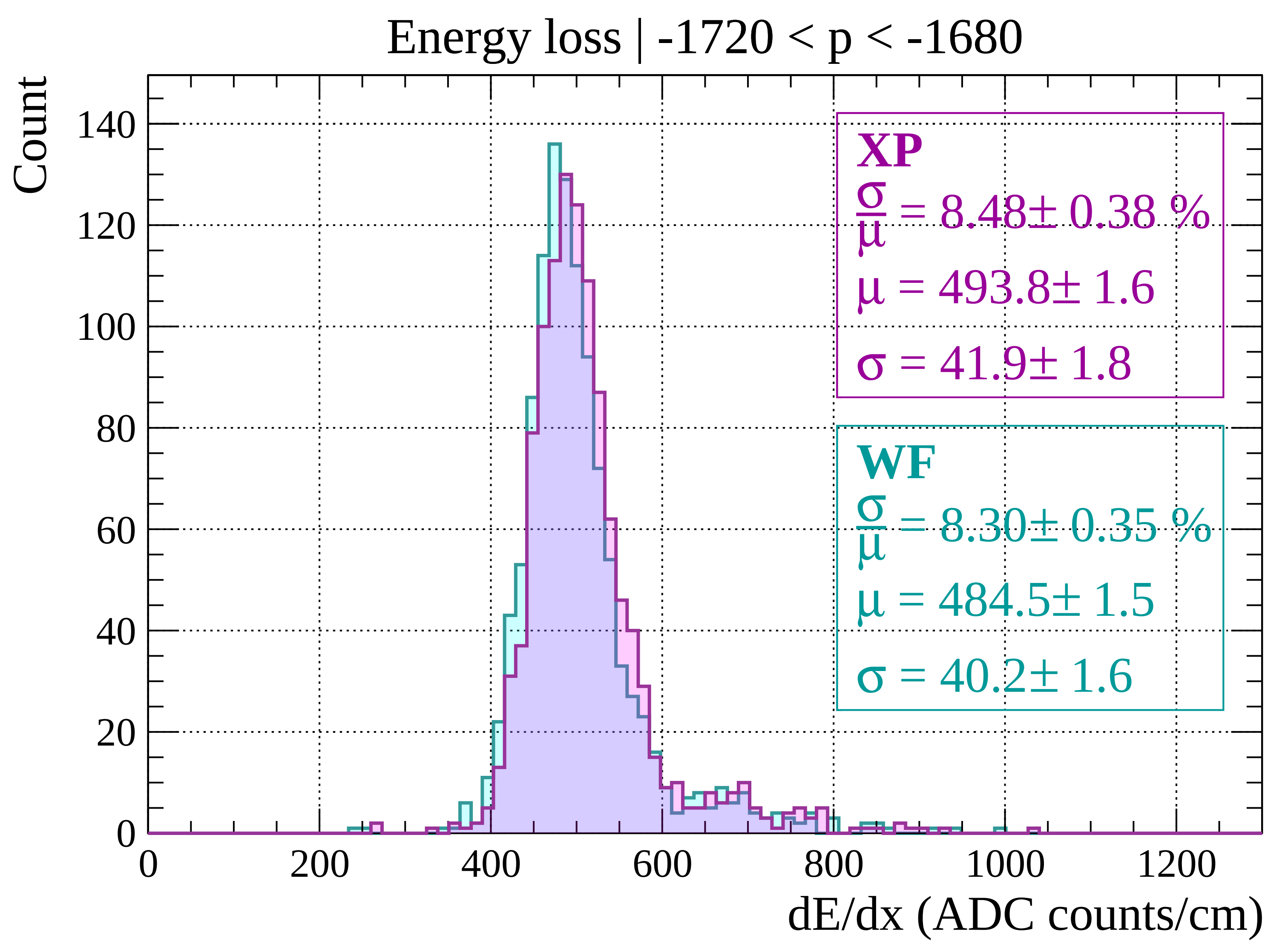


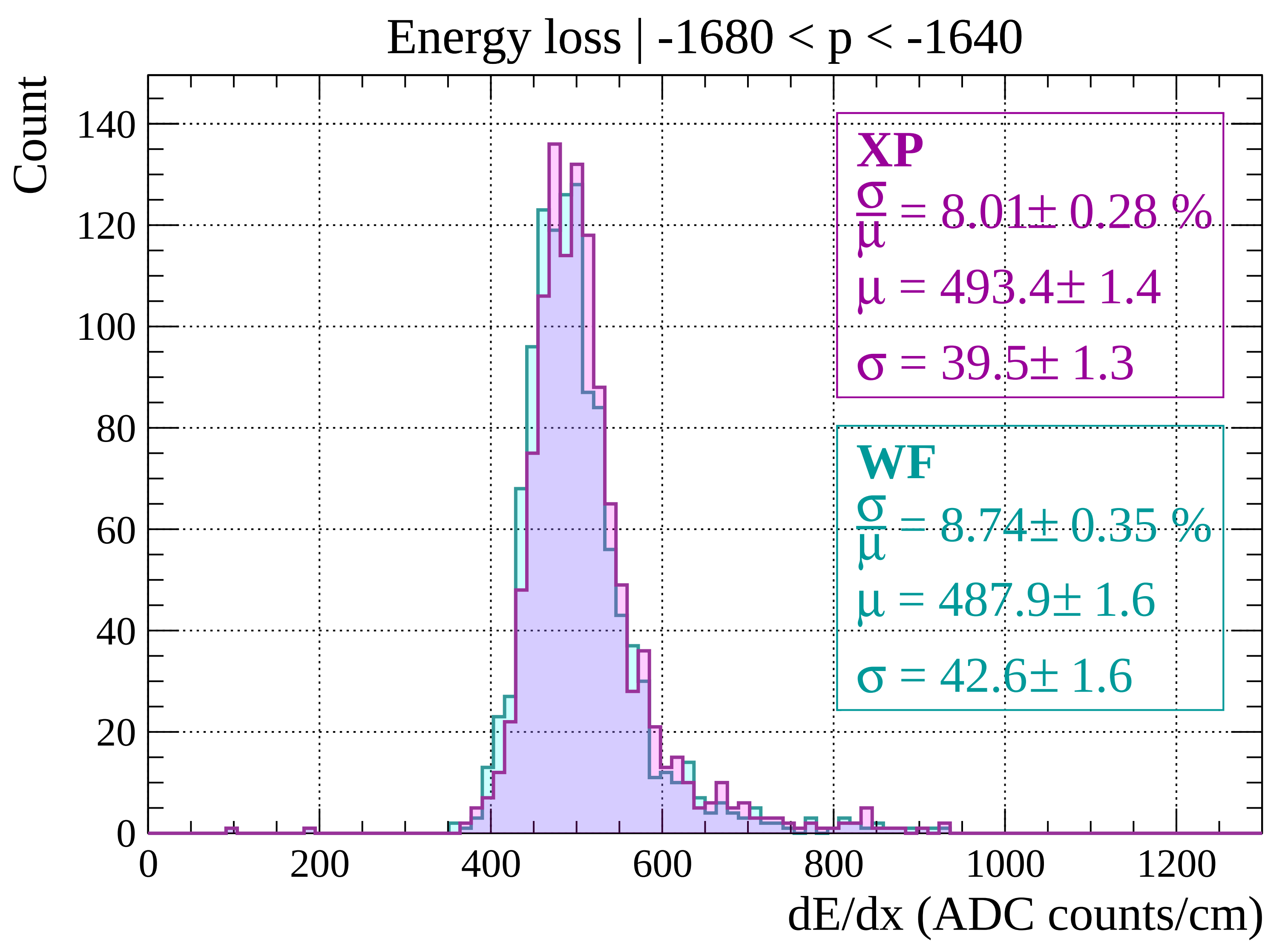
Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.02 \pm 0.34$ %
$\mu = 498.2 \pm 1.6$
$\sigma = 40.0 \pm 1.6$
WF
$\frac{\sigma}{\mu} = 7.72 \pm 0.27$ %
$\mu = 492.2 \pm 1.5$
$\sigma = 38.0 \pm 1.2$

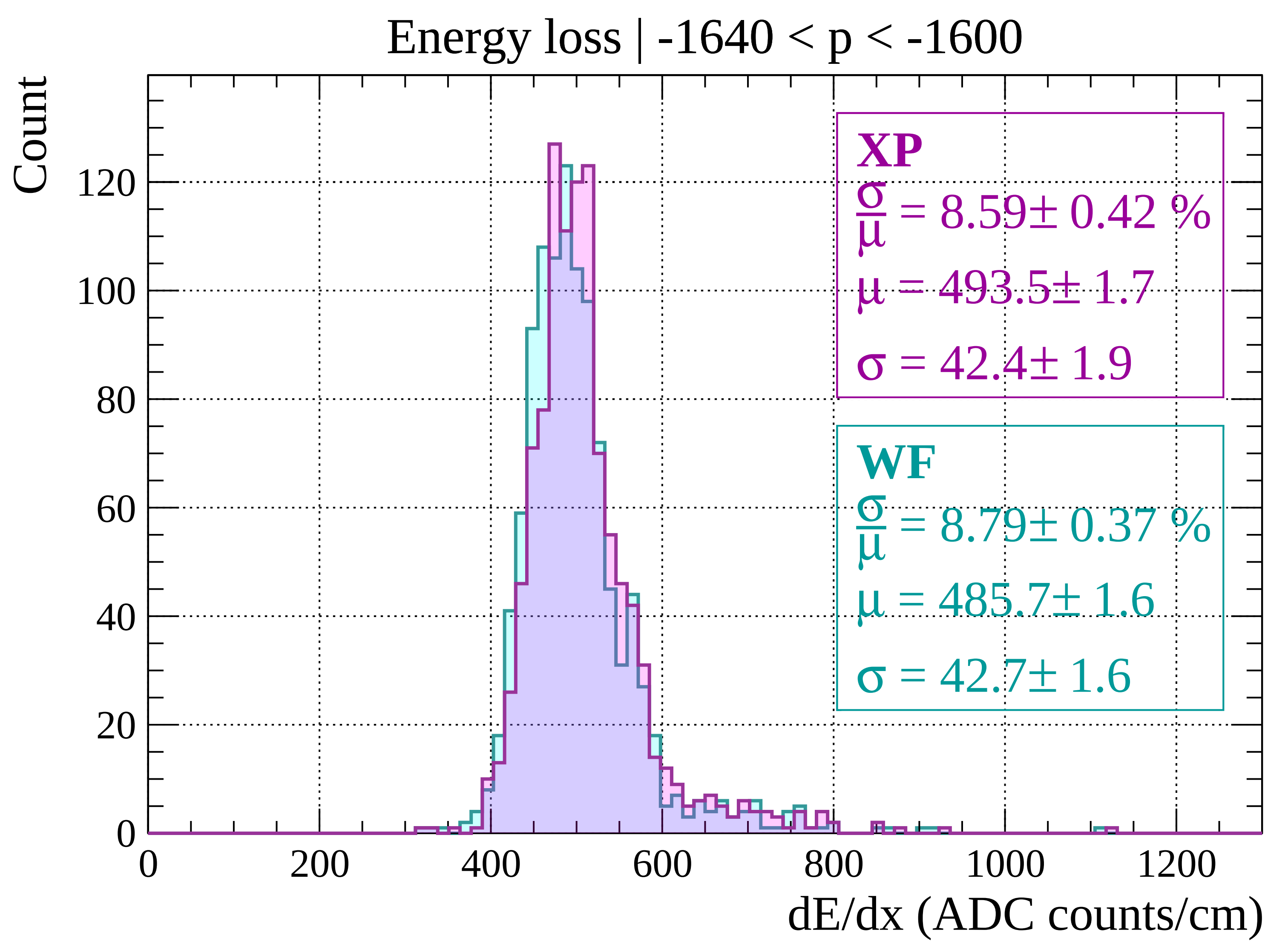

Energy loss | -1800 < p < -1760
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.61± 0.37 %
μ = 496.4± 1.6
σ = 42.7± 1.7
WF
$\frac{\sigma}{\mu}$ = 8.59± 0.34 %
μ = 490.2± 1.6
σ = 42.1± 1.5



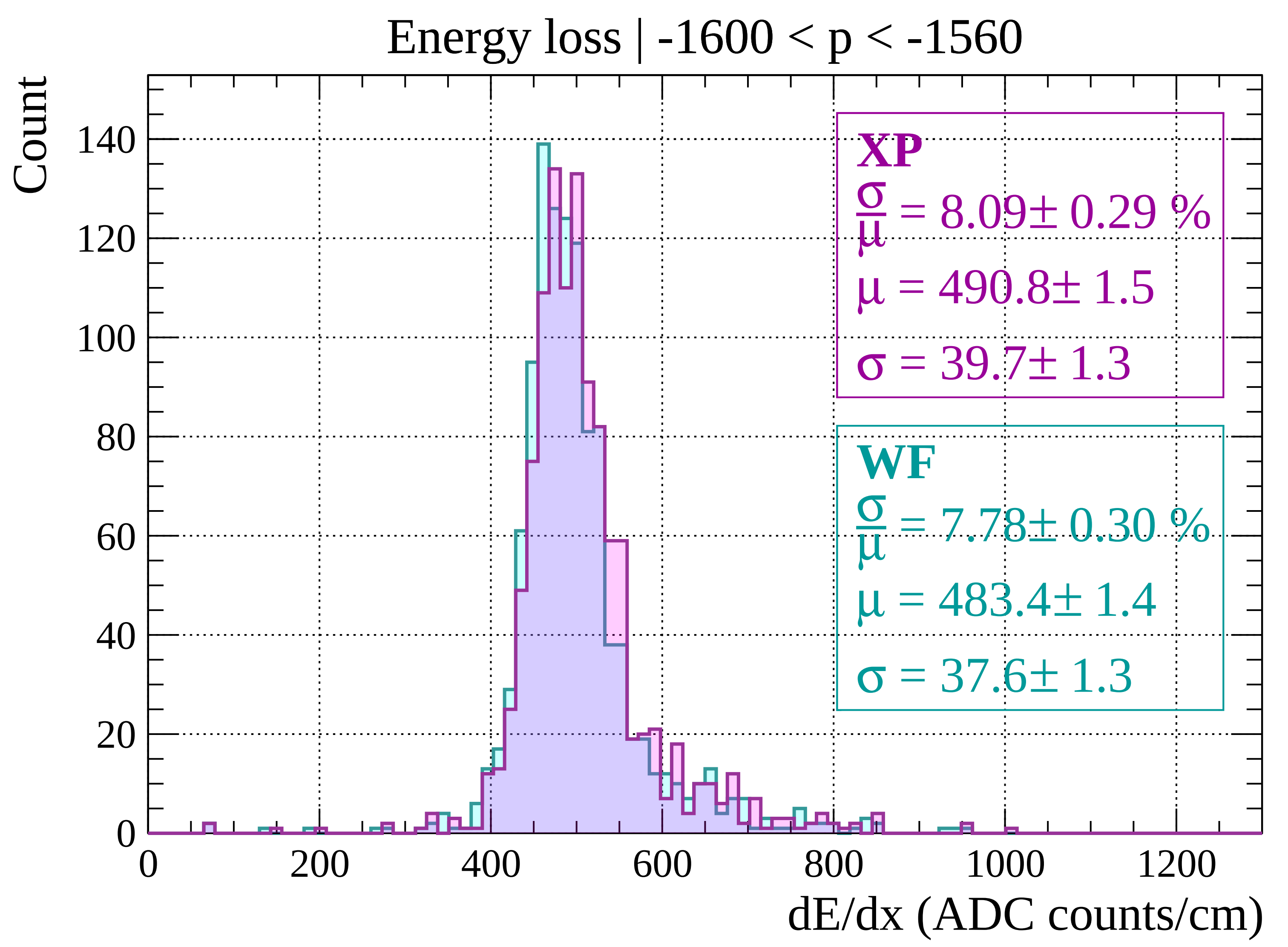
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 8.09± 0.34 %
μ = 497.5± 1.6
σ = 40.3± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.57± 0.33 %
μ = 490.5± 1.6
σ = 42.0± 1.5

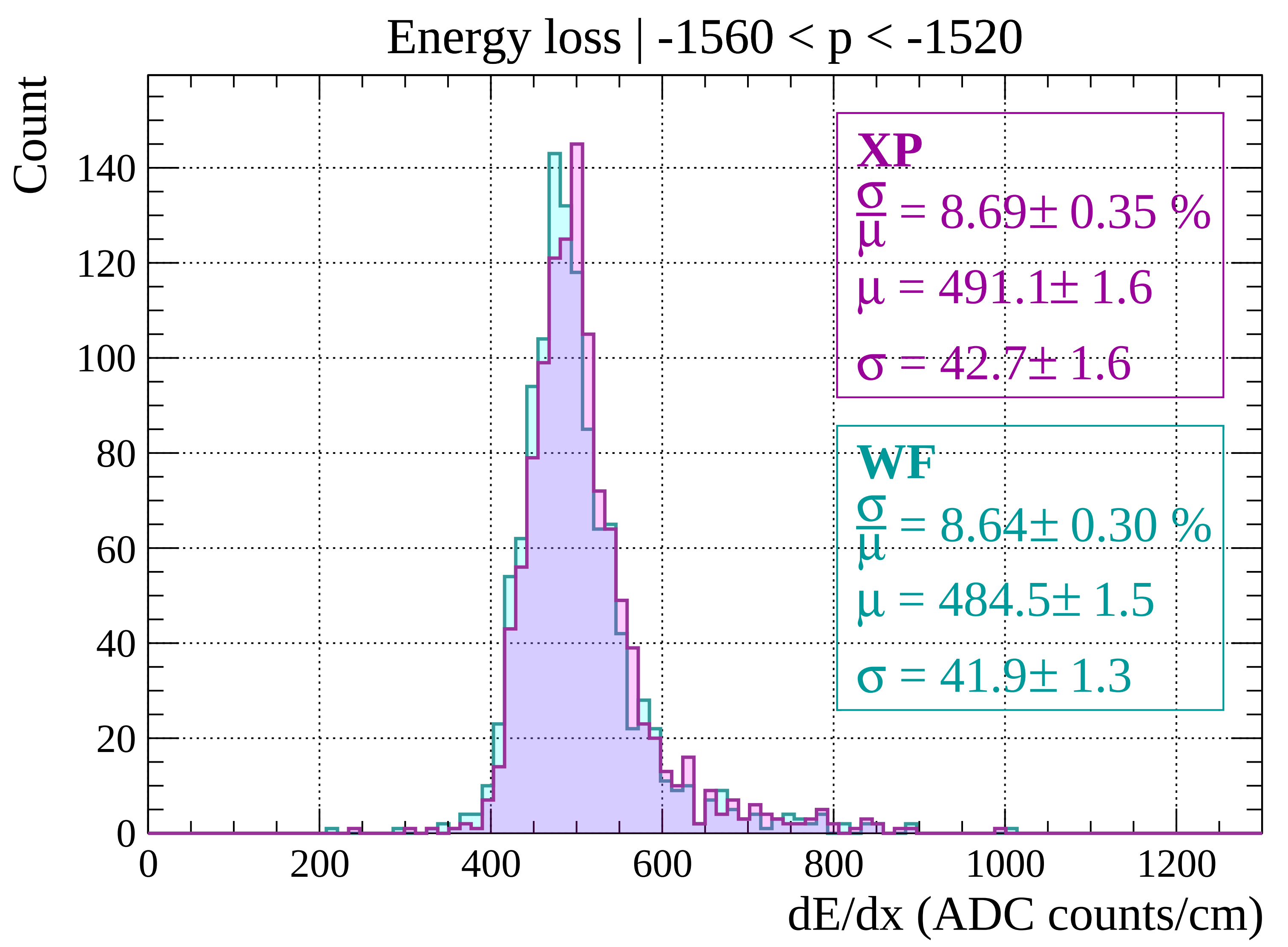

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.48± 0.38 %
μ = 493.8± 1.6
σ = 41.9± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.30± 0.35 %
μ = 484.5± 1.5
σ = 40.2± 1.6


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.01± 0.28 %
μ = 493.4± 1.4
σ = 39.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.74± 0.35 %
μ = 487.9± 1.6
σ = 42.6± 1.6

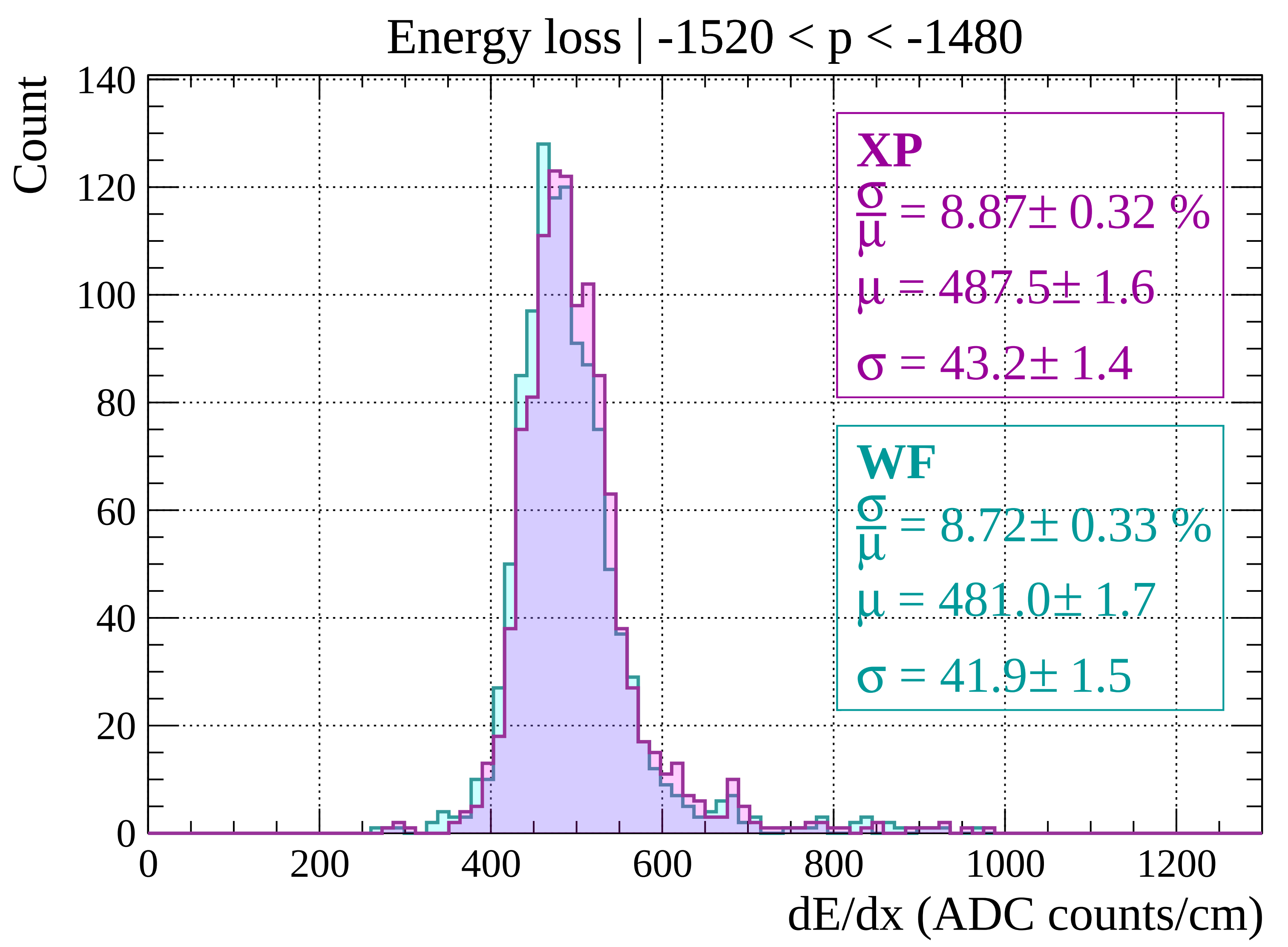


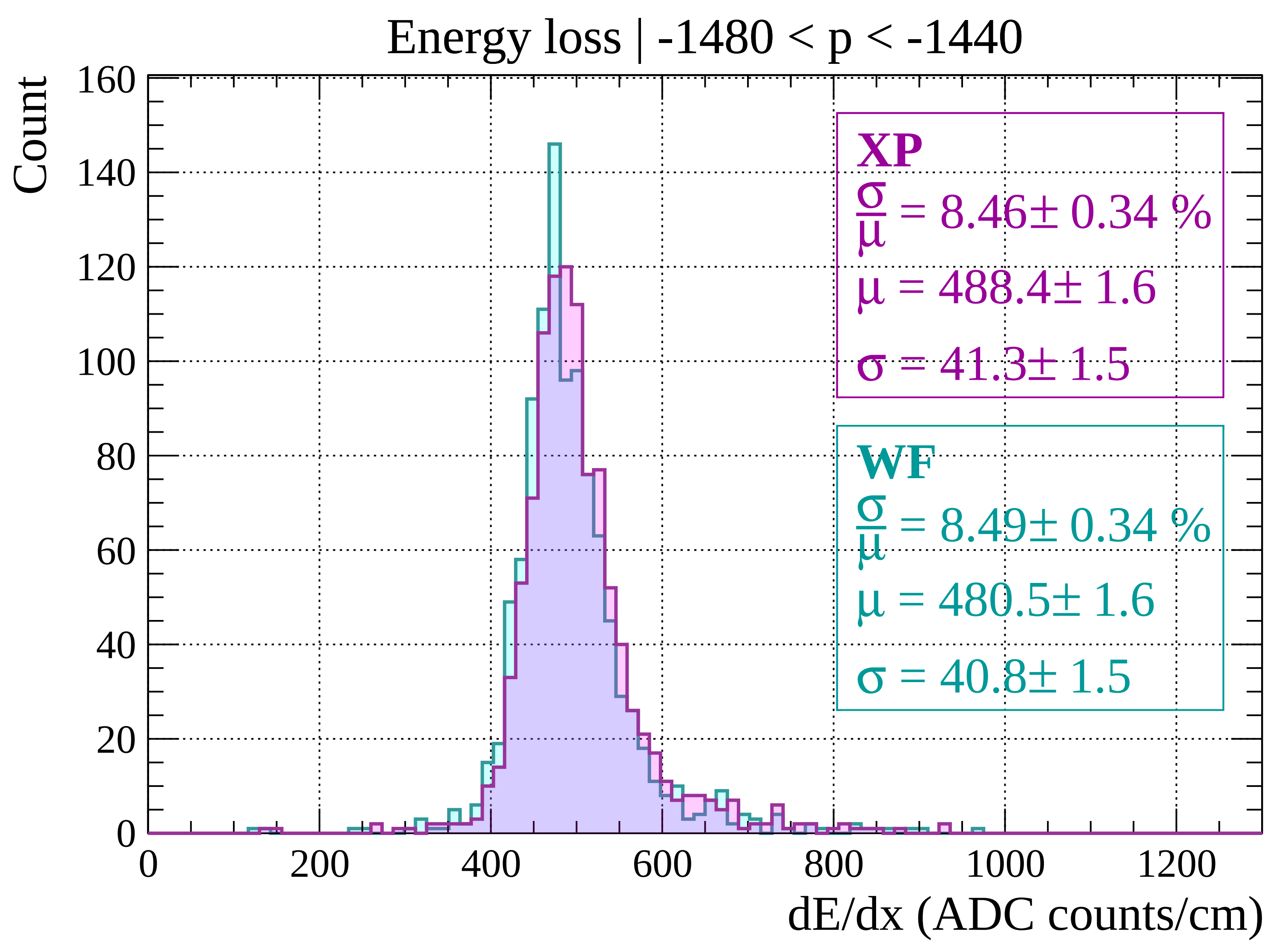
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.59 ± 0.42 %
μ = 493.5 ± 1.7
σ = 42.4 ± 1.9
WF
σ/μ = 8.79 ± 0.37 %
μ = 485.7 ± 1.6
σ = 42.7 ± 1.6


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.09± 0.29 %
μ = 490.8± 1.5
σ = 39.7± 1.3
WF
σ/μ = 7.78± 0.30 %
μ = 483.4± 1.4
σ = 37.6± 1.3

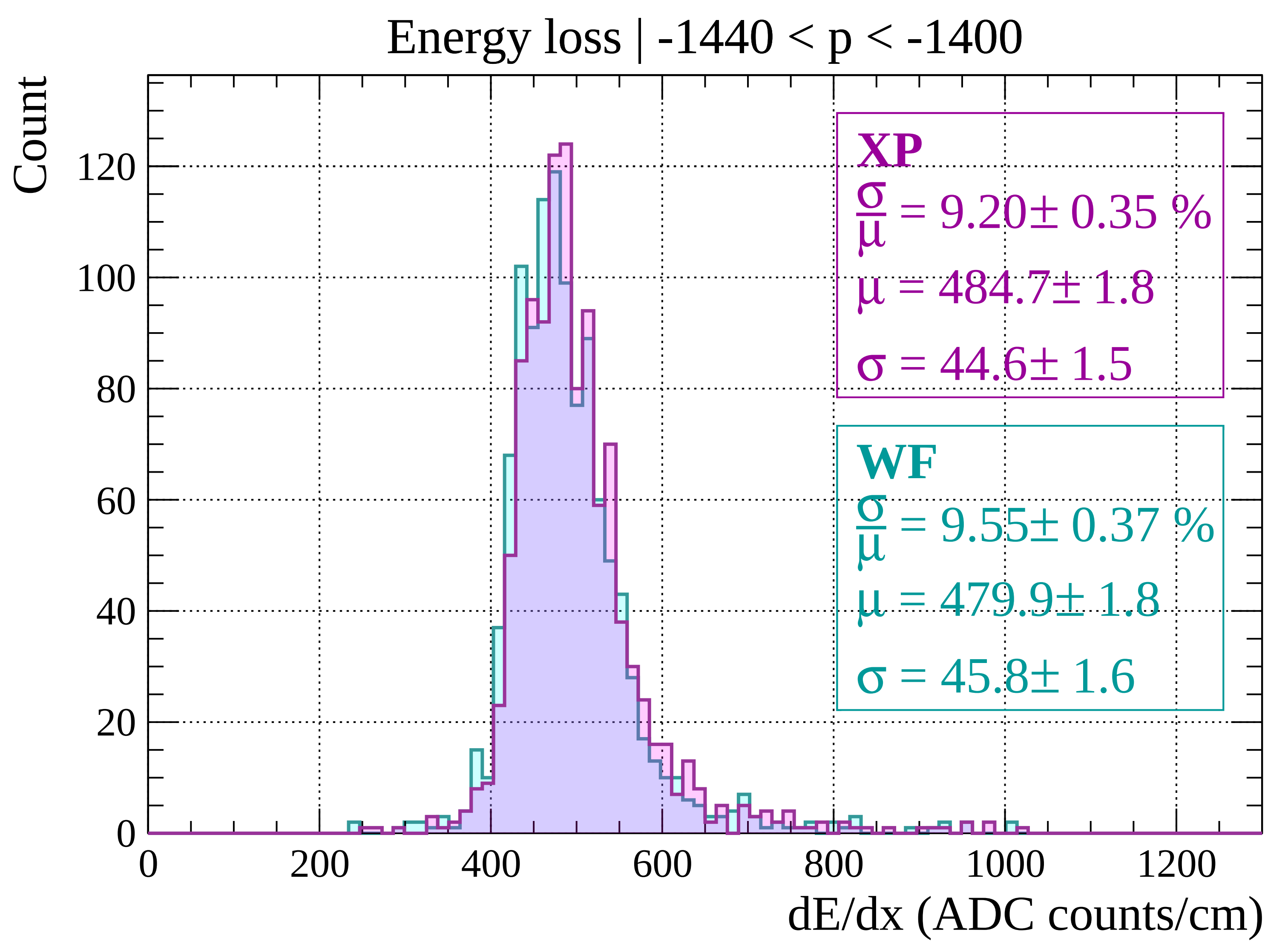


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.69± 0.35 %
μ = 491.1± 1.6
σ = 42.7± 1.6
WF
σ/μ = 8.64± 0.30 %
μ = 484.5± 1.5
σ = 41.9± 1.3
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

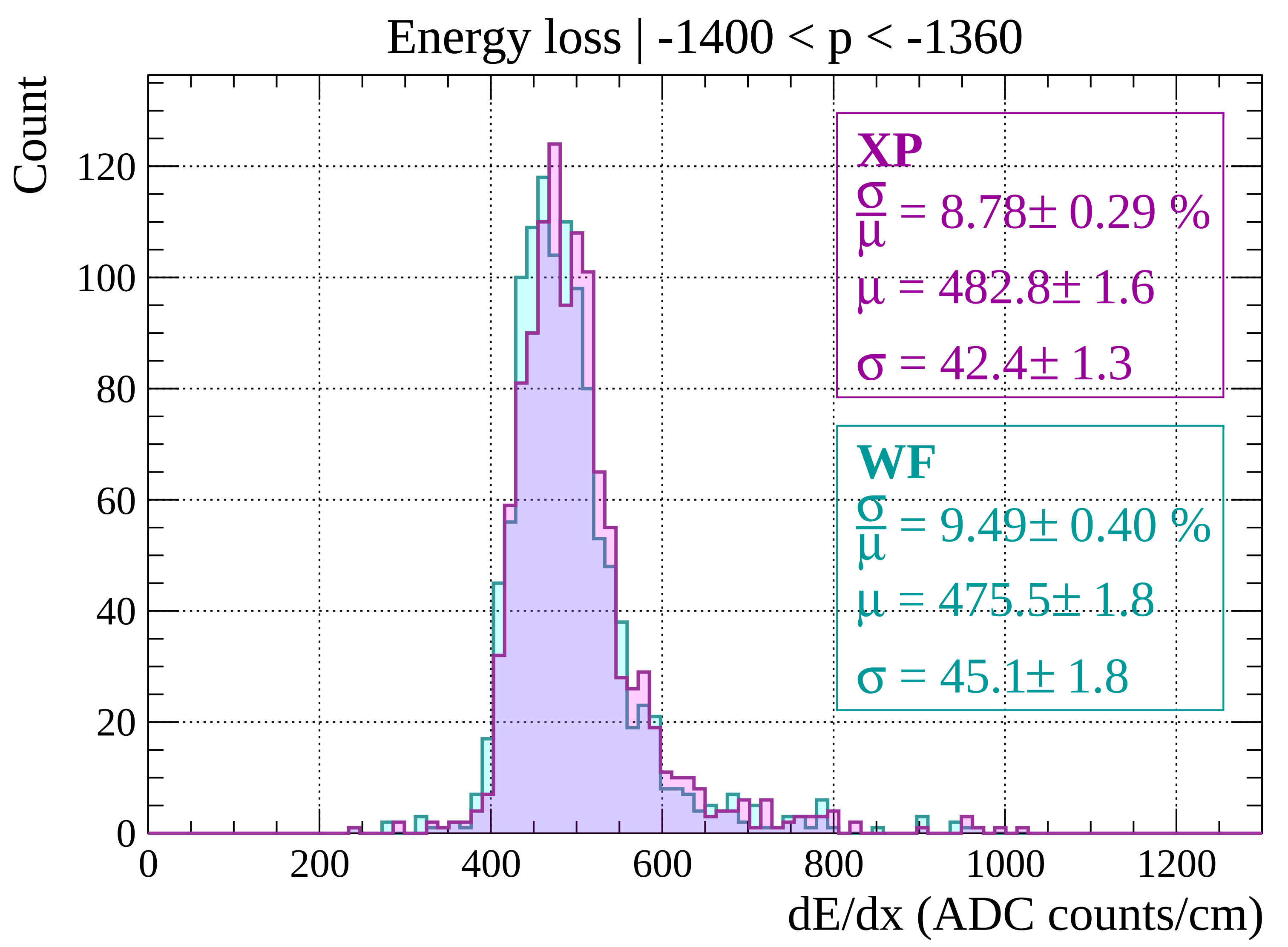

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.87± 0.32 %
μ = 487.5± 1.6
σ = 43.2± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.72± 0.33 %
μ = 481.0± 1.7
σ = 41.9± 1.5

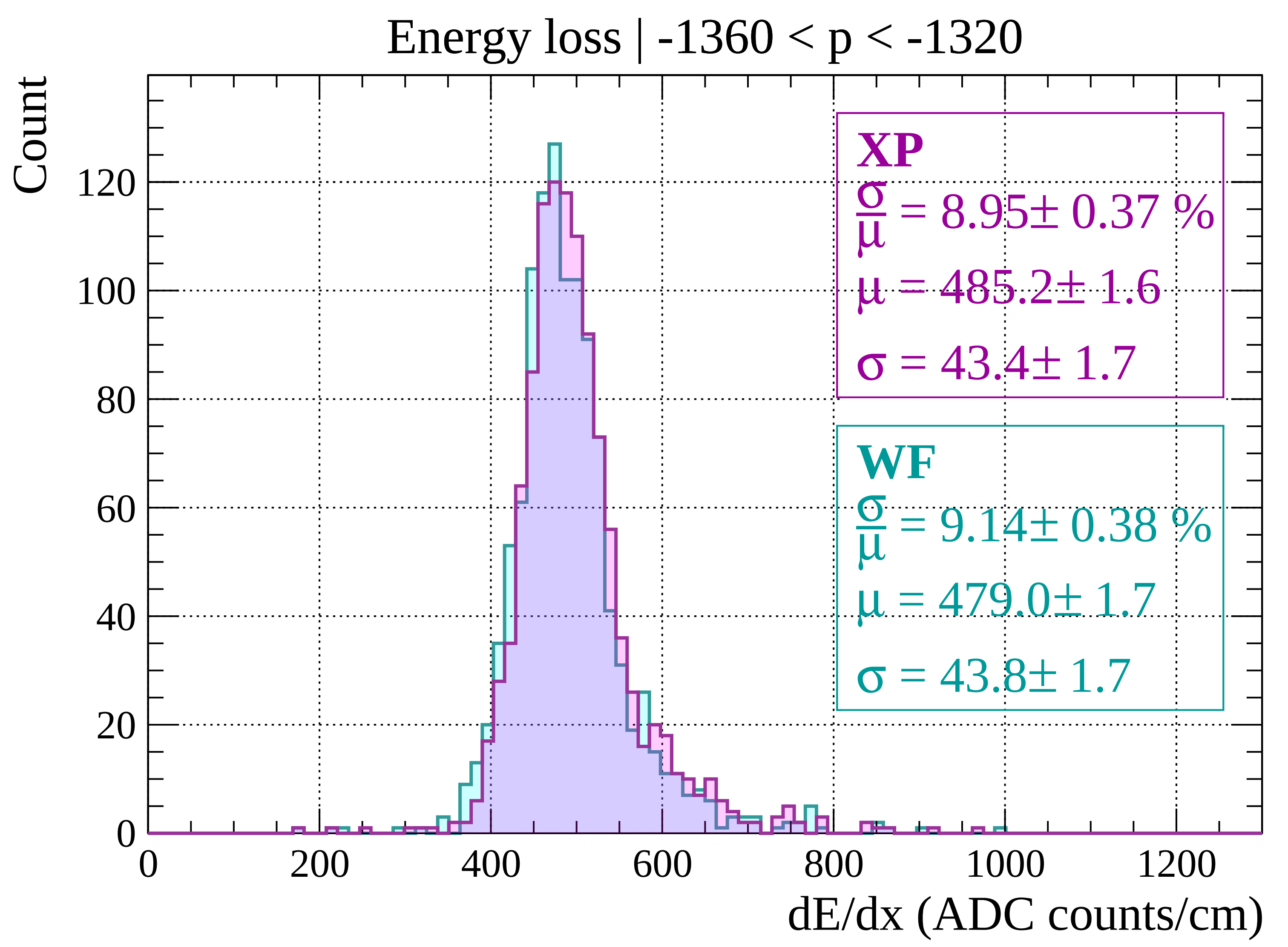


Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 8.46± 0.34 %
μ = 488.4± 1.6
σ = 41.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.49± 0.34 %
μ = 480.5± 1.6
σ = 40.8± 1.5

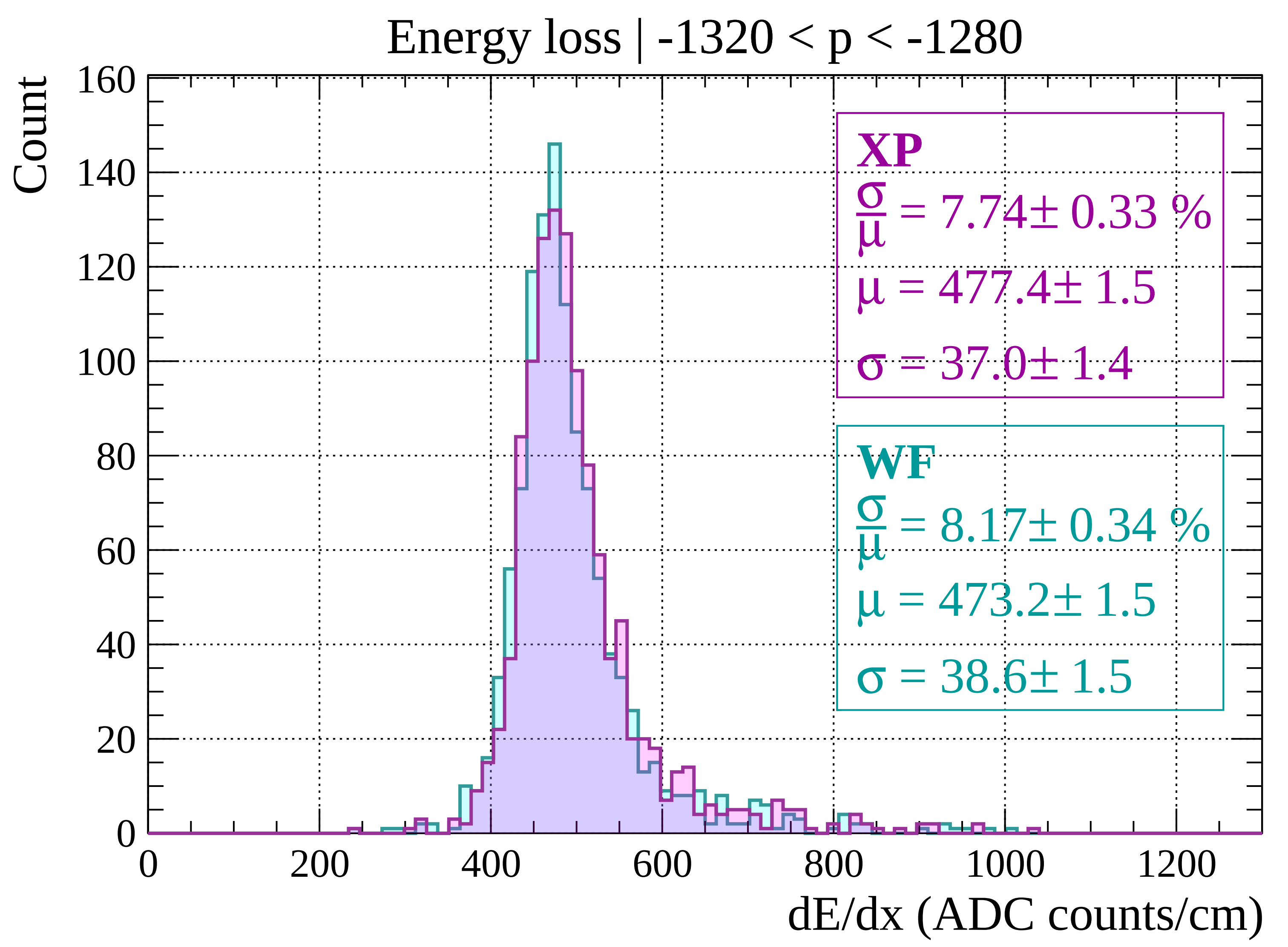


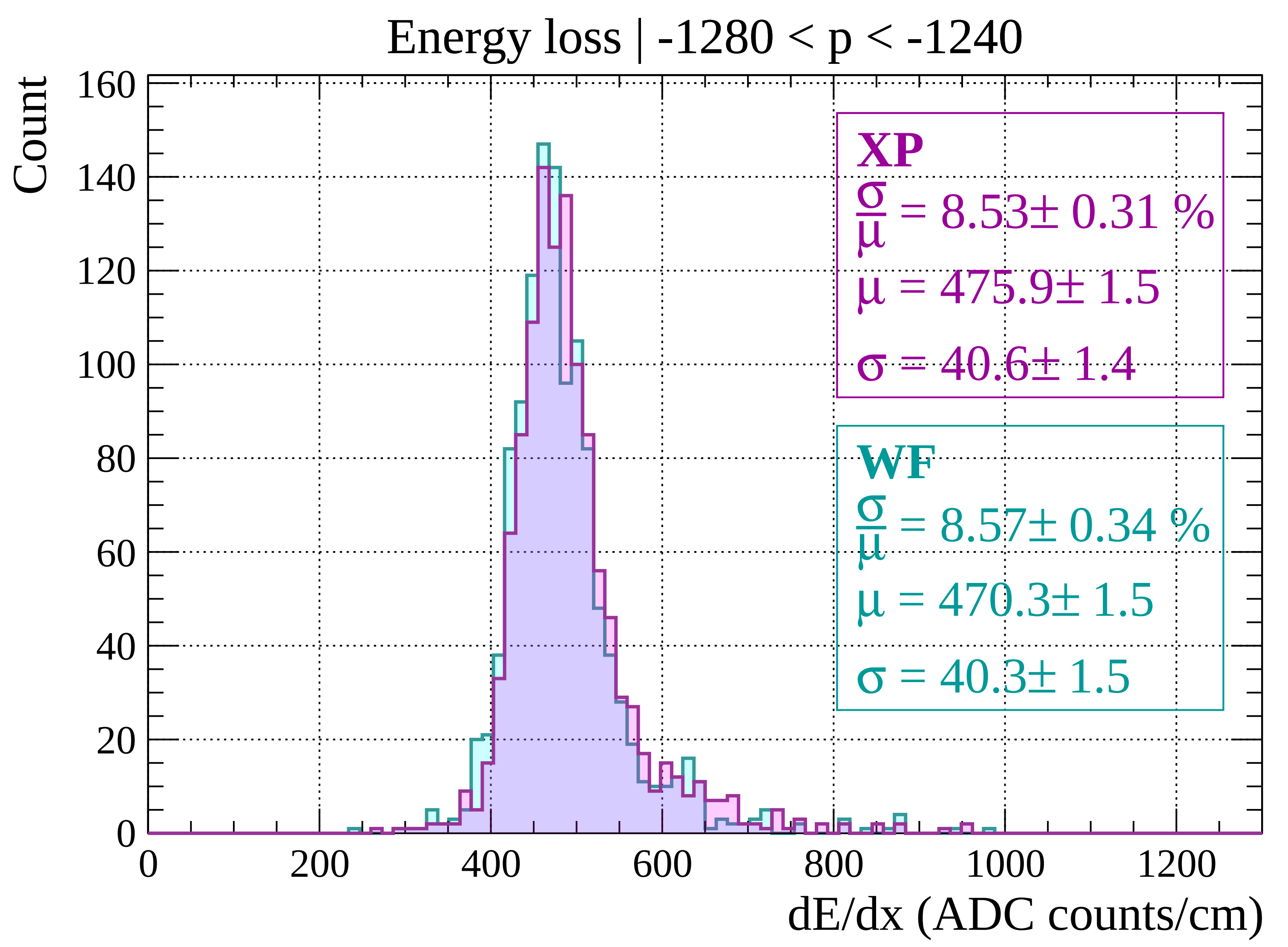
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 9.20± 0.35 %
μ = 484.7± 1.8
σ = 44.6± 1.5
WF
σ/μ = 9.55± 0.37 %
μ = 479.9± 1.8
σ = 45.8± 1.6

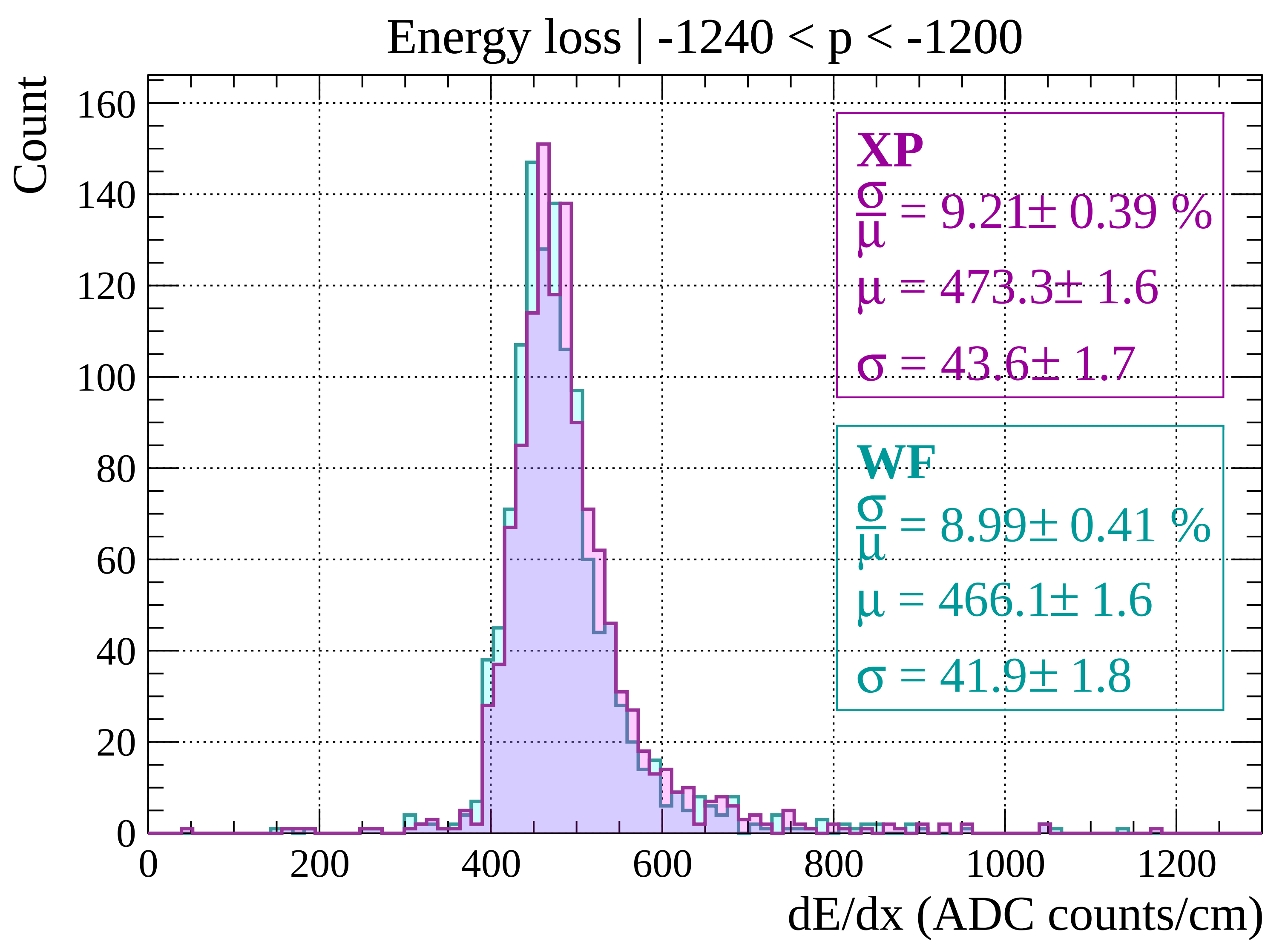

Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
σ/μ = 8.78± 0.29 %
μ = 482.8± 1.6
σ = 42.4± 1.3
WF
σ/μ = 9.49± 0.40 %
μ = 475.5± 1.8
σ = 45.1± 1.8



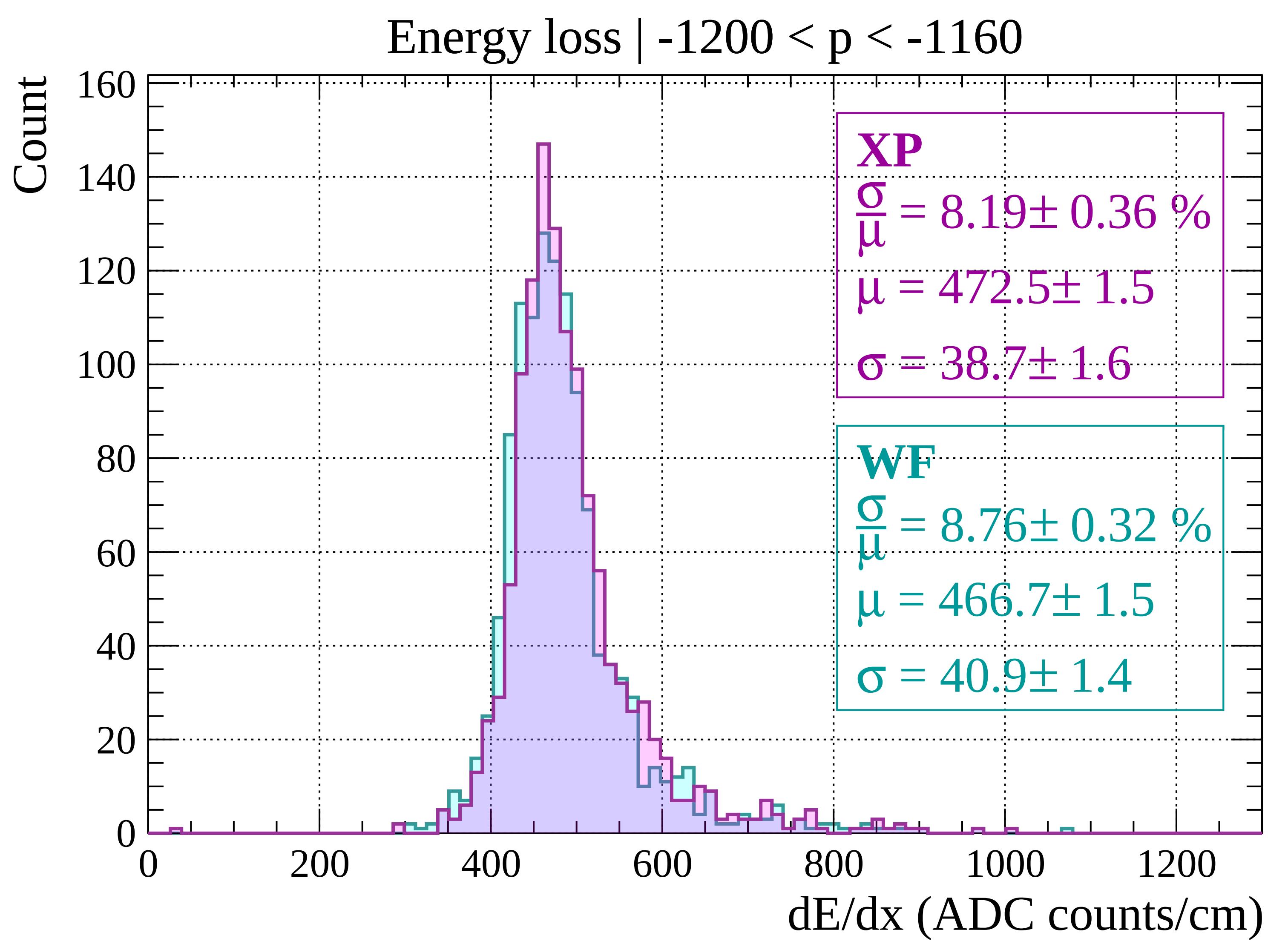
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.95± 0.37 %
μ = 485.2± 1.6
σ = 43.4± 1.7
WF
σ/μ = 9.14± 0.38 %
μ = 479.0± 1.7
σ = 43.8± 1.7


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.74± 0.33 %
μ = 477.4± 1.5
σ = 37.0± 1.4
WF
σ/μ = 8.17± 0.34 %
μ = 473.2± 1.5
σ = 38.6± 1.5

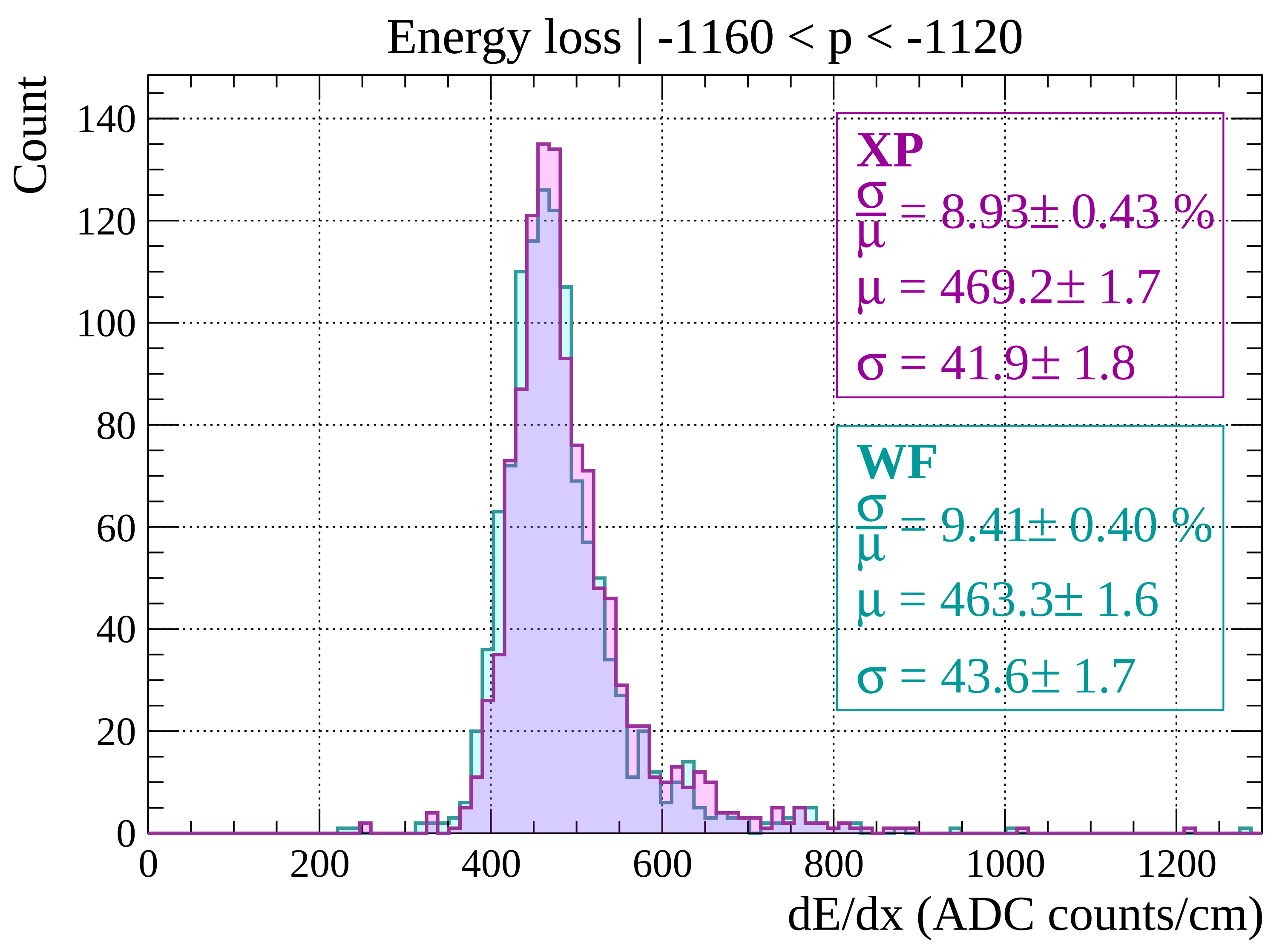

Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.53± 0.31 %
μ = 475.9± 1.5
σ = 40.6± 1.4
WF
σ/μ = 8.57± 0.34 %
μ = 470.3± 1.5
σ = 40.3± 1.5



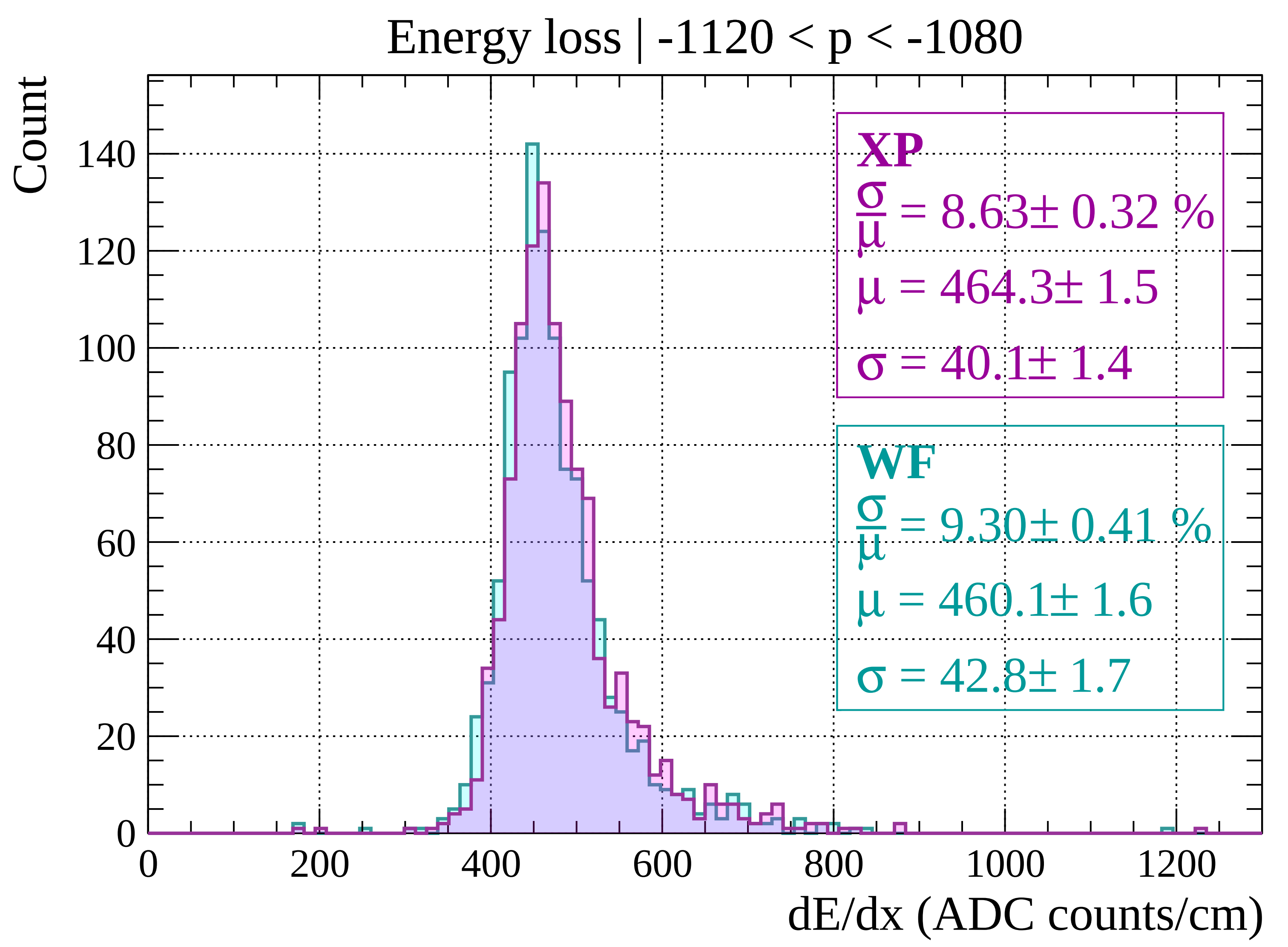
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.21± 0.39 %
μ = 473.3± 1.6
σ = 43.6± 1.7
WF
$\frac{\sigma}{\mu}$ = 8.99± 0.41 %
μ = 466.1± 1.6
σ = 41.9± 1.8
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



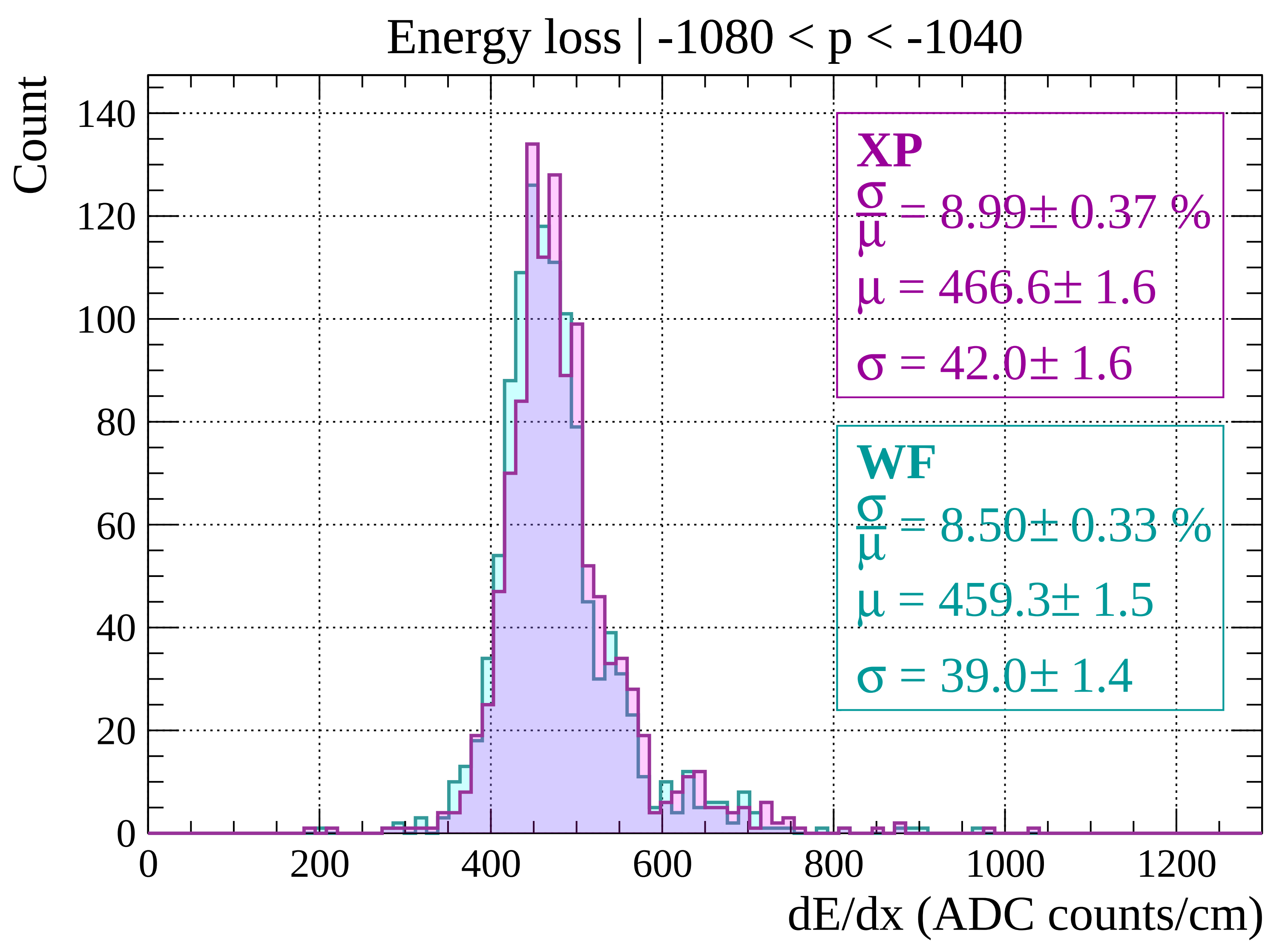
Energy loss | -1200 < p < -1160
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.19± 0.36 %
μ = 472.5± 1.5
σ = 38.7± 1.6
WF
σ/μ = 8.76± 0.32 %
μ = 466.7± 1.5
σ = 40.9± 1.4

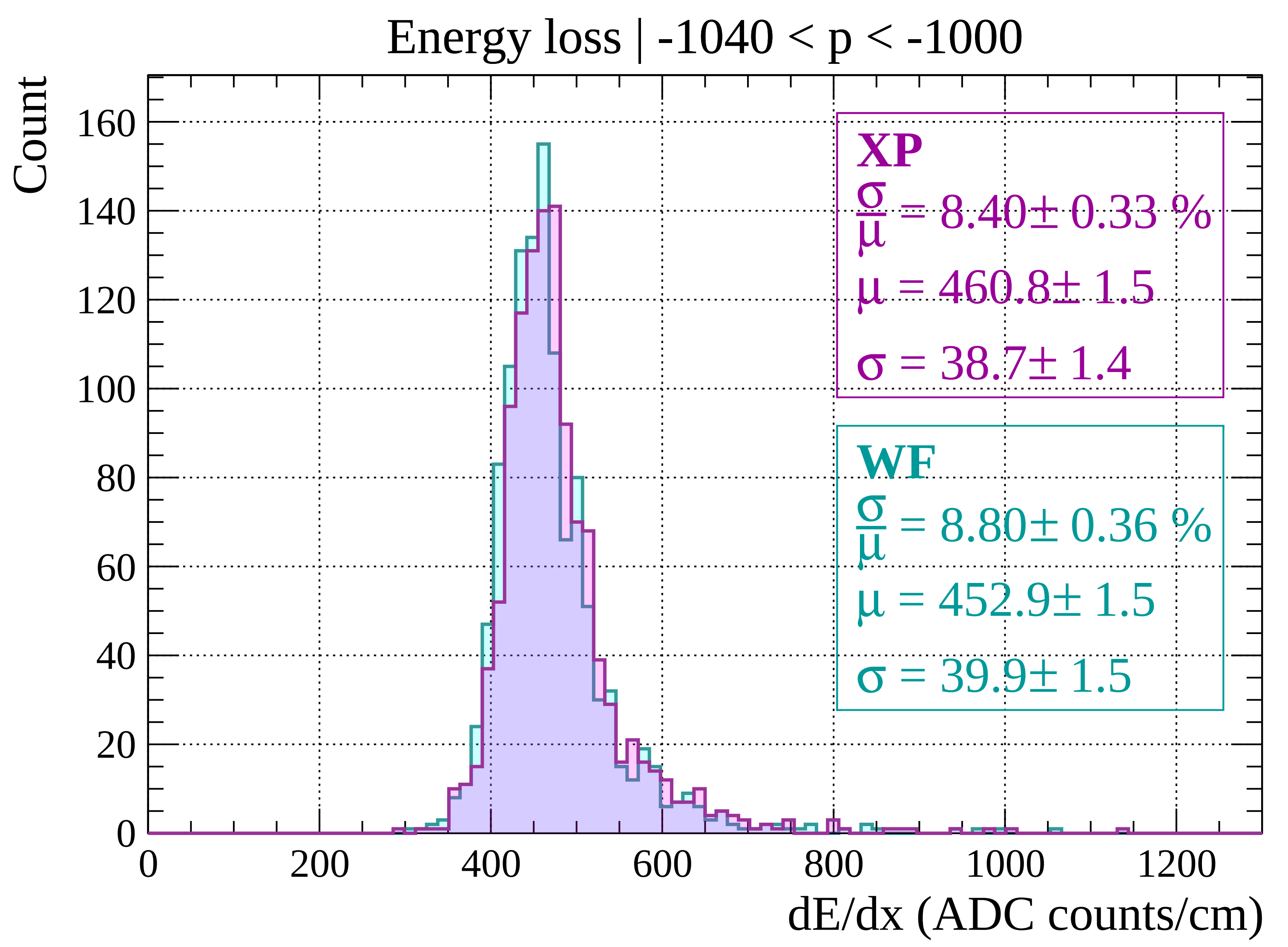


Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.93± 0.43 %
μ = 469.2± 1.7
σ = 41.9± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.41± 0.40 %
μ = 463.3± 1.6
σ = 43.6± 1.7

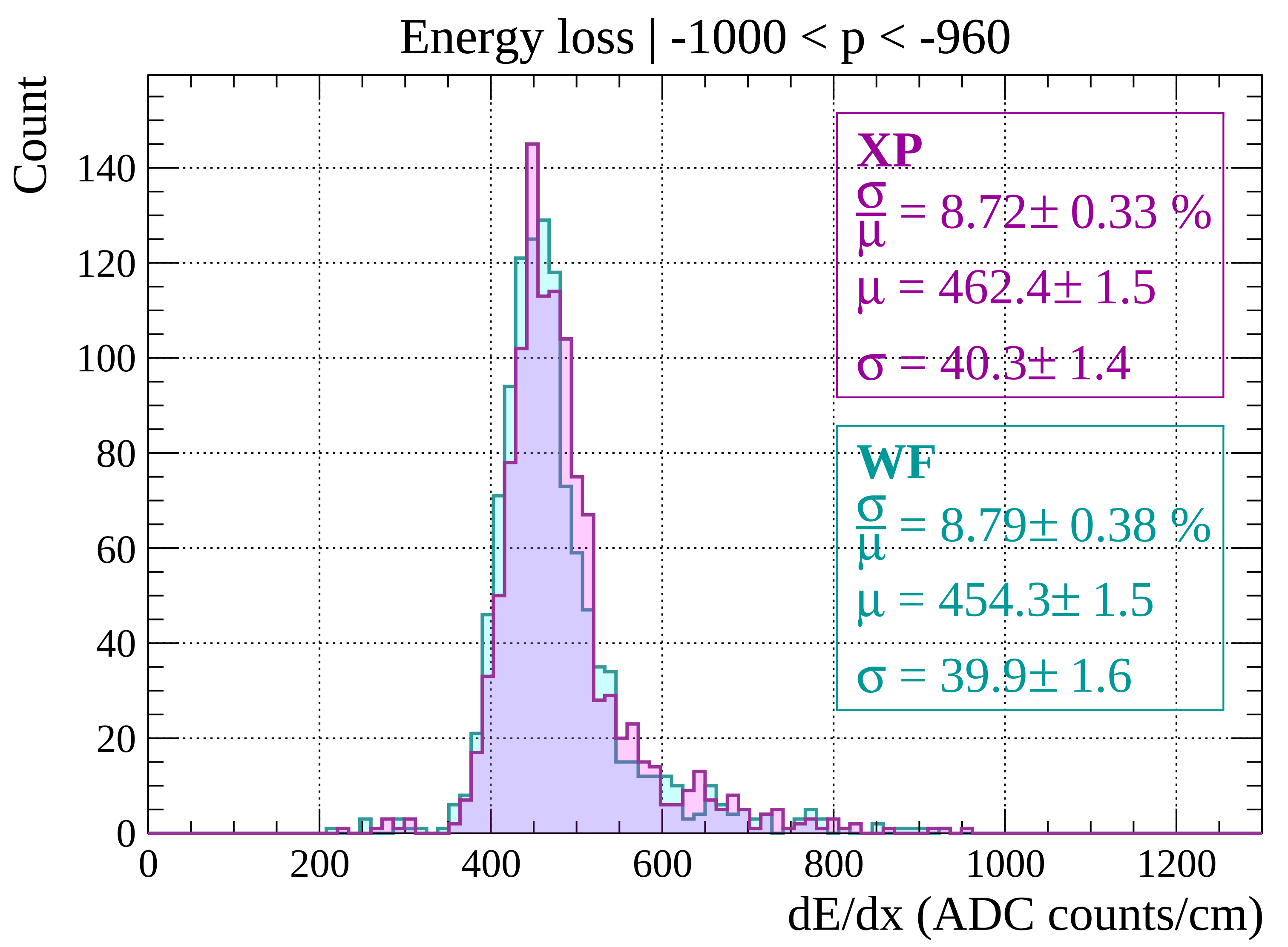


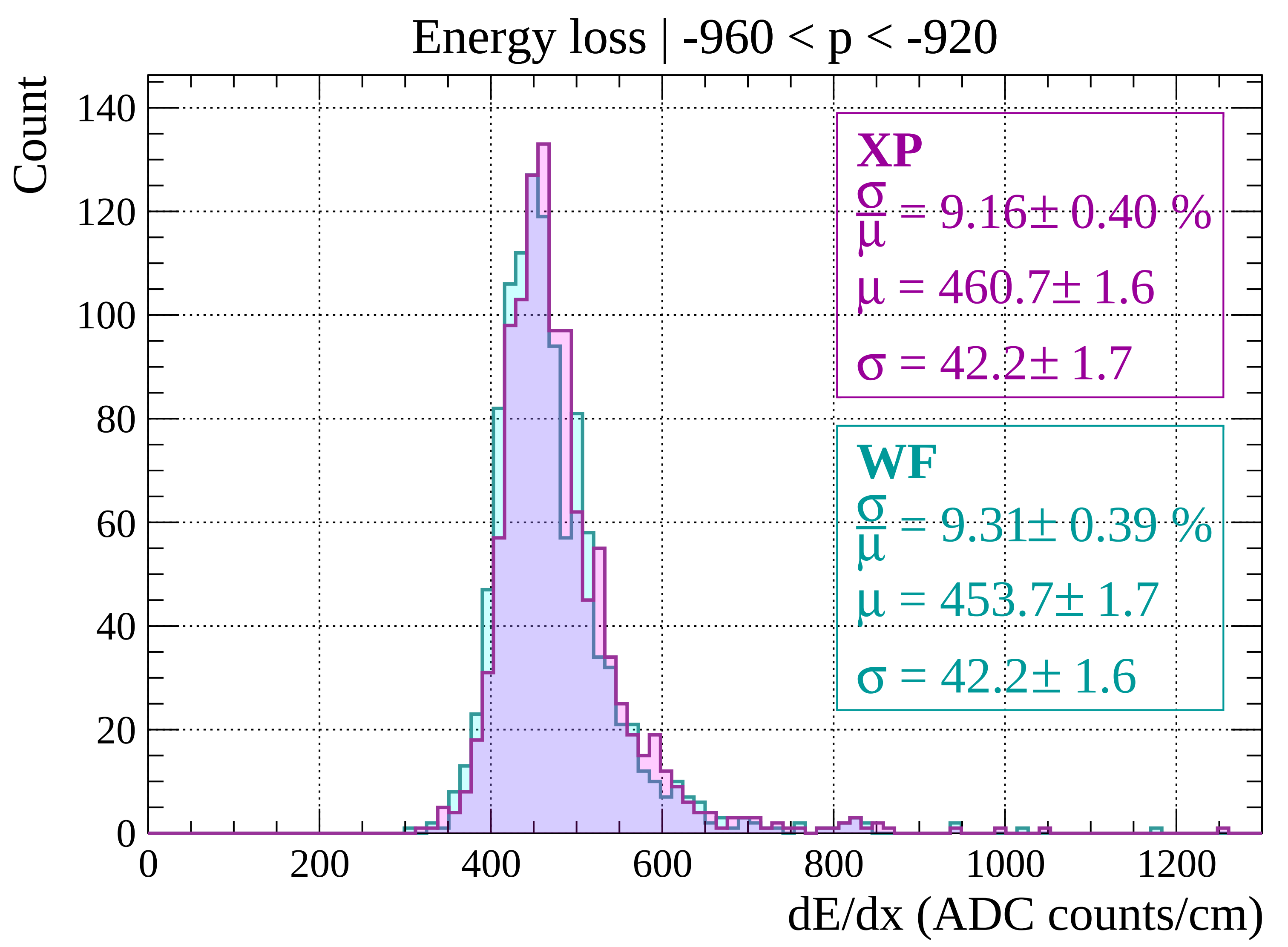
Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.63± 0.32 %
μ = 464.3± 1.5
σ = 40.1± 1.4
WF
$\frac{\sigma}{\mu}$ = 9.30± 0.41 %
μ = 460.1± 1.6
σ = 42.8± 1.7


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.99± 0.37 %
μ = 466.6± 1.6
σ = 42.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.33 %
μ = 459.3± 1.5
σ = 39.0± 1.4
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

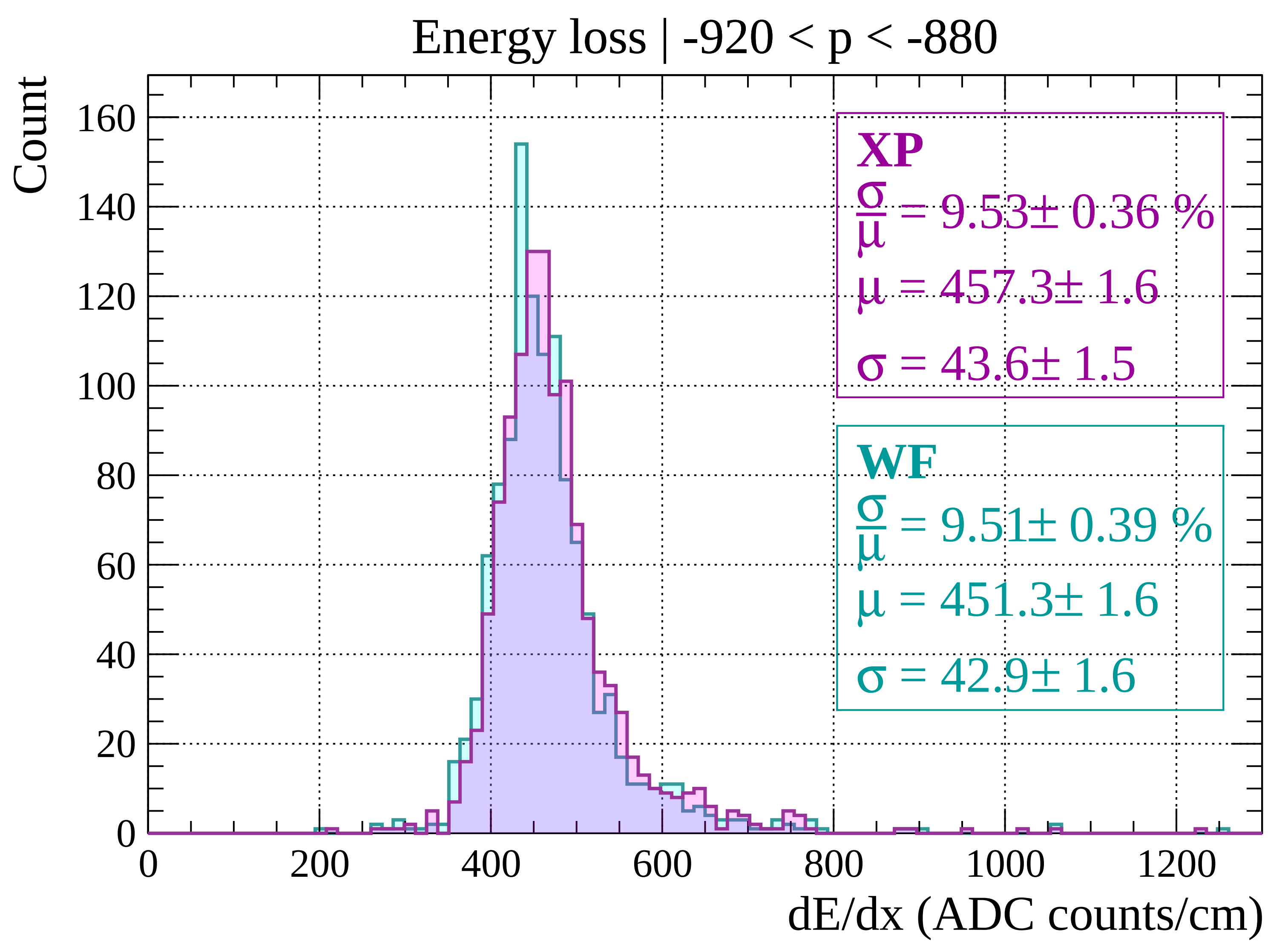


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.40± 0.33 %
μ = 460.8± 1.5
σ = 38.7± 1.4
WF
σ/μ = 8.80± 0.36 %
μ = 452.9± 1.5
σ = 39.9± 1.5

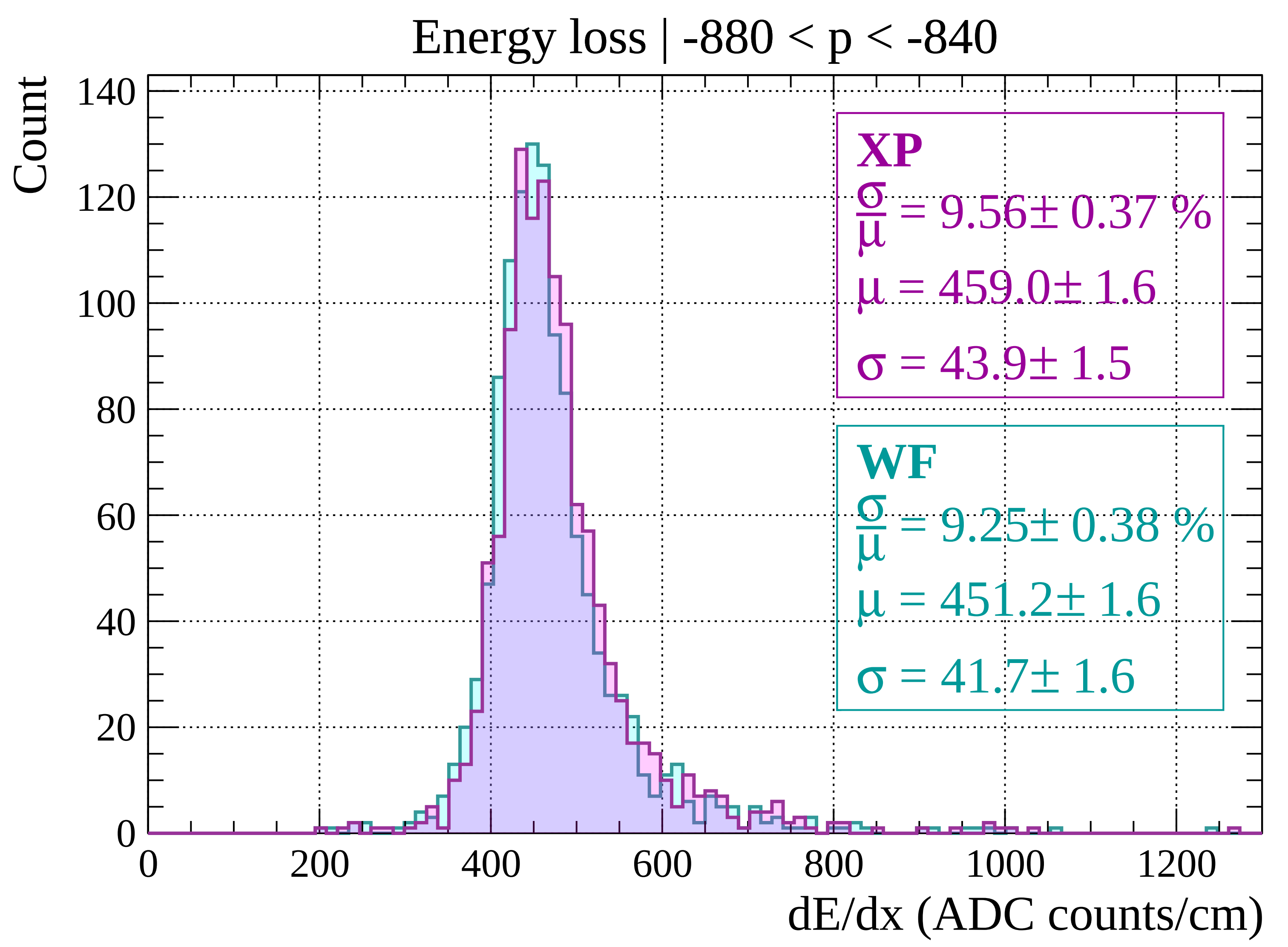

Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 8.72± 0.33 %
μ = 462.4± 1.5
σ = 40.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.79± 0.38 %
μ = 454.3± 1.5
σ = 39.9± 1.6



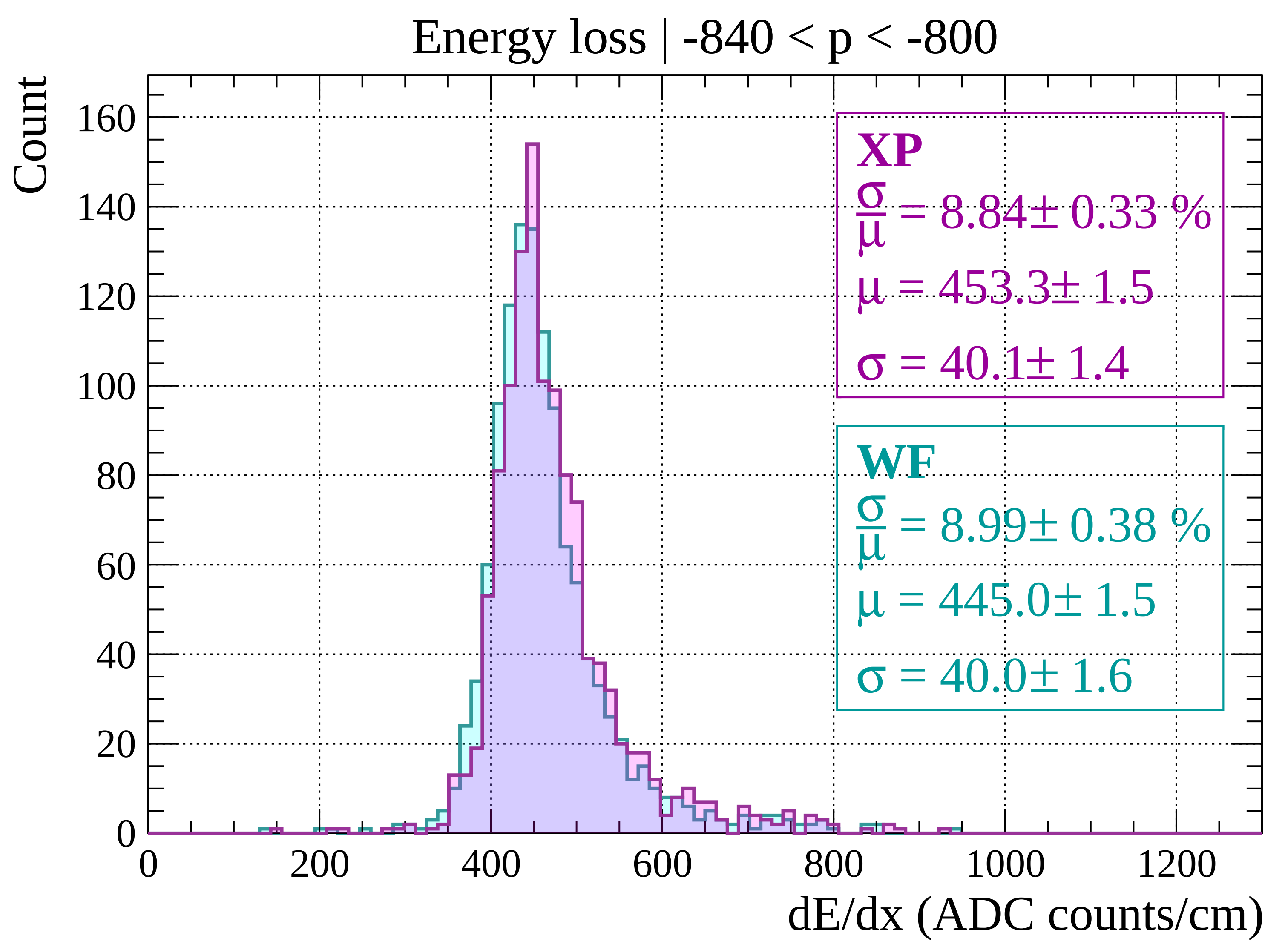
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.16 \pm 0.40$ %
$\mu = 460.7 \pm 1.6$
$\sigma = 42.2 \pm 1.7$
WF
$\frac{\sigma}{\mu} = 9.31 \pm 0.39$ %
$\mu = 453.7 \pm 1.7$
$\sigma = 42.2 \pm 1.6$
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140

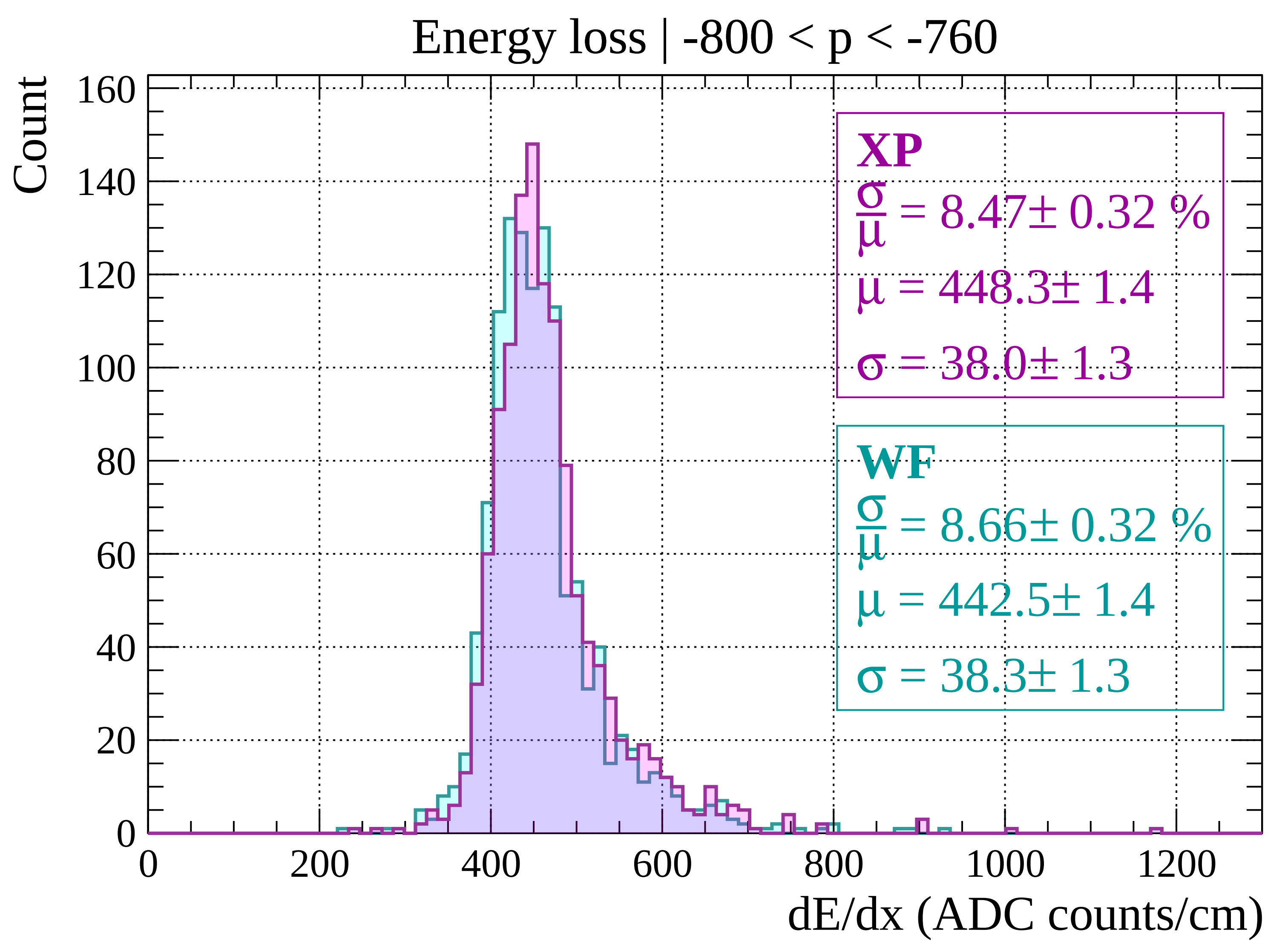


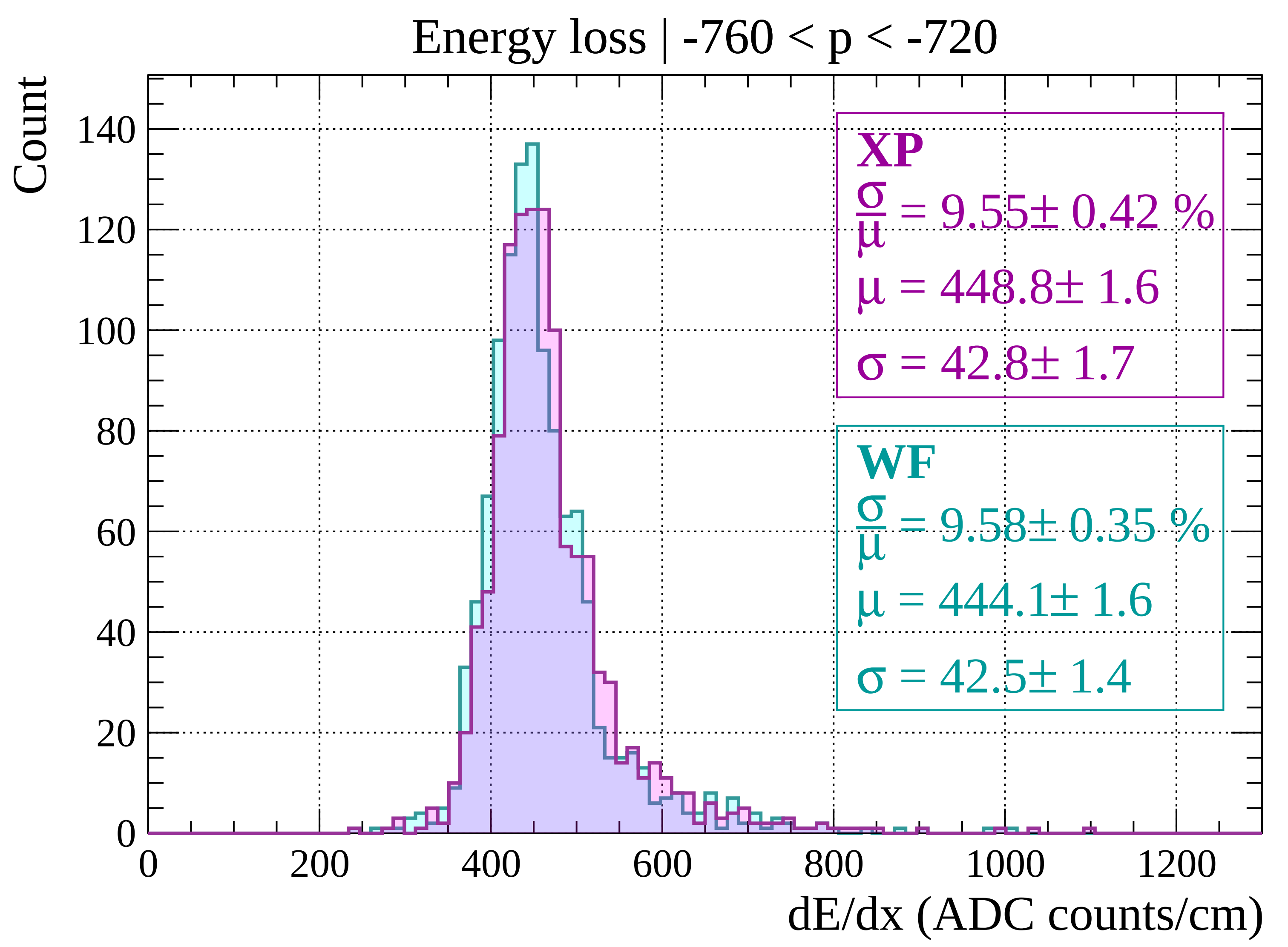
Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.53± 0.36 %
μ = 457.3± 1.6
σ = 43.6± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.51± 0.39 %
μ = 451.3± 1.6
σ = 42.9± 1.6

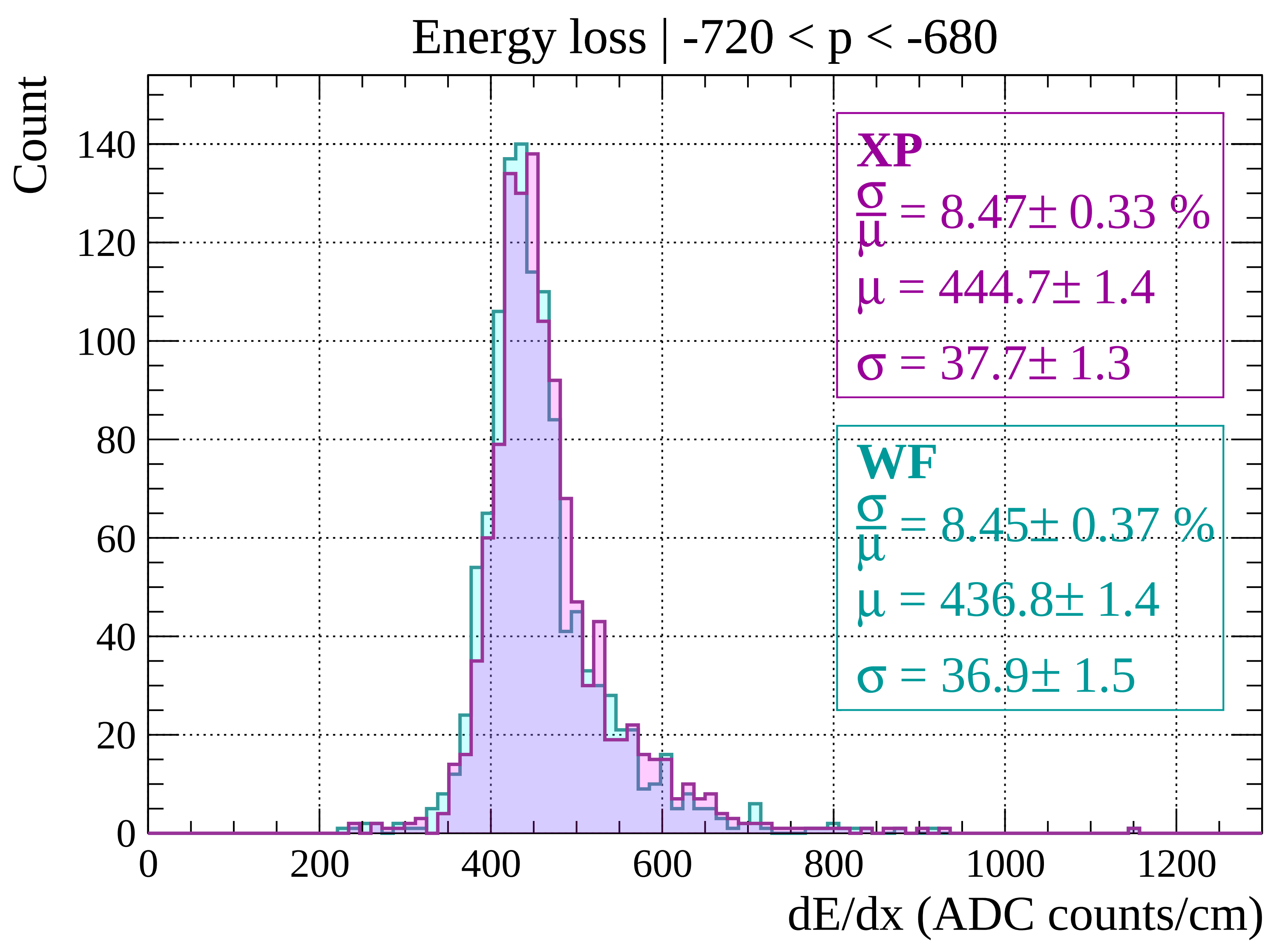


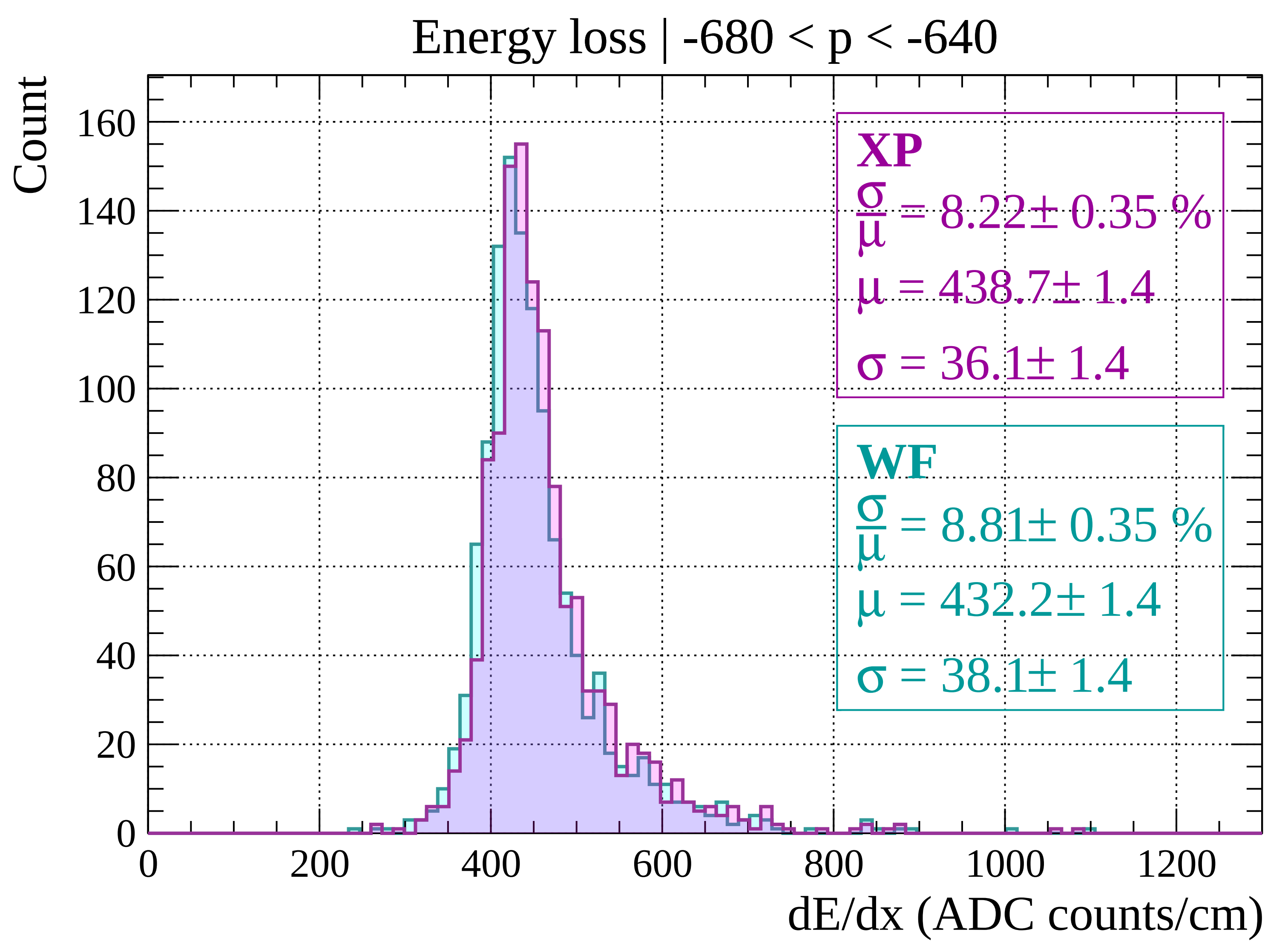
Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 9.56± 0.37 %
μ = 459.0± 1.6
σ = 43.9± 1.5
WF
σ/μ = 9.25± 0.38 %
μ = 451.2± 1.6
σ = 41.7± 1.6

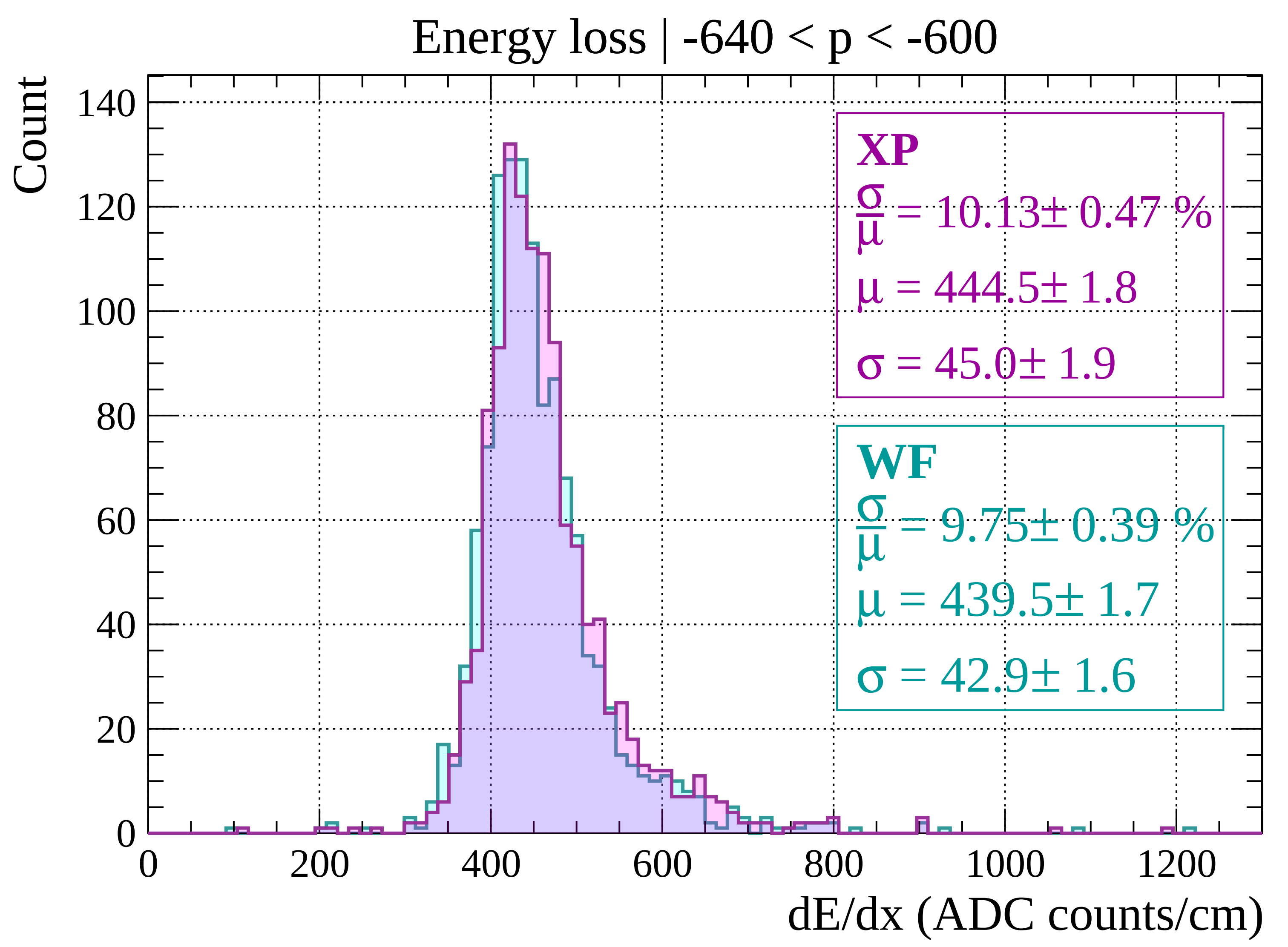

Energy loss | -840 < p < -800
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.84± 0.33 %
μ = 453.3± 1.5
σ = 40.1± 1.4
WF
σ/μ = 8.99± 0.38 %
μ = 445.0± 1.5
σ = 40.0± 1.6


Energy loss | -800 < p < -760
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.47± 0.32 %
μ = 448.3± 1.4
σ = 38.0± 1.3
WF
σ/μ = 8.66± 0.32 %
μ = 442.5± 1.4
σ = 38.3± 1.3

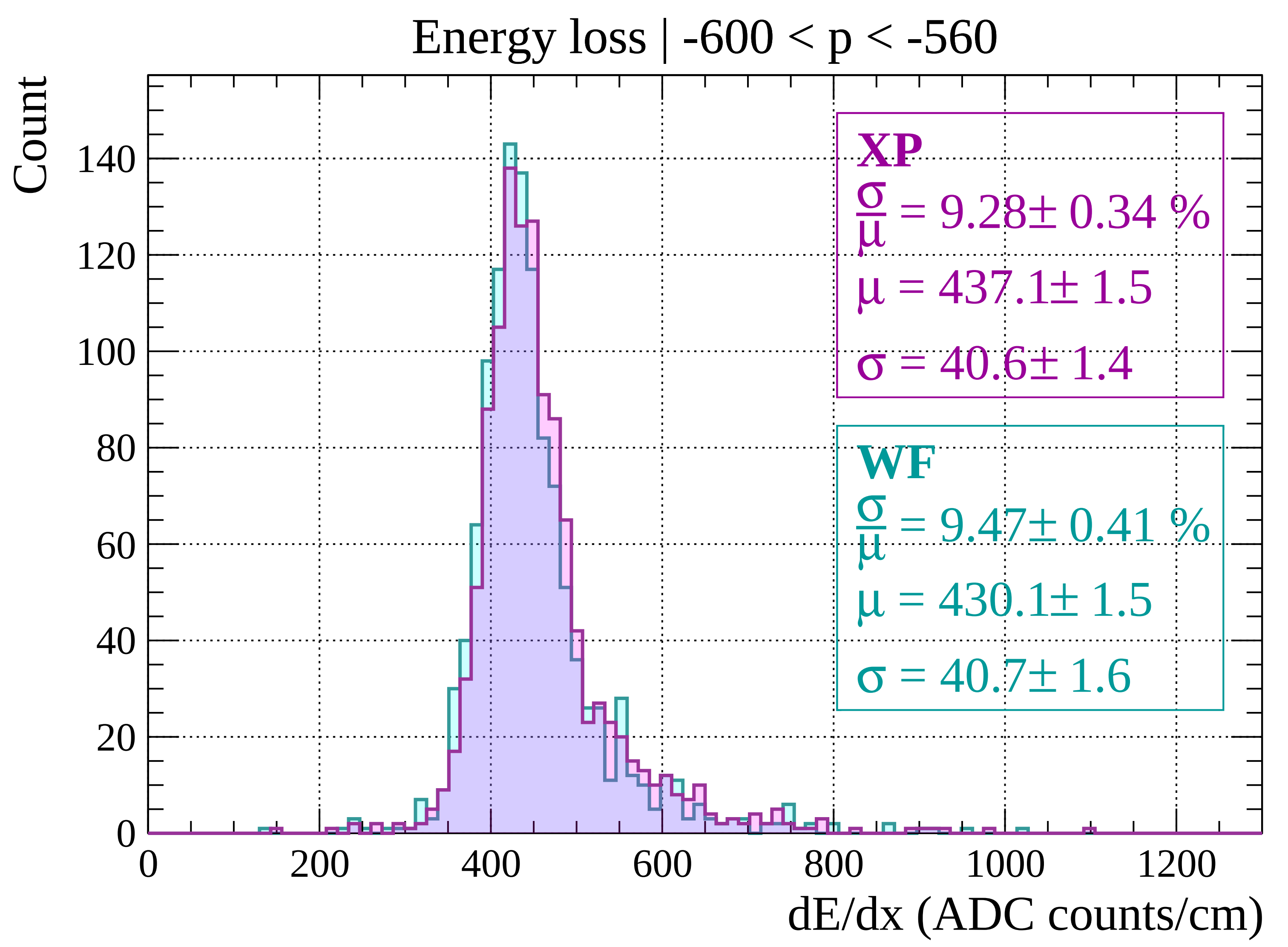

Energy loss | -760 < p < -720
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.55± 0.42 %
μ = 448.8± 1.6
σ = 42.8± 1.7
WF
σ/μ = 9.58± 0.35 %
μ = 444.1± 1.6
σ = 42.5± 1.4


Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.47± 0.33 %
μ = 444.7± 1.4
σ = 37.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.45± 0.37 %
μ = 436.8± 1.4
σ = 36.9± 1.5

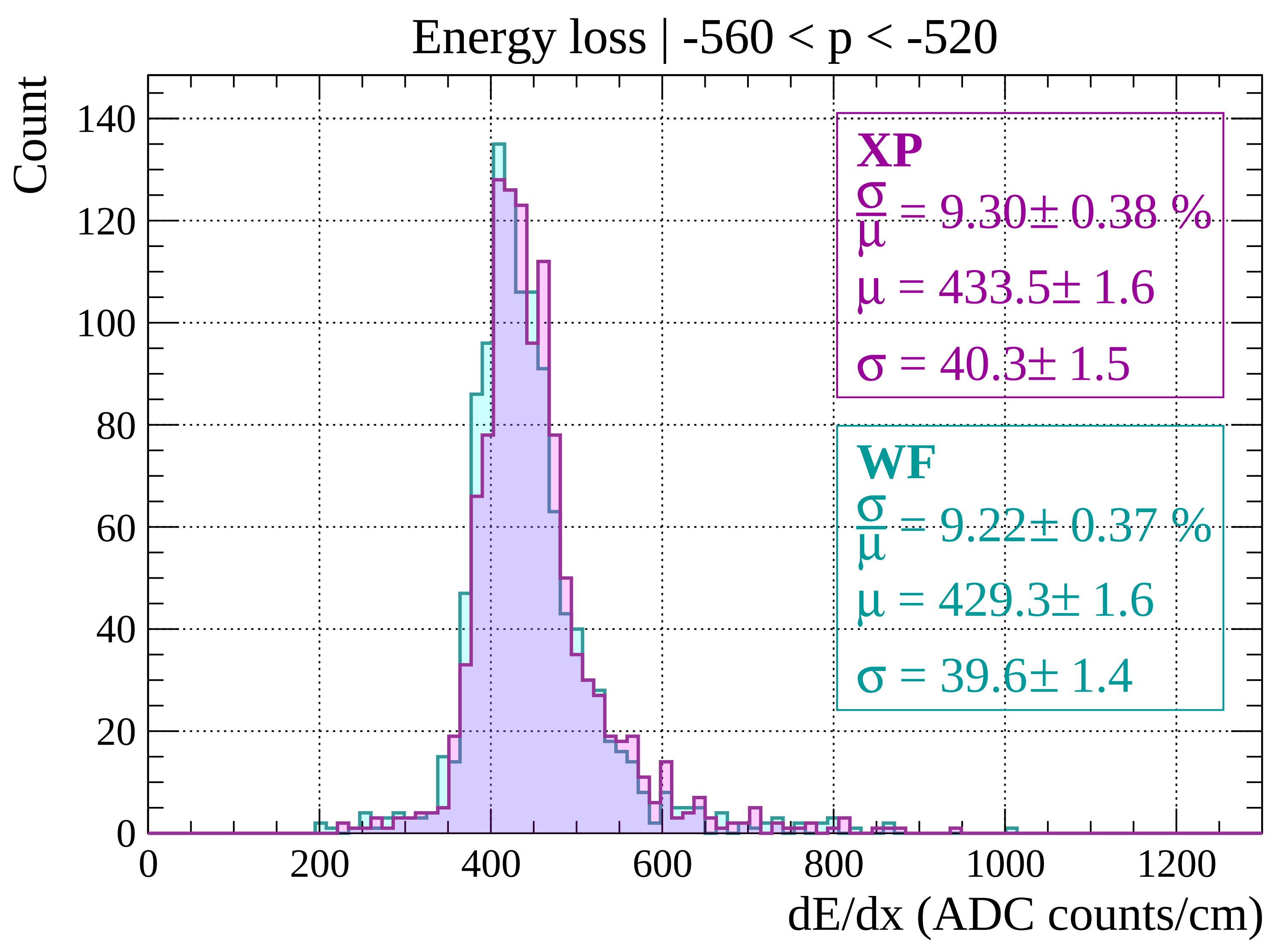

Energy loss | -680 < p < -640
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.22± 0.35 %
μ = 438.7± 1.4
σ = 36.1± 1.4
WF
σ/μ = 8.81± 0.35 %
μ = 432.2± 1.4
σ = 38.1± 1.4



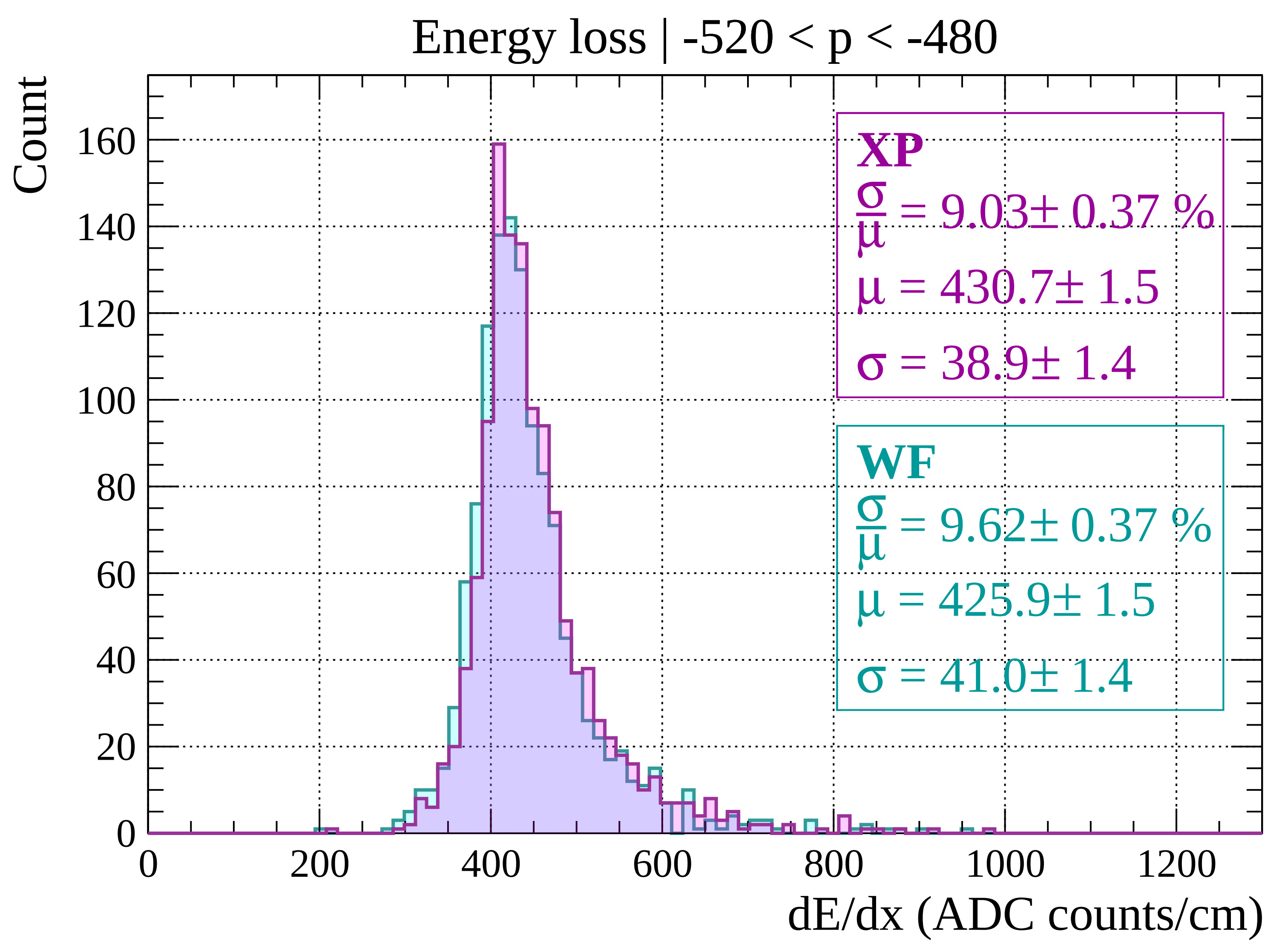
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 10.13± 0.47 %
μ = 444.5± 1.8
σ = 45.0± 1.9
WF
σ/μ = 9.75± 0.39 %
μ = 439.5± 1.7
σ = 42.9± 1.6



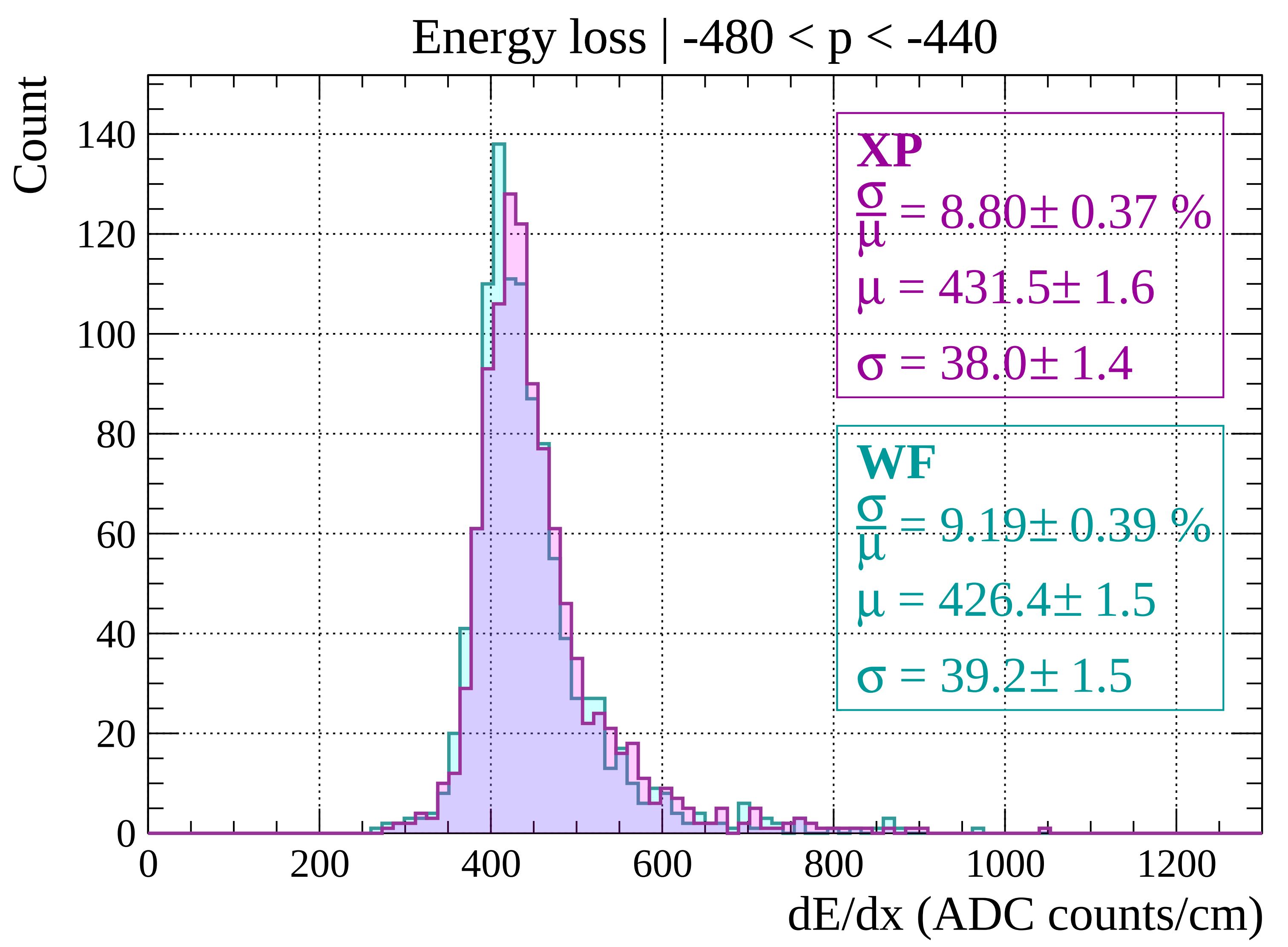
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.28± 0.34 %
μ = 437.1± 1.5
σ = 40.6± 1.4
WF
σ/μ = 9.47± 0.41 %
μ = 430.1± 1.5
σ = 40.7± 1.6
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



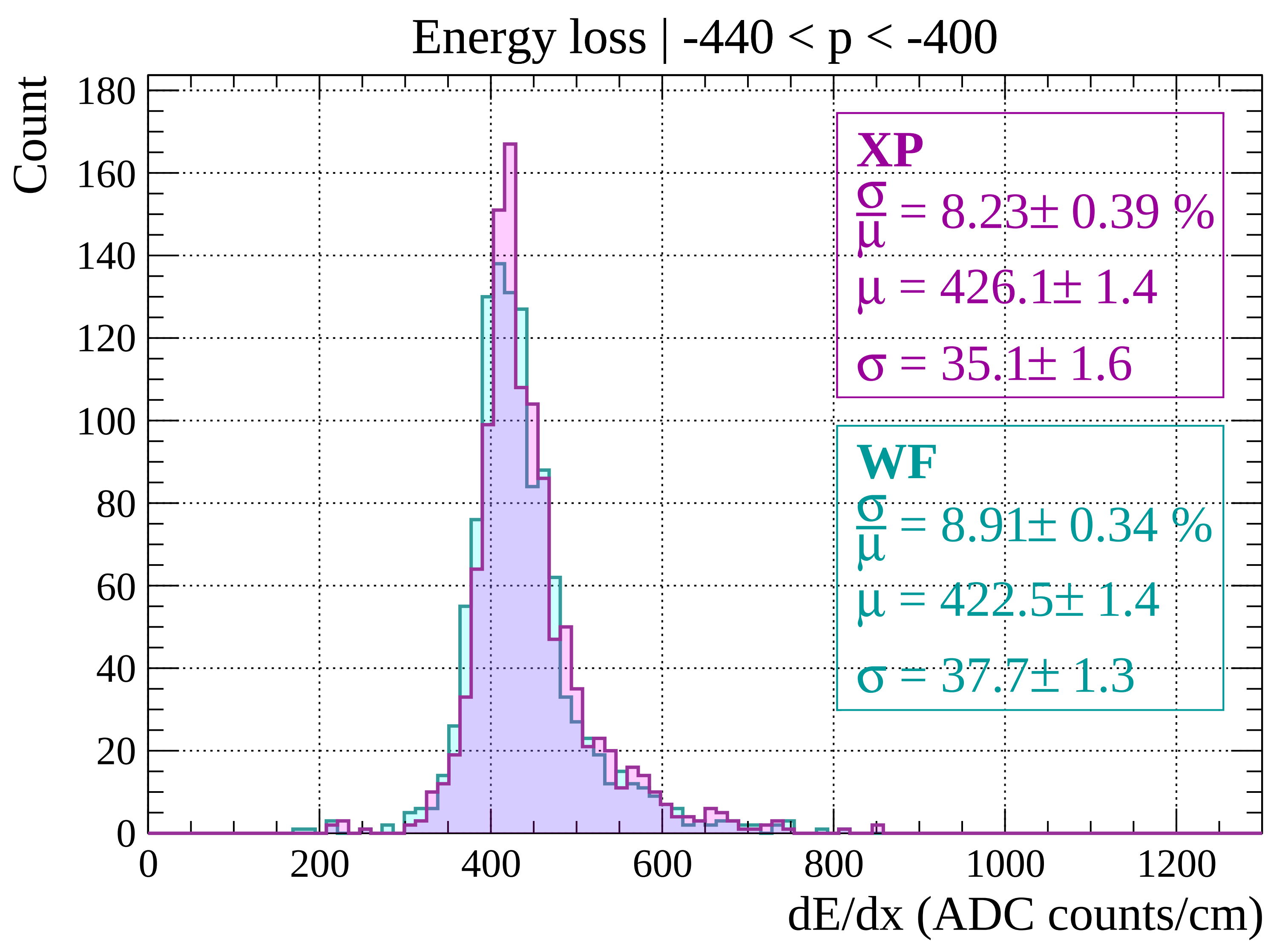
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.30± 0.38 %
μ = 433.5± 1.6
σ = 40.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.22± 0.37 %
μ = 429.3± 1.6
σ = 39.6± 1.4



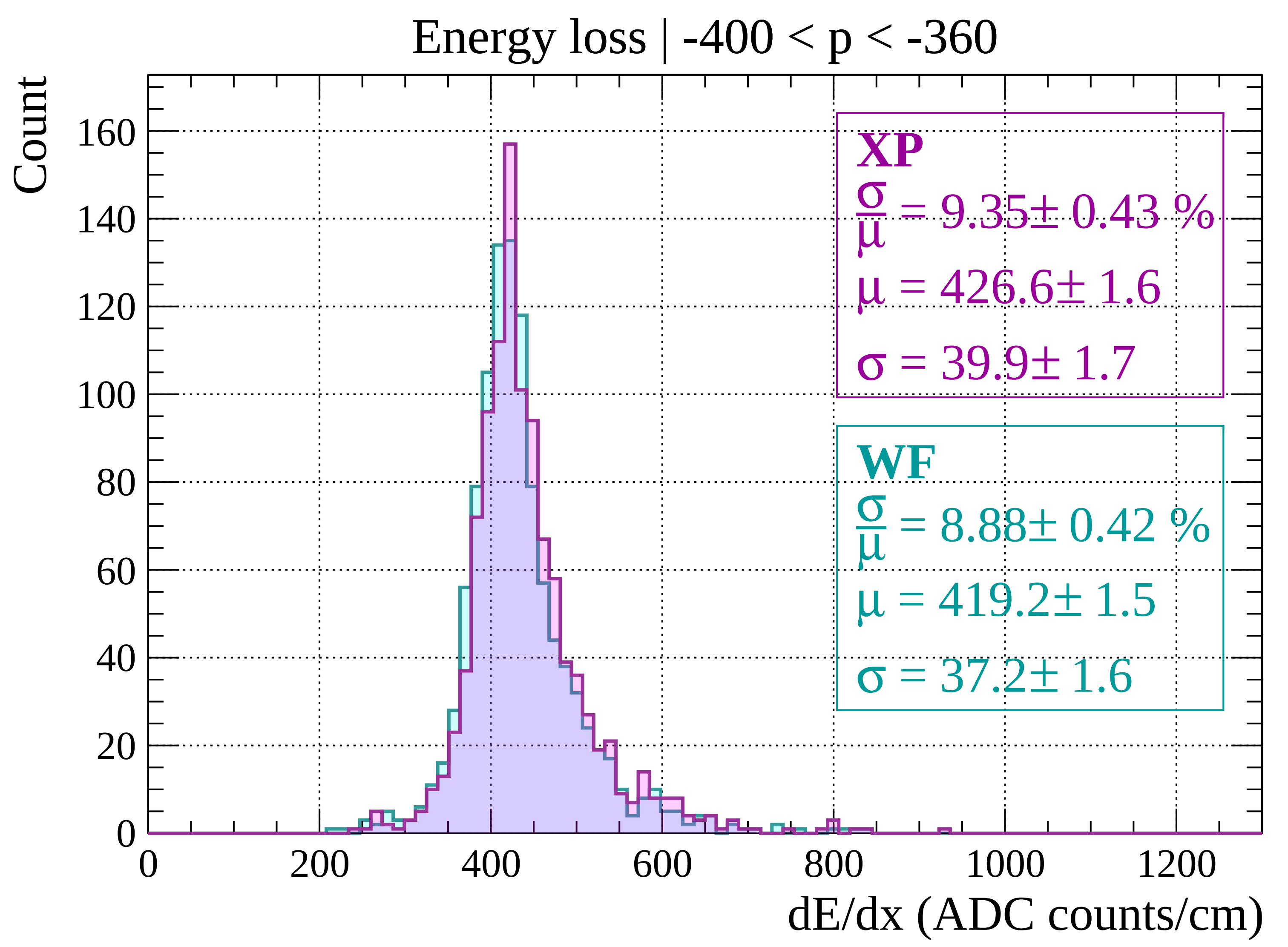
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 9.03± 0.37 %
μ = 430.7± 1.5
σ = 38.9± 1.4
WF
σ/μ = 9.62± 0.37 %
μ = 425.9± 1.5
σ = 41.0± 1.4

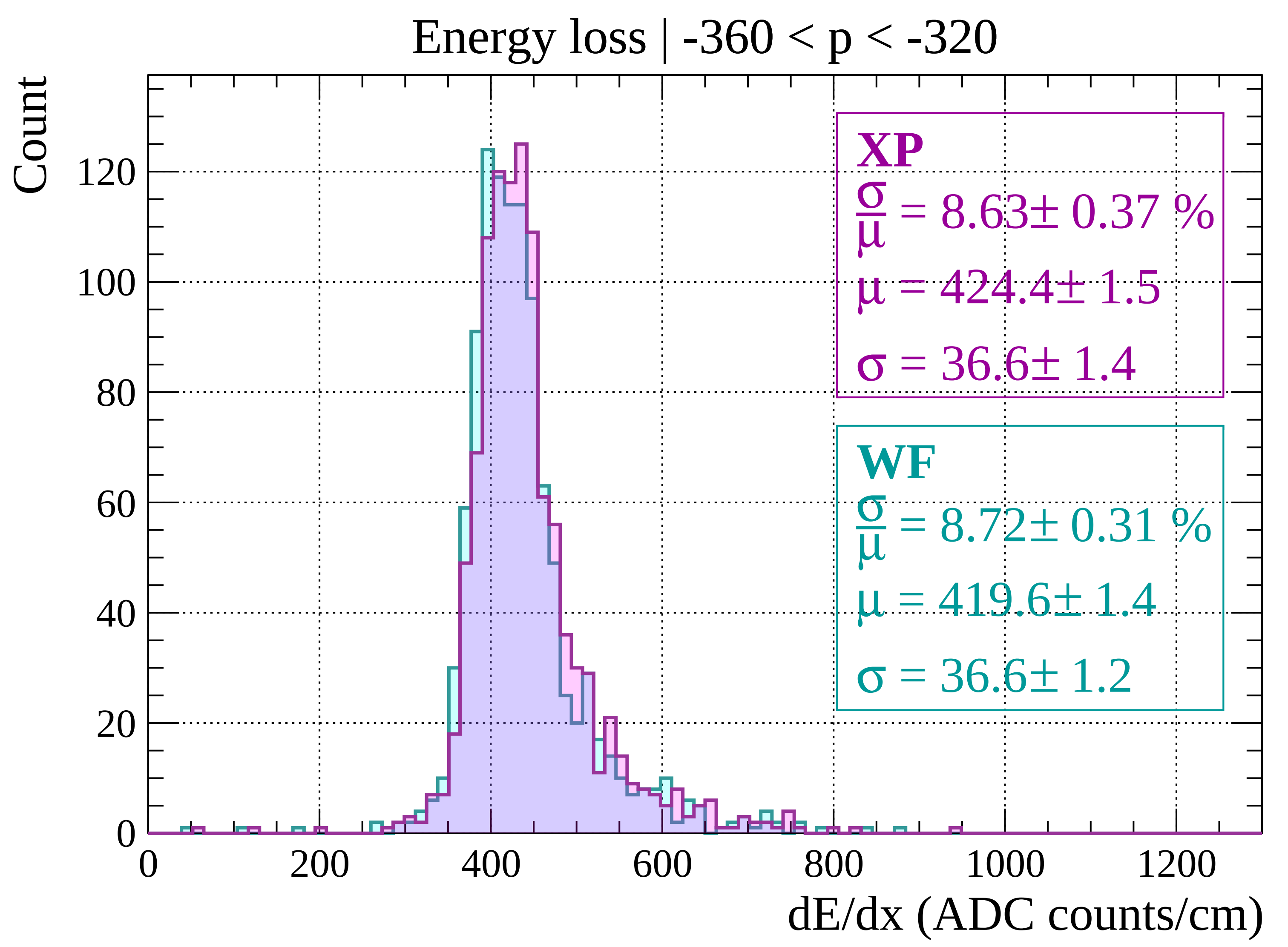


Energy loss | -480 < p < -440
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.80± 0.37 %
μ = 431.5± 1.6
σ = 38.0± 1.4
WF
σ/μ = 9.19± 0.39 %
μ = 426.4± 1.5
σ = 39.2± 1.5

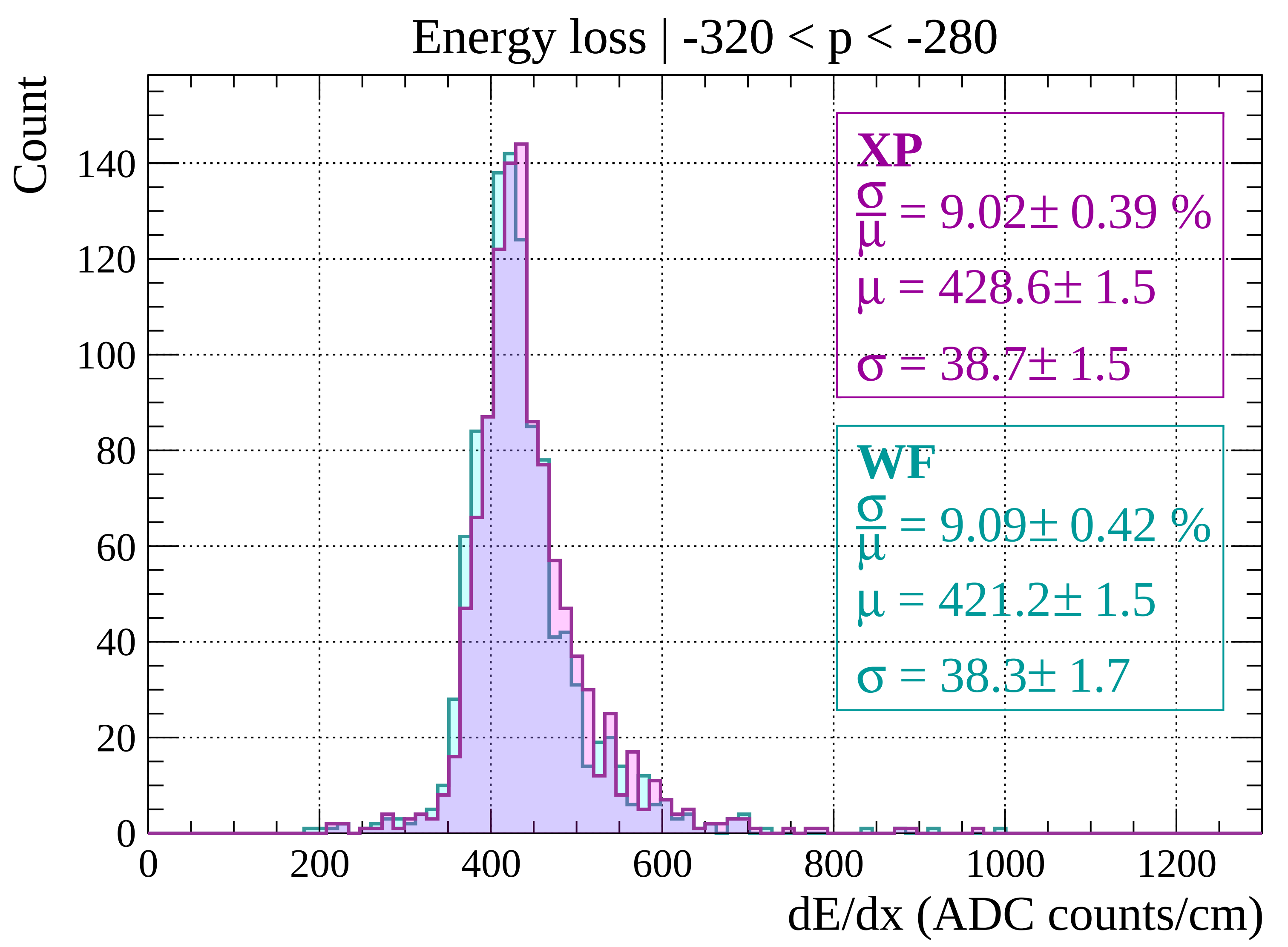


Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.23± 0.39 %
μ = 426.1± 1.4
σ = 35.1± 1.6
WF
σ/μ = 8.91± 0.34 %
μ = 422.5± 1.4
σ = 37.7± 1.3

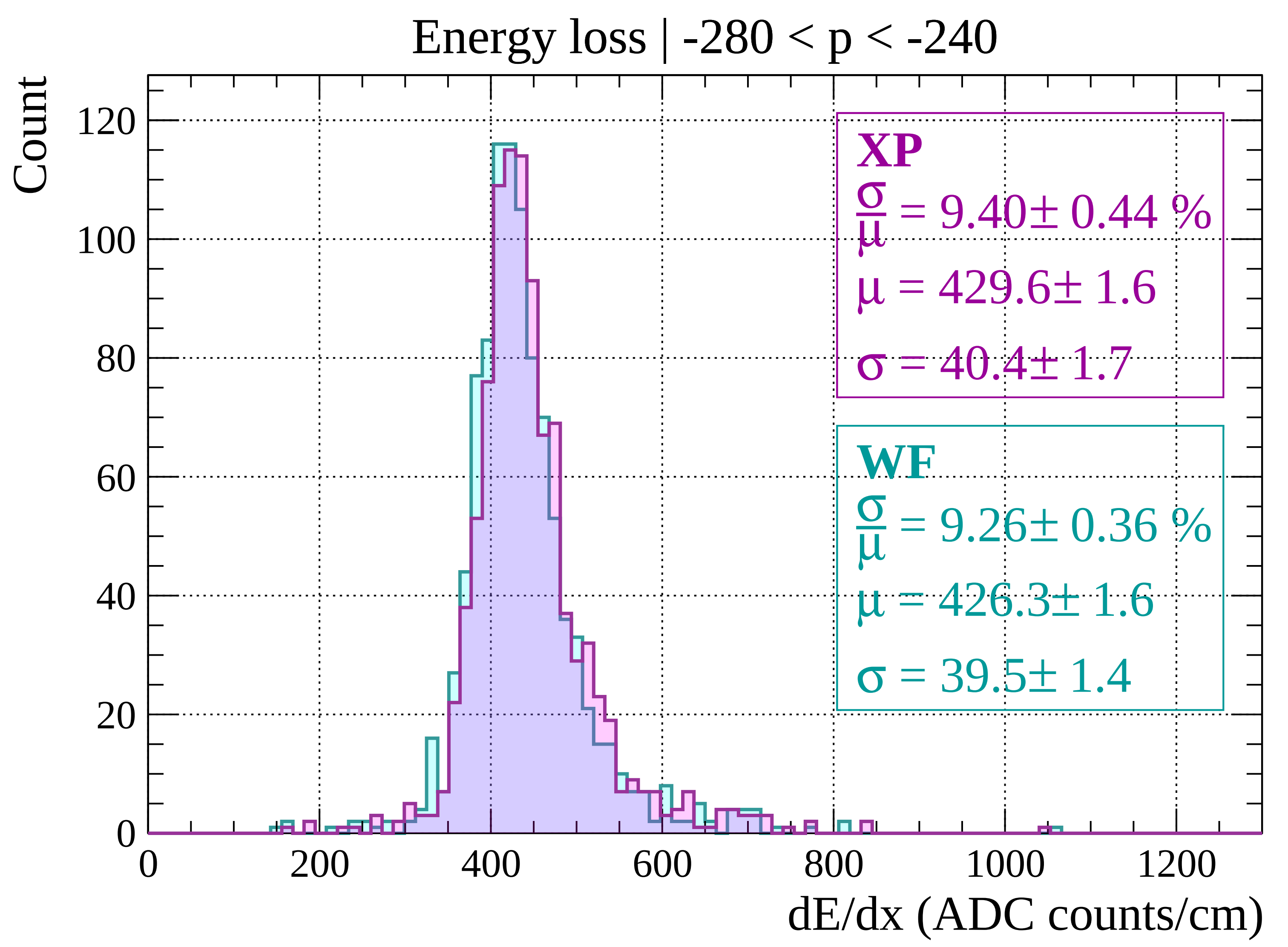

Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.35± 0.43 %
μ = 426.6± 1.6
σ = 39.9± 1.7
WF
$\frac{\sigma}{\mu}$ = 8.88± 0.42 %
μ = 419.2± 1.5
σ = 37.2± 1.6



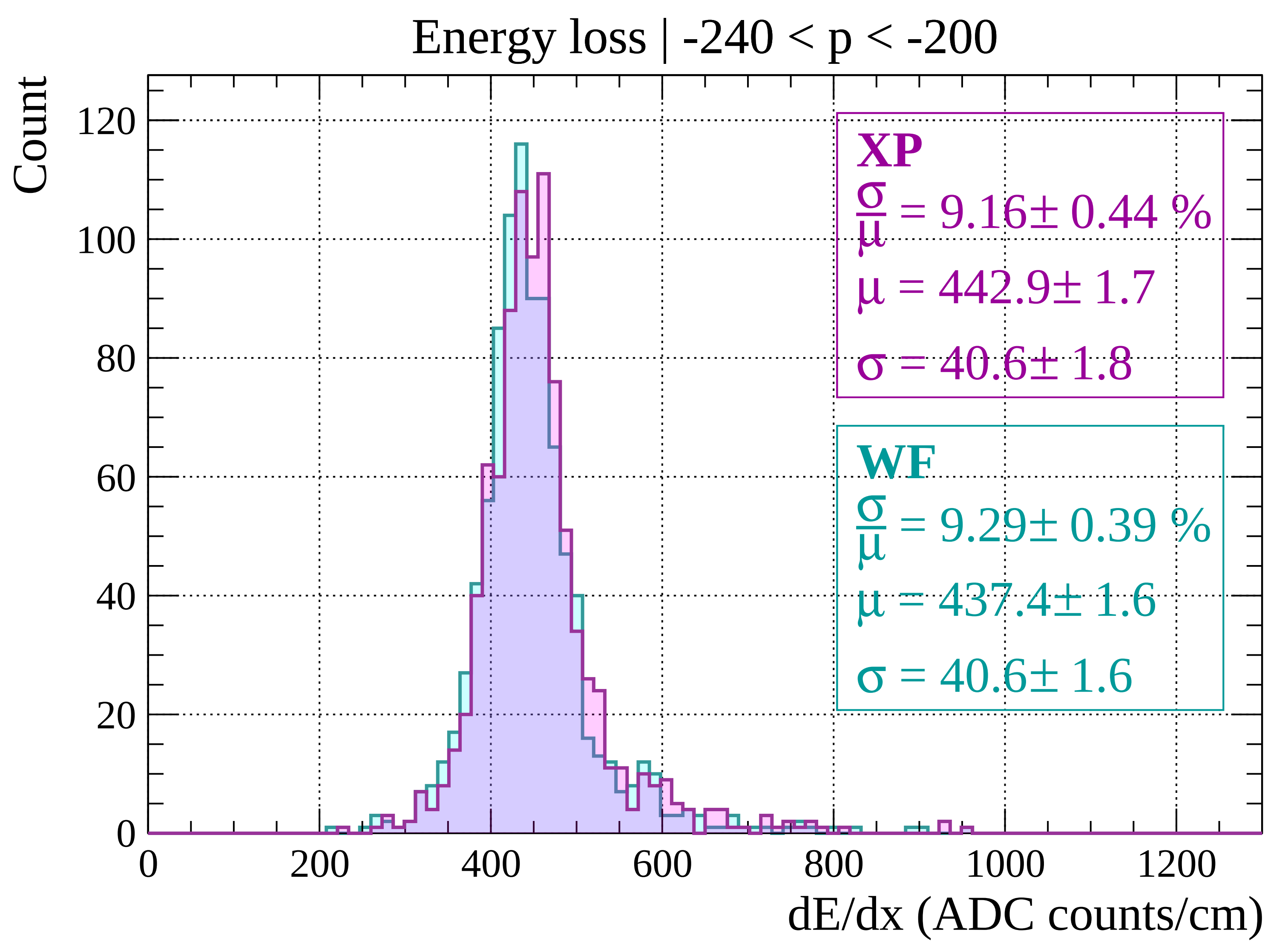
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.63± 0.37 %
μ = 424.4± 1.5
σ = 36.6± 1.4
WF
σ/μ = 8.72± 0.31 %
μ = 419.6± 1.4
σ = 36.6± 1.2
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



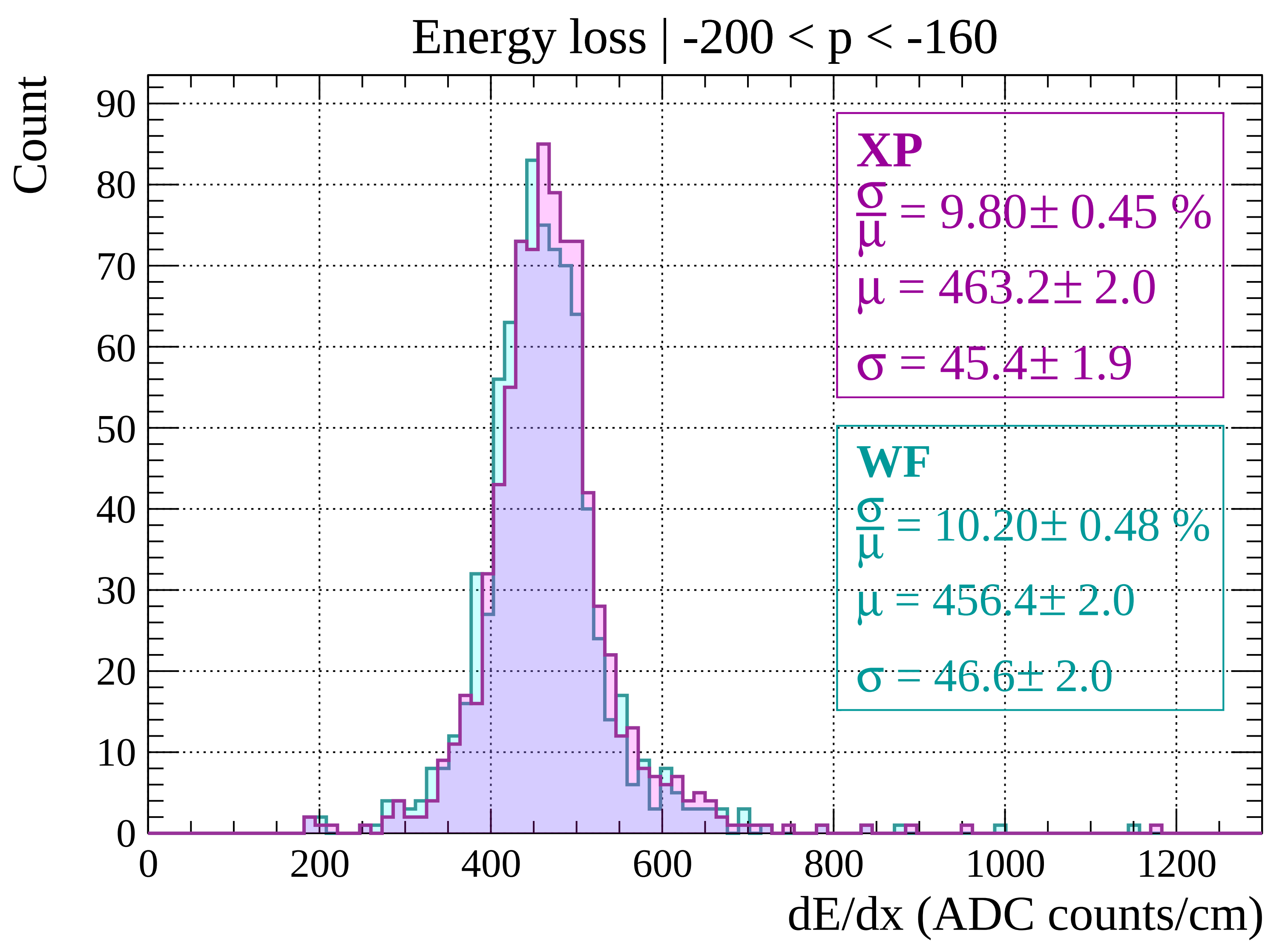
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.02± 0.39 %
μ = 428.6± 1.5
σ = 38.7± 1.5
WF
σ/μ = 9.09± 0.42 %
μ = 421.2± 1.5
σ = 38.3± 1.7
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

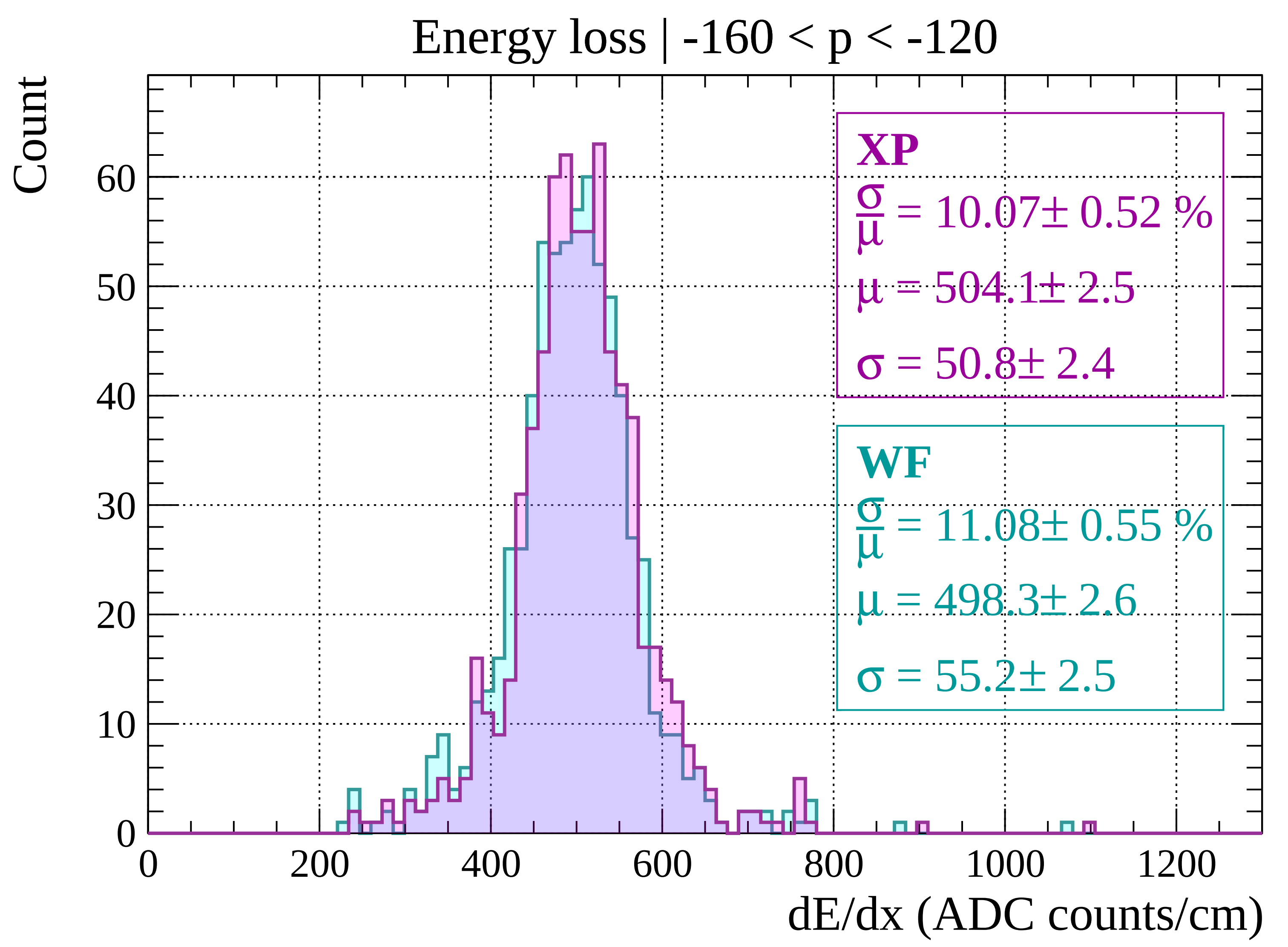


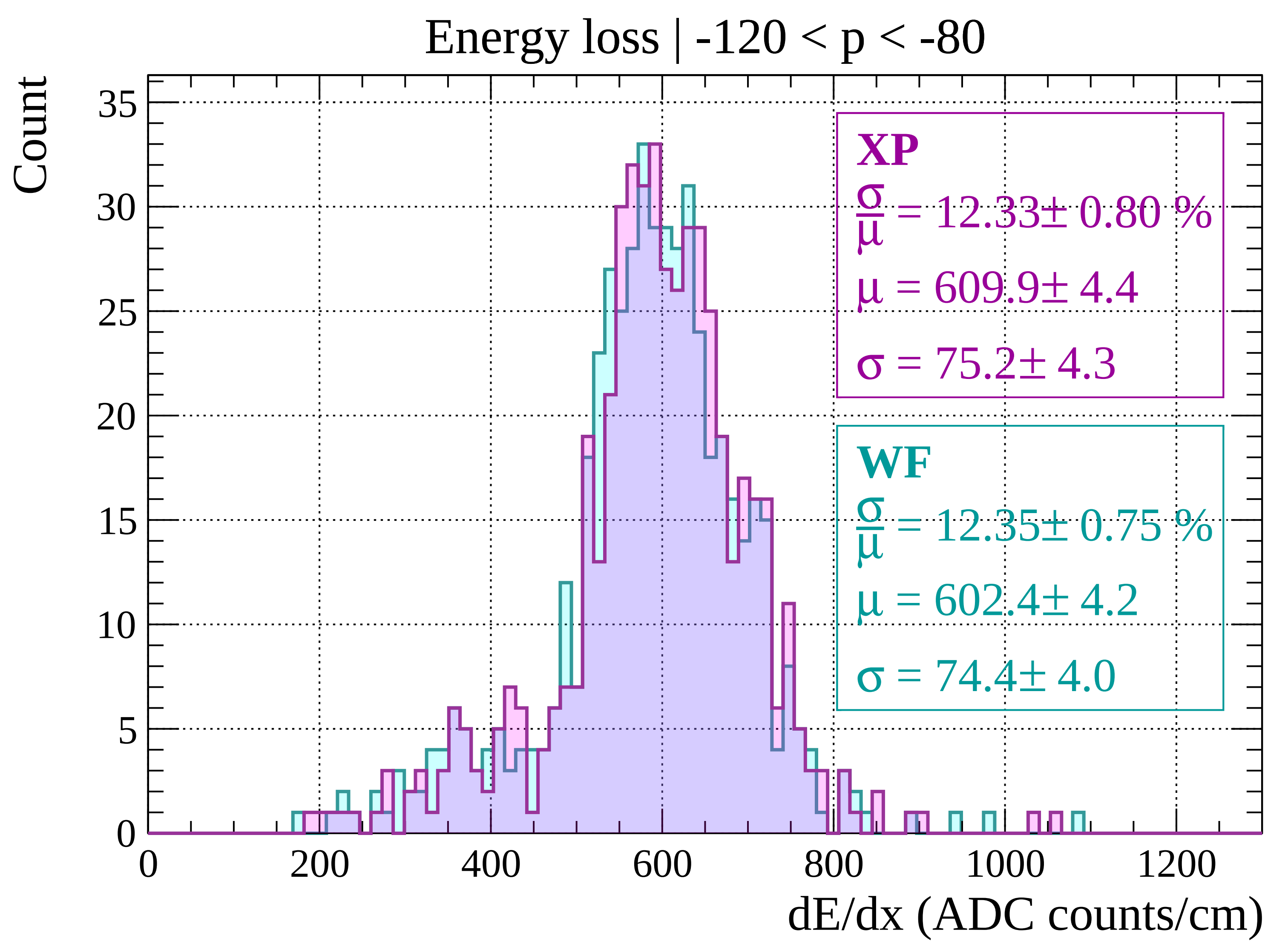
Energy loss | -280 < p < -240
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.40± 0.44 %
μ = 429.6± 1.6
σ = 40.4± 1.7
WF
σ/μ = 9.26± 0.36 %
μ = 426.3± 1.6
σ = 39.5± 1.4

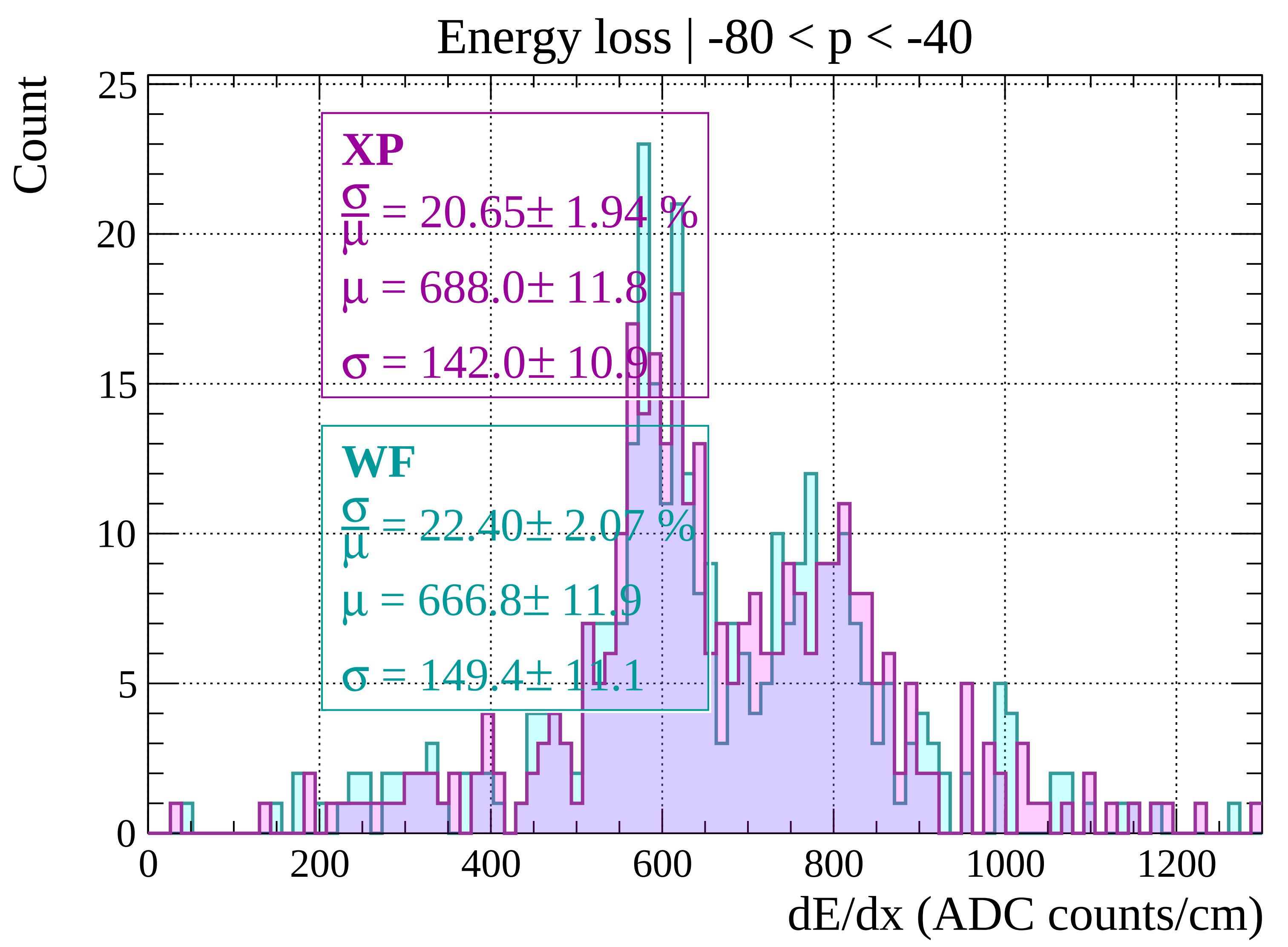


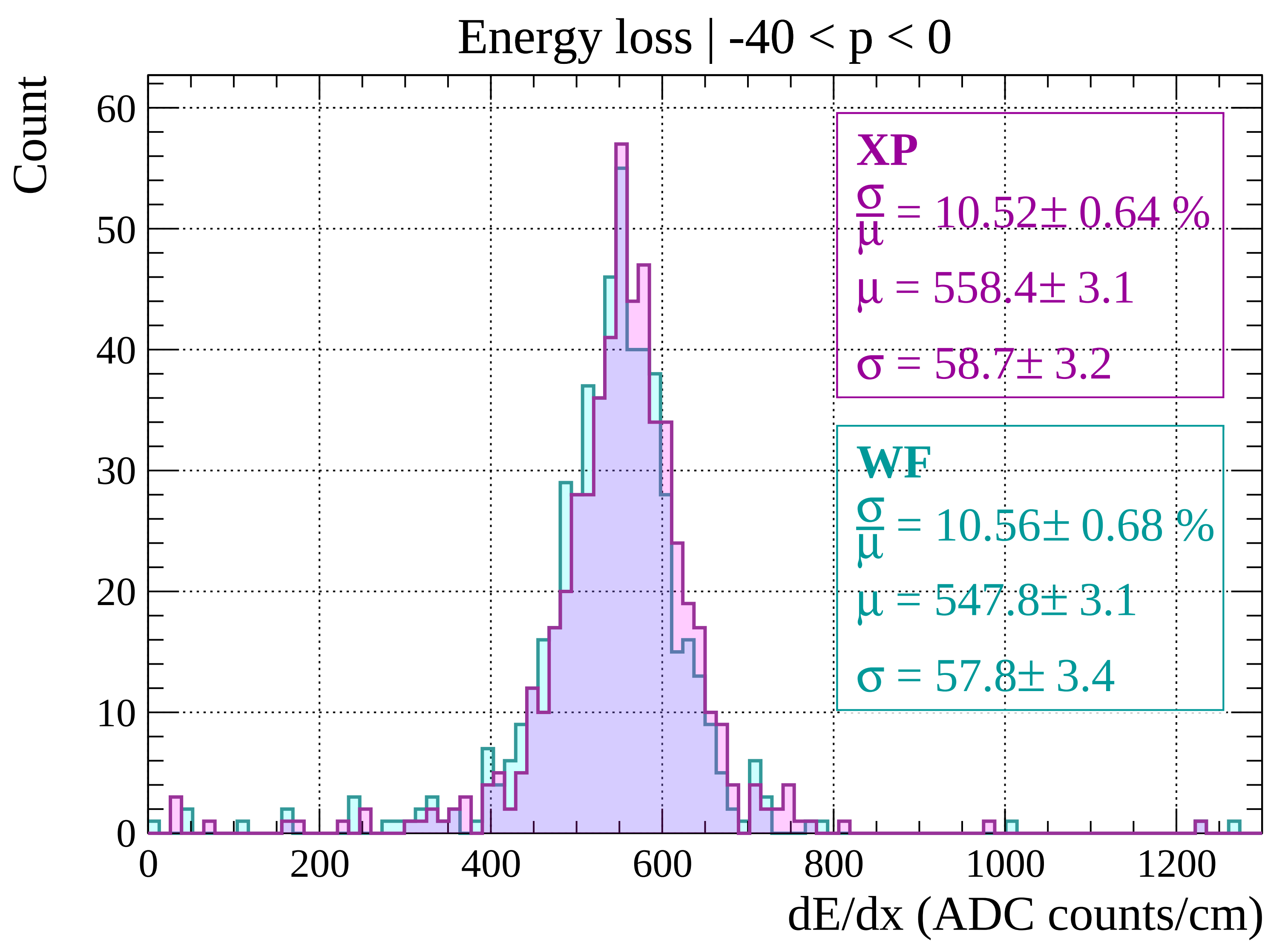
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 9.16± 0.44 %
μ = 442.9± 1.7
σ = 40.6± 1.8
WF
σ/μ = 9.29± 0.39 %
μ = 437.4± 1.6
σ = 40.6± 1.6

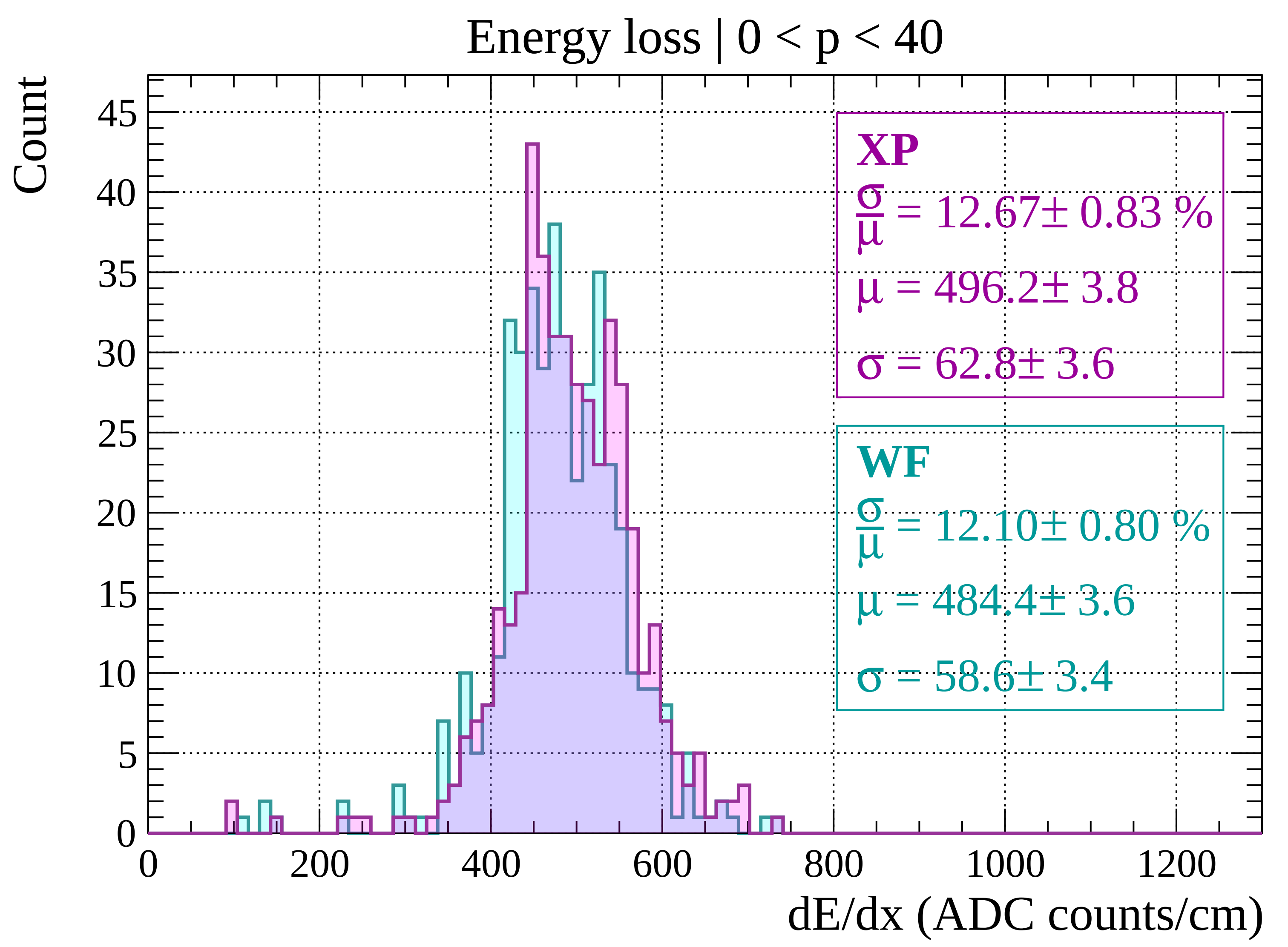

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.80± 0.45 %
μ = 463.2± 2.0
σ = 45.4± 1.9
WF
σ/μ = 10.20± 0.48 %
μ = 456.4± 2.0
σ = 46.6± 2.0
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.07± 0.52 %
μ = 504.1± 2.5
σ = 50.8± 2.4
WF
σ/μ = 11.08± 0.55 %
μ = 498.3± 2.6
σ = 55.2± 2.5
0
10
20
30
40
50
60
200
400
600
800
1000
1200

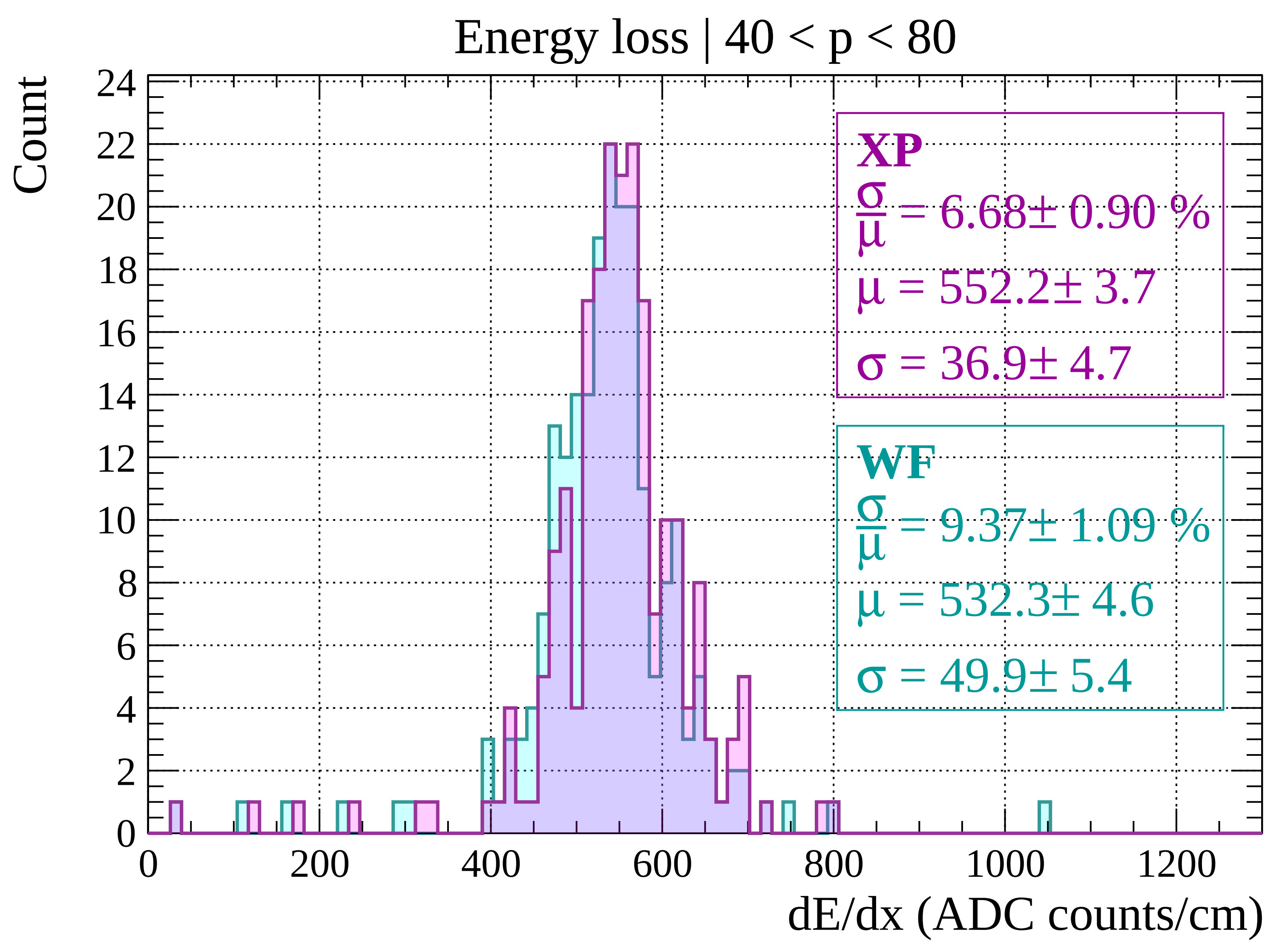

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.33± 0.80 %
μ = 609.9± 4.4
σ = 75.2± 4.3
WF
σ/μ = 12.35± 0.75 %
μ = 602.4± 4.2
σ = 74.4± 4.0

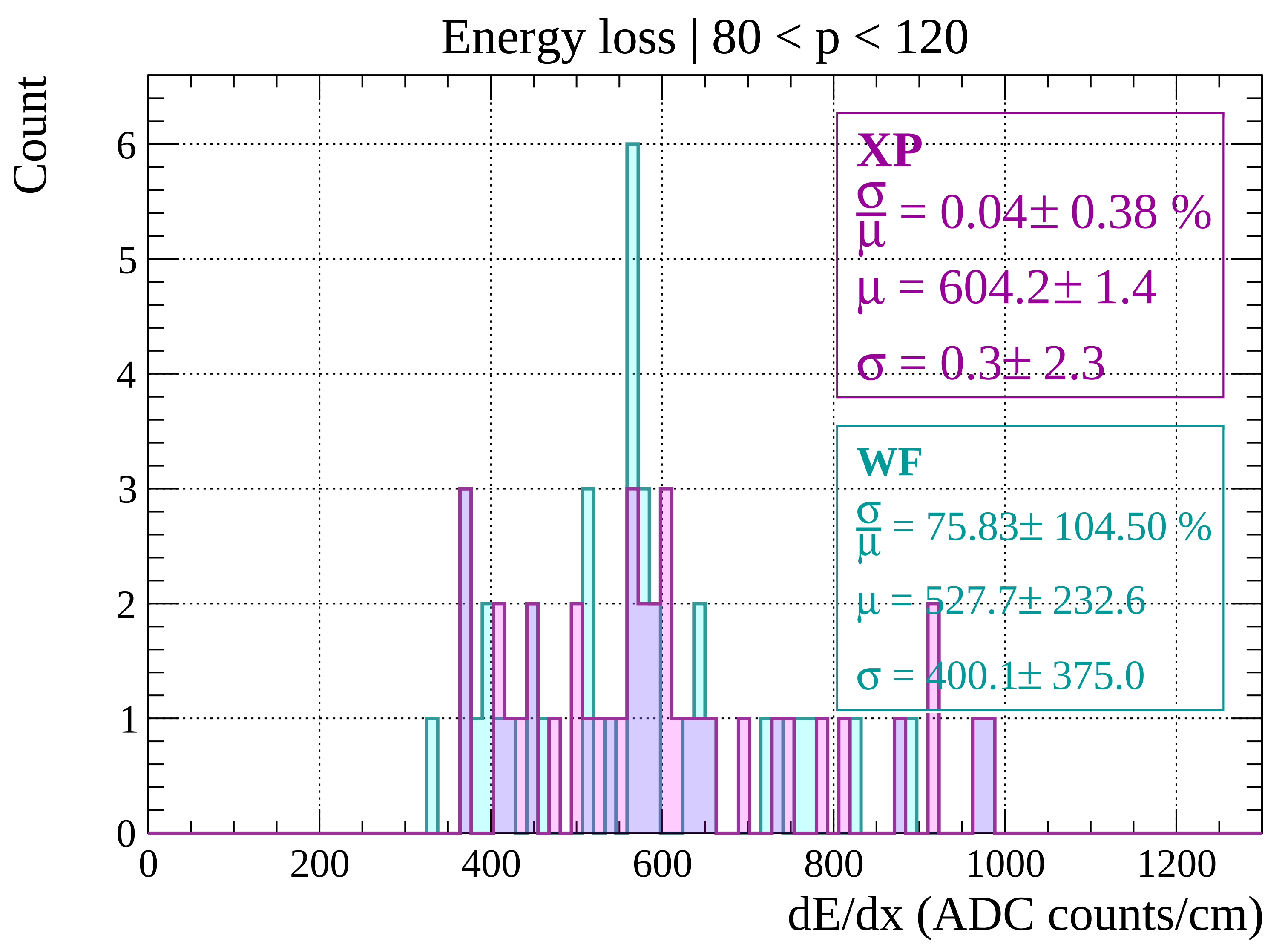

Energy loss | -80 < p < -40
Count
XP
$\frac{\sigma}{\mu}$ = 20.65± 1.94 %
μ = 688.0± 11.8
σ = 142.0± 10.9
WF
$\frac{\sigma}{\mu}$ = 22.40± 2.07 %
μ = 666.8± 11.9
σ = 149.4± 11.1
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200

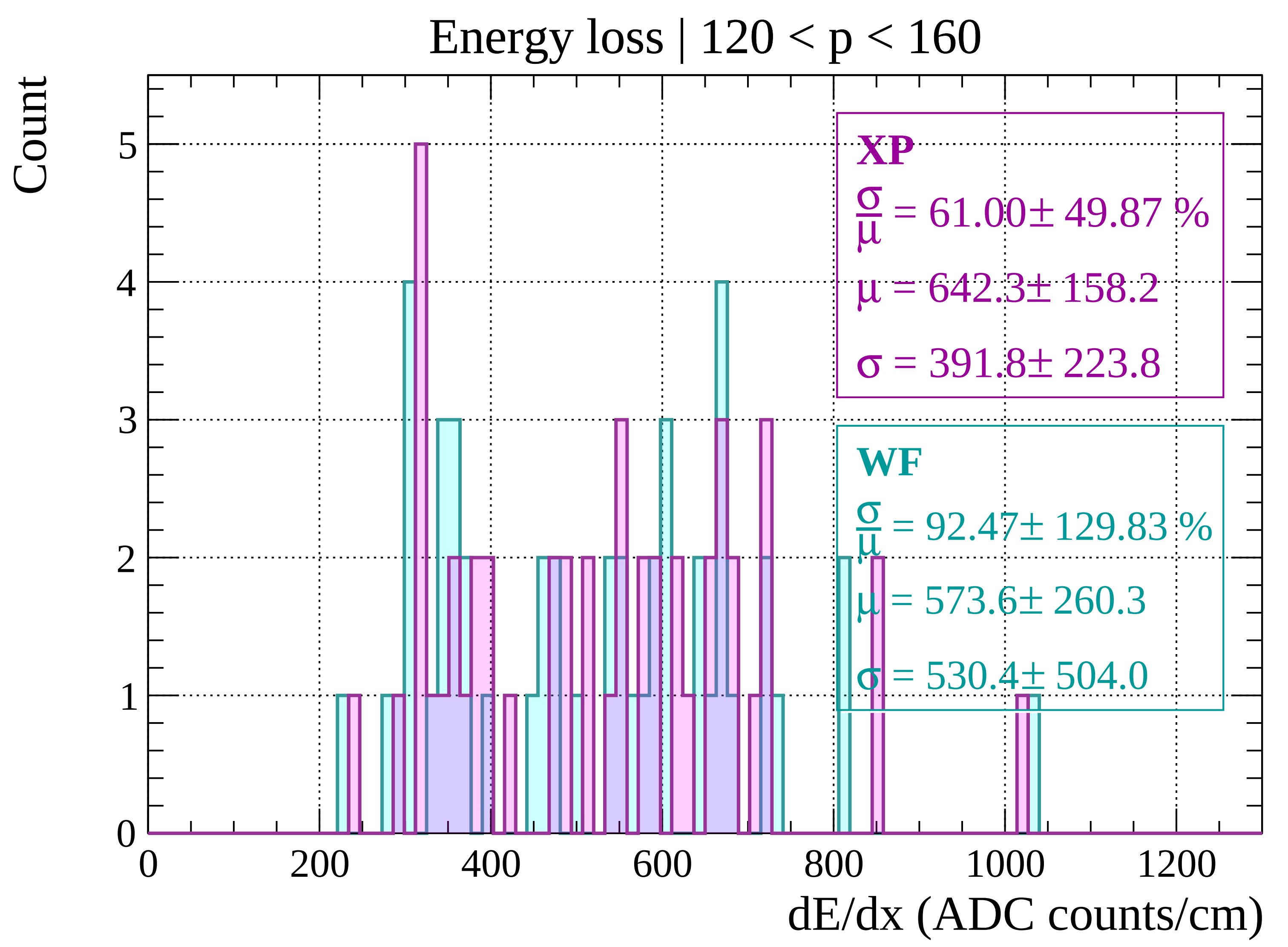

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 10.52± 0.64 %
μ = 558.4± 3.1
σ = 58.7± 3.2
WF
σ/μ = 10.56± 0.68 %
μ = 547.8± 3.1
σ = 57.8± 3.4



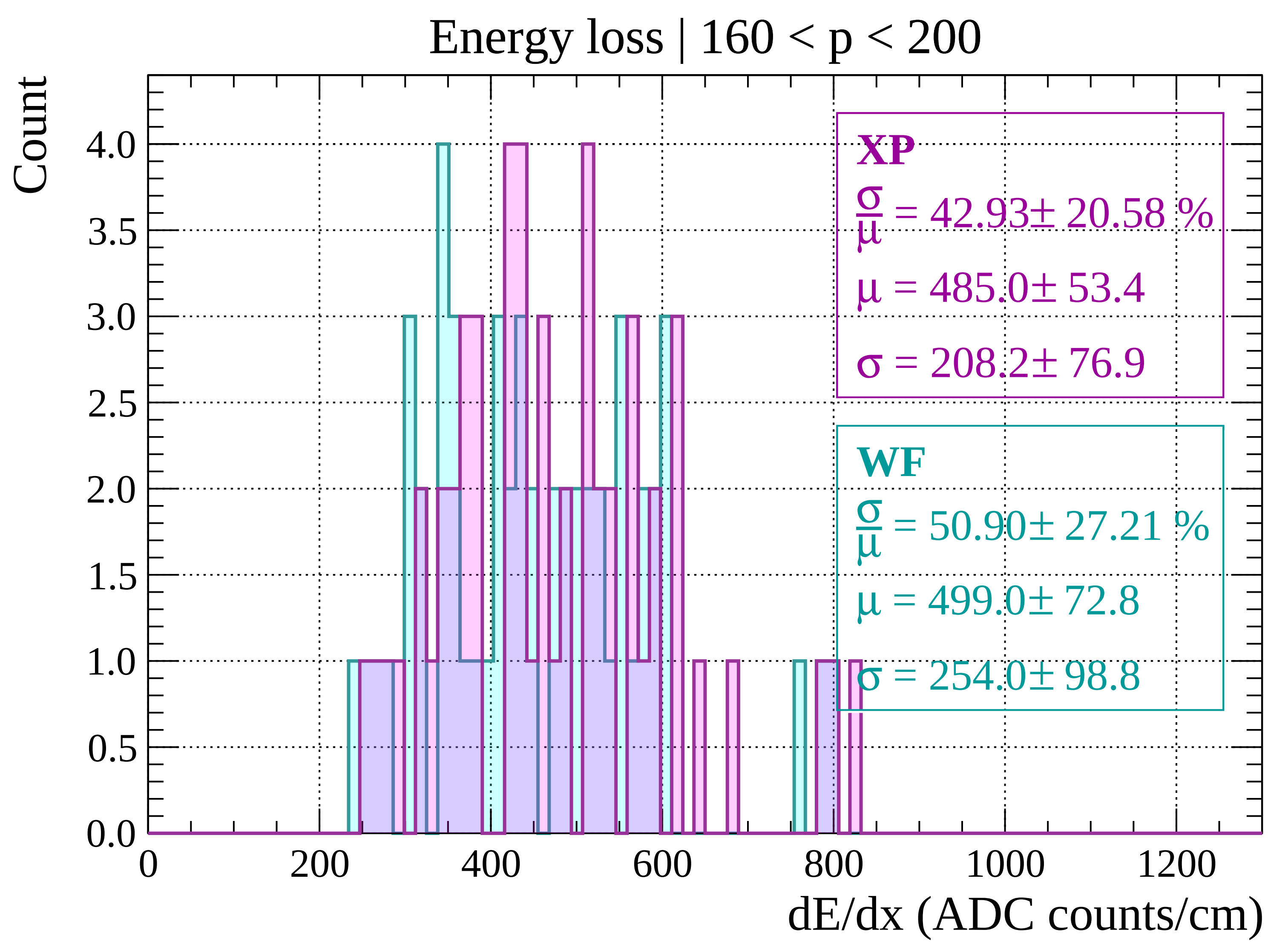
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 12.67± 0.83 %
μ = 496.2± 3.8
σ = 62.8± 3.6
WF
σ/μ = 12.10± 0.80 %
μ = 484.4± 3.6
σ = 58.6± 3.4


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.68± 0.90 %
μ = 552.2± 3.7
σ = 36.9± 4.7
WF
σ/μ = 9.37± 1.09 %
μ = 532.3± 4.6
σ = 49.9± 5.4

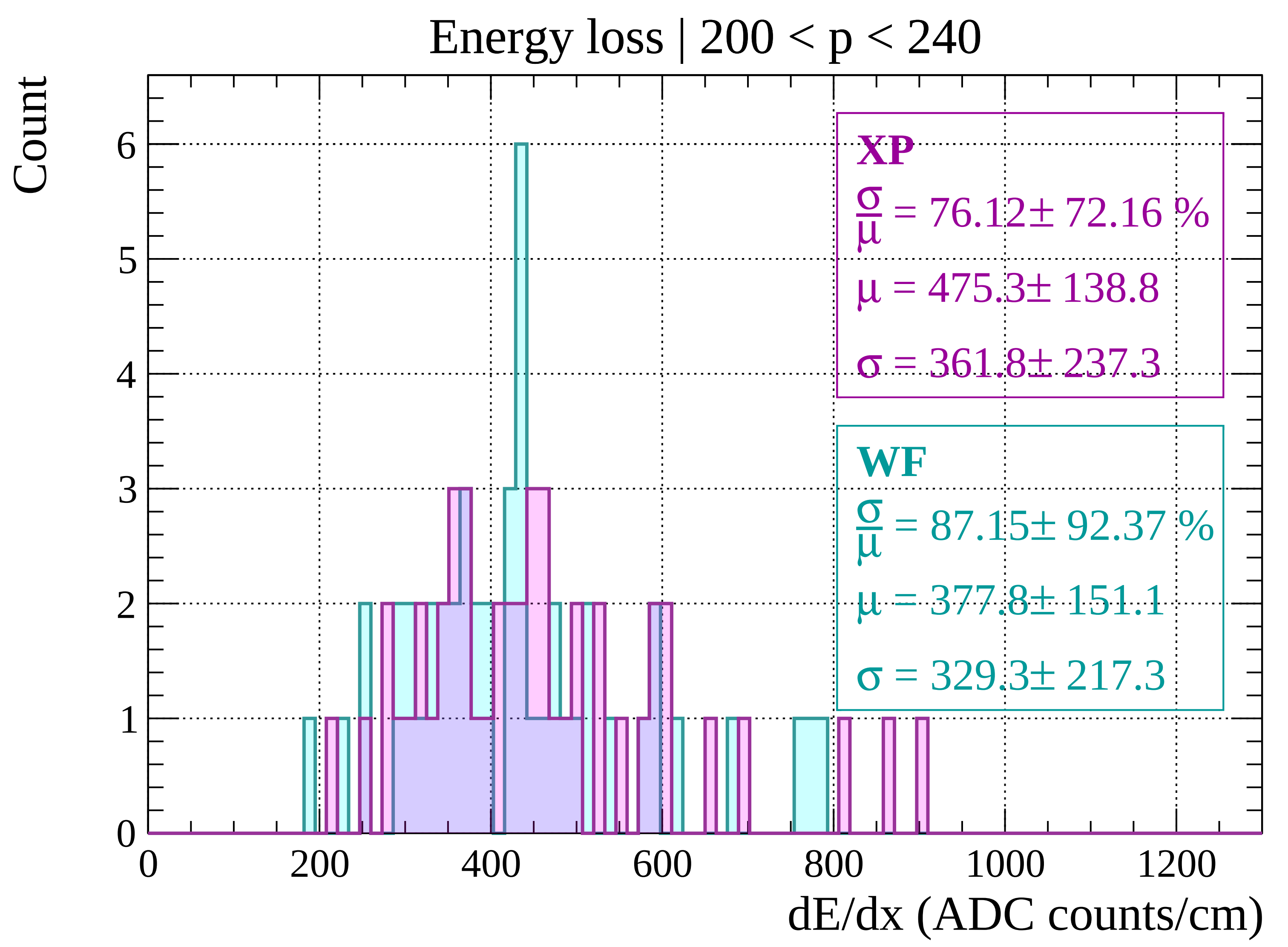

Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
$\frac{\sigma}{\mu}$ = 0.04± 0.38 %
μ = 604.2± 1.4
σ = 0.3± 2.3
WF
$\frac{\sigma}{\mu}$ = 75.83± 104.50 %
μ = 527.7± 232.6
σ = 400.1± 375.0



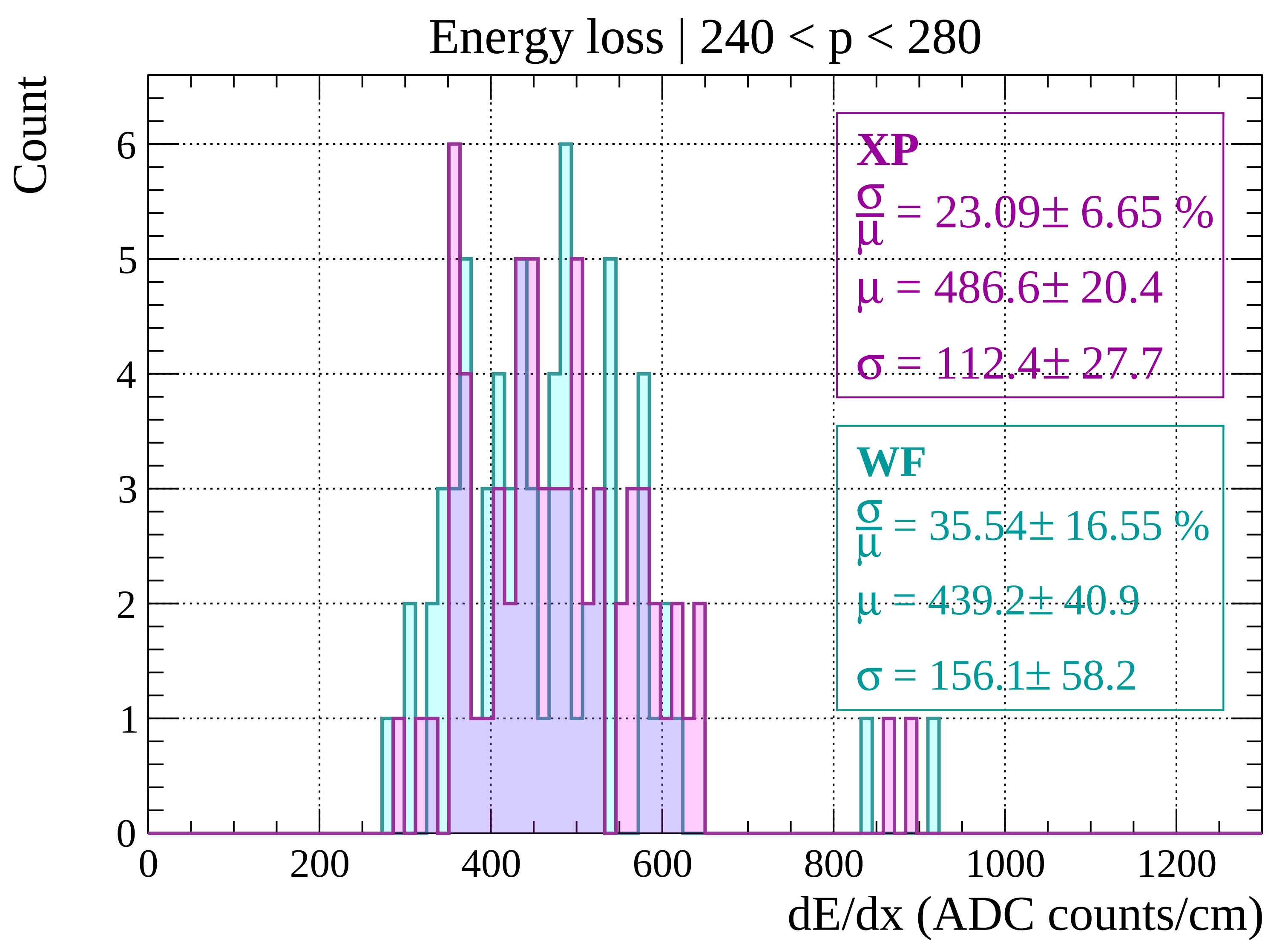
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 61.00± 49.87 %
μ = 642.3± 158.2
σ = 391.8± 223.8
WF
σ/μ = 92.47± 129.83 %
μ = 573.6± 260.3
σ = 530.4± 504.0



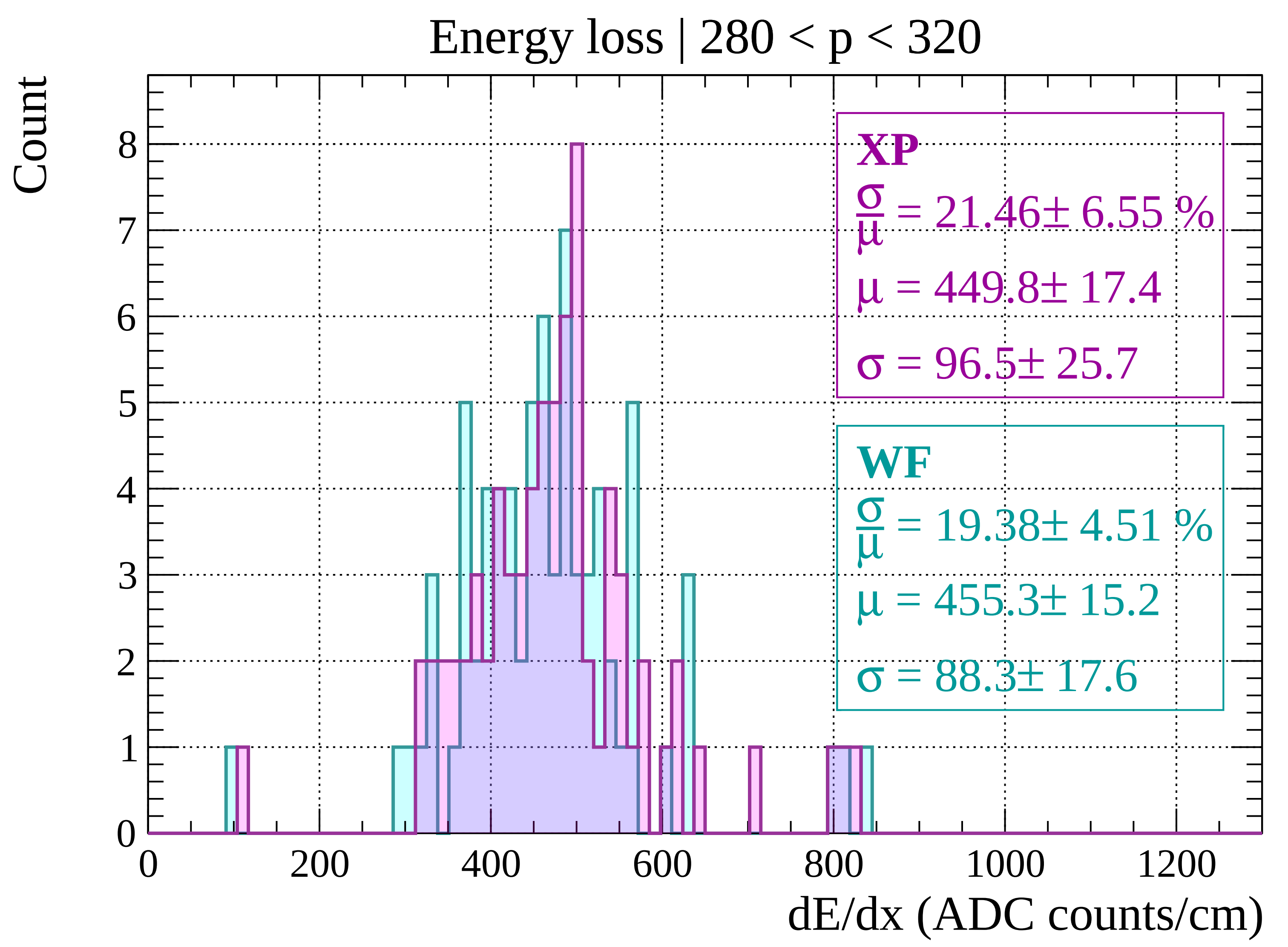
Energy loss | 160 < p < 200
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 42.93± 20.58 %
μ = 485.0± 53.4
σ = 208.2± 76.9
WF
σ/μ = 50.90± 27.21 %
μ = 499.0± 72.8
σ = 254.0± 98.8

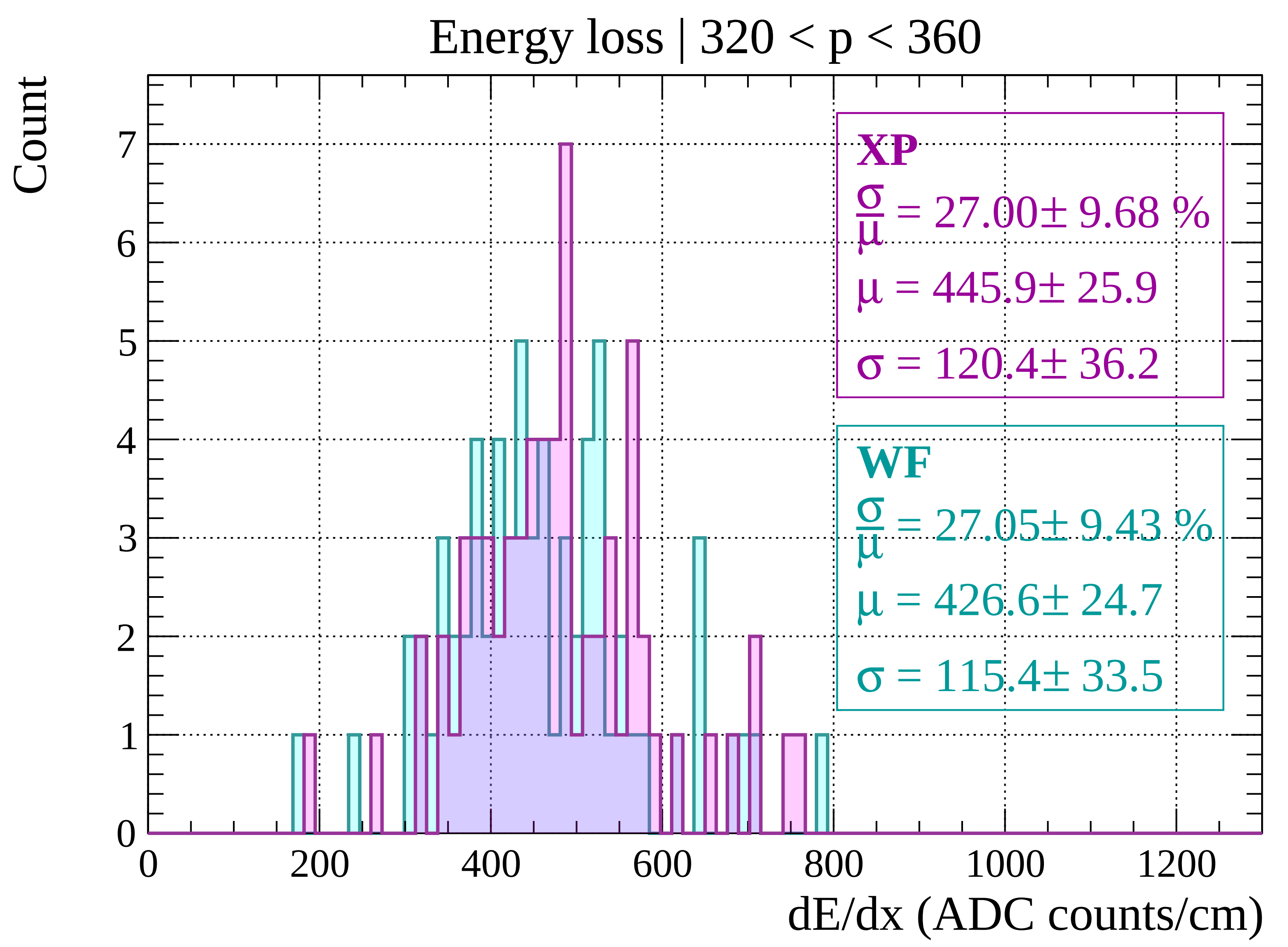


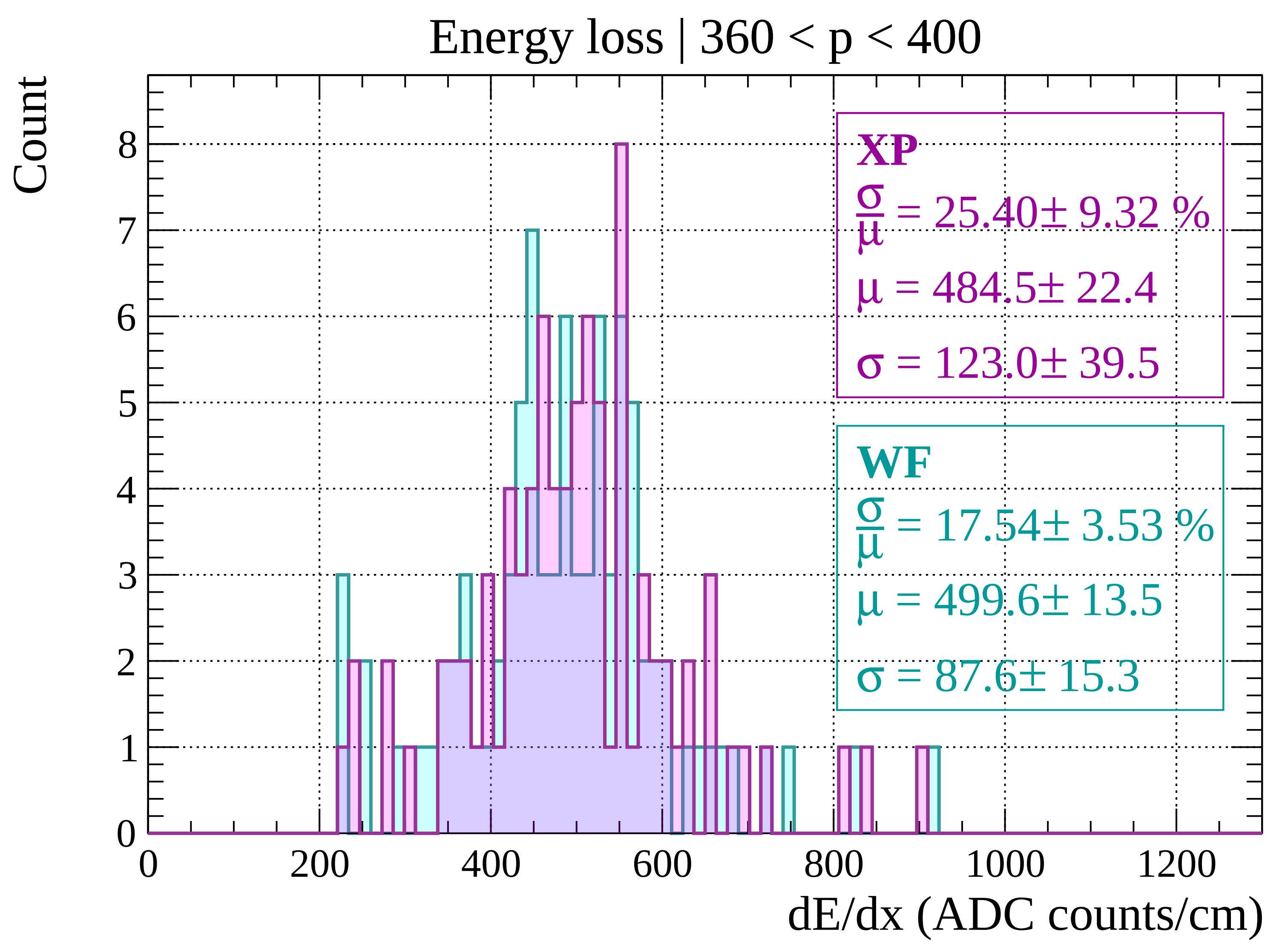
Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 76.12± 72.16 %
μ = 475.3± 138.8
σ = 361.8± 237.3
WF
$\frac{\sigma}{\mu}$ = 87.15± 92.37 %
μ = 377.8± 151.1
σ = 329.3± 217.3



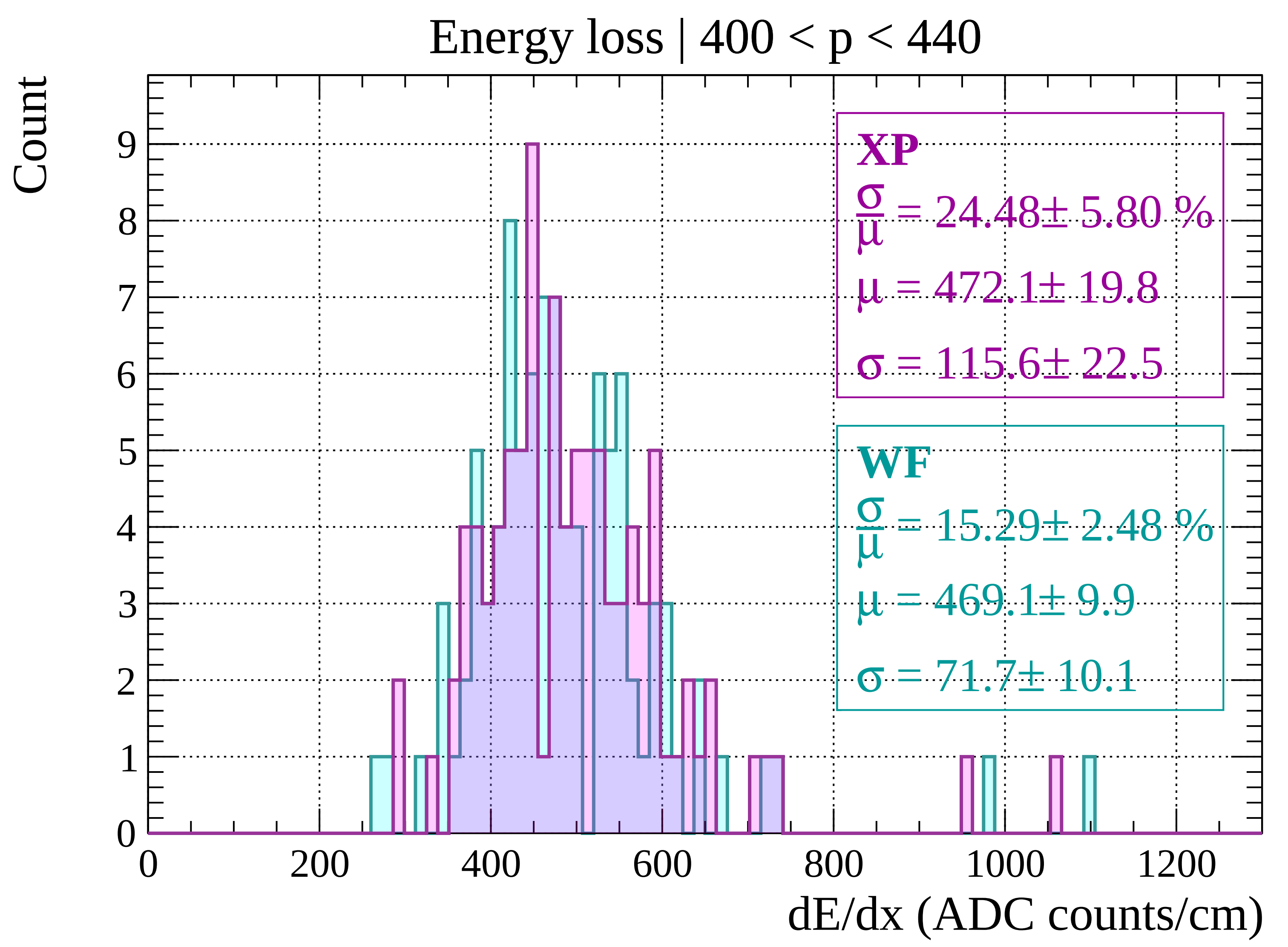
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 23.09± 6.65 %
μ = 486.6± 20.4
σ = 112.4± 27.7
WF
σ/μ = 35.54± 16.55 %
μ = 439.2± 40.9
σ = 156.1± 58.2


Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
XP
σ/μ = 21.46± 6.55 %
μ = 449.8± 17.4
σ = 96.5± 25.7
WF
σ/μ = 19.38± 4.51 %
μ = 455.3± 15.2
σ = 88.3± 17.6

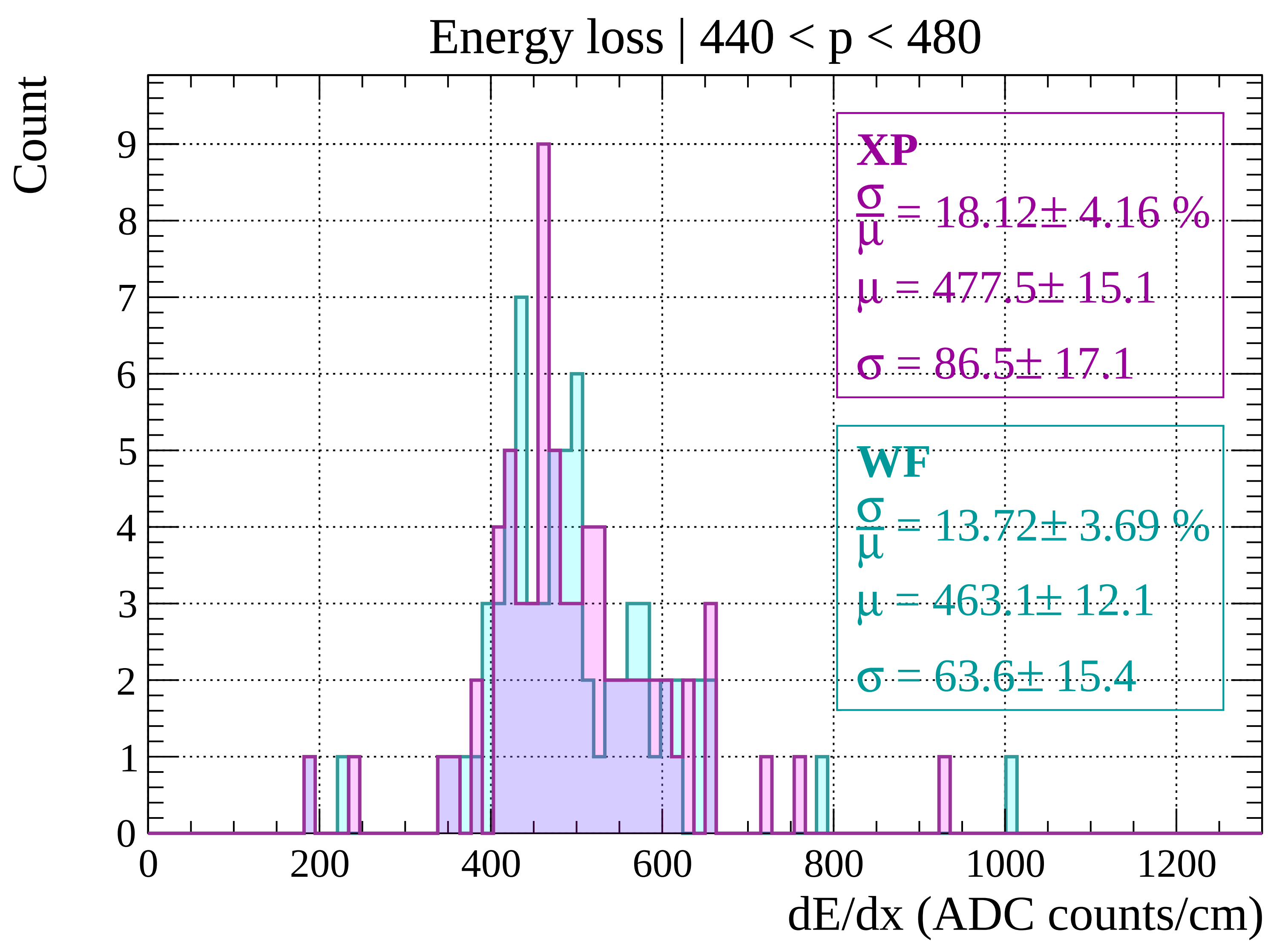

Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 27.00± 9.68 %
μ = 445.9± 25.9
σ = 120.4± 36.2
WF
$\frac{\sigma}{\mu}$ = 27.05± 9.43 %
μ = 426.6± 24.7
σ = 115.4± 33.5



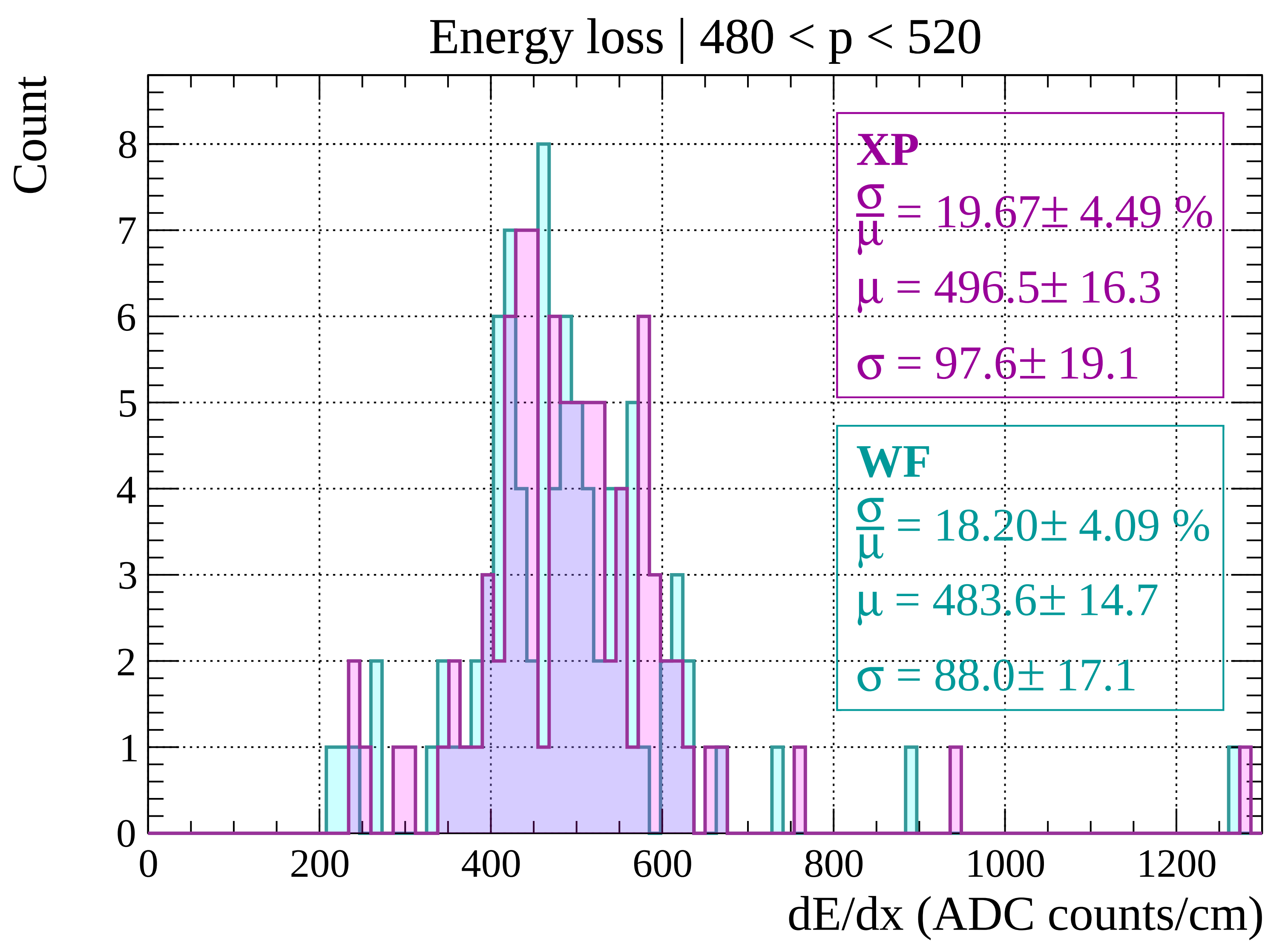
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 25.40± 9.32 %
μ = 484.5± 22.4
σ = 123.0± 39.5
WF
σ/μ = 17.54± 3.53 %
μ = 499.6± 13.5
σ = 87.6± 15.3



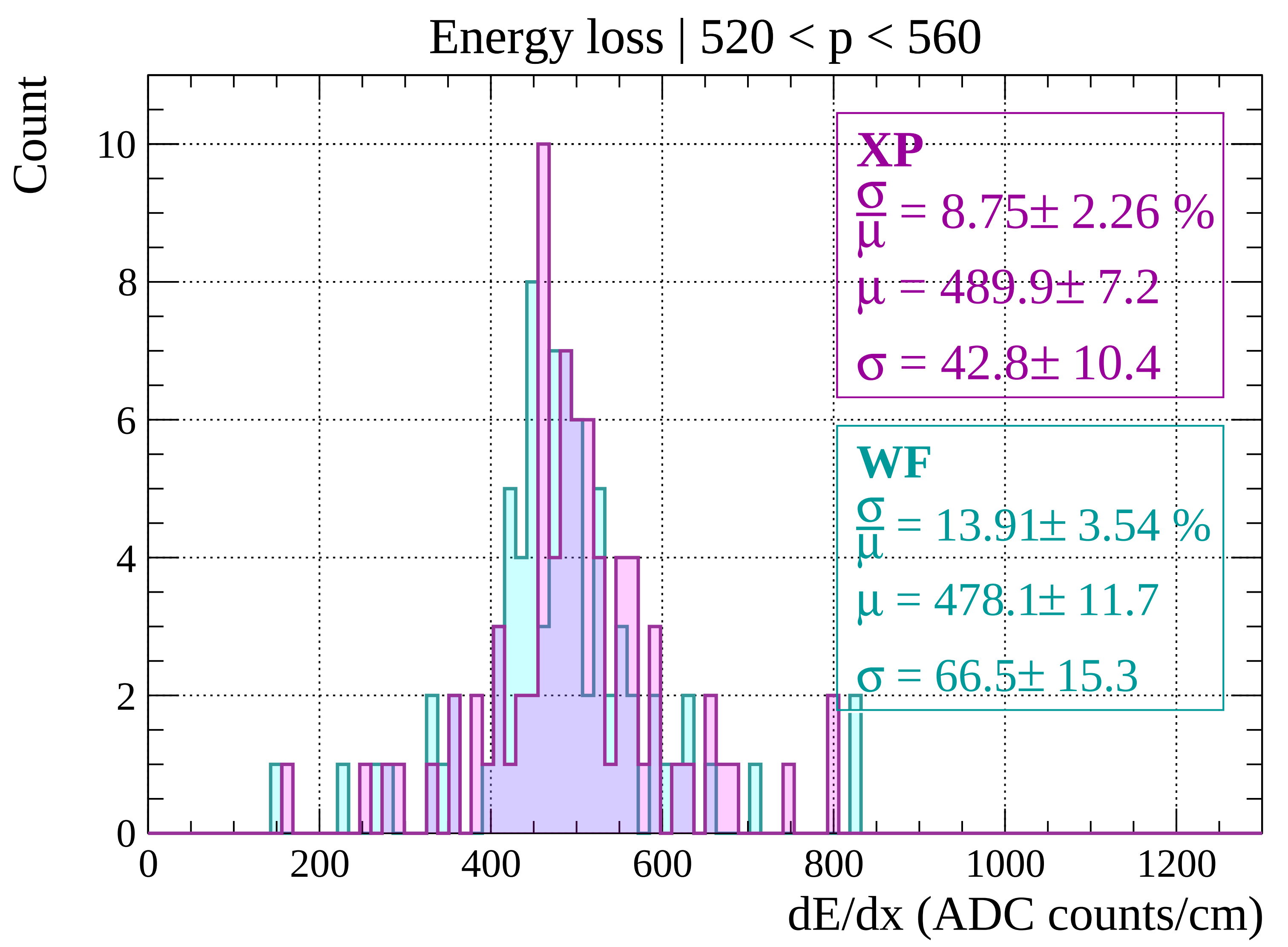
Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 24.48± 5.80 %
μ = 472.1± 19.8
σ = 115.6± 22.5
WF
σ/μ = 15.29± 2.48 %
μ = 469.1± 9.9
σ = 71.7± 10.1

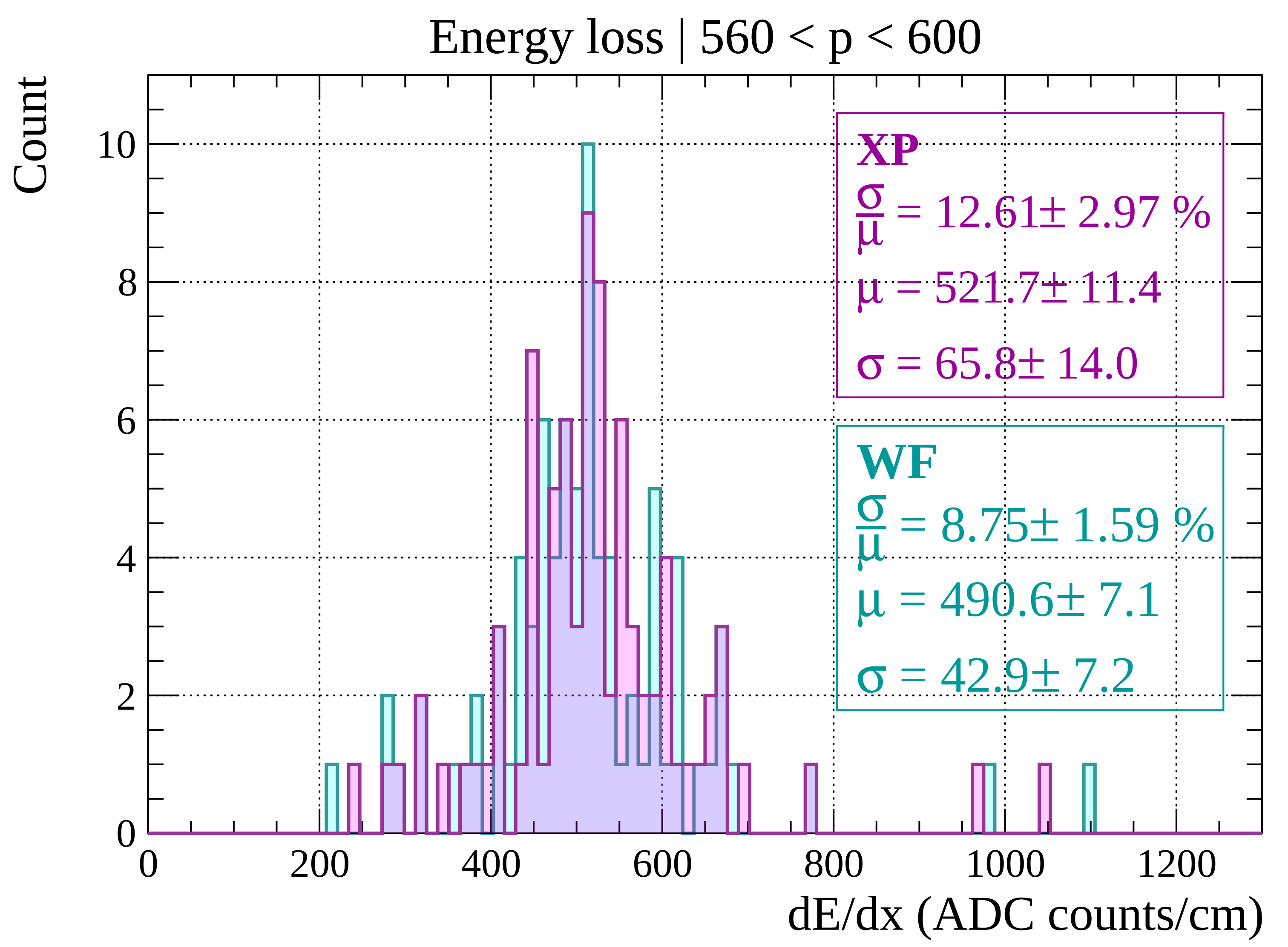


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 18.12± 4.16 %
μ = 477.5± 15.1
σ = 86.5± 17.1
WF
$\frac{\sigma}{\mu}$ = 13.72± 3.69 %
μ = 463.1± 12.1
σ = 63.6± 15.4

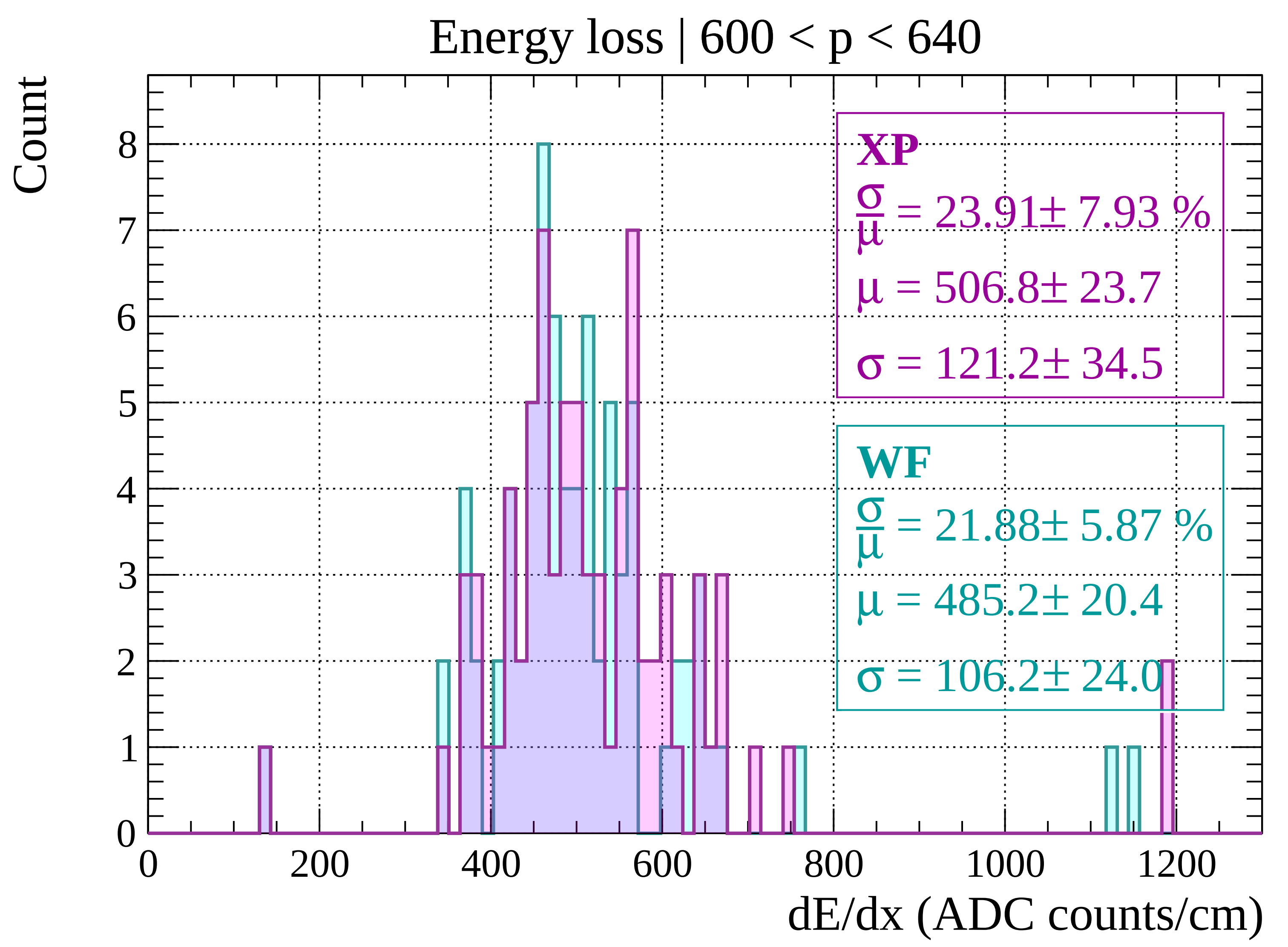


Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 19.67± 4.49 %
μ = 496.5± 16.3
σ = 97.6± 19.1
WF
$\frac{\sigma}{\mu}$ = 18.20± 4.09 %
μ = 483.6± 14.7
σ = 88.0± 17.1

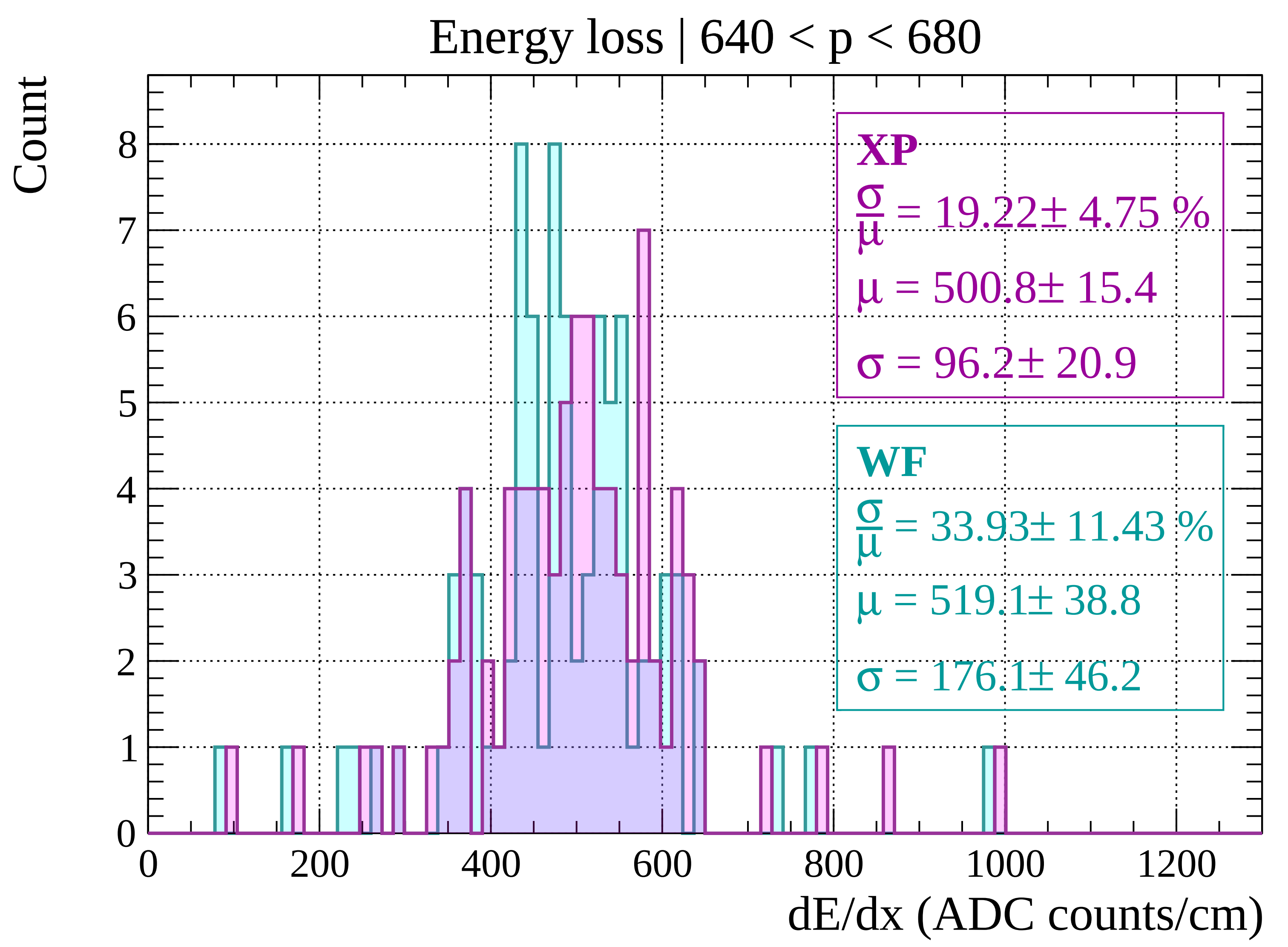

Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.75± 2.26 %
μ = 489.9± 7.2
σ = 42.8± 10.4
WF
$\frac{\sigma}{\mu}$ = 13.91± 3.54 %
μ = 478.1± 11.7
σ = 66.5± 15.3

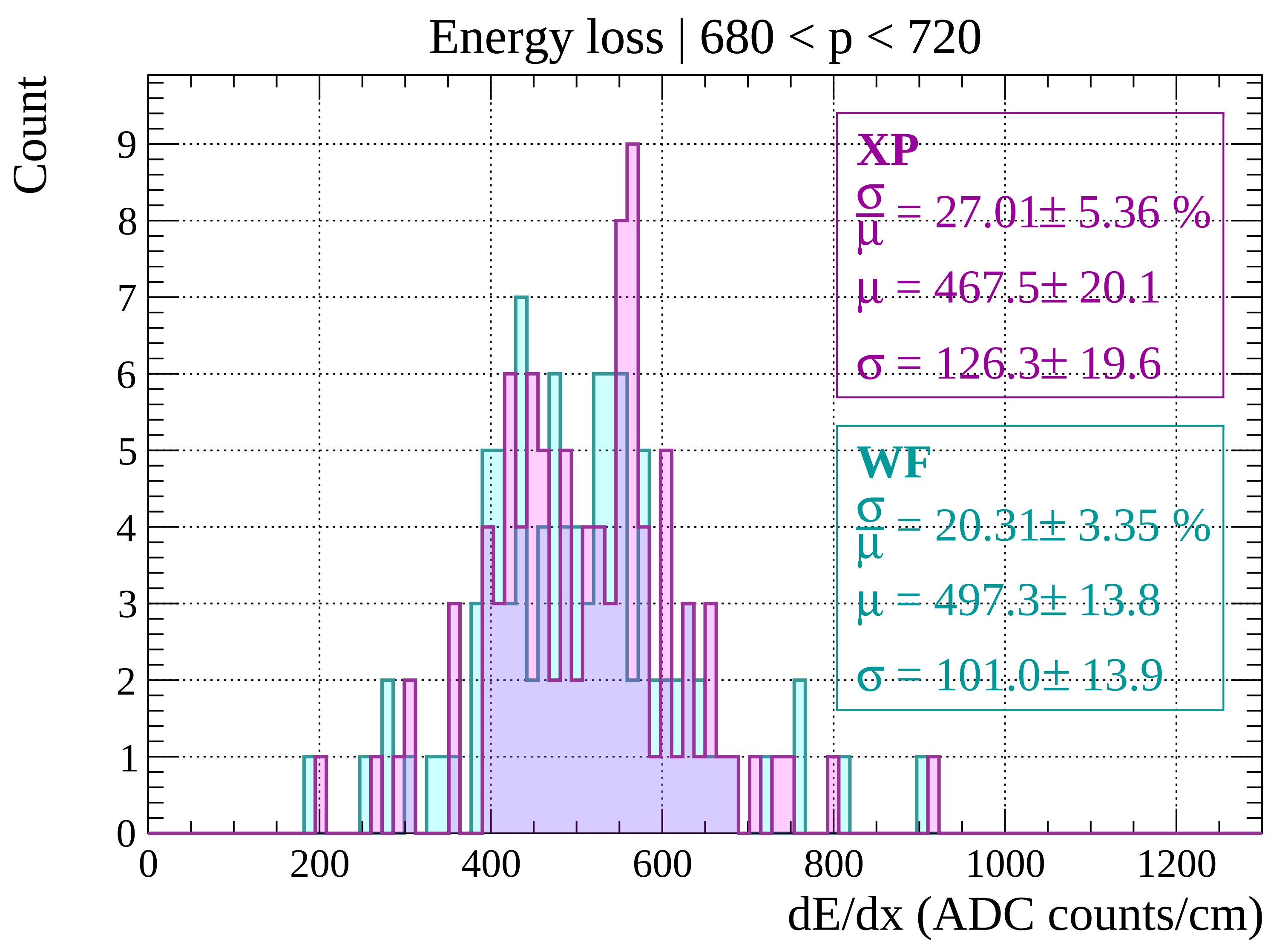


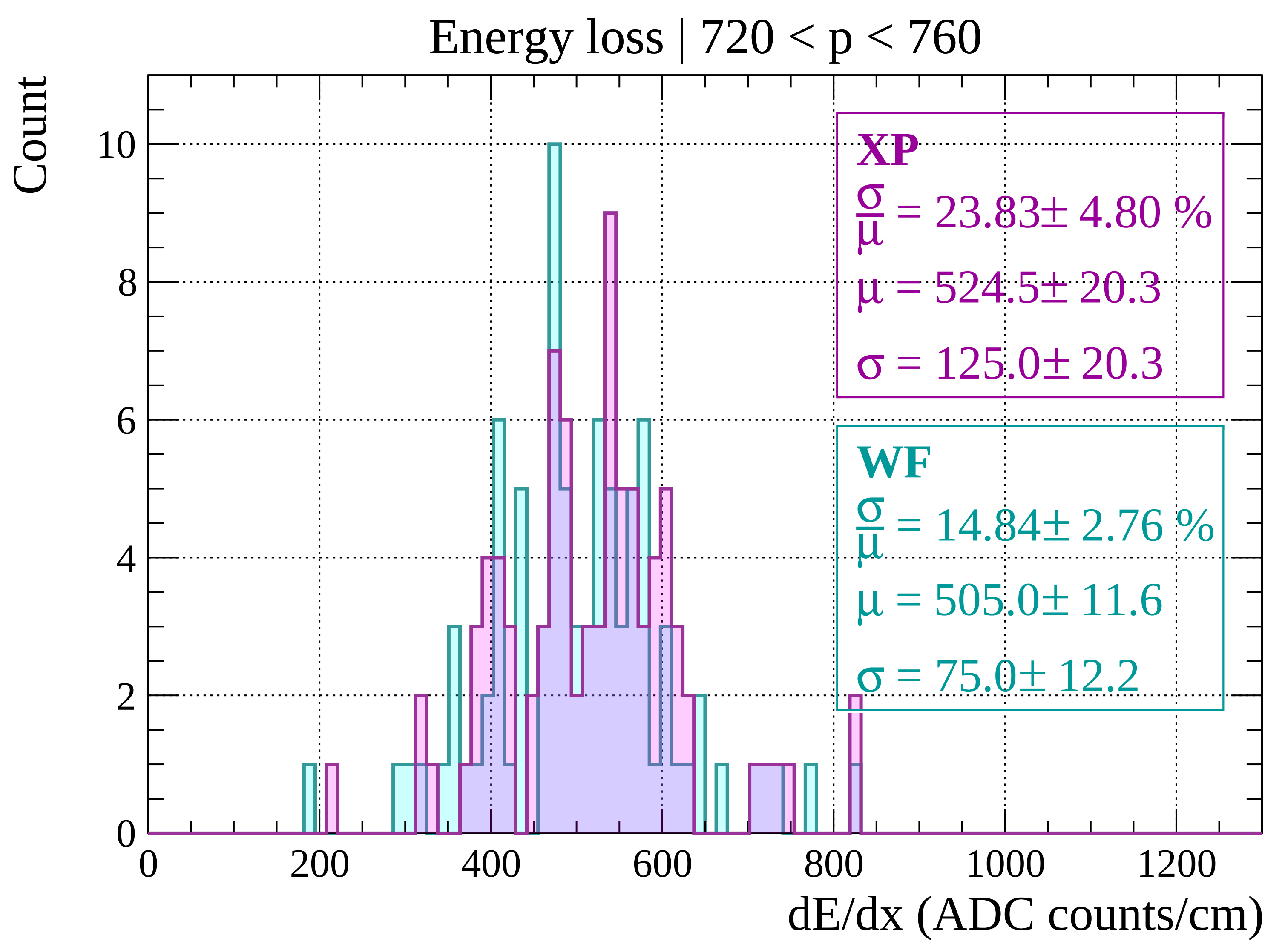
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.61± 2.97 %
μ = 521.7± 11.4
σ = 65.8± 14.0
WF
$\frac{\sigma}{\mu}$ = 8.75± 1.59 %
μ = 490.6± 7.1
σ = 42.9± 7.2

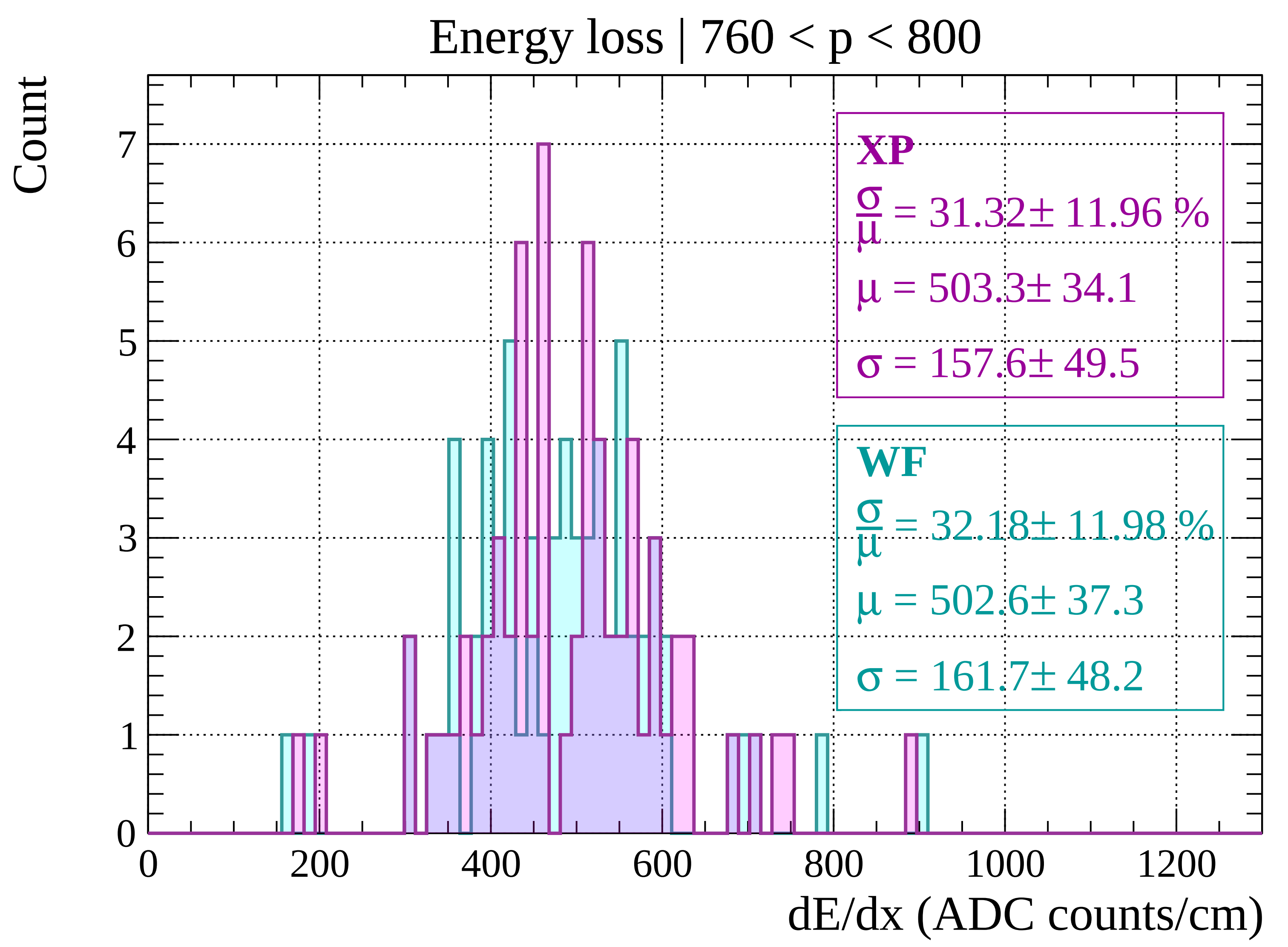


Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 23.91± 7.93 %
μ = 506.8± 23.7
σ = 121.2± 34.5
WF
σ/μ = 21.88± 5.87 %
μ = 485.2± 20.4
σ = 106.2± 24.0

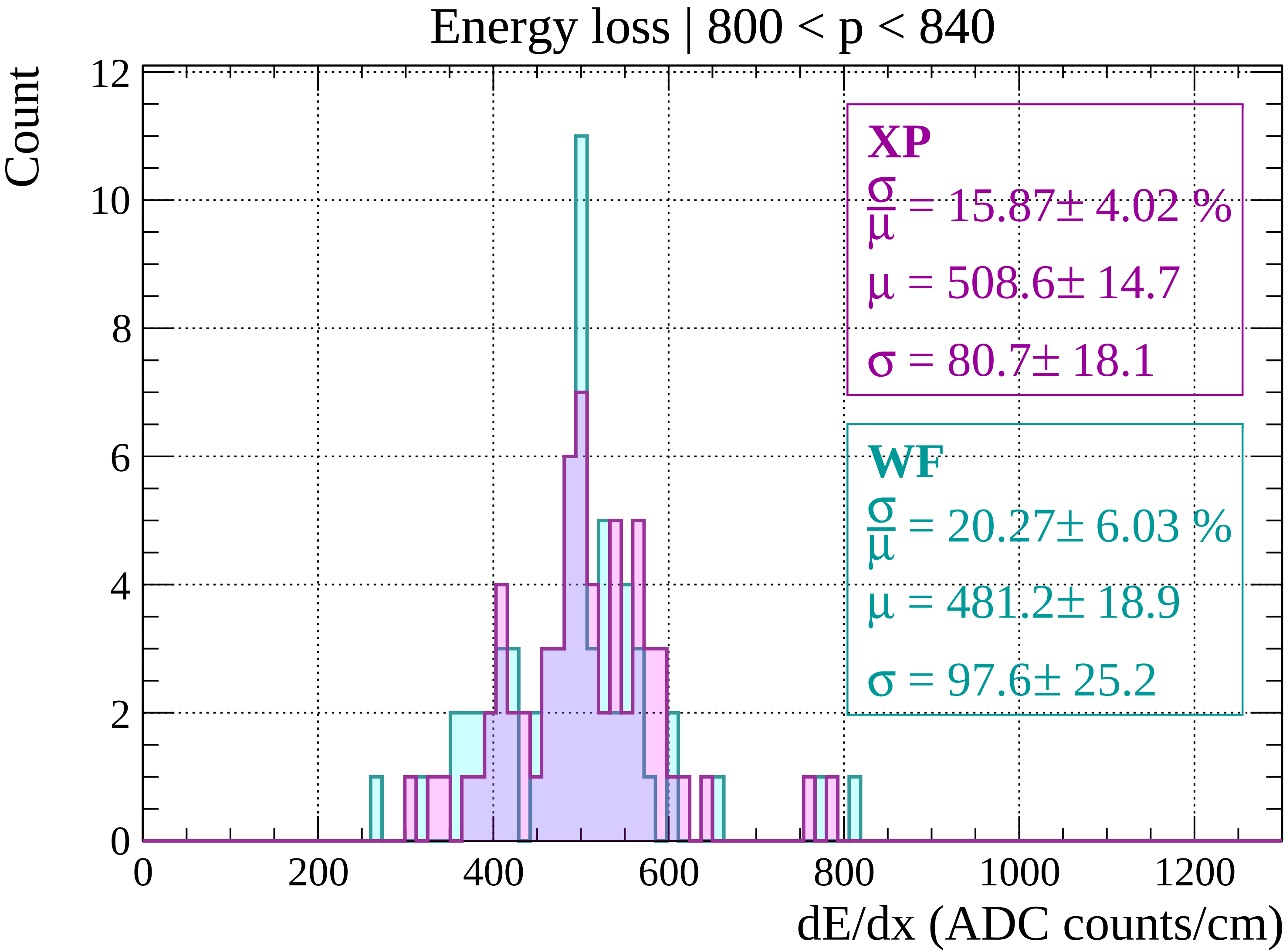

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 19.22± 4.75 %
μ = 500.8± 15.4
σ = 96.2± 20.9
WF
σ/μ = 33.93± 11.43 %
μ = 519.1± 38.8
σ = 176.1± 46.2


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 27.01± 5.36 %
μ = 467.5± 20.1
σ = 126.3± 19.6
WF
$\frac{\sigma}{\mu}$ = 20.31± 3.35 %
μ = 497.3± 13.8
σ = 101.0± 13.9

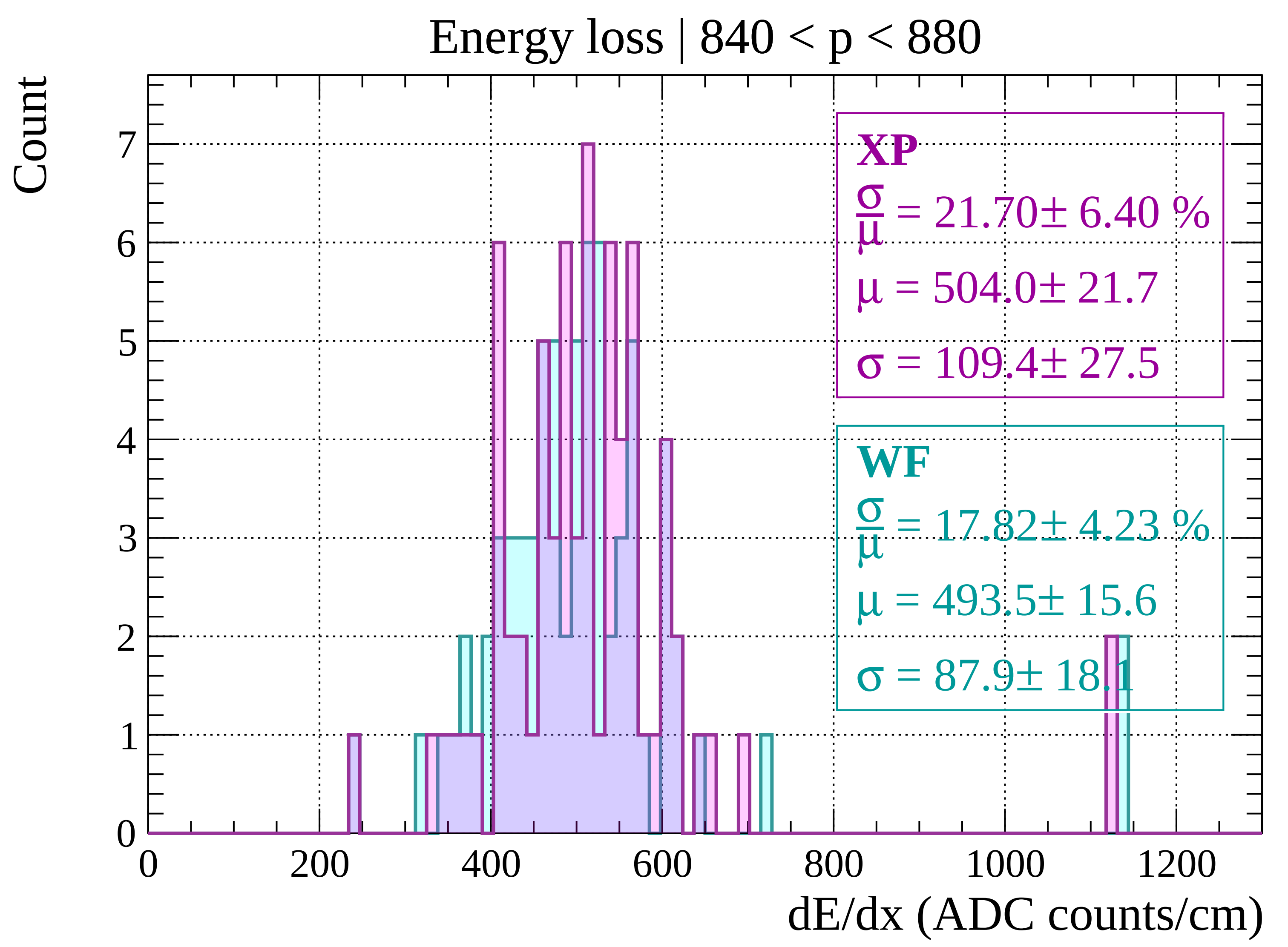

Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP
$\frac{\sigma}{\mu}$ = 23.83± 4.80 %
μ = 524.5± 20.3
σ = 125.0± 20.3
WF
$\frac{\sigma}{\mu}$ = 14.84± 2.76 %
μ = 505.0± 11.6
σ = 75.0± 12.2



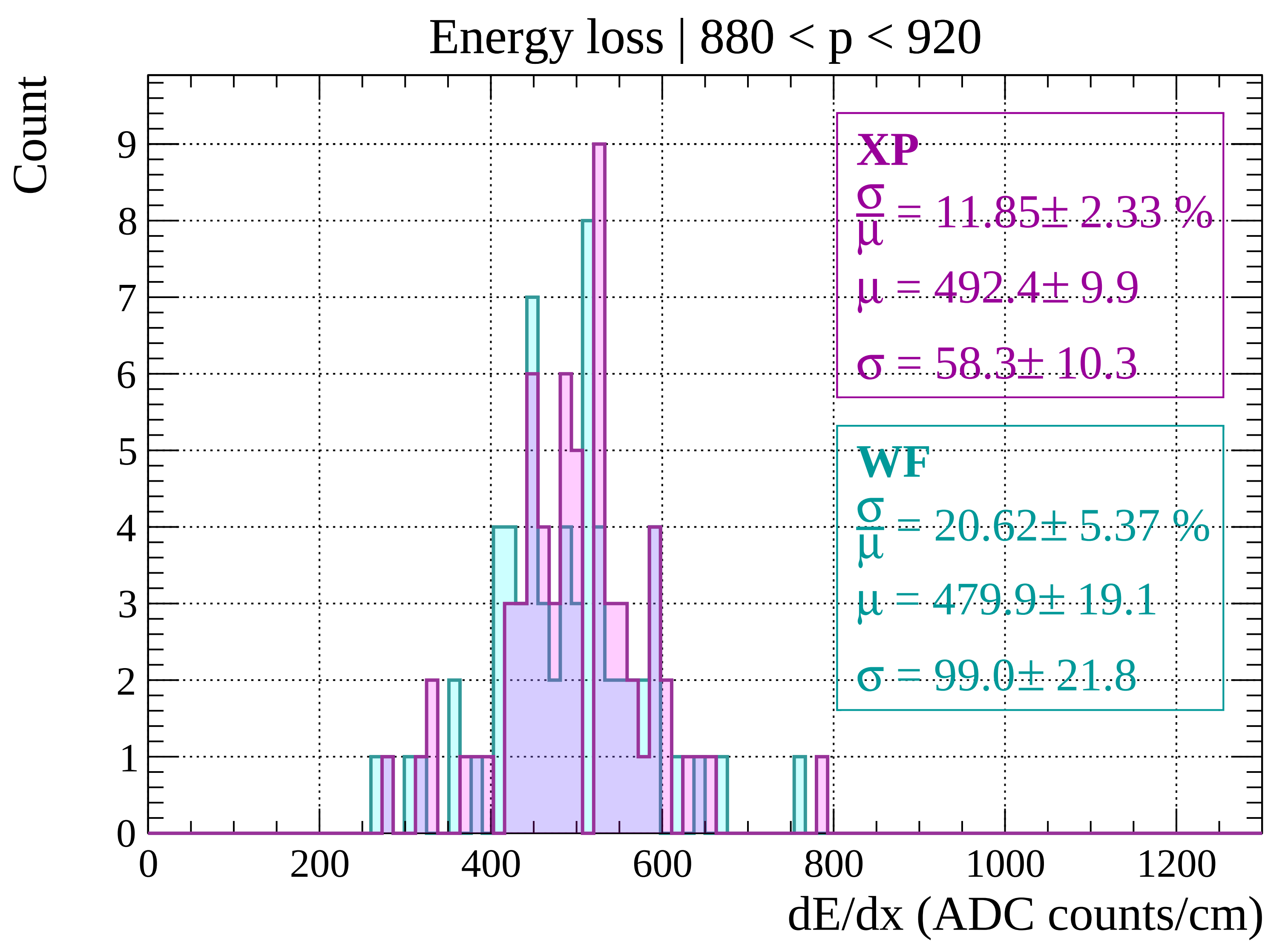
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 31.32± 11.96 %
μ = 503.3± 34.1
σ = 157.6± 49.5
WF
$\frac{\sigma}{\mu}$ = 32.18± 11.98 %
μ = 502.6± 37.3
σ = 161.7± 48.2
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7

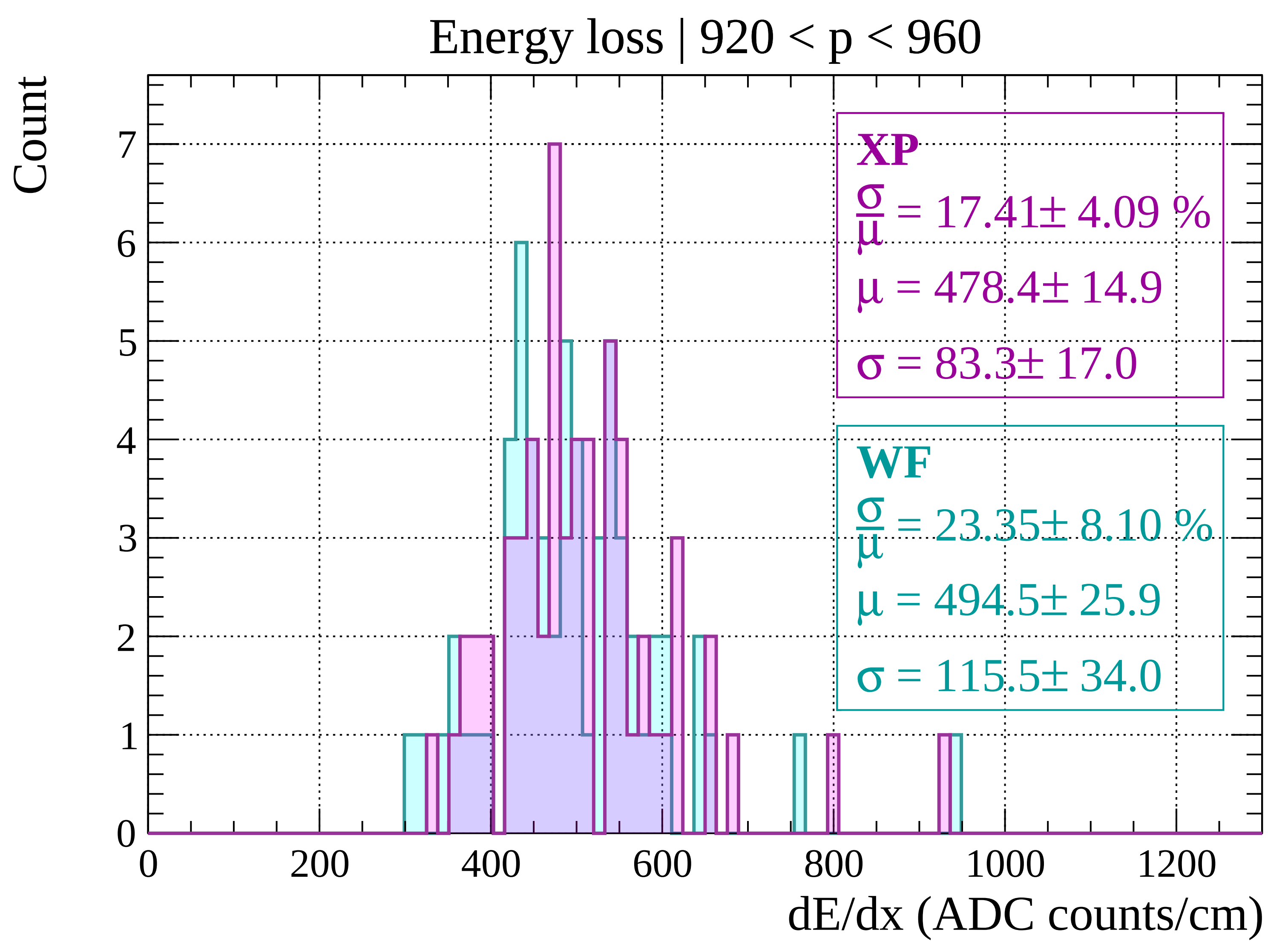


Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.87 ± 4.02 %
μ = 508.6 ± 14.7
σ = 80.7 ± 18.1
WF
$\frac{\sigma}{\mu}$ = 20.27 ± 6.03 %
μ = 481.2 ± 18.9
σ = 97.6 ± 25.2

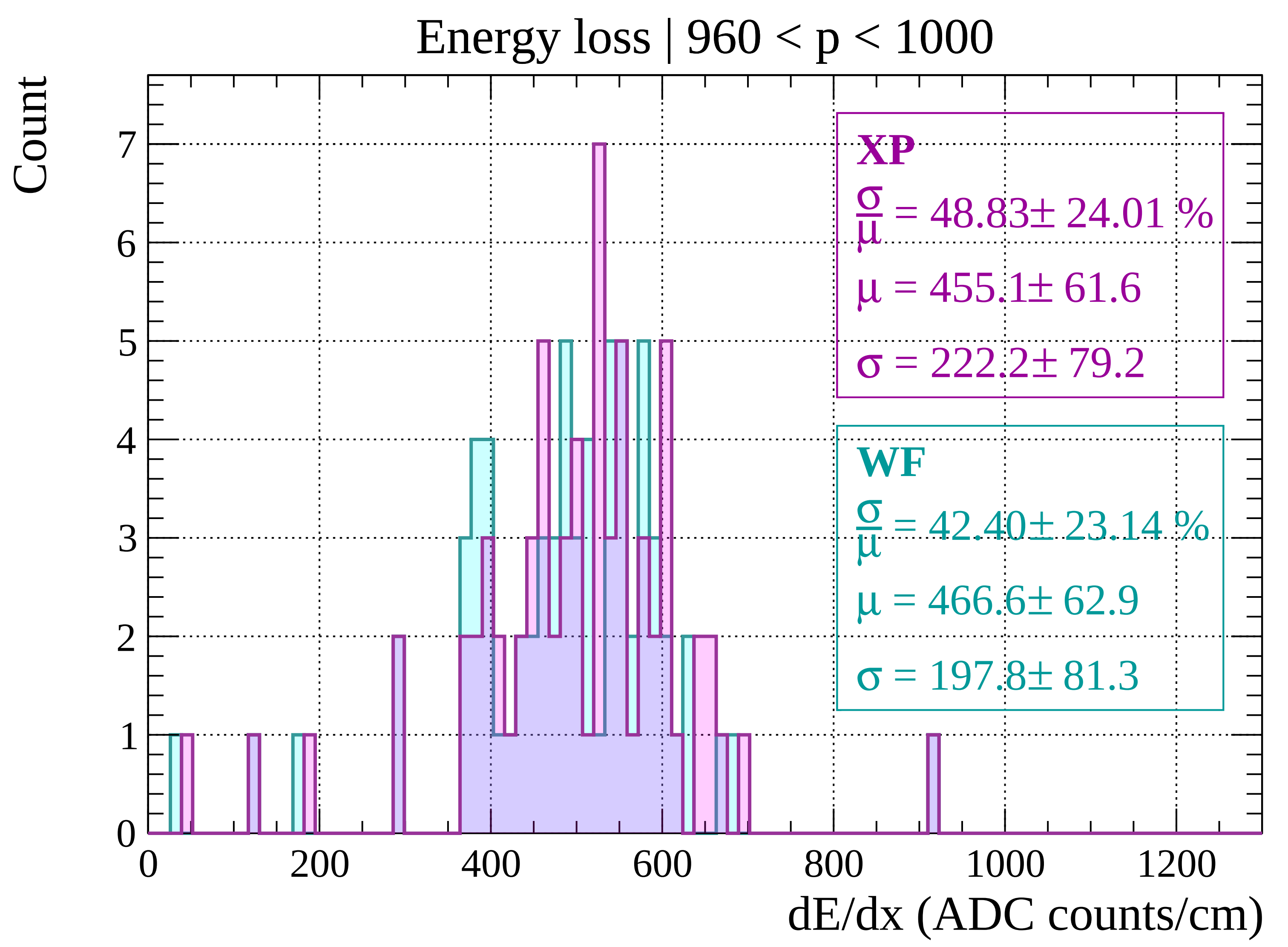


Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 21.70± 6.40 %
μ = 504.0± 21.7
σ = 109.4± 27.5
WF
σ/μ = 17.82± 4.23 %
μ = 493.5± 15.6
σ = 87.9± 18.1

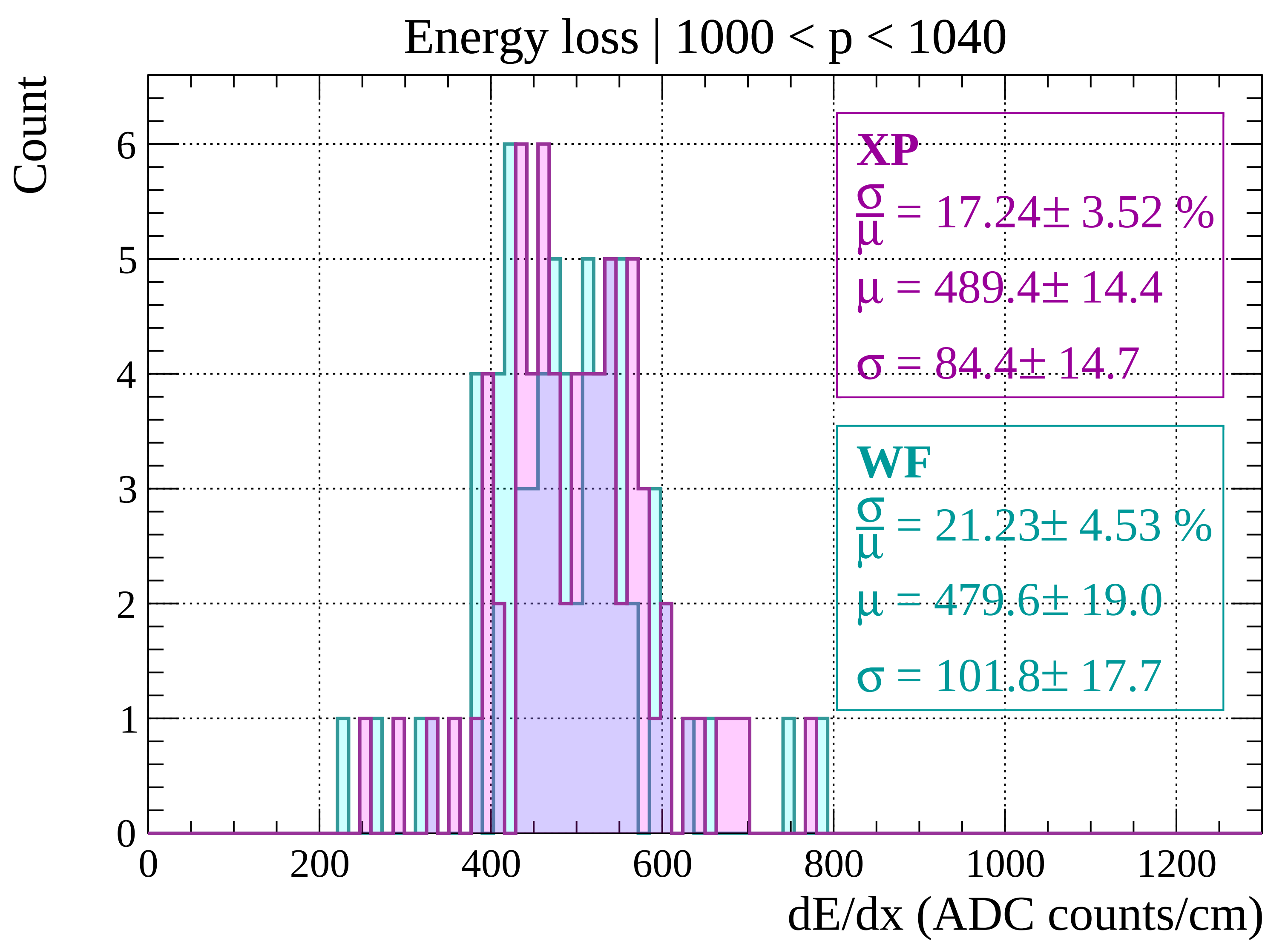

Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.85± 2.33 %
μ = 492.4± 9.9
σ = 58.3± 10.3
WF
σ/μ = 20.62± 5.37 %
μ = 479.9± 19.1
σ = 99.0± 21.8
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200

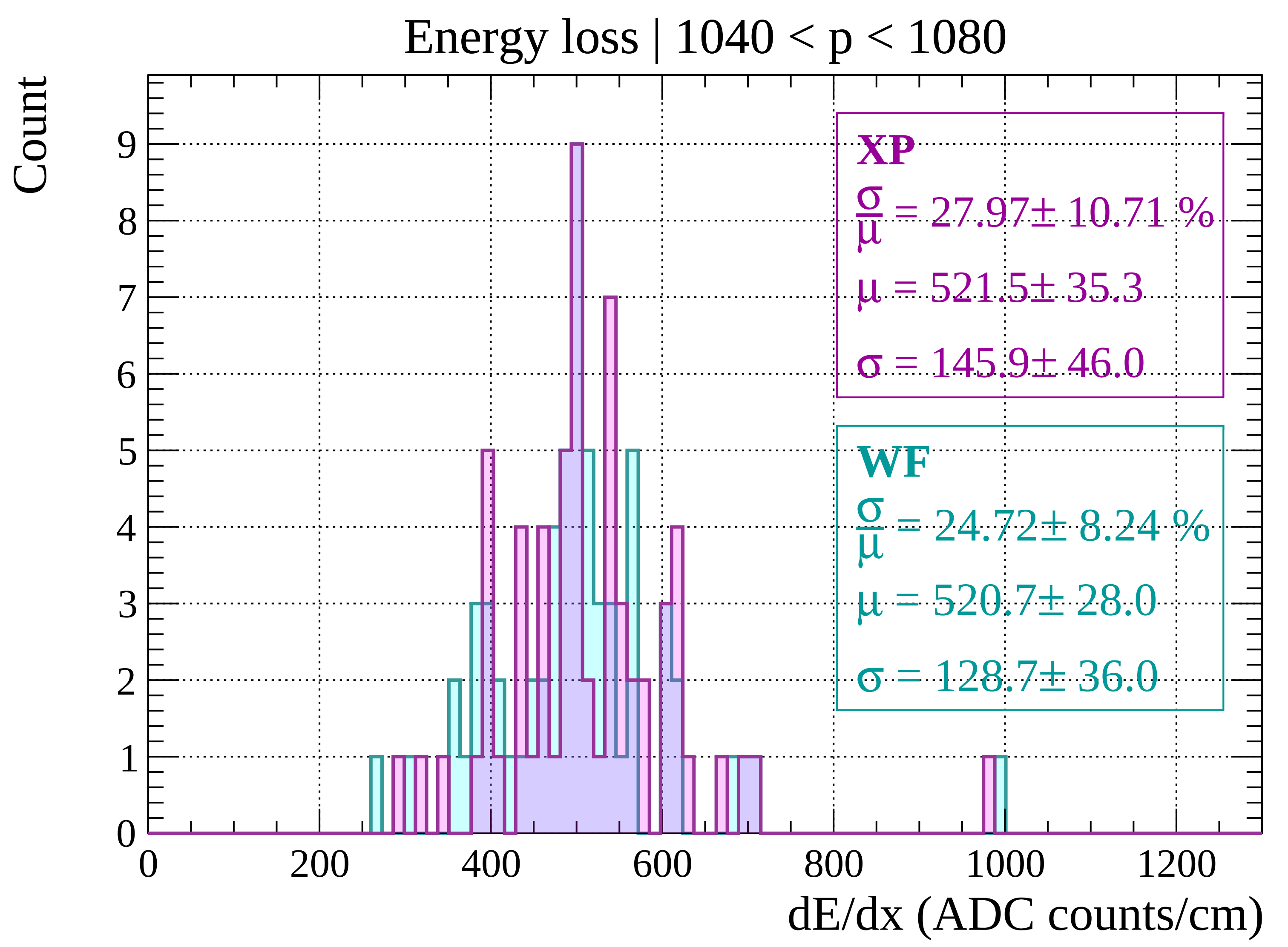


Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 17.41± 4.09 %
μ = 478.4± 14.9
σ = 83.3± 17.0
WF
σ/μ = 23.35± 8.10 %
μ = 494.5± 25.9
σ = 115.5± 34.0

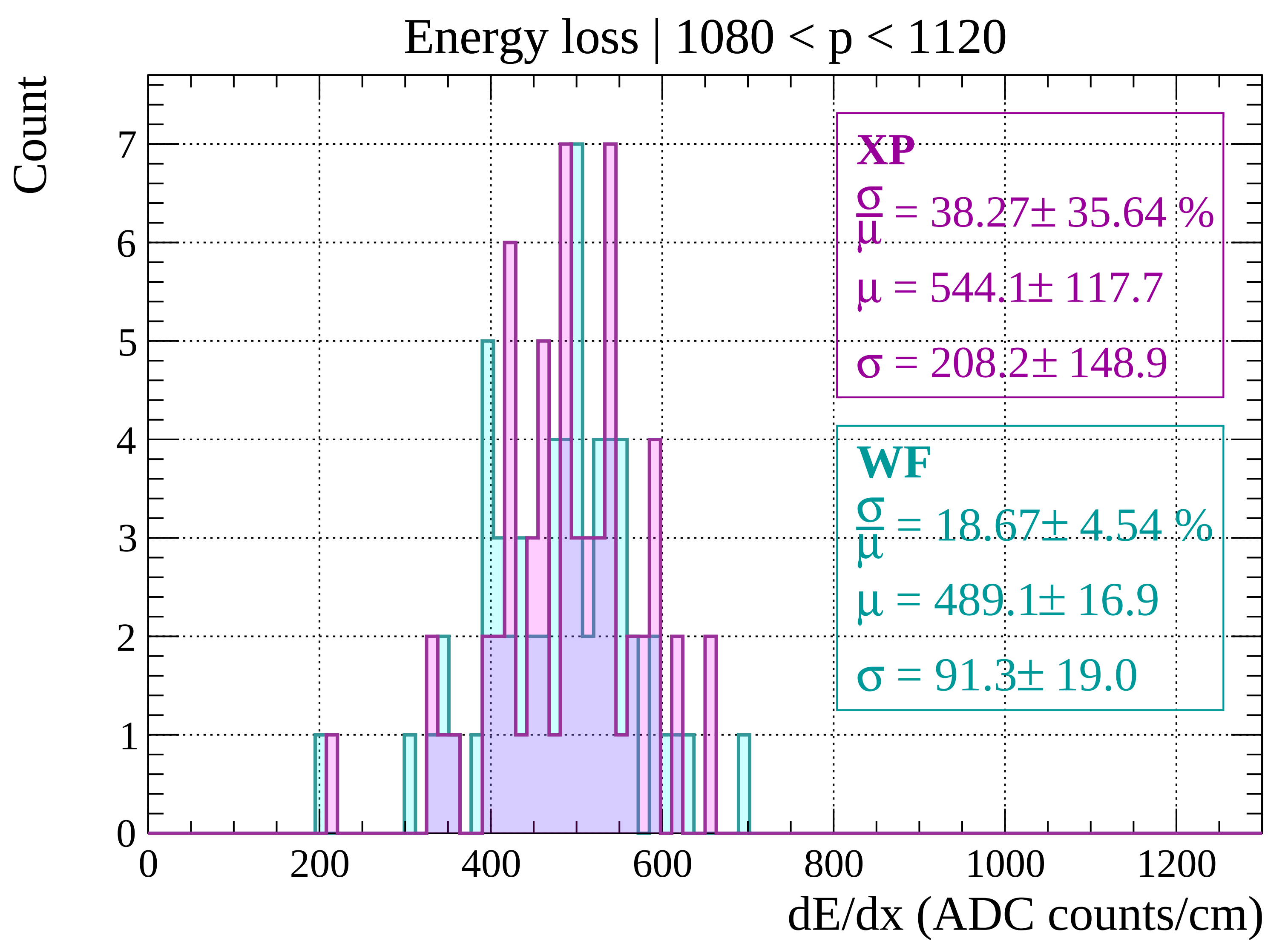


Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 48.83± 24.01 %
μ = 455.1± 61.6
σ = 222.2± 79.2
WF
$\frac{\sigma}{\mu}$ = 42.40± 23.14 %
μ = 466.6± 62.9
σ = 197.8± 81.3

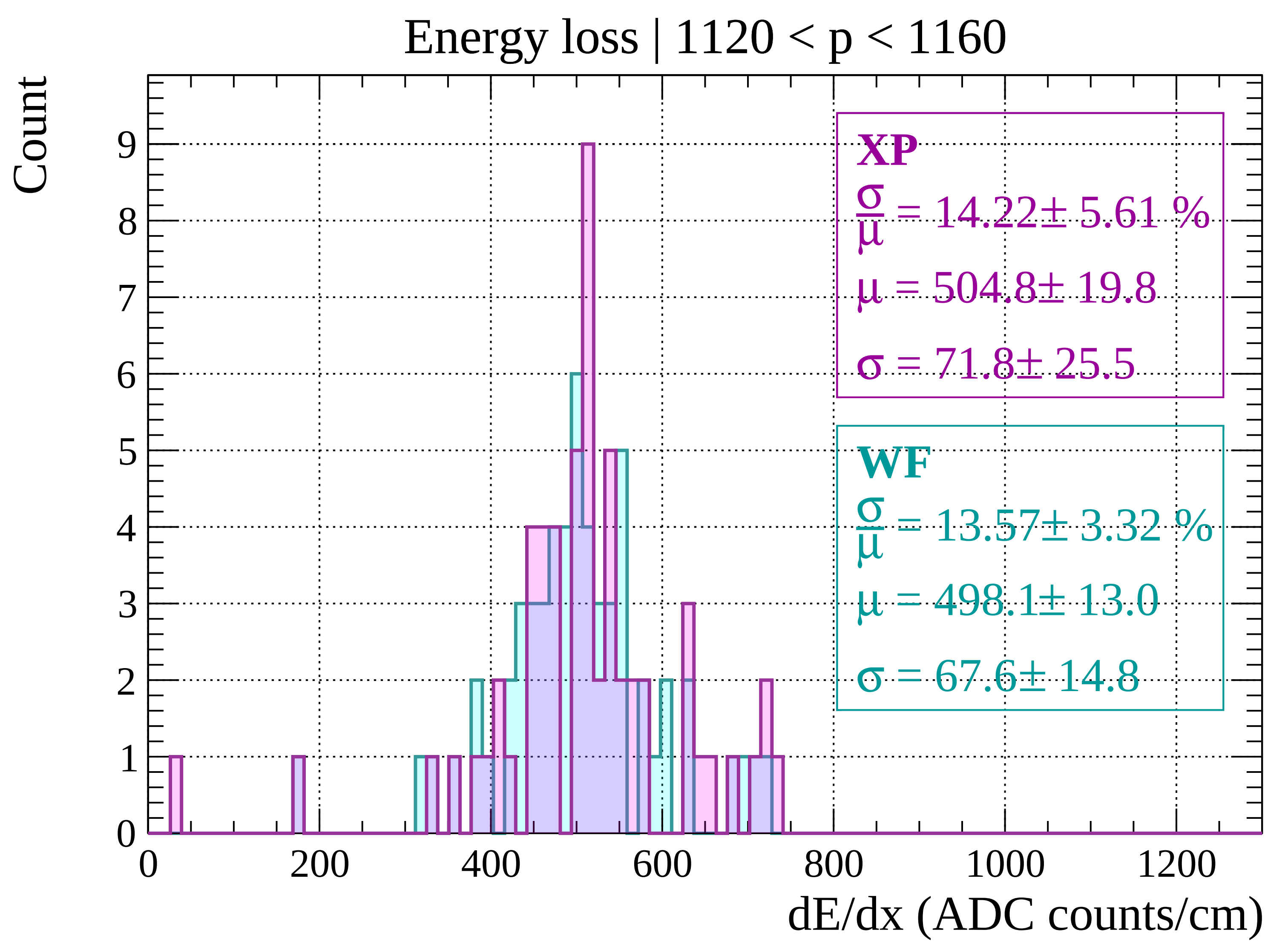

Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP
σ/μ = 17.24± 3.52 %
μ = 489.4± 14.4
σ = 84.4± 14.7
WF
σ/μ = 21.23± 4.53 %
μ = 479.6± 19.0
σ = 101.8± 17.7

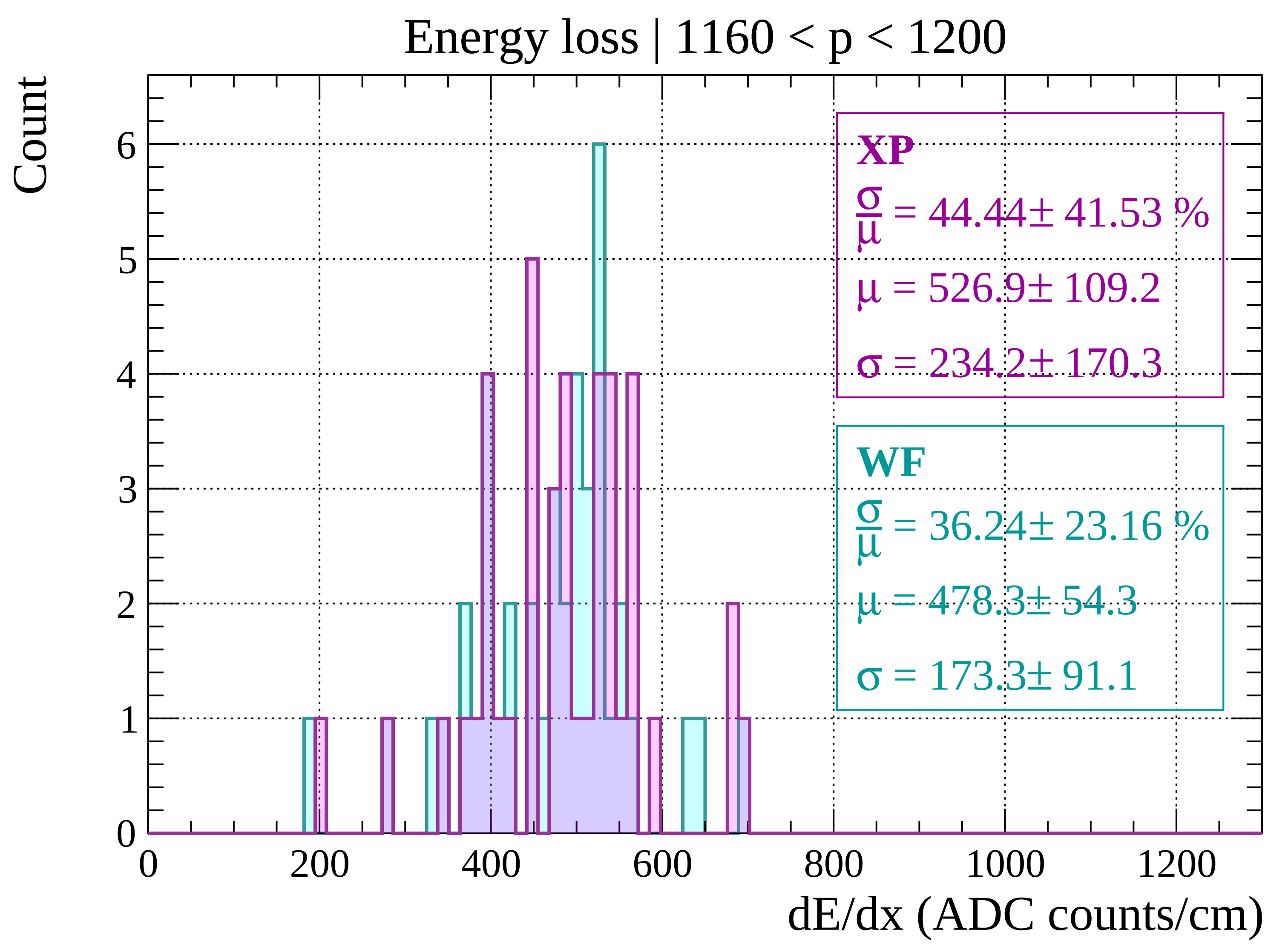


Energy loss | 1040 < p < 1080
Count
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 27.97± 10.71 %
μ = 521.5± 35.3
σ = 145.9± 46.0
WF
σ/μ = 24.72± 8.24 %
μ = 520.7± 28.0
σ = 128.7± 36.0

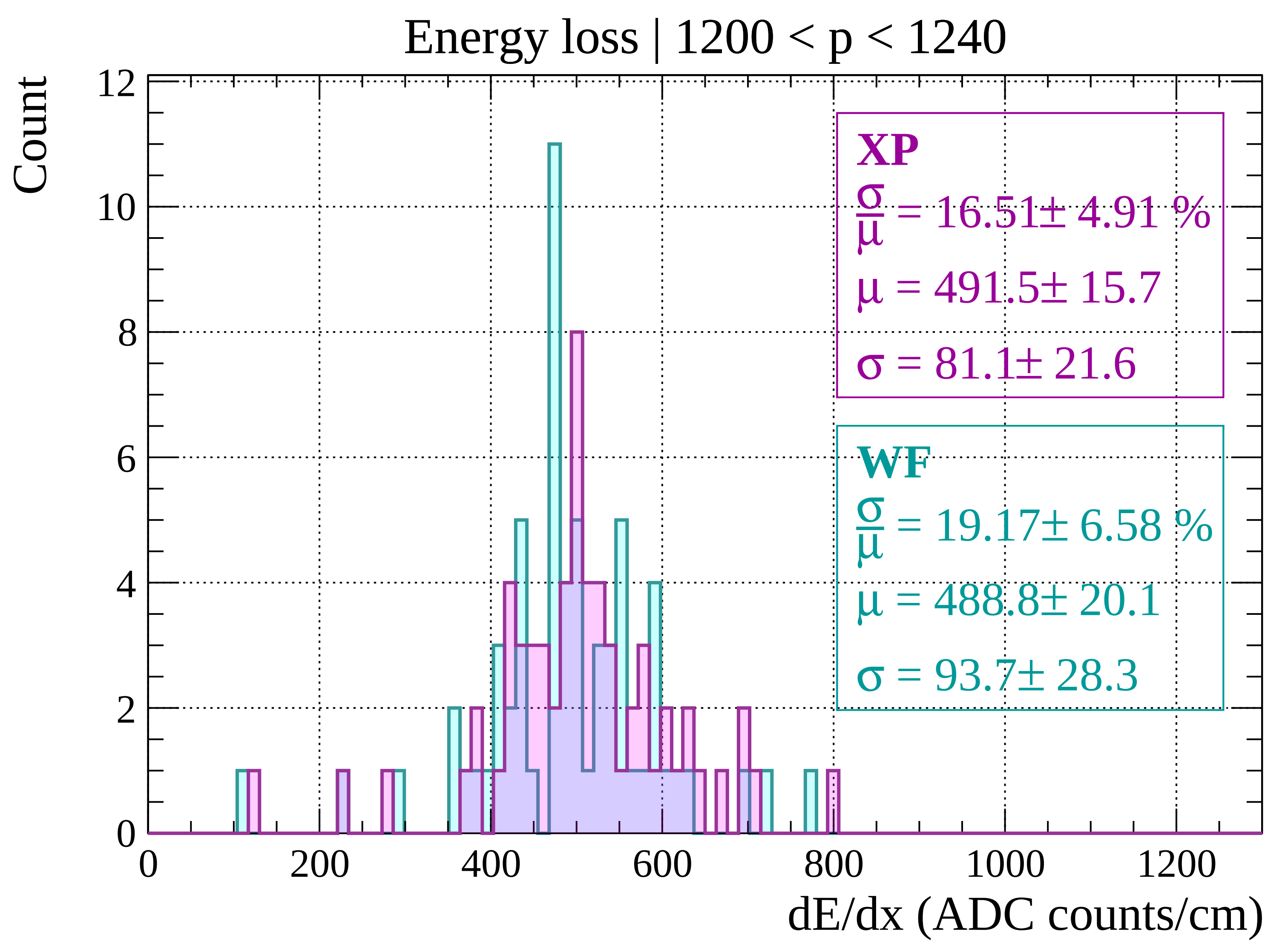


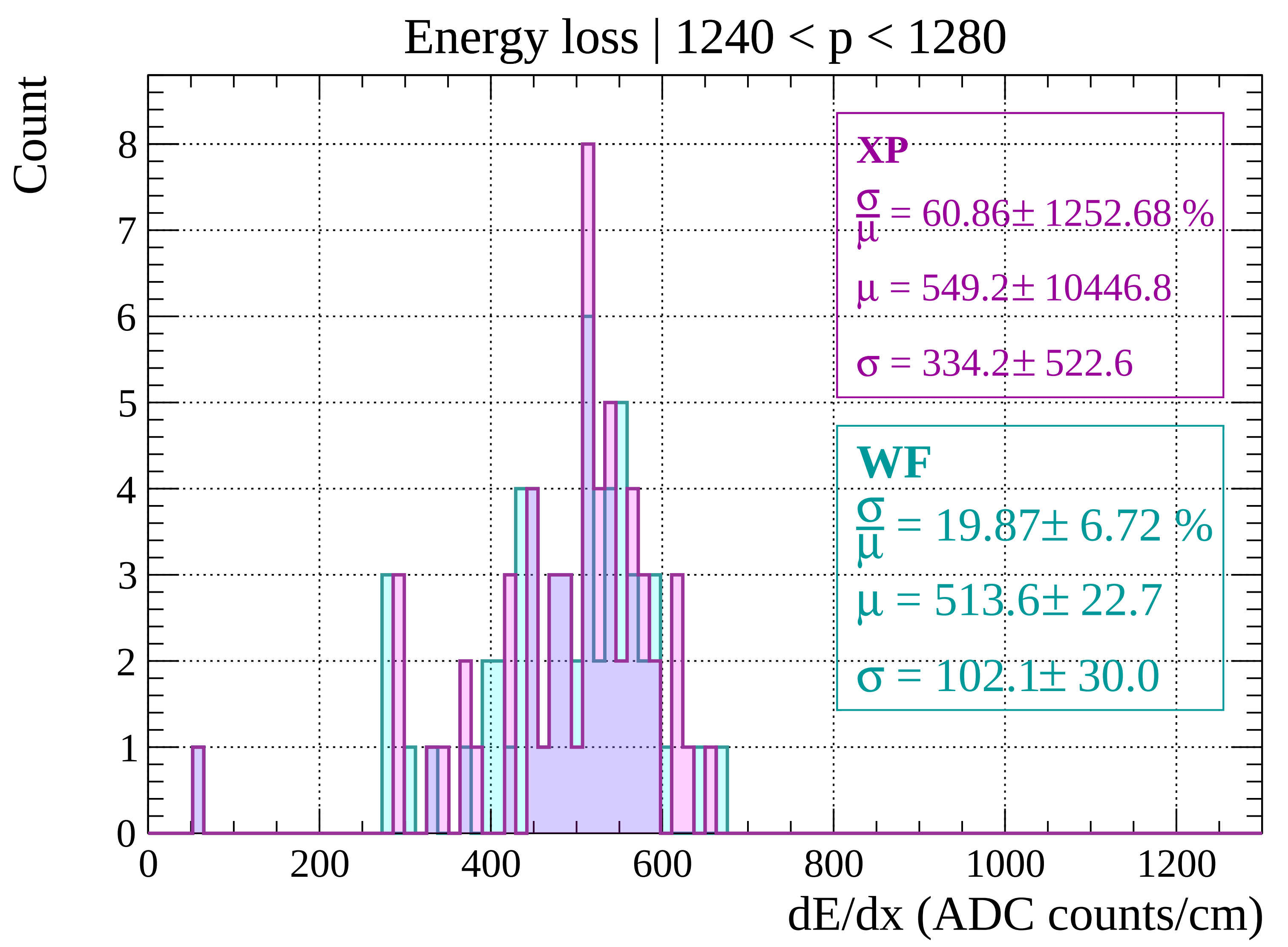
Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
$\frac{\sigma}{\mu}$ = 38.27± 35.64 %
μ = 544.1± 117.7
σ = 208.2± 148.9
WF
$\frac{\sigma}{\mu}$ = 18.67± 4.54 %
μ = 489.1± 16.9
σ = 91.3± 19.0


Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 14.22 ± 5.61 %
μ = 504.8 ± 19.8
σ = 71.8 ± 25.5
WF
$\frac{\sigma}{\mu}$ = 13.57 ± 3.32 %
μ = 498.1 ± 13.0
σ = 67.6 ± 14.8

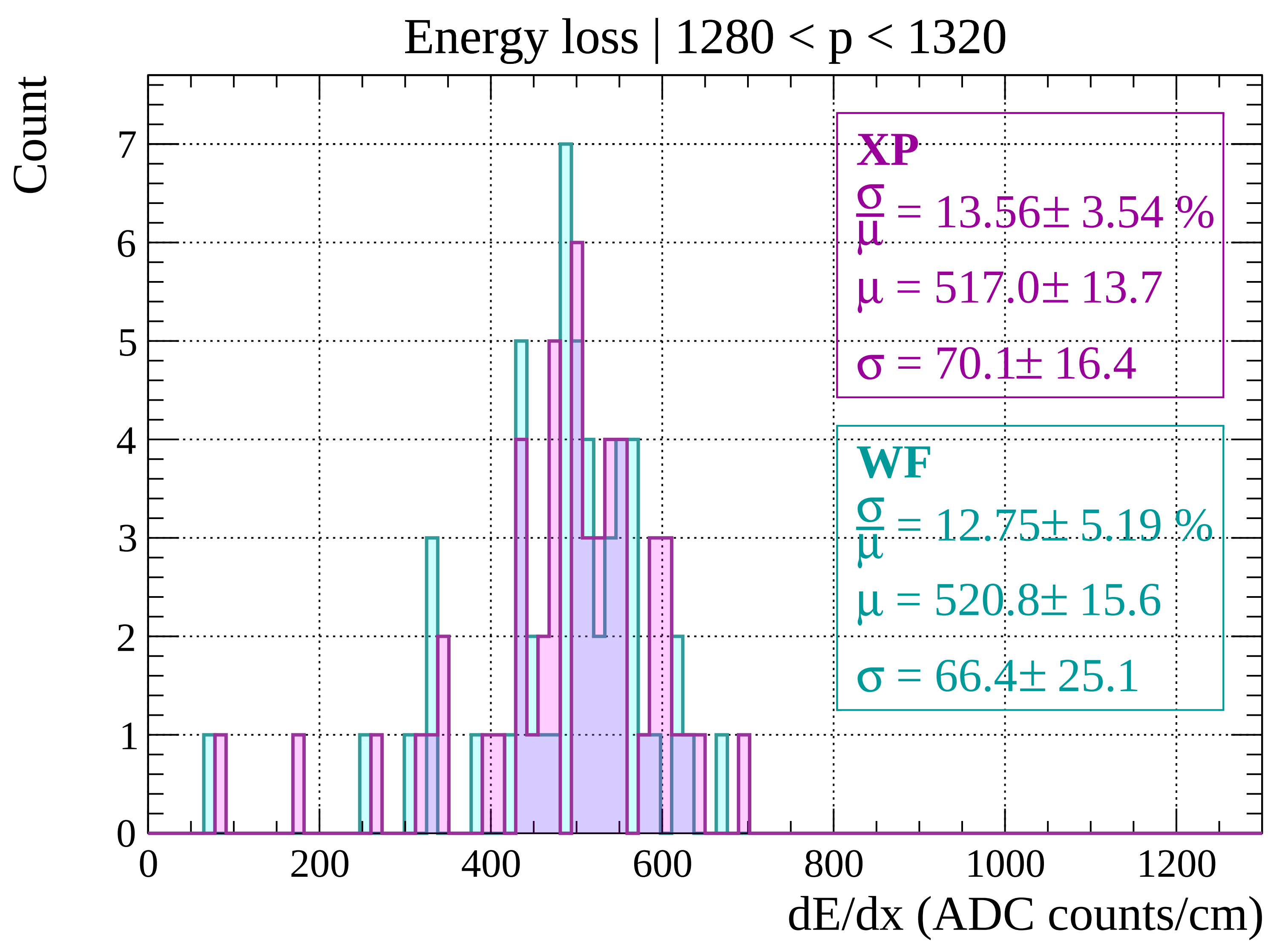


Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
$\frac{\sigma}{\mu}$ = 44.44± 41.53 %
μ = 526.9± 109.2
σ = 234.2± 170.3
WF
$\frac{\sigma}{\mu}$ = 36.24± 23.16 %
μ = 478.3± 54.3
σ = 173.3± 91.1

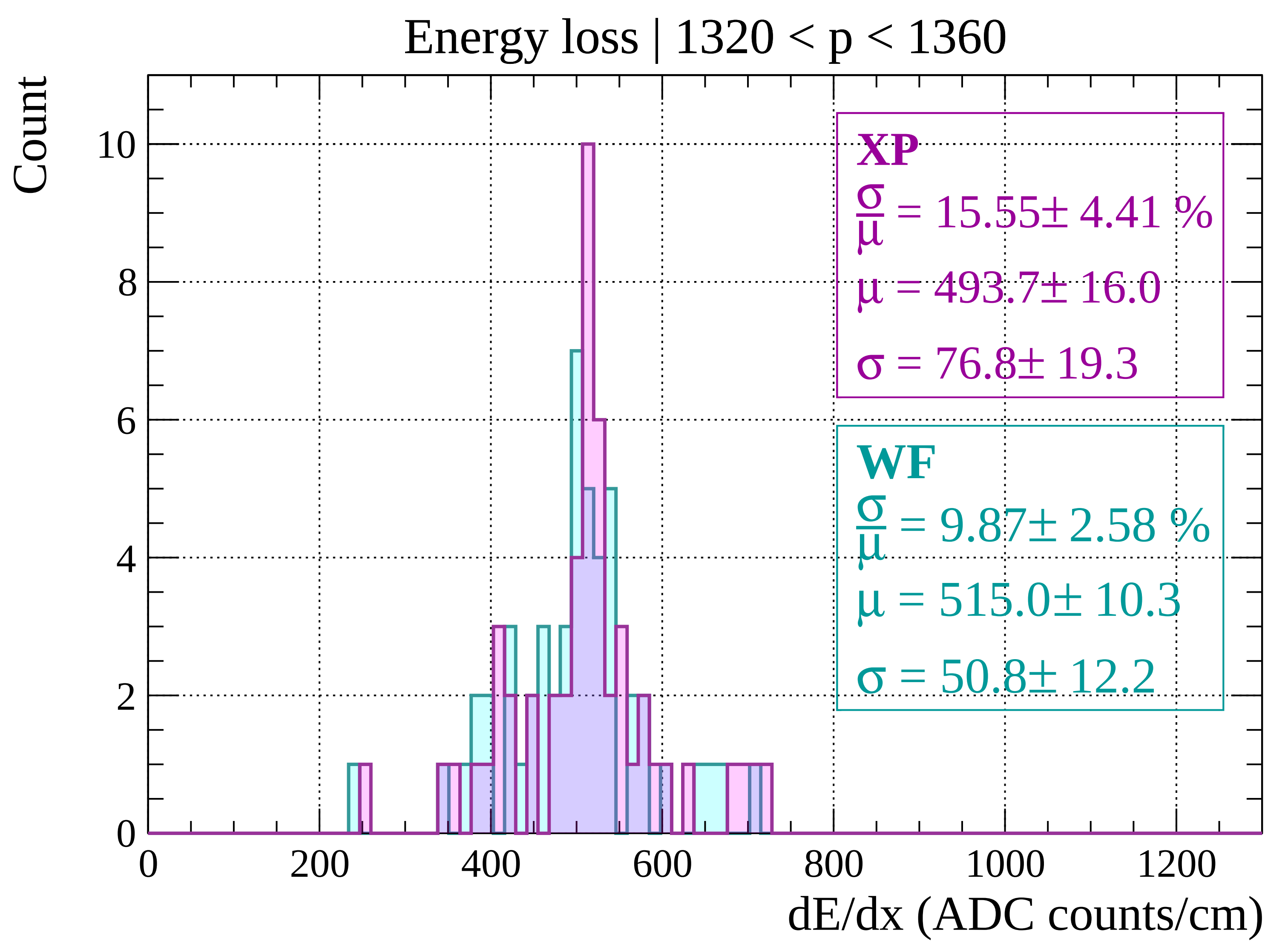

Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
XP
σ/μ = 16.51± 4.91 %
μ = 491.5± 15.7
σ = 81.1± 21.6
WF
σ/μ = 19.17± 6.58 %
μ = 488.8± 20.1
σ = 93.7± 28.3



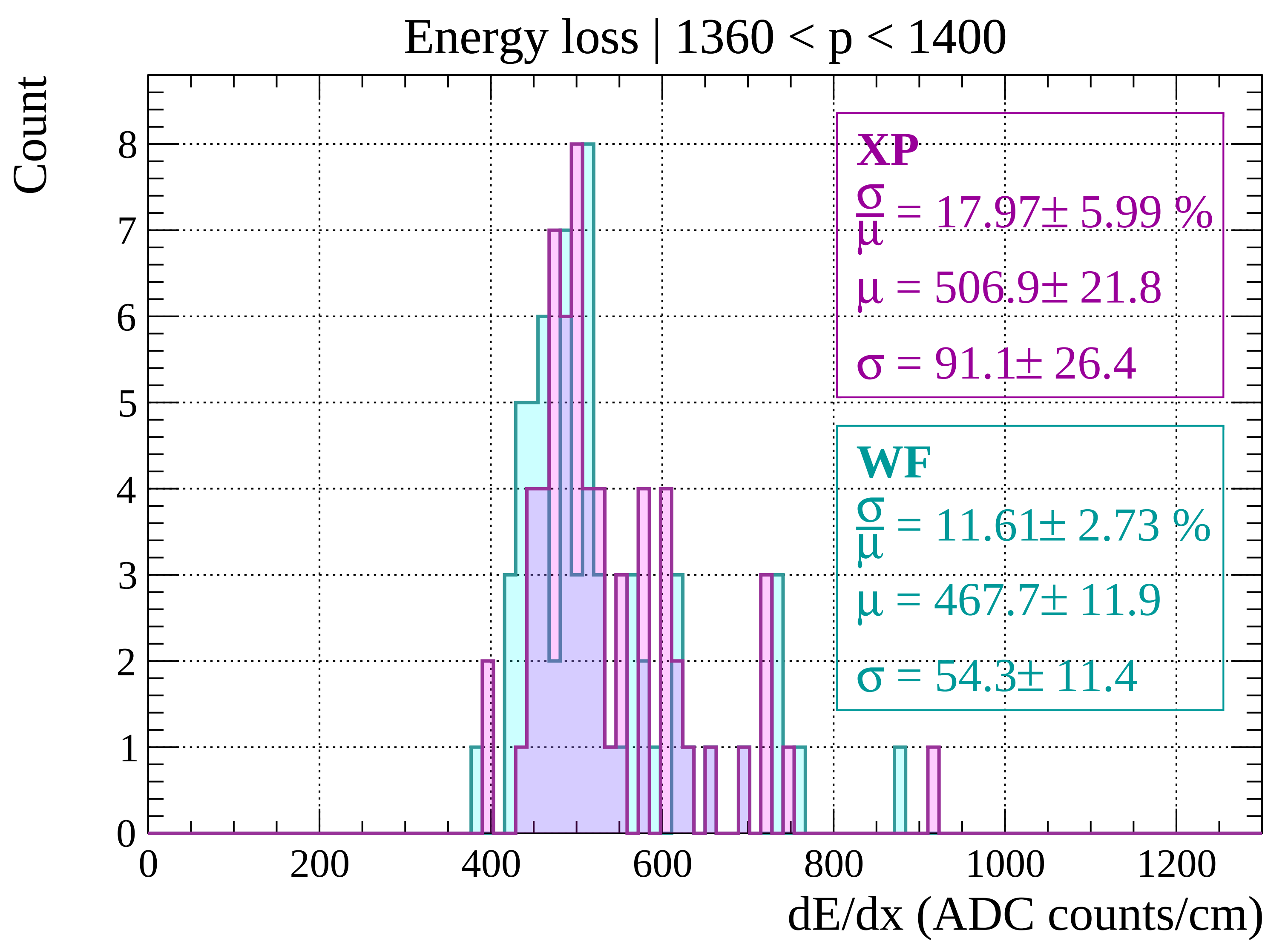
Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 60.86± 1252.68 %
μ = 549.2± 10446.8
σ = 334.2± 522.6
WF
σ/μ = 19.87± 6.72 %
μ = 513.6± 22.7
σ = 102.1± 30.0



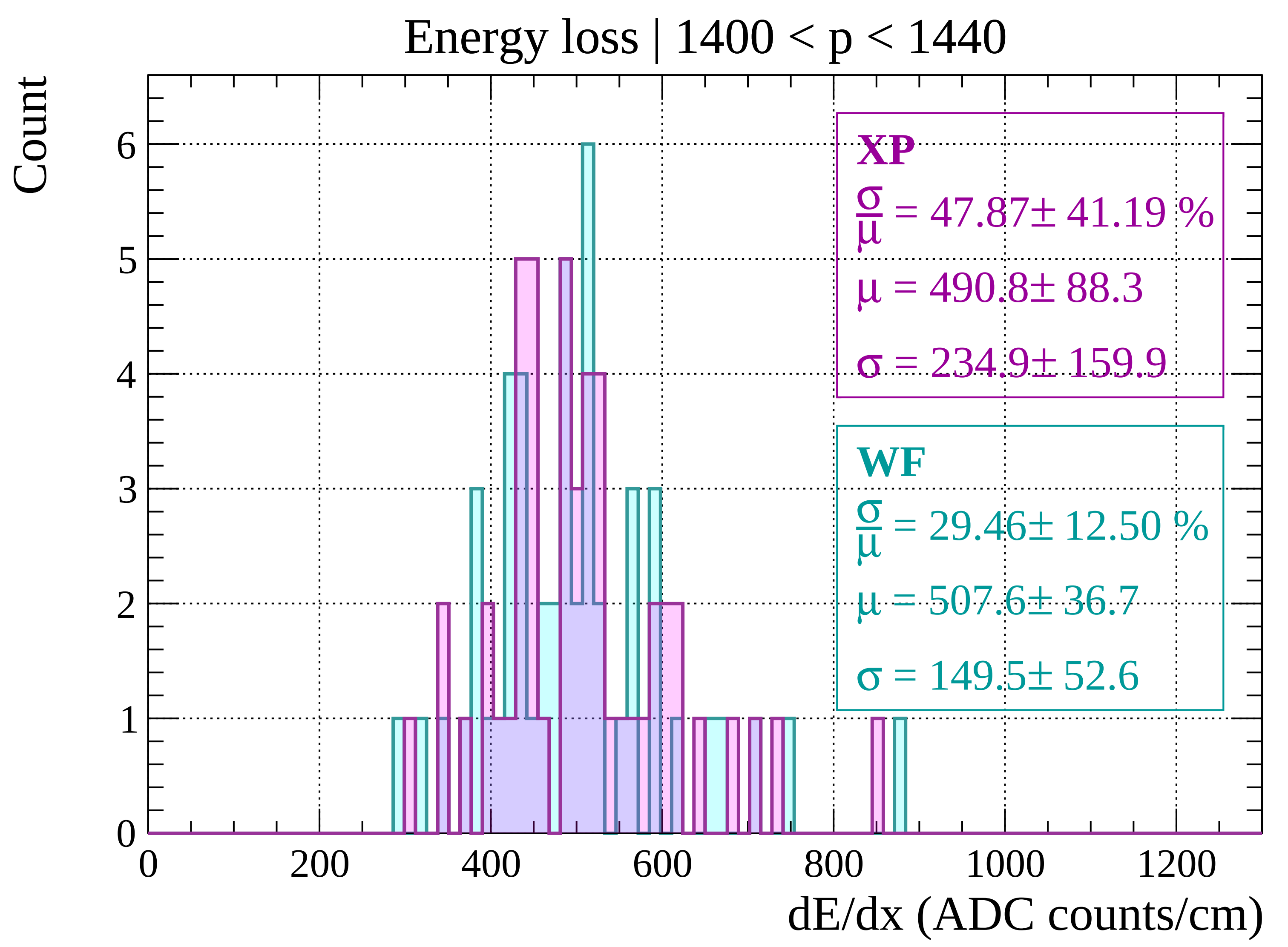
Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 13.56± 3.54 %
μ = 517.0± 13.7
σ = 70.1± 16.4
WF
σ/μ = 12.75± 5.19 %
μ = 520.8± 15.6
σ = 66.4± 25.1

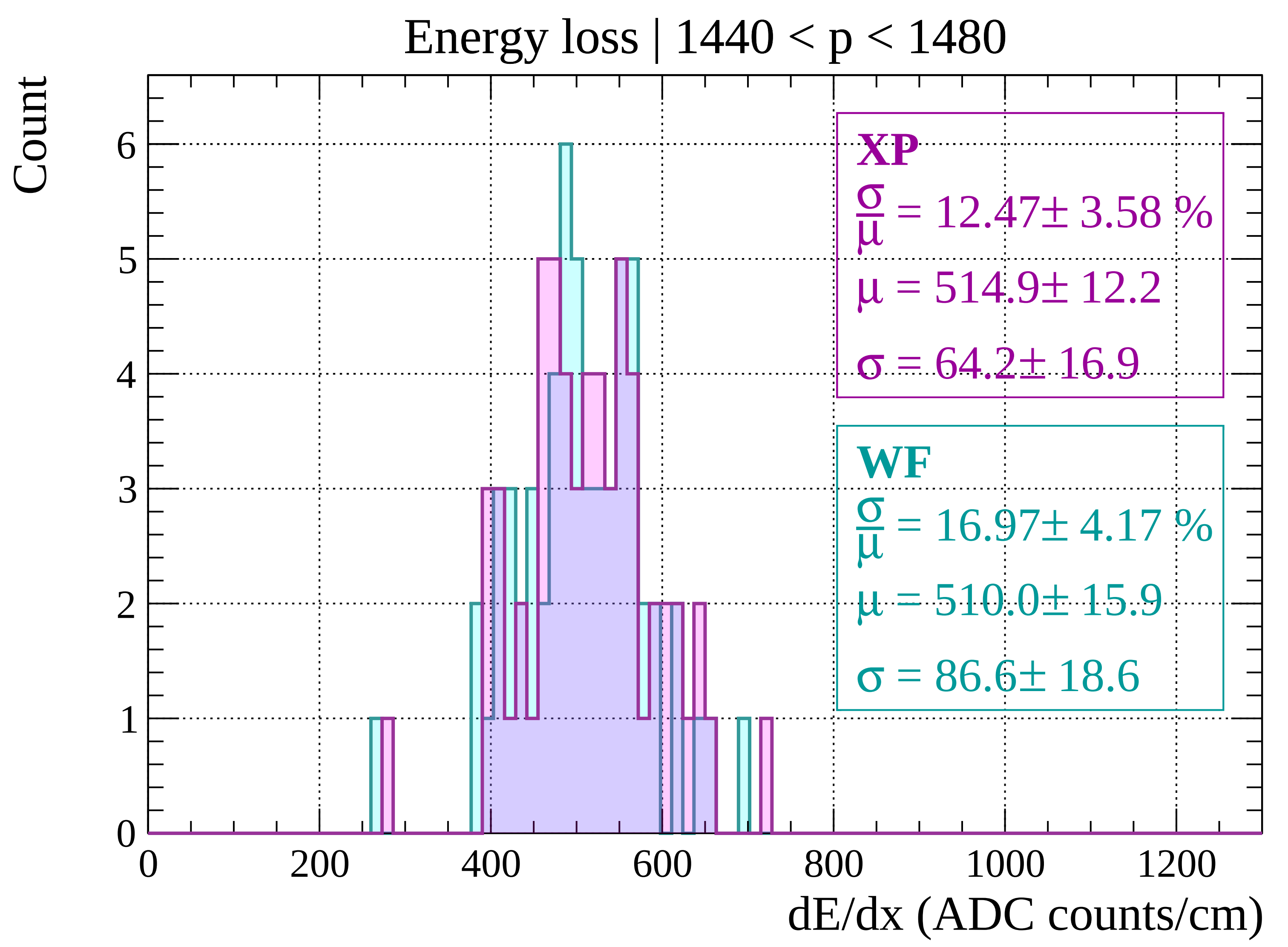


Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP
$\frac{\sigma}{\mu}$ = 15.55± 4.41 %
μ = 493.7± 16.0
σ = 76.8± 19.3
WF
$\frac{\sigma}{\mu}$ = 9.87± 2.58 %
μ = 515.0± 10.3
σ = 50.8± 12.2

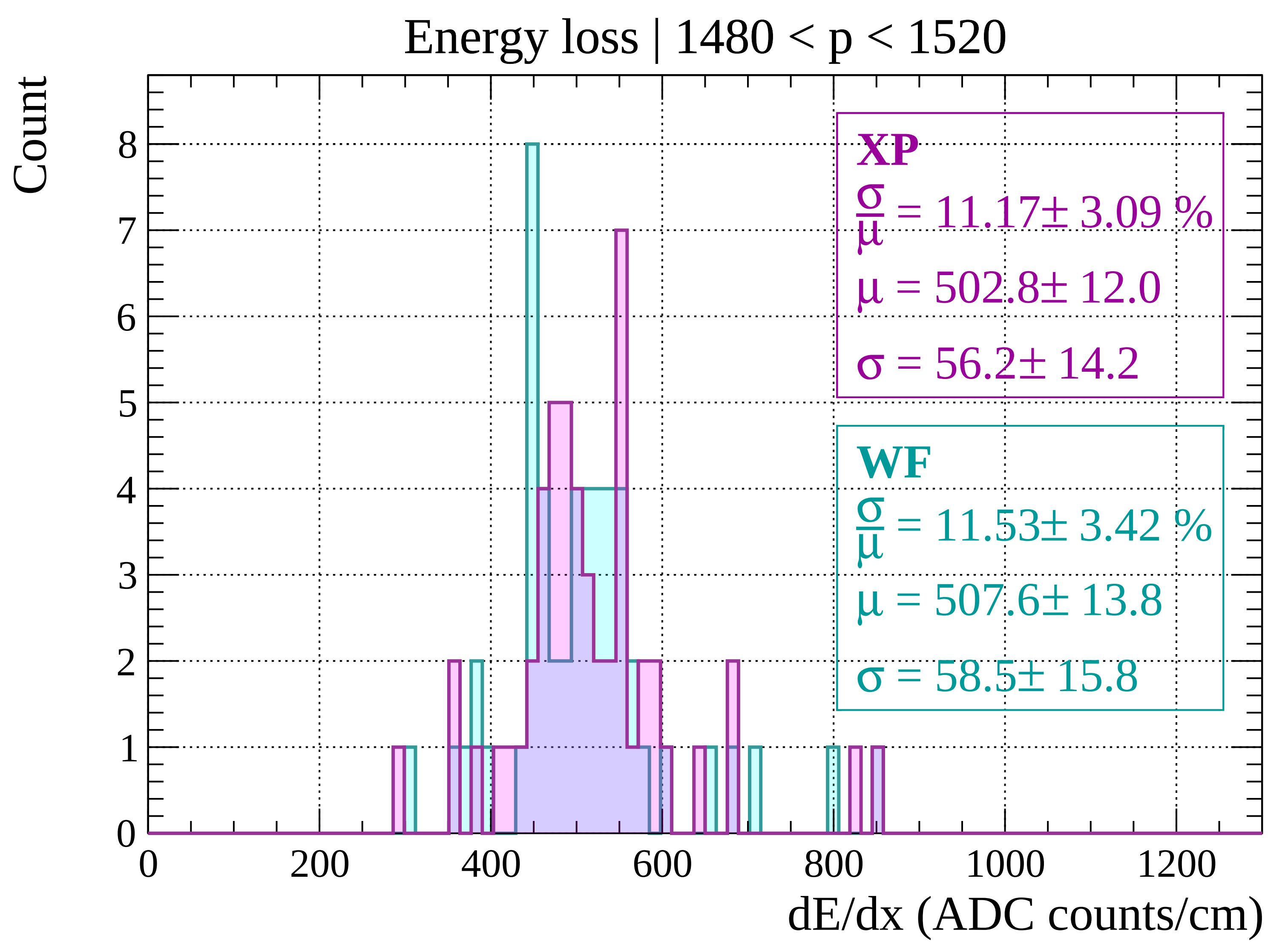


Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 17.97± 5.99 %
μ = 506.9± 21.8
σ = 91.1± 26.4
WF
σ/μ = 11.61± 2.73 %
μ = 467.7± 11.9
σ = 54.3± 11.4

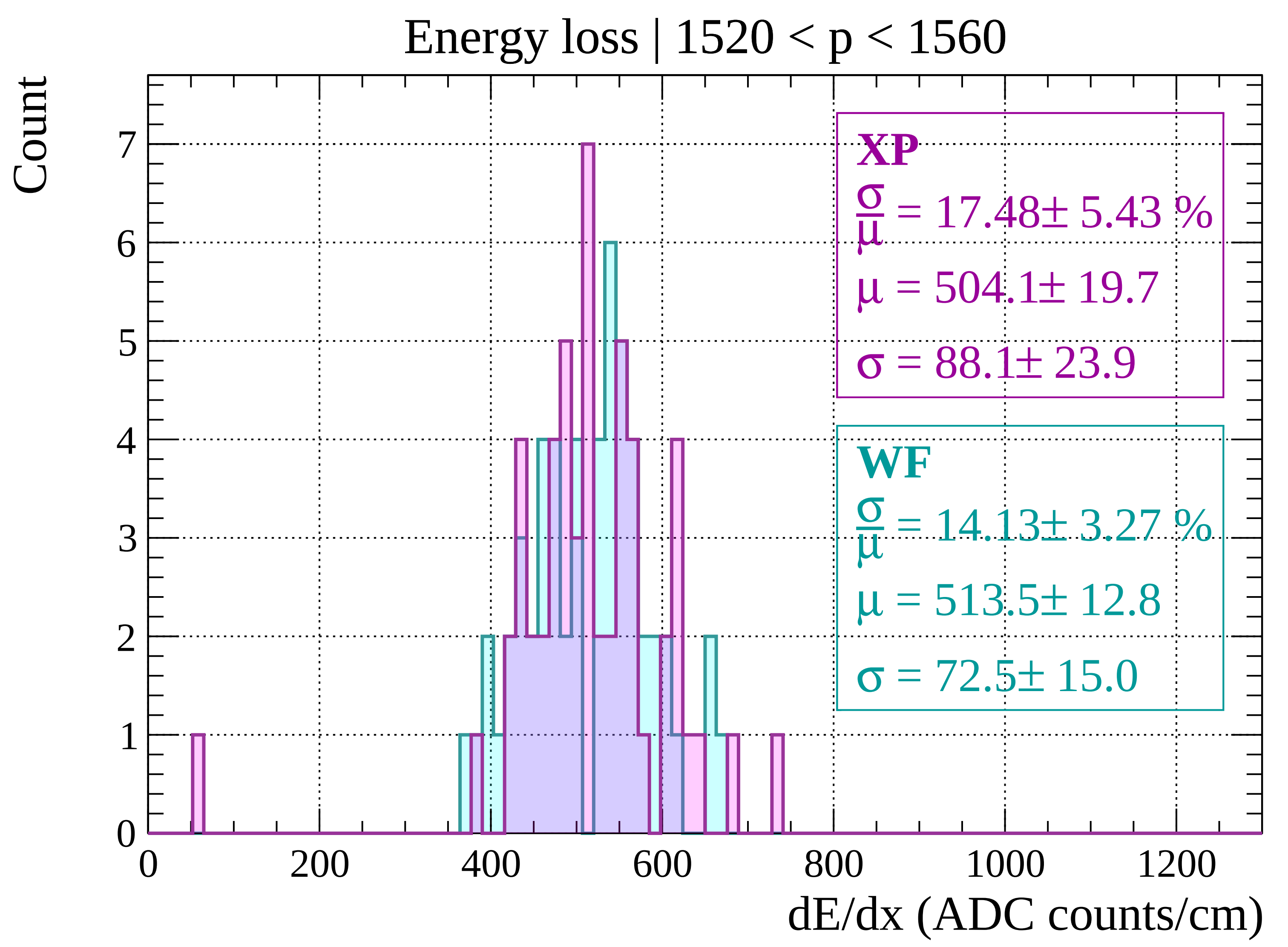

Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 47.87± 41.19 %
μ = 490.8± 88.3
σ = 234.9± 159.9
WF
$\frac{\sigma}{\mu}$ = 29.46± 12.50 %
μ = 507.6± 36.7
σ = 149.5± 52.6

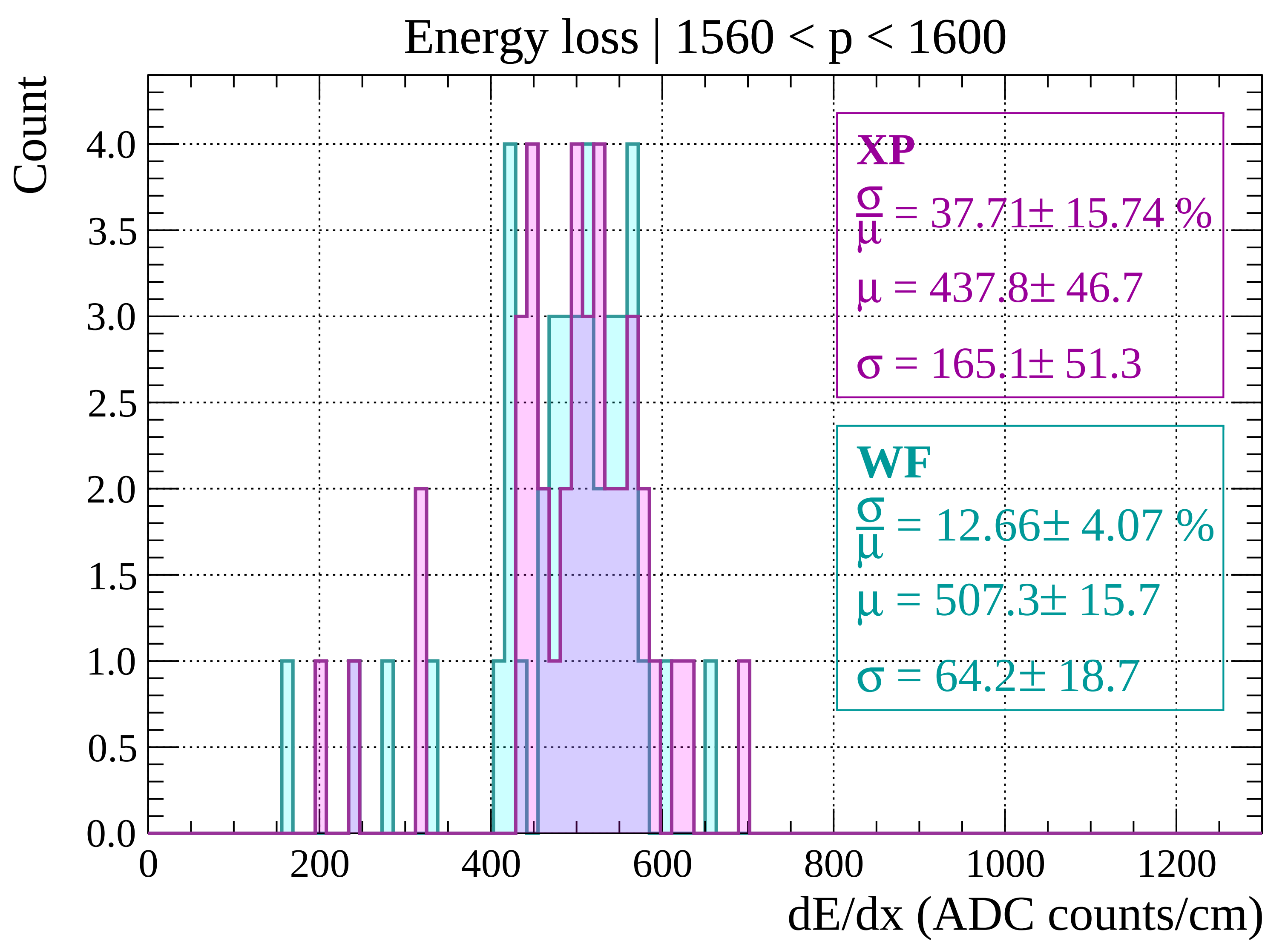


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP
σ/μ = 12.47± 3.58 %
μ = 514.9± 12.2
σ = 64.2± 16.9
WF
σ/μ = 16.97± 4.17 %
μ = 510.0± 15.9
σ = 86.6± 18.6

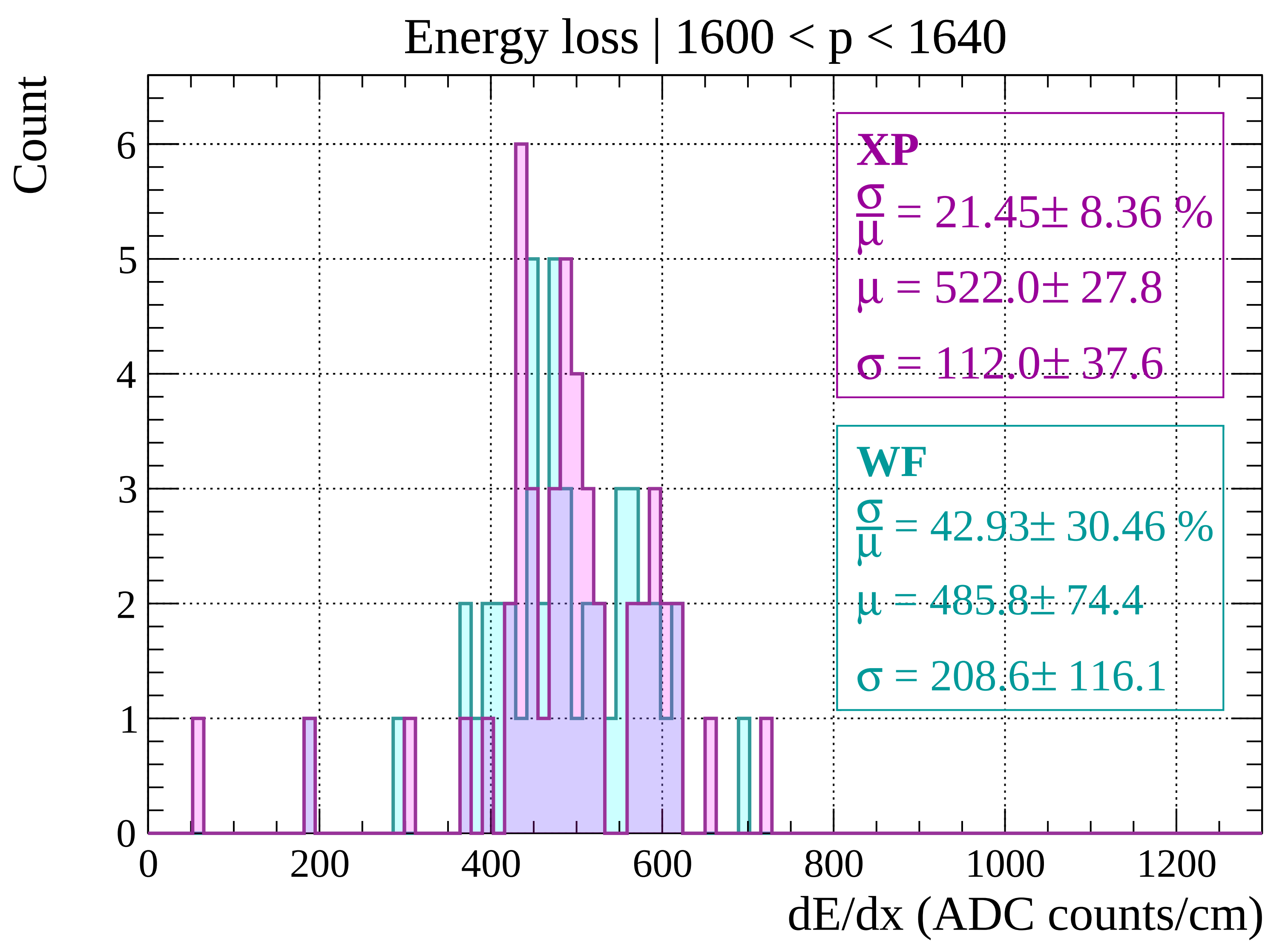


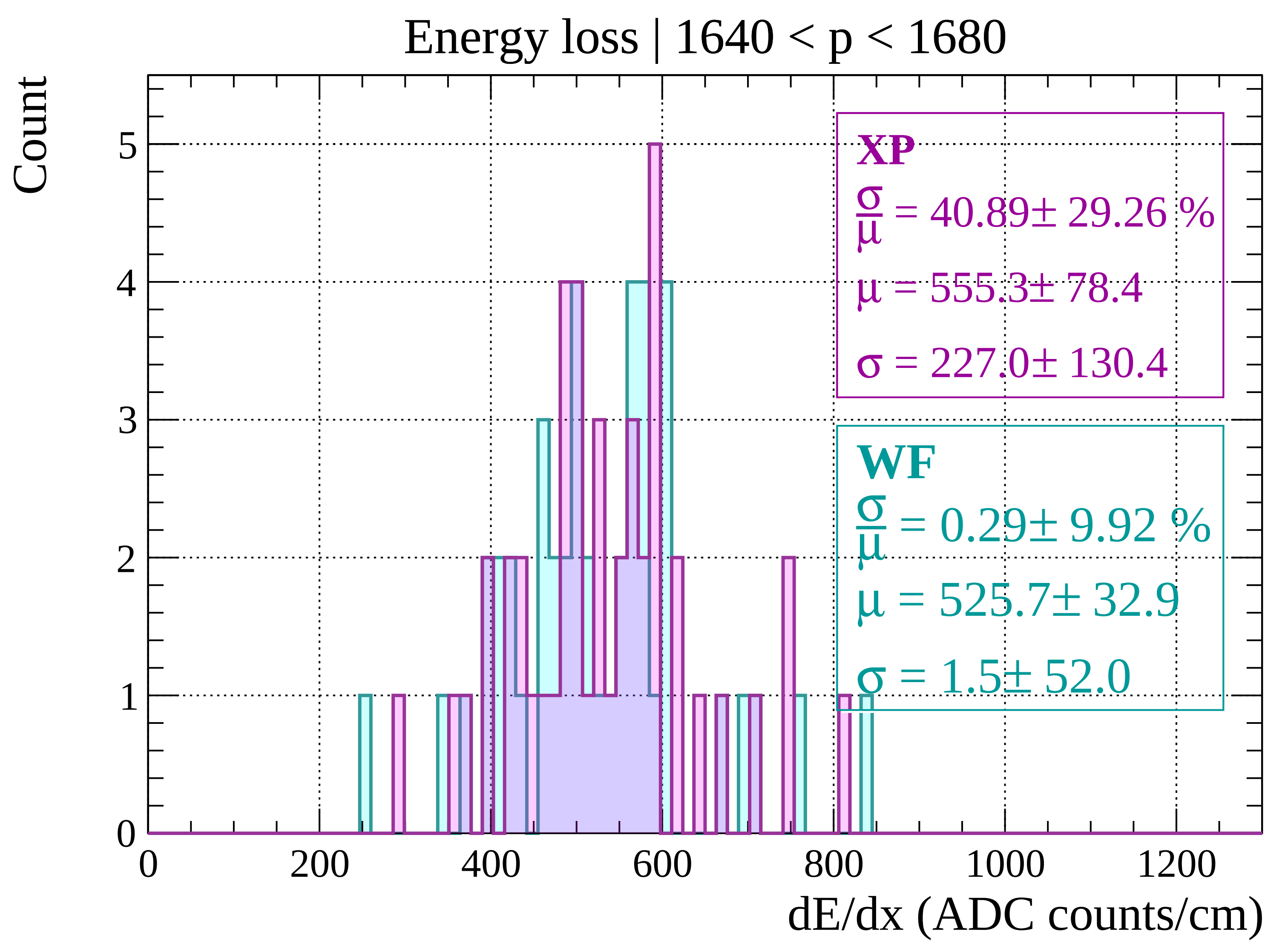
Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 11.17± 3.09 %
μ = 502.8± 12.0
σ = 56.2± 14.2
WF
$\frac{\sigma}{\mu}$ = 11.53± 3.42 %
μ = 507.6± 13.8
σ = 58.5± 15.8


Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 17.48± 5.43 %
μ = 504.1± 19.7
σ = 88.1± 23.9
WF
σ/μ = 14.13± 3.27 %
μ = 513.5± 12.8
σ = 72.5± 15.0

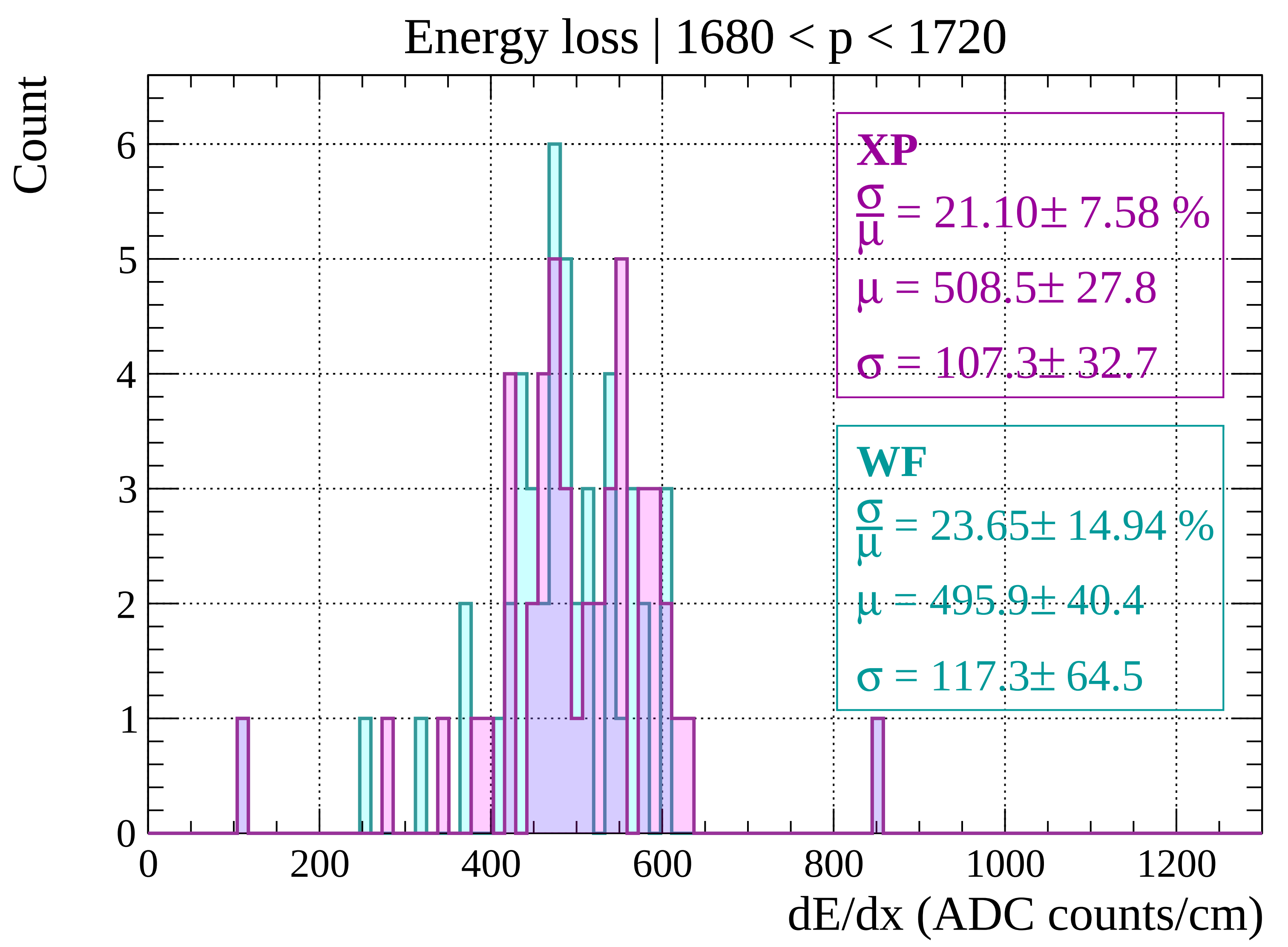


Energy loss | 1560 < p < 1600
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 37.71± 15.74 %
μ = 437.8± 46.7
σ = 165.1± 51.3
WF
$\frac{\sigma}{\mu}$ = 12.66± 4.07 %
μ = 507.3± 15.7
σ = 64.2± 18.7

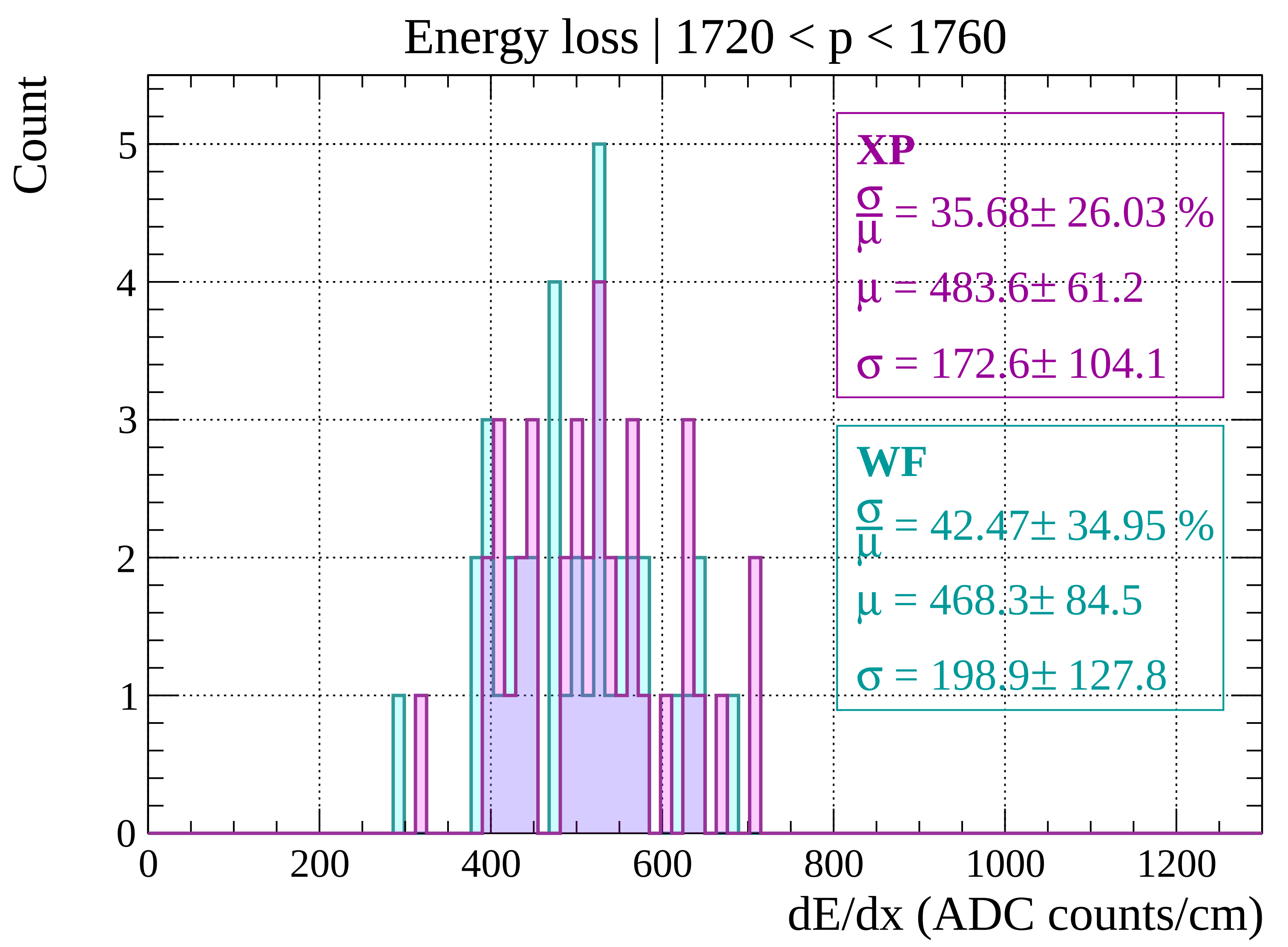

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
$\frac{\sigma}{\mu}$ = 21.45± 8.36 %
μ = 522.0± 27.8
σ = 112.0± 37.6
WF
$\frac{\sigma}{\mu}$ = 42.93± 30.46 %
μ = 485.8± 74.4
σ = 208.6± 116.1



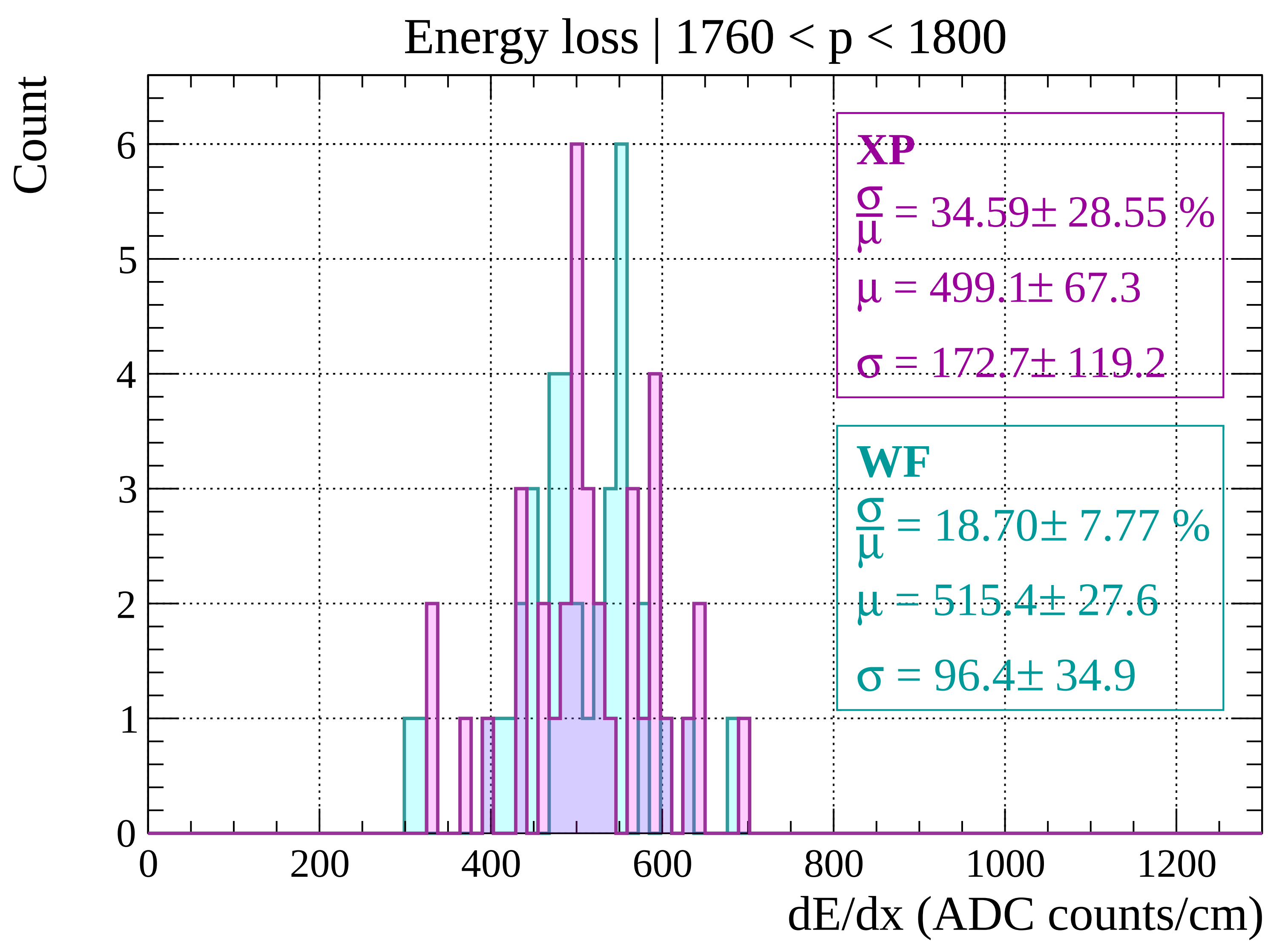
Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 40.89± 29.26 %
μ = 555.3± 78.4
σ = 227.0± 130.4
WF
$\frac{\sigma}{\mu}$ = 0.29± 9.92 %
μ = 525.7± 32.9
σ = 1.5± 52.0

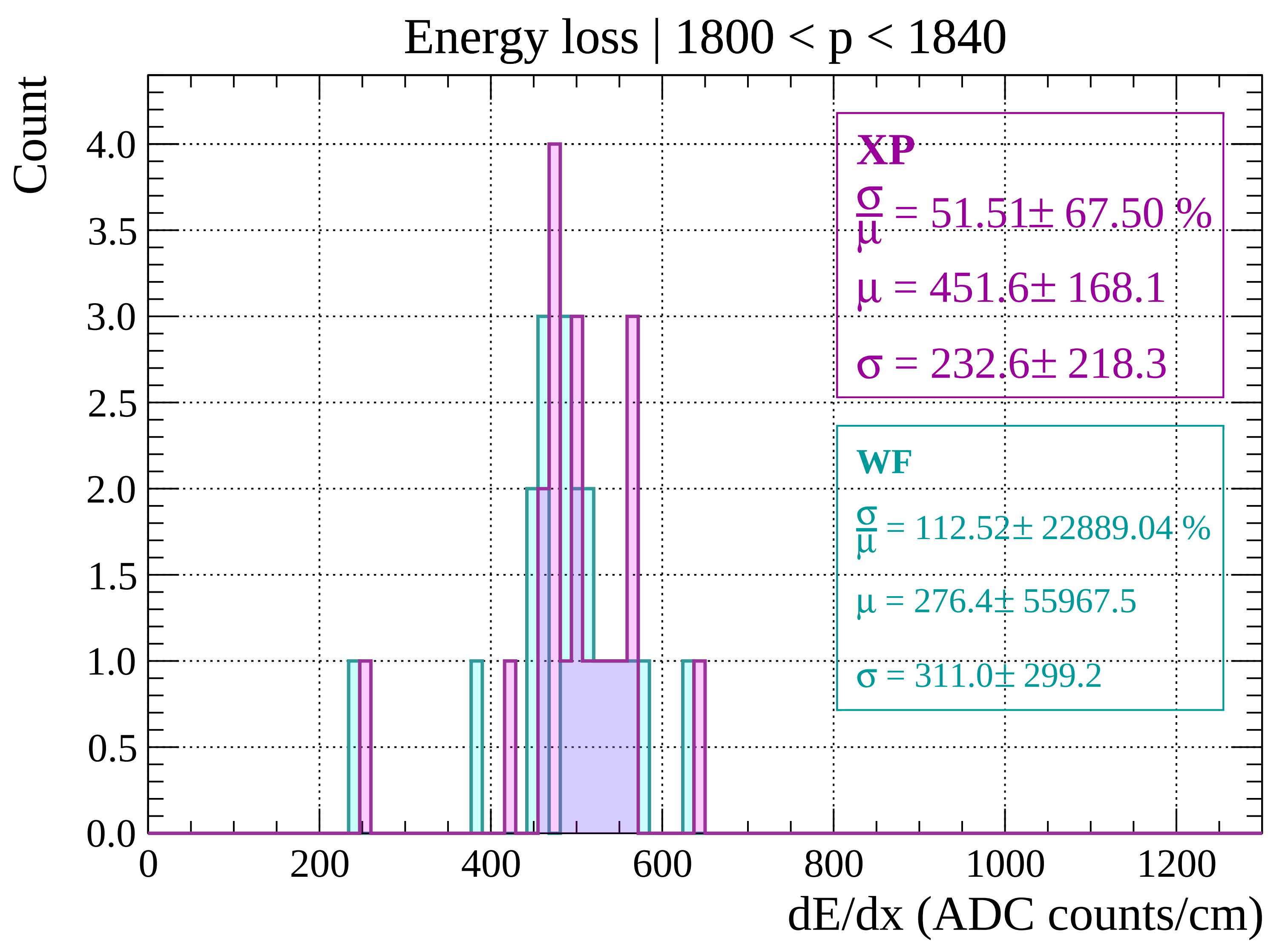


Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 21.10± 7.58 %
μ = 508.5± 27.8
σ = 107.3± 32.7
WF
σ/μ = 23.65± 14.94 %
μ = 495.9± 40.4
σ = 117.3± 64.5

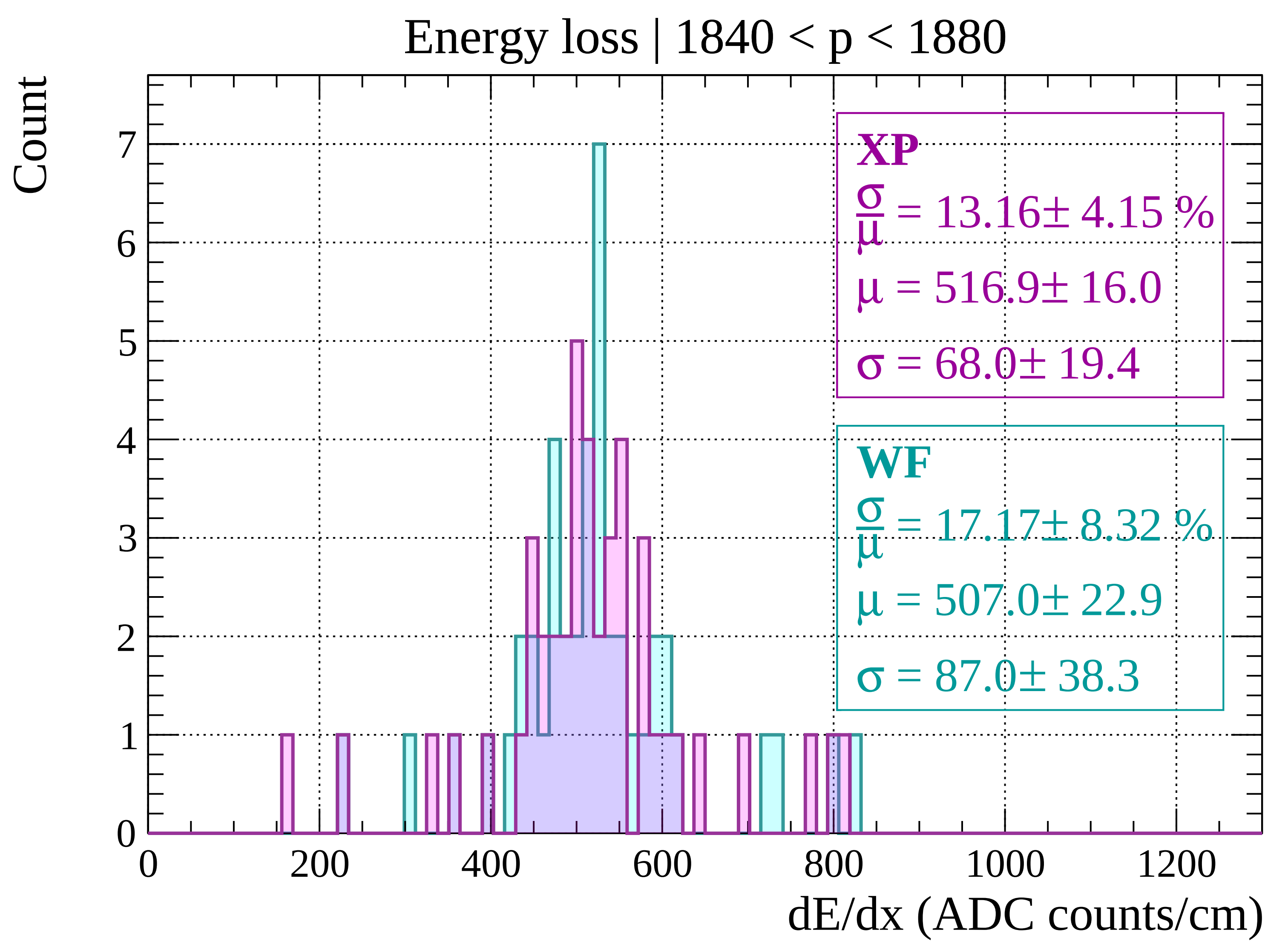


Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 35.68± 26.03 %
μ = 483.6± 61.2
σ = 172.6± 104.1
WF
$\frac{\sigma}{\mu}$ = 42.47± 34.95 %
μ = 468.3± 84.5
σ = 198.9± 127.8

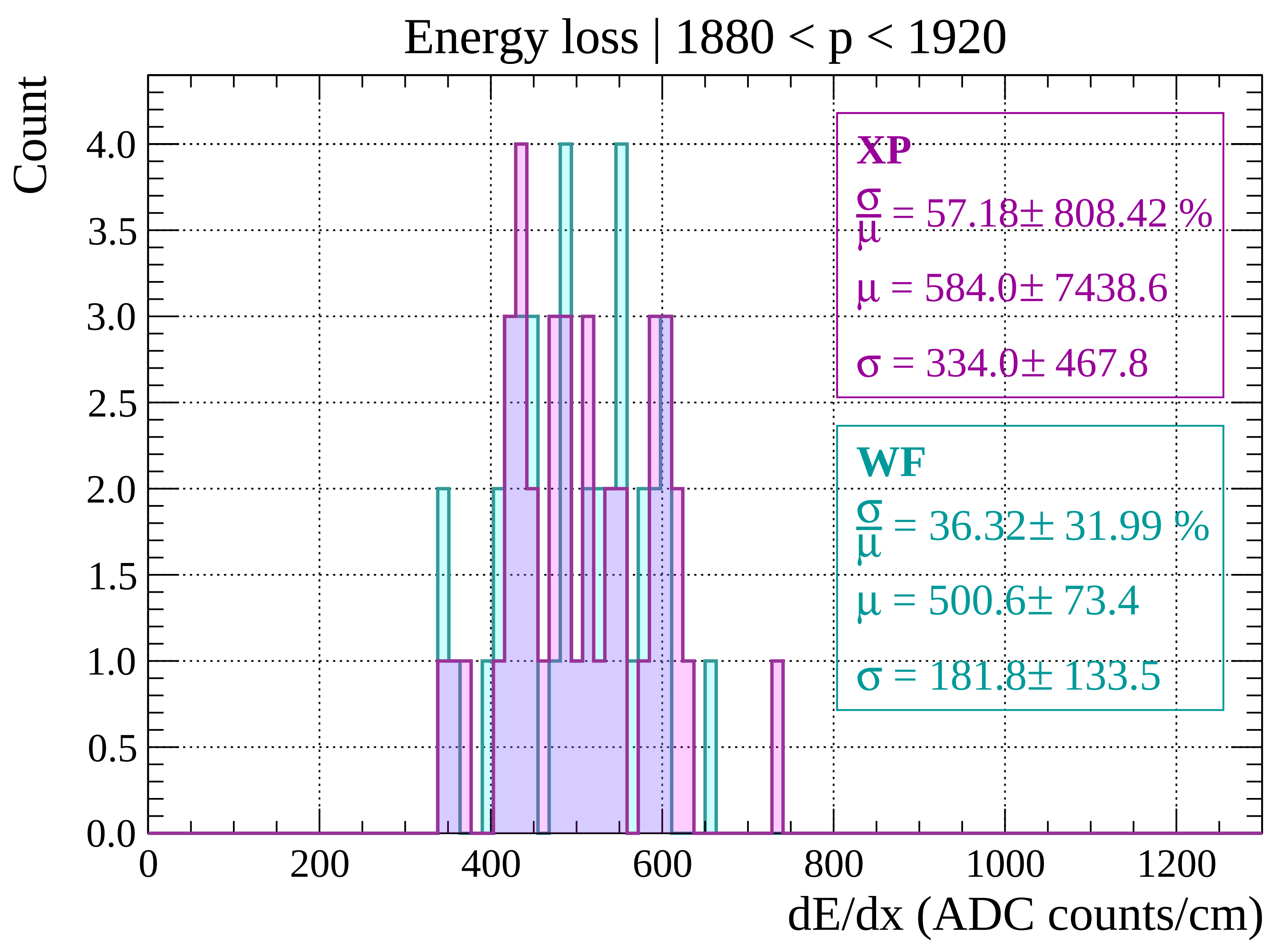

Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 34.59± 28.55 %
μ = 499.1± 67.3
σ = 172.7± 119.2
WF
σ/μ = 18.70± 7.77 %
μ = 515.4± 27.6
σ = 96.4± 34.9

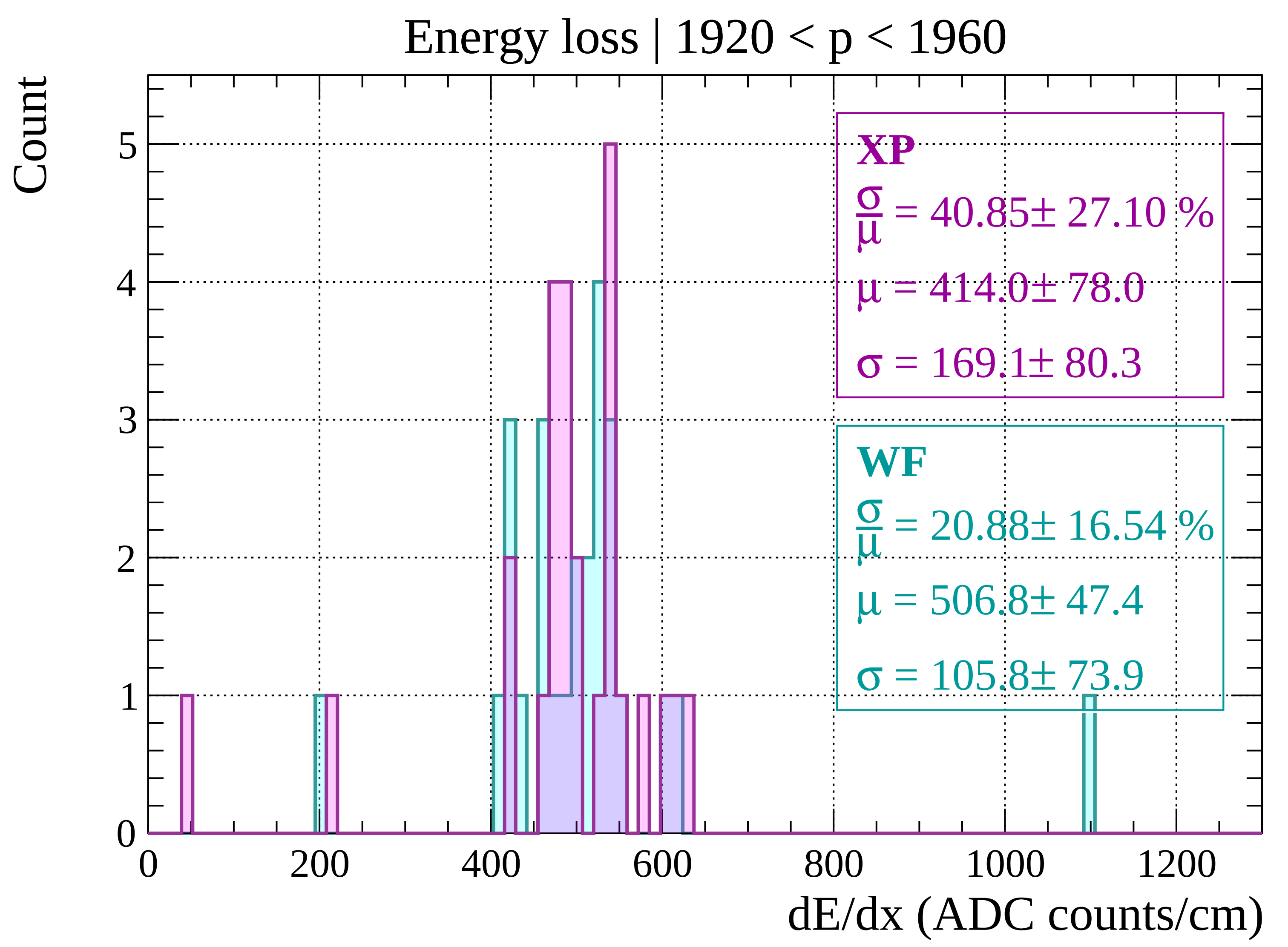


Energy loss | 1800 < p < 1840
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 51.51± 67.50 %
μ = 451.6± 168.1
σ = 232.6± 218.3
WF
σ/μ = 112.52± 22889.04 %
μ = 276.4± 55967.5
σ = 311.0± 299.2

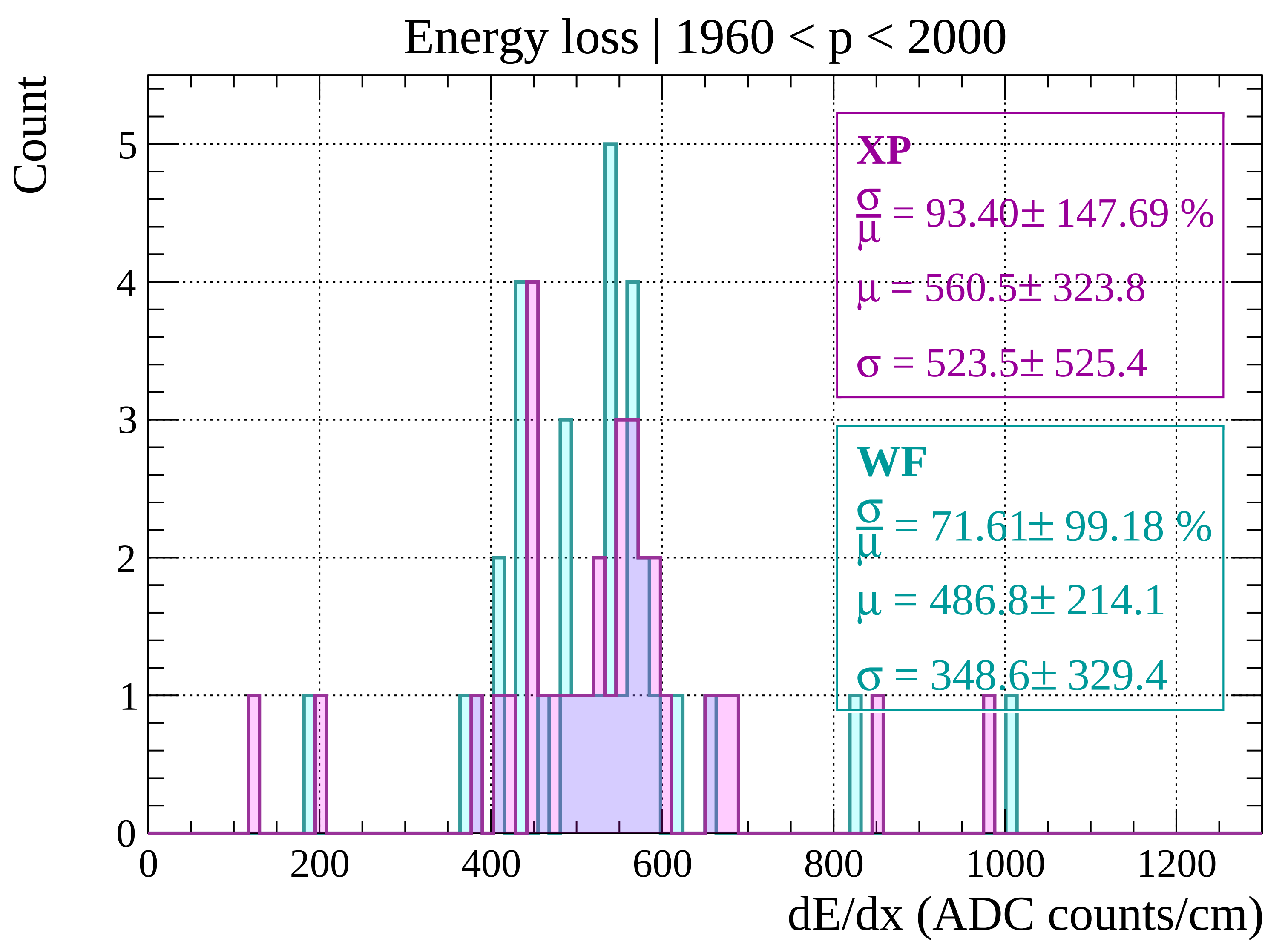


Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 13.16± 4.15 %
μ = 516.9± 16.0
σ = 68.0± 19.4
WF
σ/μ = 17.17± 8.32 %
μ = 507.0± 22.9
σ = 87.0± 38.3

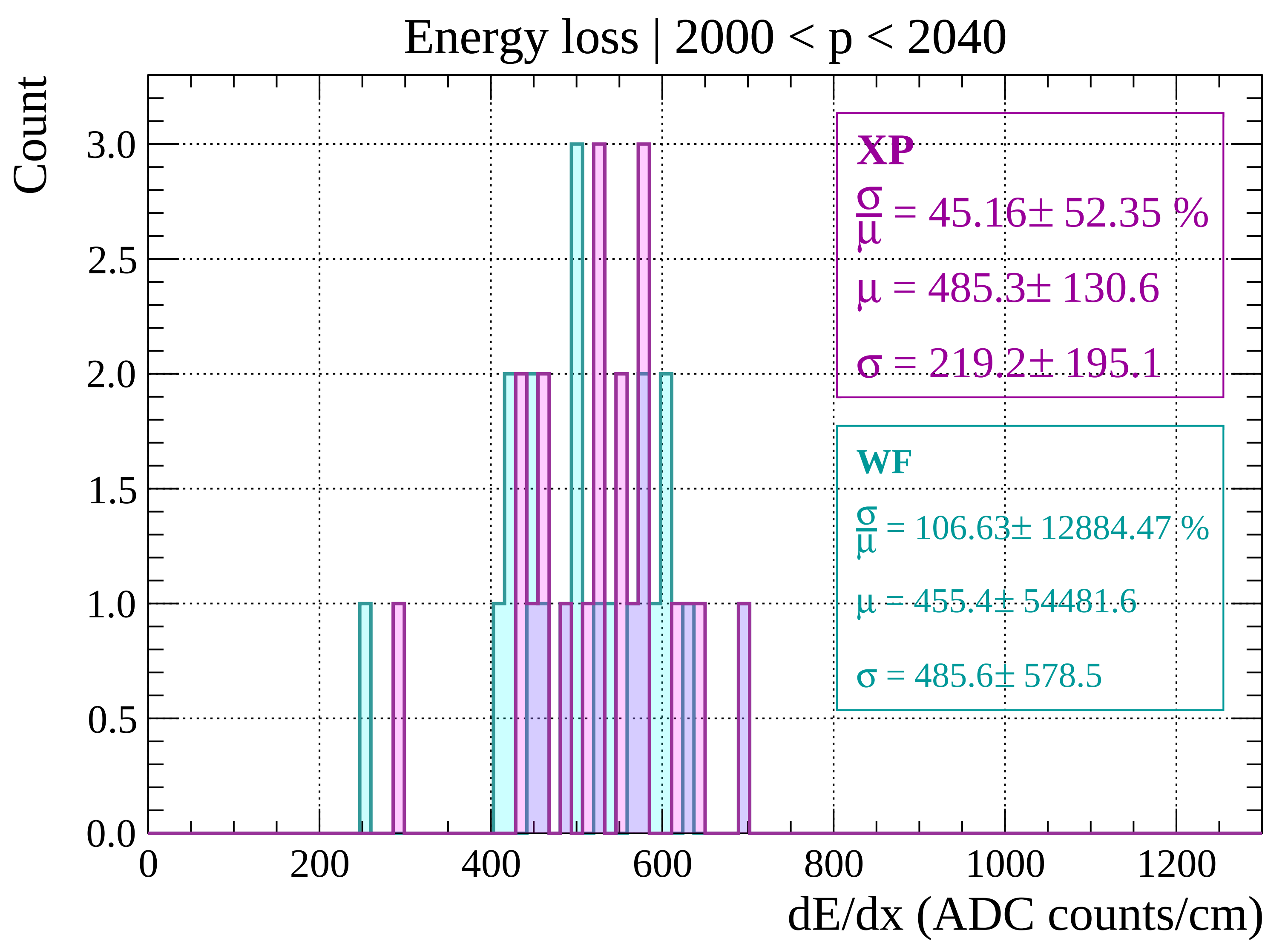

Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 57.18± 808.42 %
μ = 584.0± 7438.6
σ = 334.0± 467.8
WF
σ/μ = 36.32± 31.99 %
μ = 500.6± 73.4
σ = 181.8± 133.5



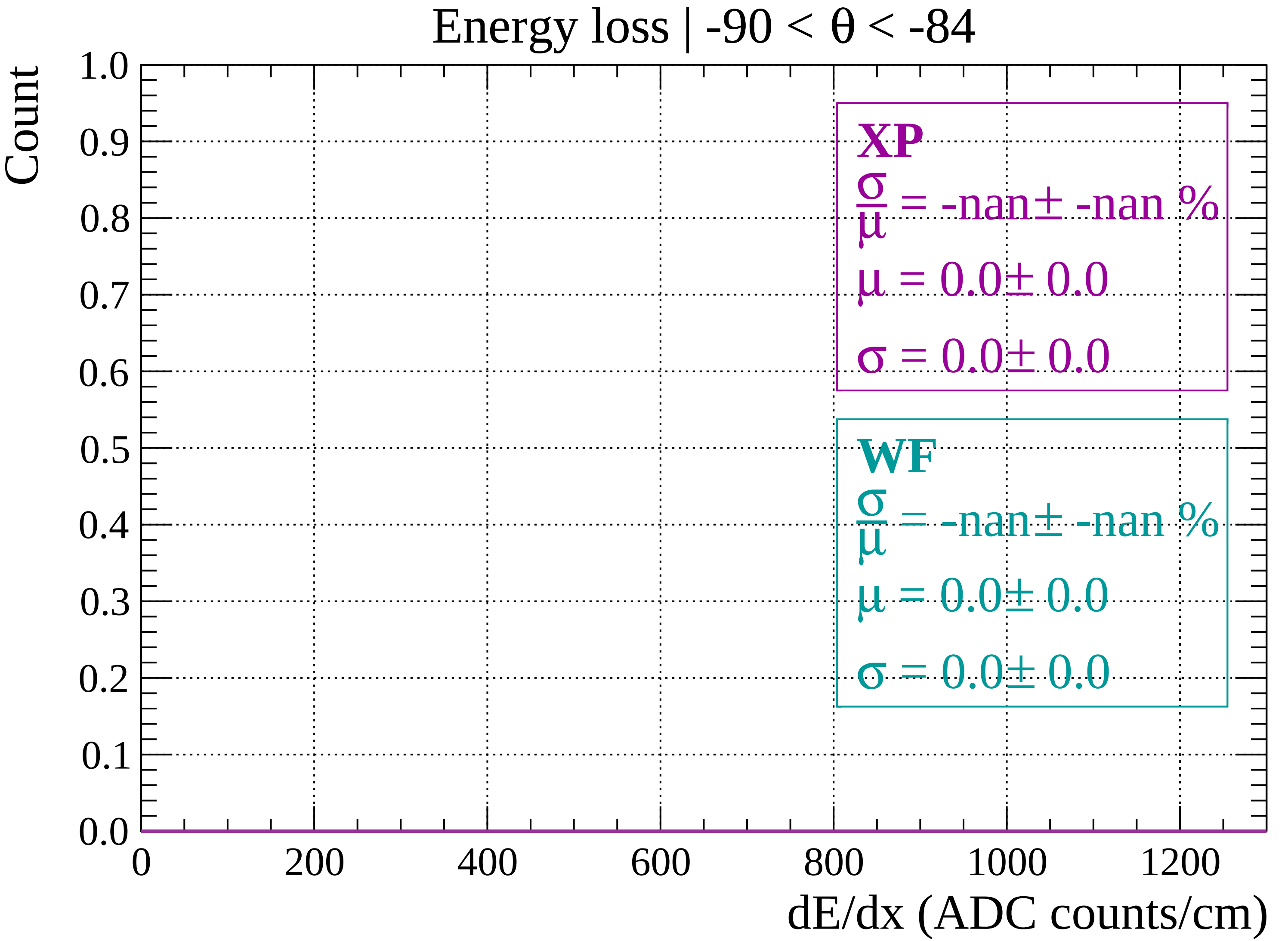
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 40.85± 27.10 %
μ = 414.0± 78.0
σ = 169.1± 80.3
WF
σ/μ = 20.88± 16.54 %
μ = 506.8± 47.4
σ = 105.8± 73.9



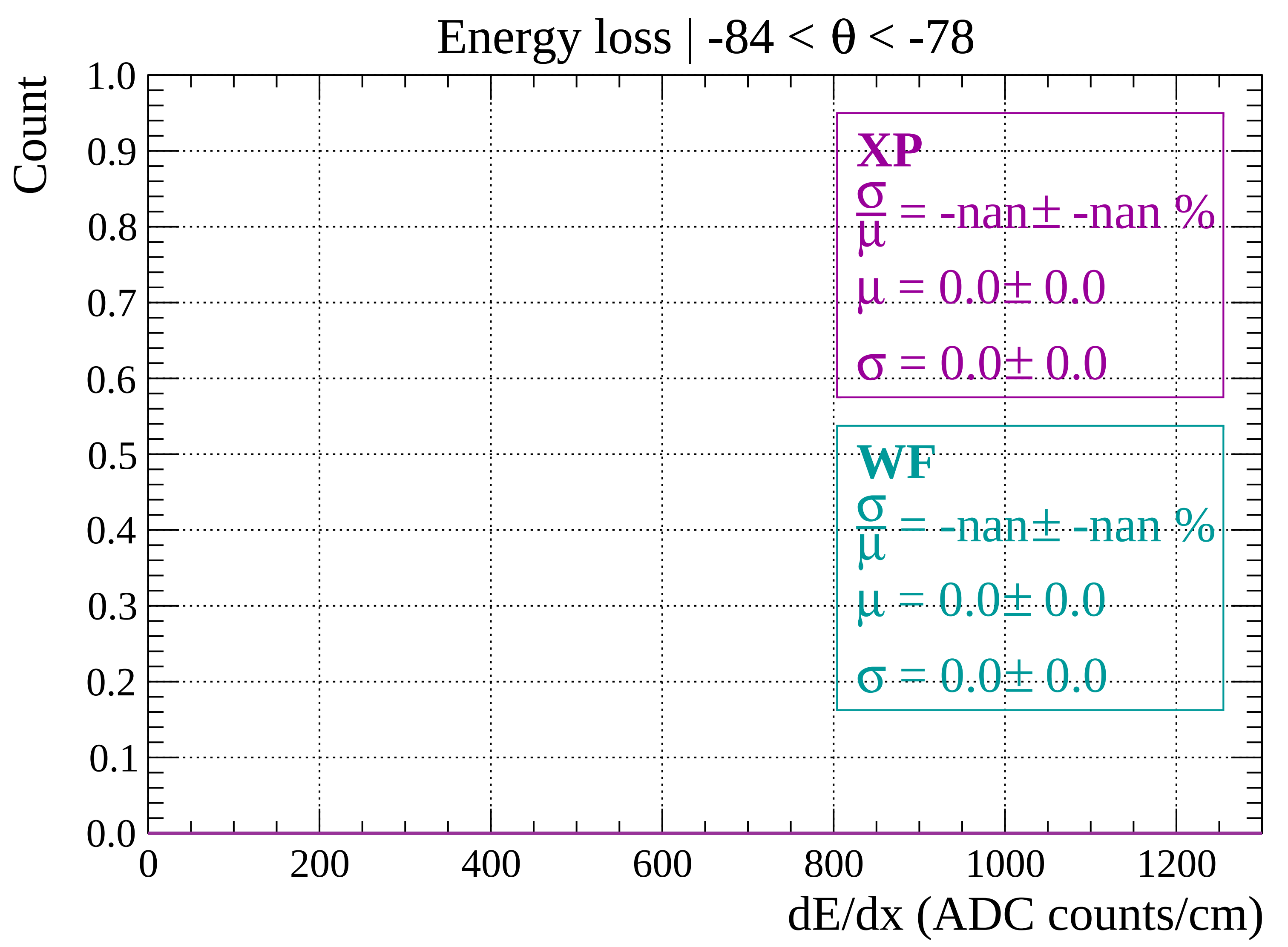
Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 93.40± 147.69 %
μ = 560.5± 323.8
σ = 523.5± 525.4
WF
$\frac{\sigma}{\mu}$ = 71.61± 99.18 %
μ = 486.8± 214.1
σ = 348.6± 329.4

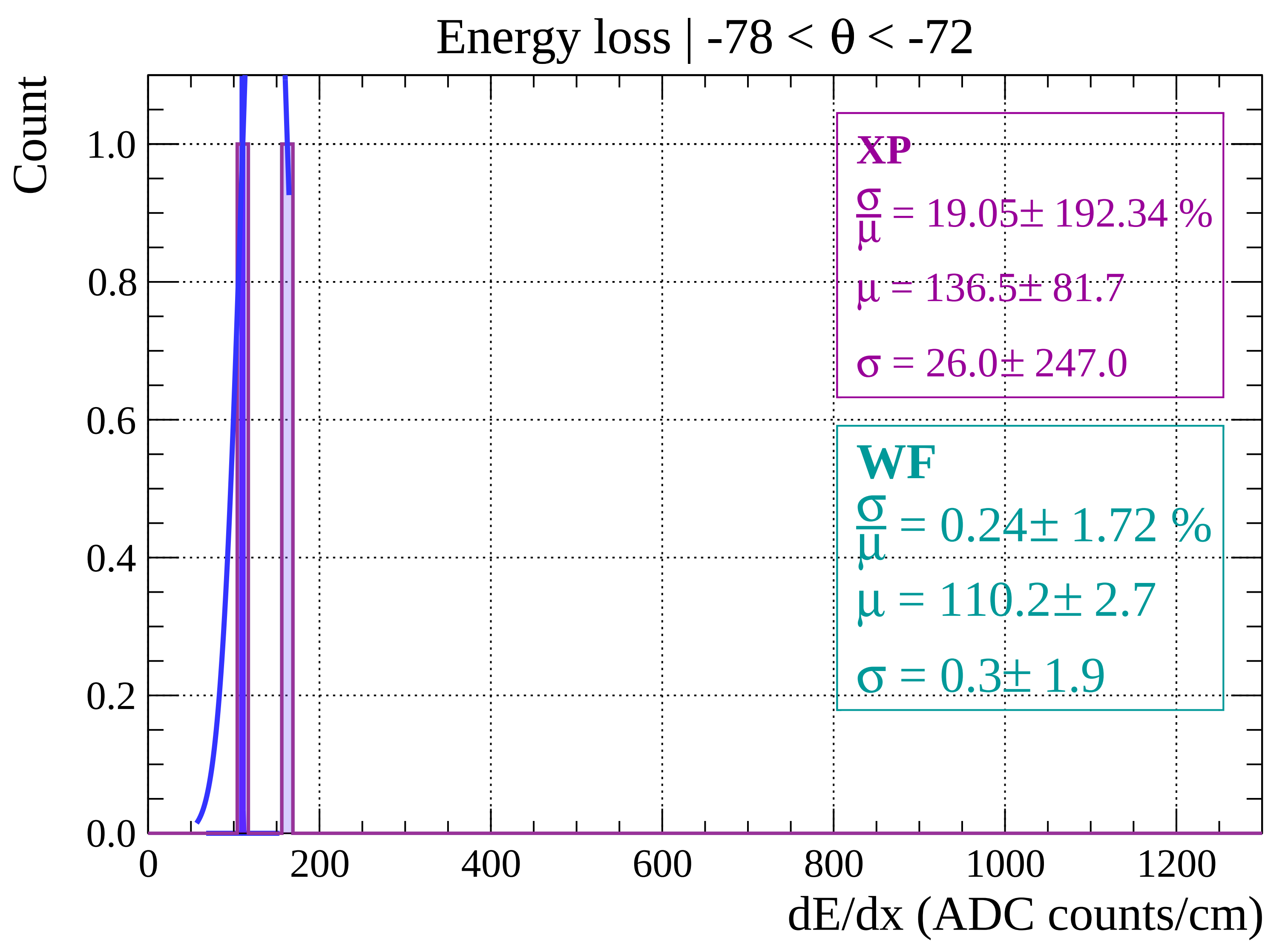


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 45.16± 52.35 %
μ = 485.3± 130.6
σ = 219.2± 195.1
WF
$\frac{\sigma}{\mu}$ = 106.63± 12884.47 %
μ = 455.4± 54481.6
σ = 485.6± 578.5

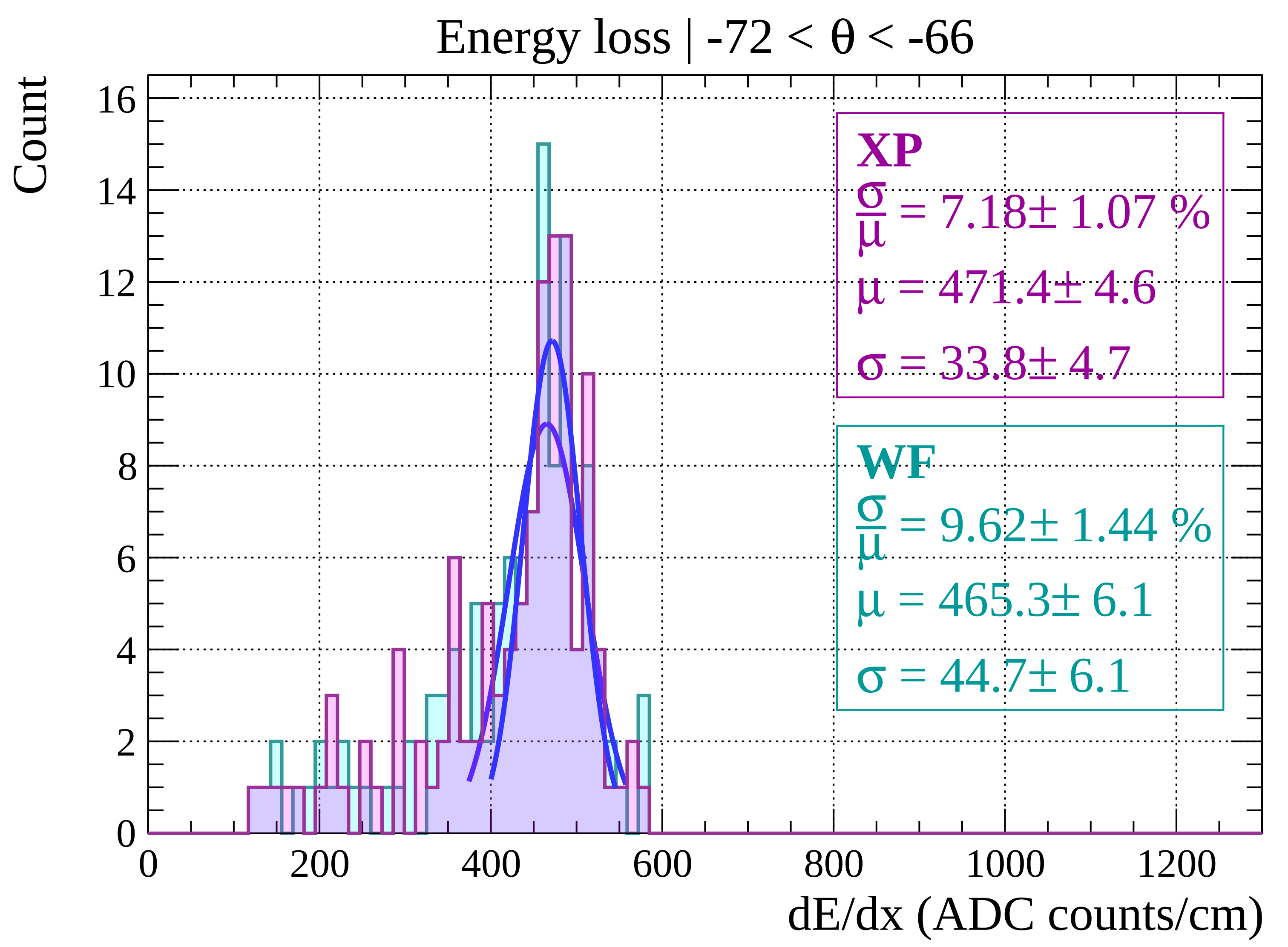


Energy loss | -90 < θ < -84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

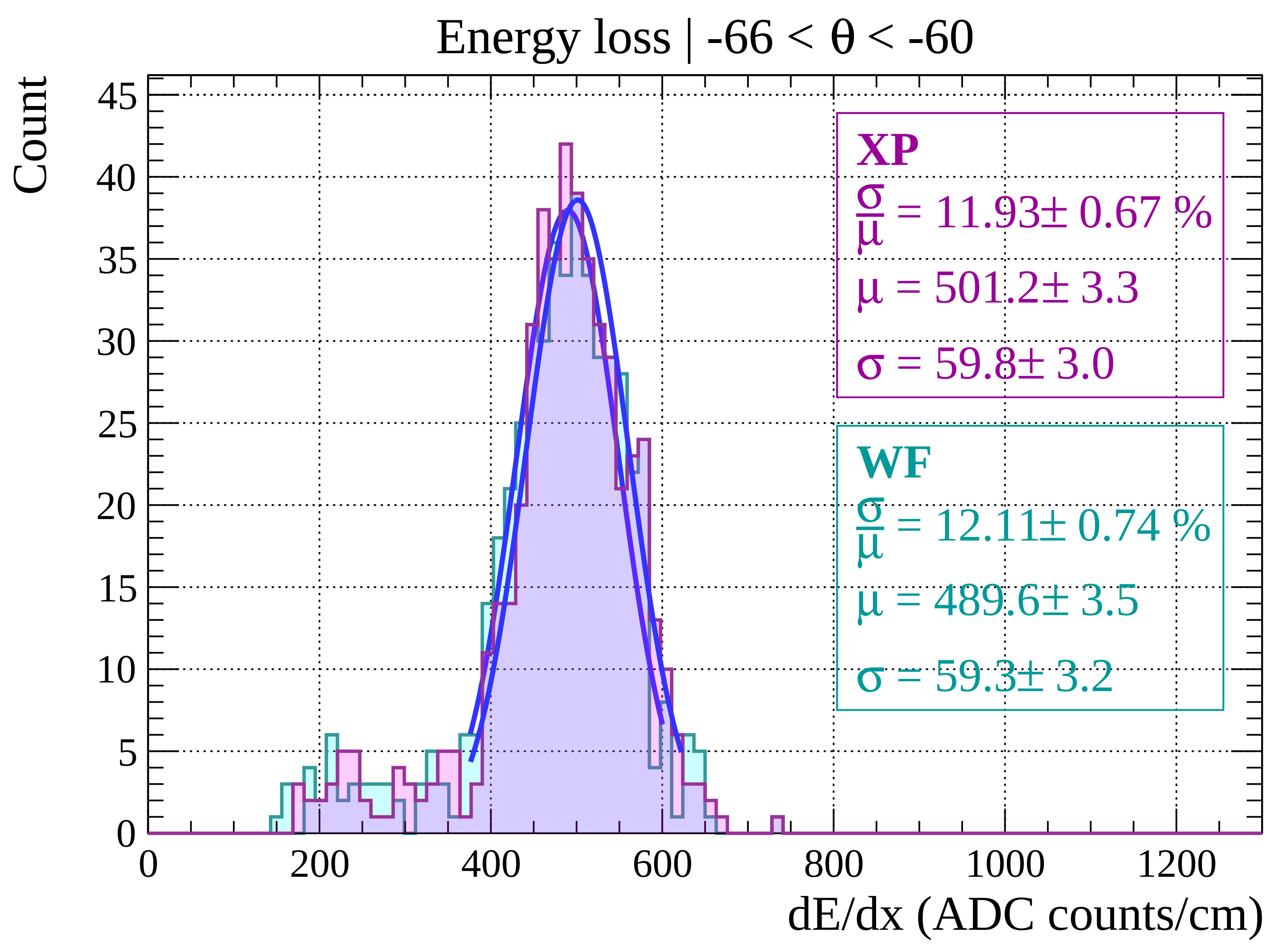

Energy loss | -84 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



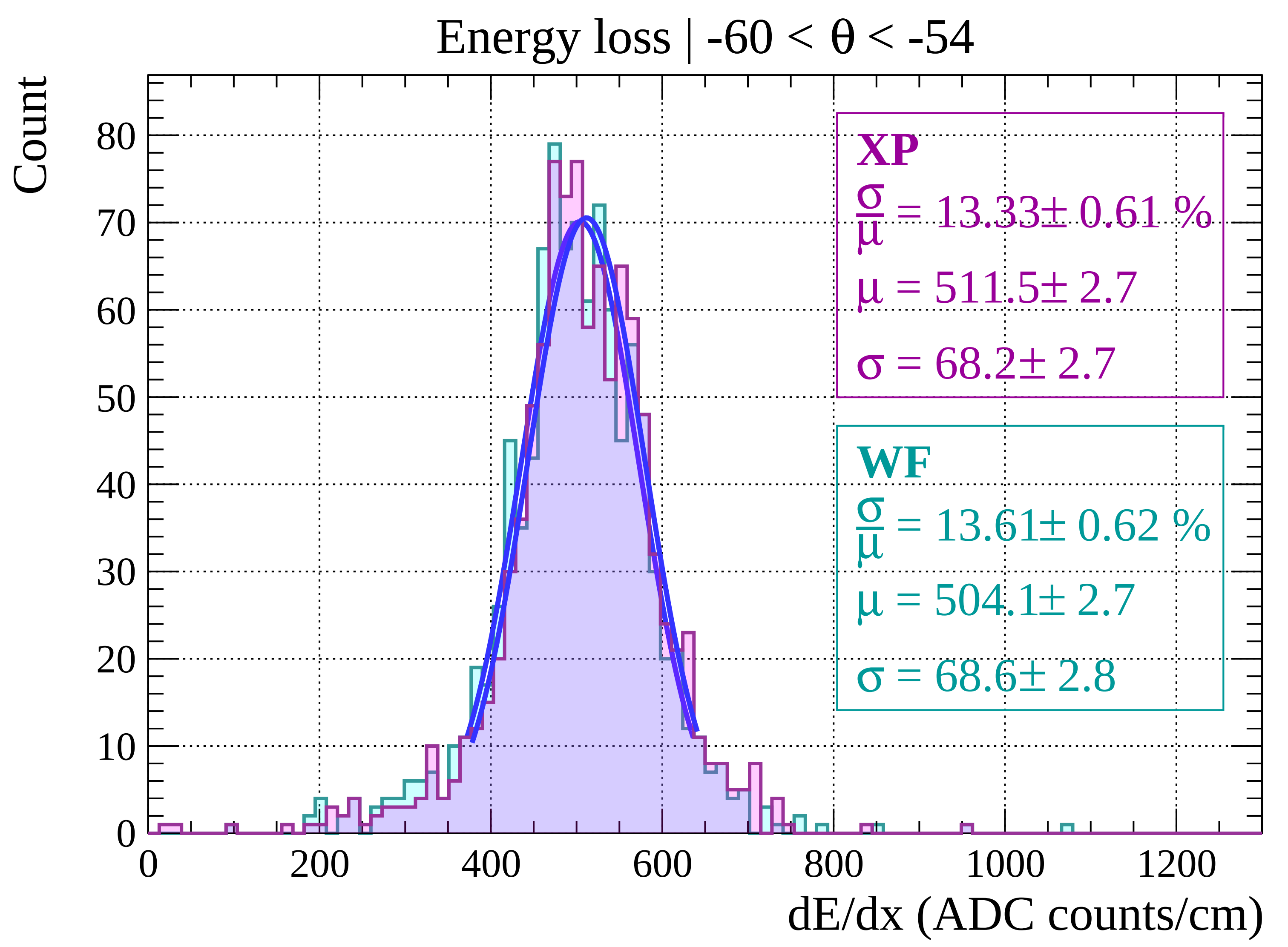
Energy loss | -78 < θ < -72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 19.05± 192.34 %
μ = 136.5± 81.7
σ = 26.0± 247.0
WF
σ/μ = 0.24± 1.72 %
μ = 110.2± 2.7
σ = 0.3± 1.9



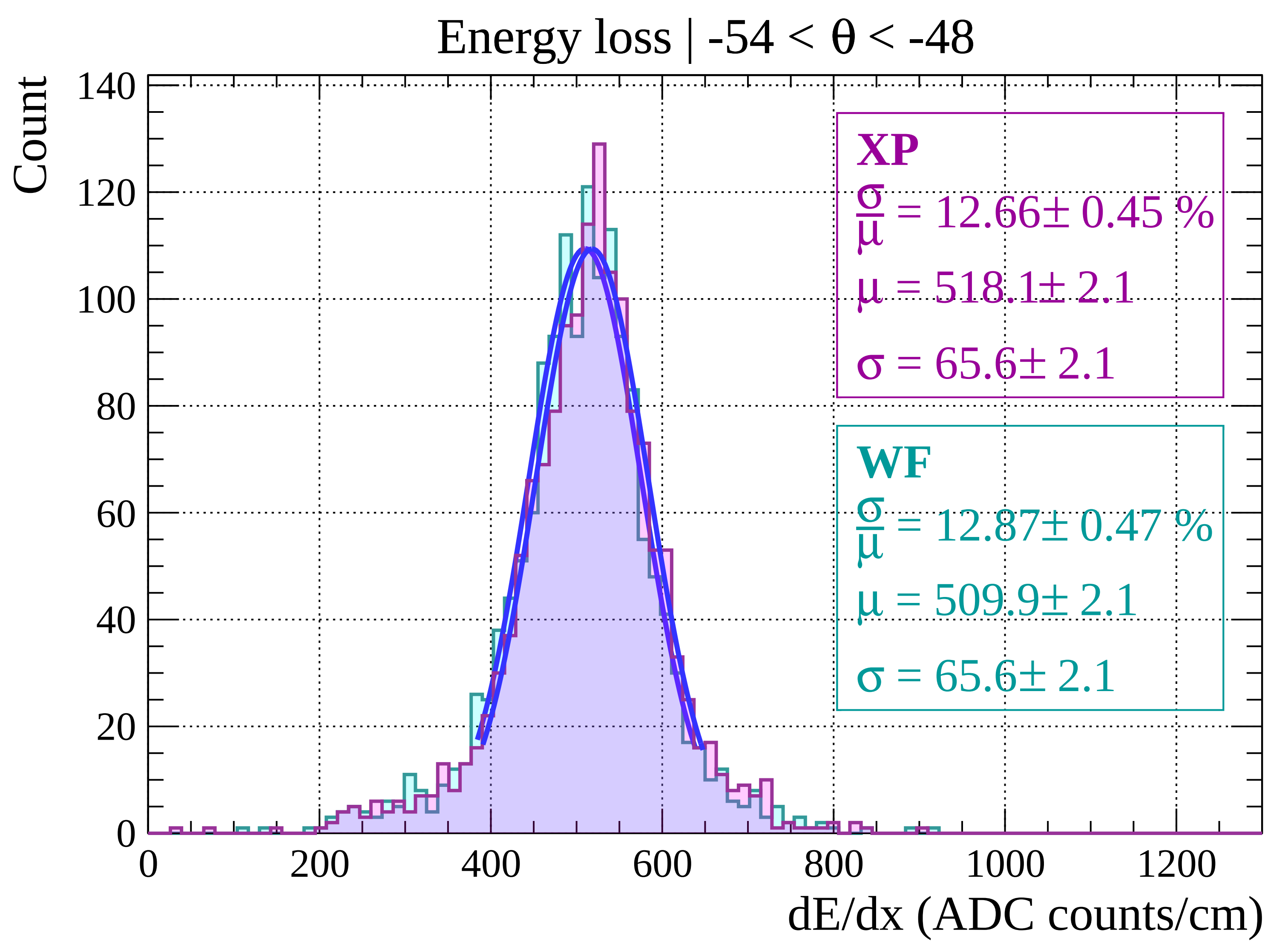
Energy loss | -72 < θ < -66
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.18± 1.07 %
μ = 471.4± 4.6
σ = 33.8± 4.7
WF
σ/μ = 9.62± 1.44 %
μ = 465.3± 6.1
σ = 44.7± 6.1



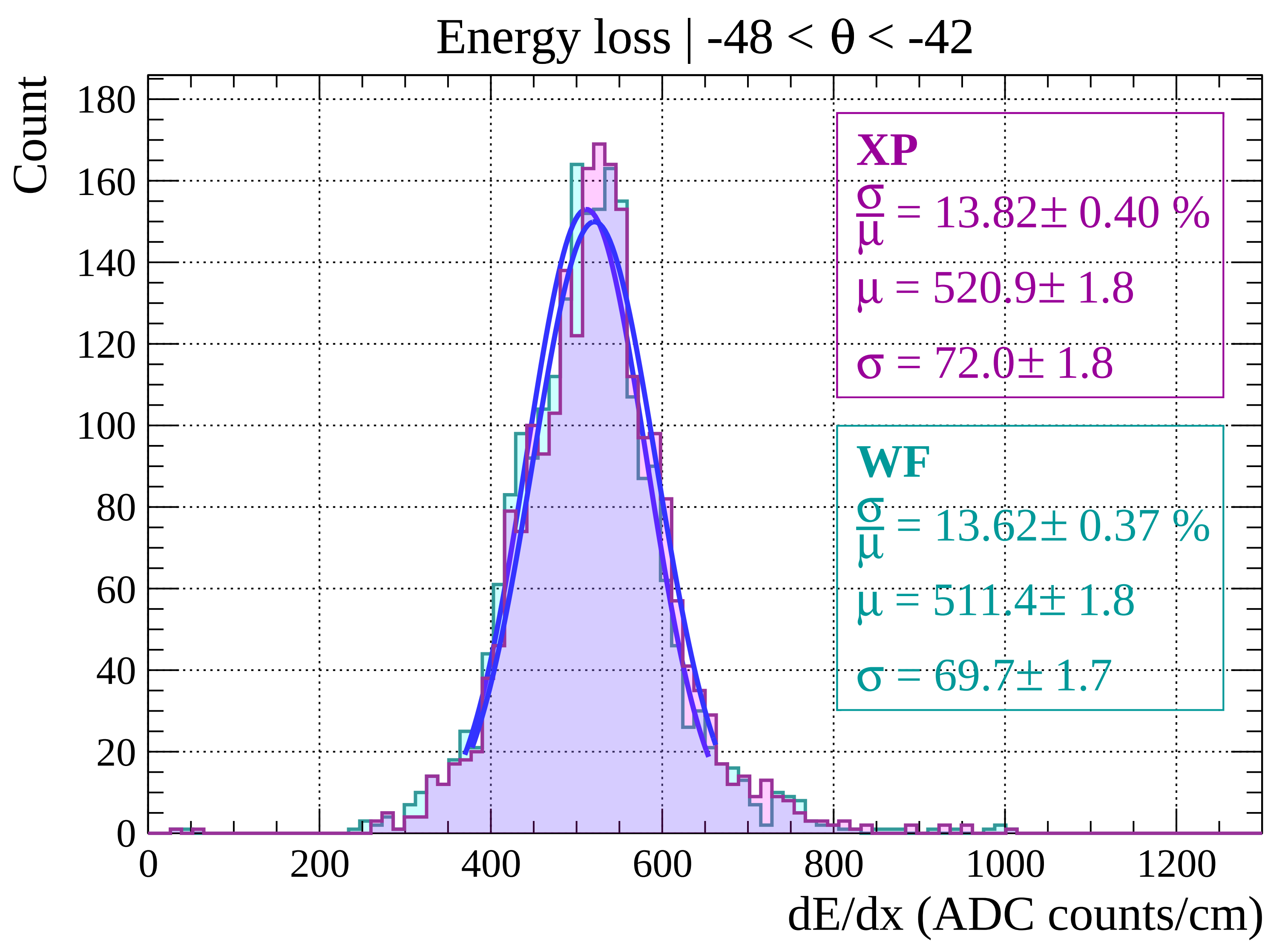
Energy loss | -66 < θ < -60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 11.93± 0.67 %
μ = 501.2± 3.3
σ = 59.8± 3.0
WF
σ/μ = 12.11± 0.74 %
μ = 489.6± 3.5
σ = 59.3± 3.2



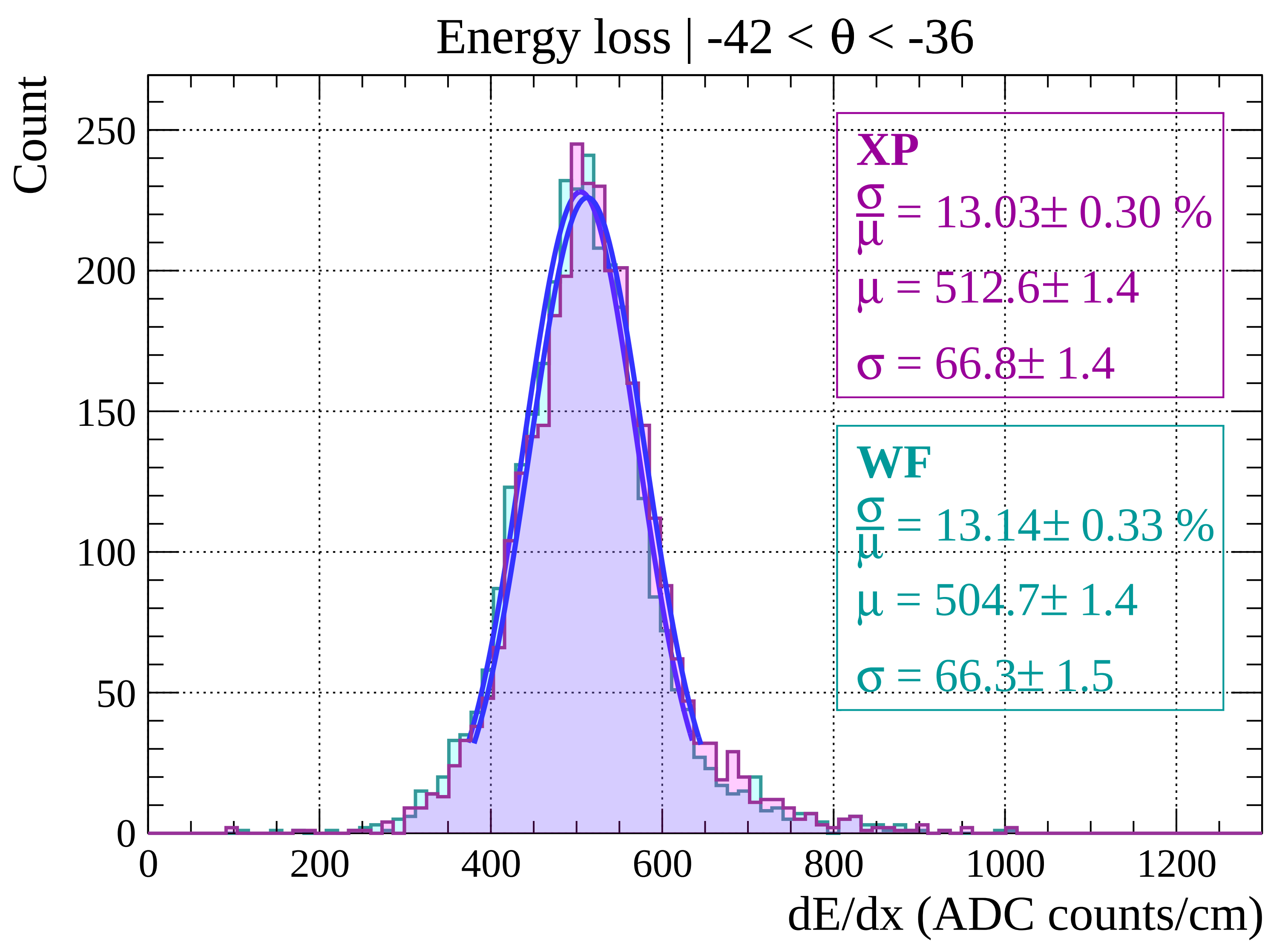
Energy loss | -60 < θ < -54
Count
XP
σ/μ = 13.33 ± 0.61 %
μ = 511.5 ± 2.7
σ = 68.2 ± 2.7
WF
σ/μ = 13.61 ± 0.62 %
μ = 504.1 ± 2.7
σ = 68.6 ± 2.8
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



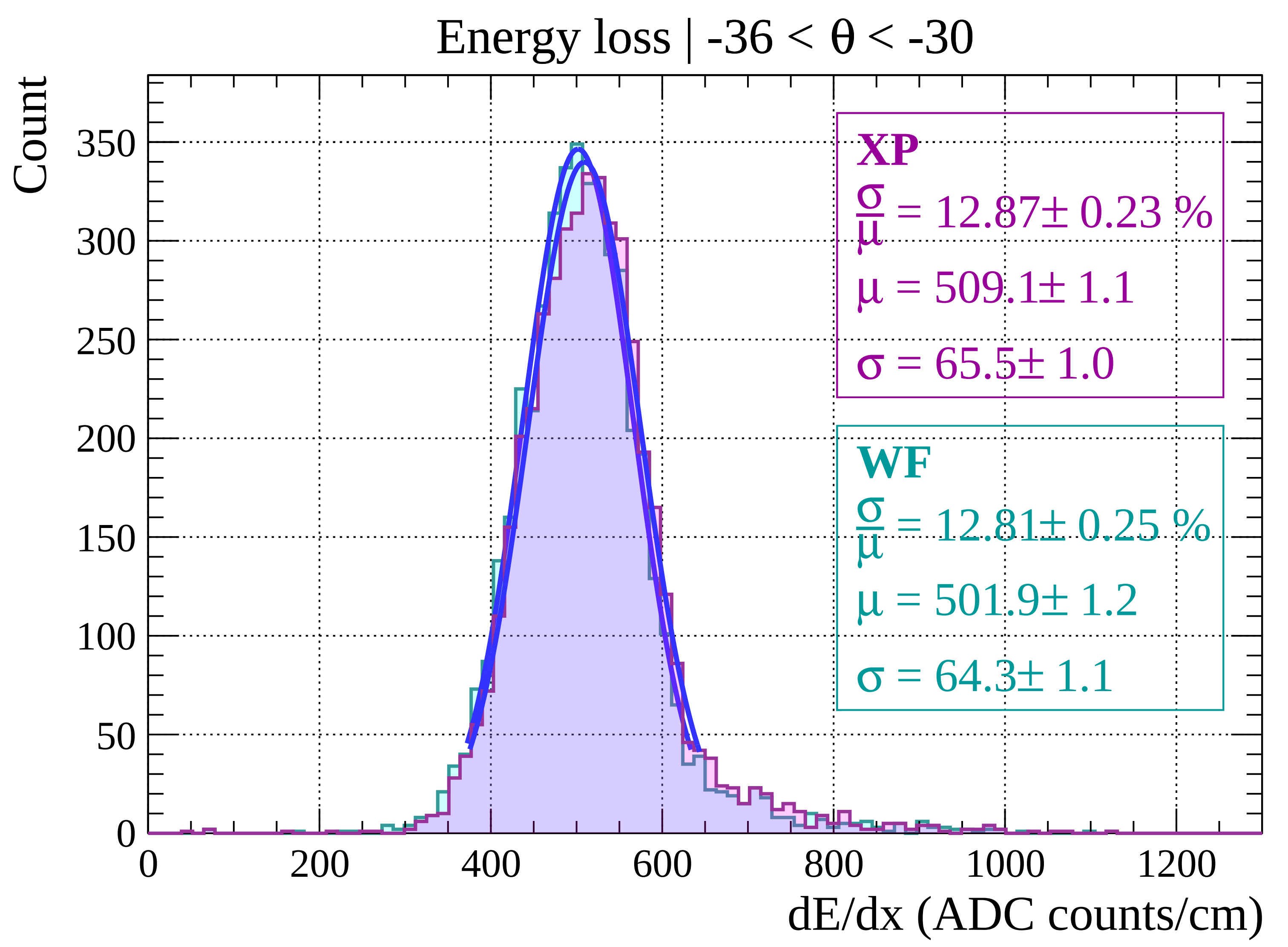
Energy loss | -54 < θ < -48
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.66± 0.45 %
μ = 518.1± 2.1
σ = 65.6± 2.1
WF
σ/μ = 12.87± 0.47 %
μ = 509.9± 2.1
σ = 65.6± 2.1



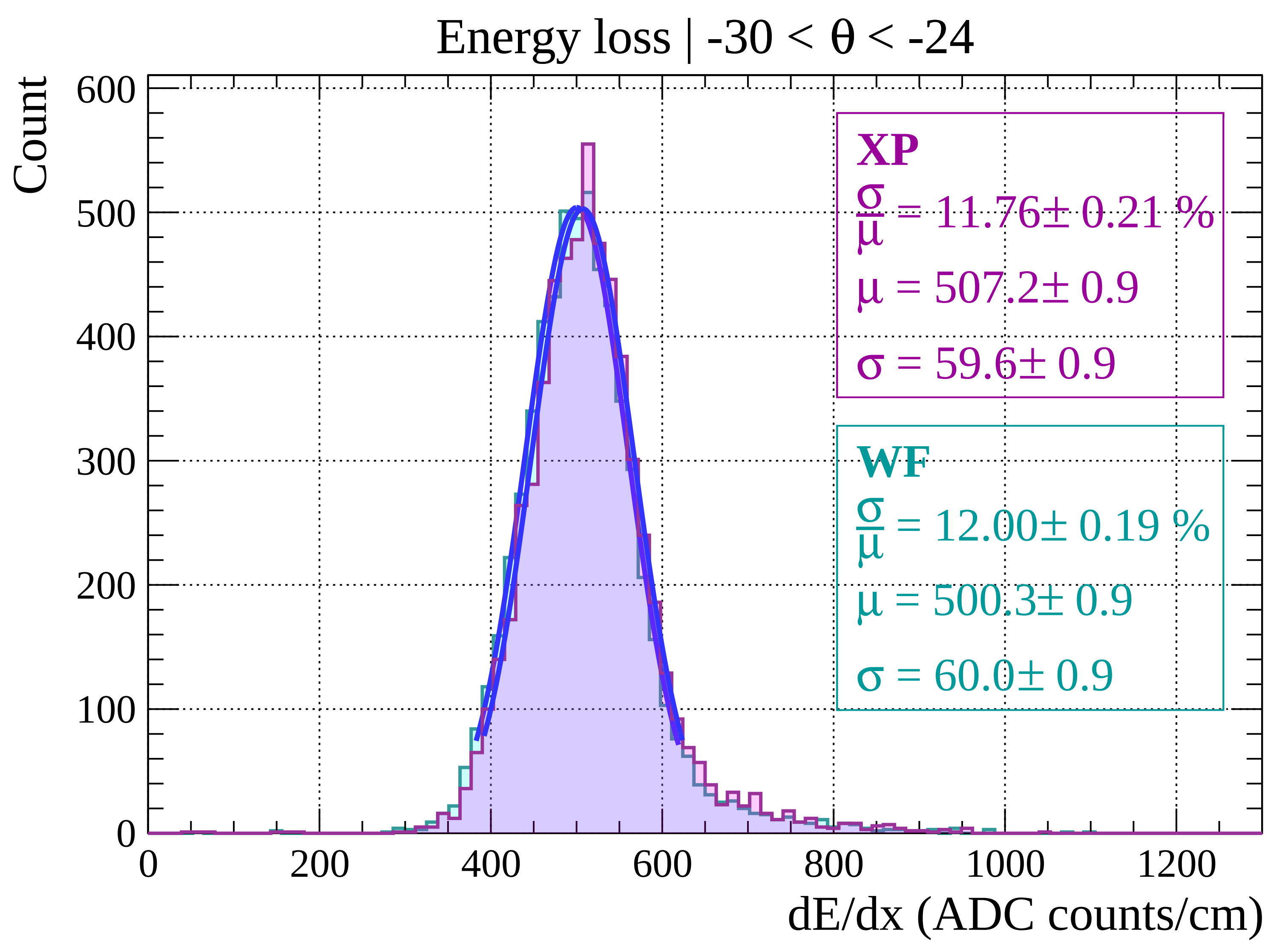
Energy loss | -48 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.82± 0.40 %
μ = 520.9± 1.8
σ = 72.0± 1.8
WF
$\frac{\sigma}{\mu}$ = 13.62± 0.37 %
μ = 511.4± 1.8
σ = 69.7± 1.7

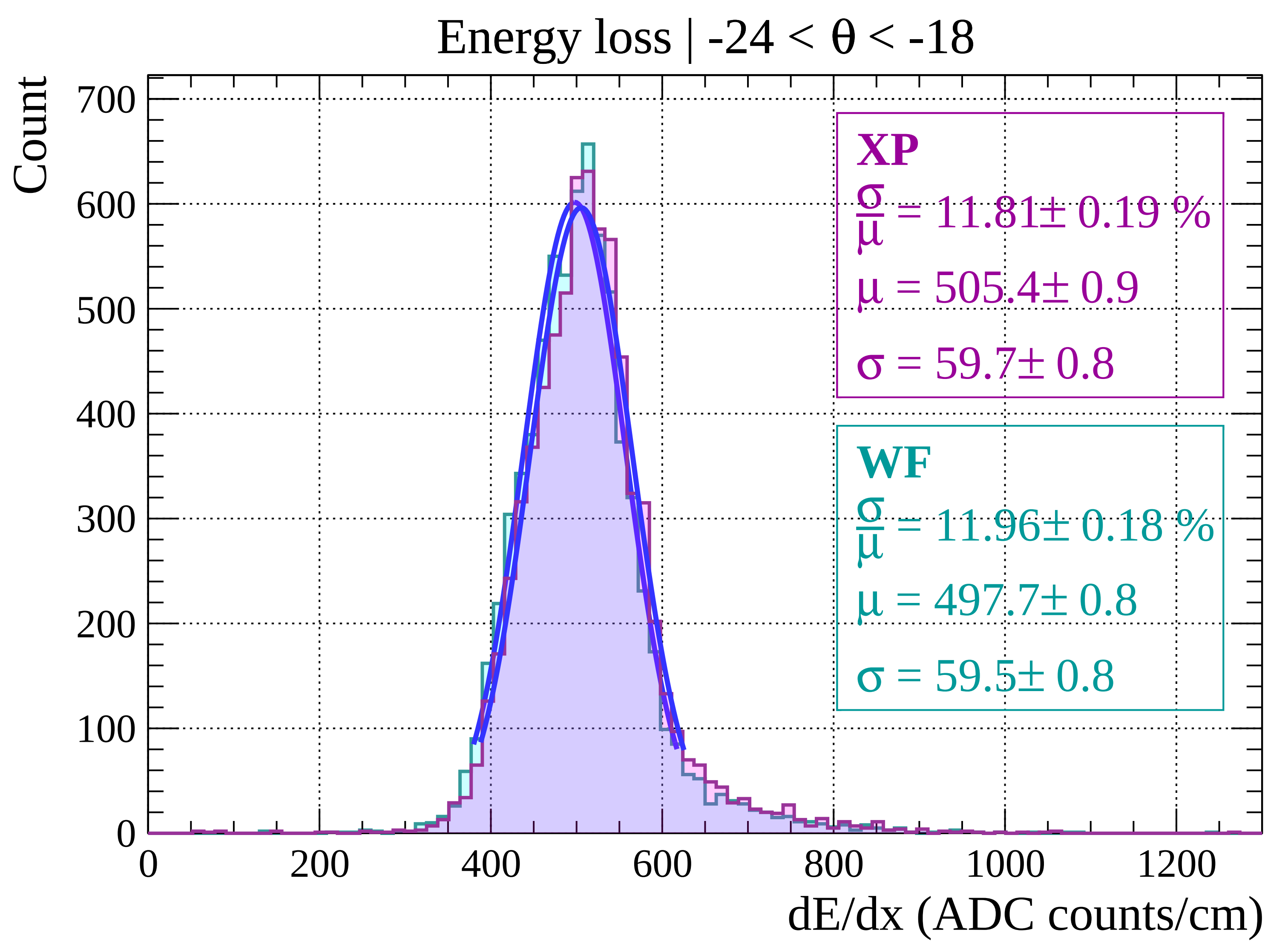


Energy loss | -42 < θ < -36
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.03± 0.30 %
μ = 512.6± 1.4
σ = 66.8± 1.4
WF
$\frac{\sigma}{\mu}$ = 13.14± 0.33 %
μ = 504.7± 1.4
σ = 66.3± 1.5

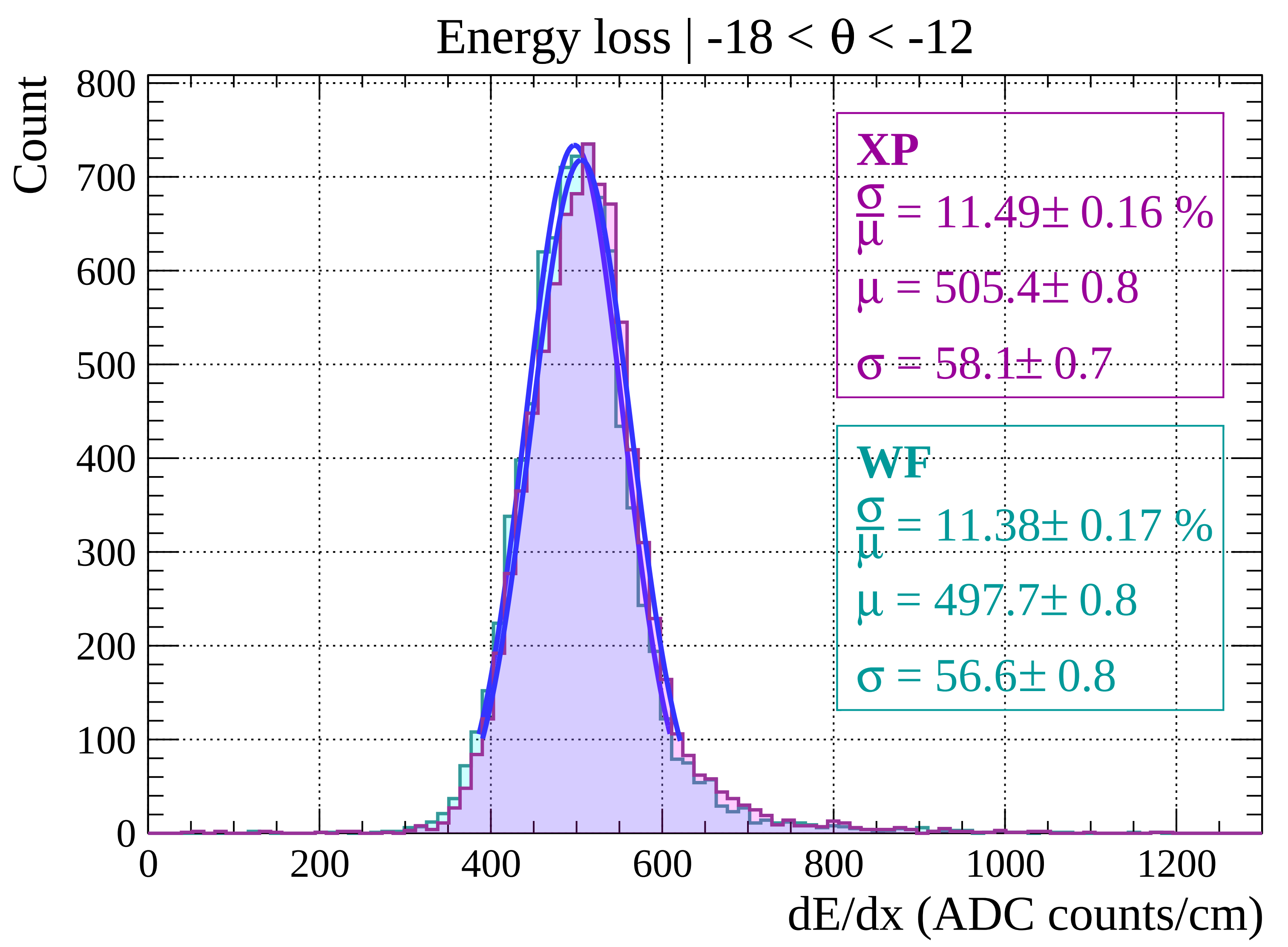


Energy loss | -36 < θ < -30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.87± 0.23 %
μ = 509.1± 1.1
σ = 65.5± 1.0
WF
σ/μ = 12.81± 0.25 %
μ = 501.9± 1.2
σ = 64.3± 1.1

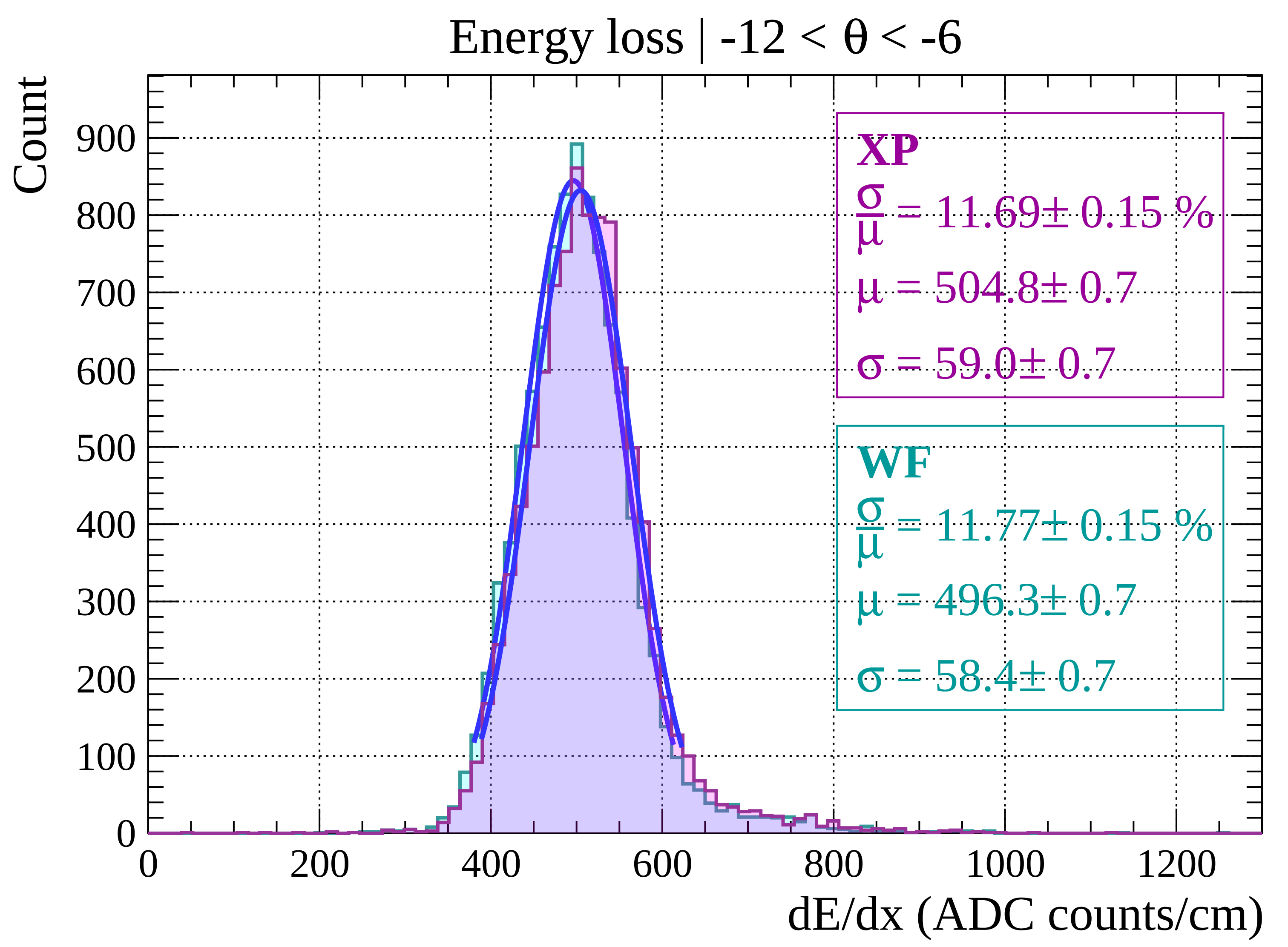

Energy loss | -30 < θ < -24
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 11.76± 0.21 %
μ = 507.2± 0.9
σ = 59.6± 0.9
WF
σ/μ = 12.00± 0.19 %
μ = 500.3± 0.9
σ = 60.0± 0.9



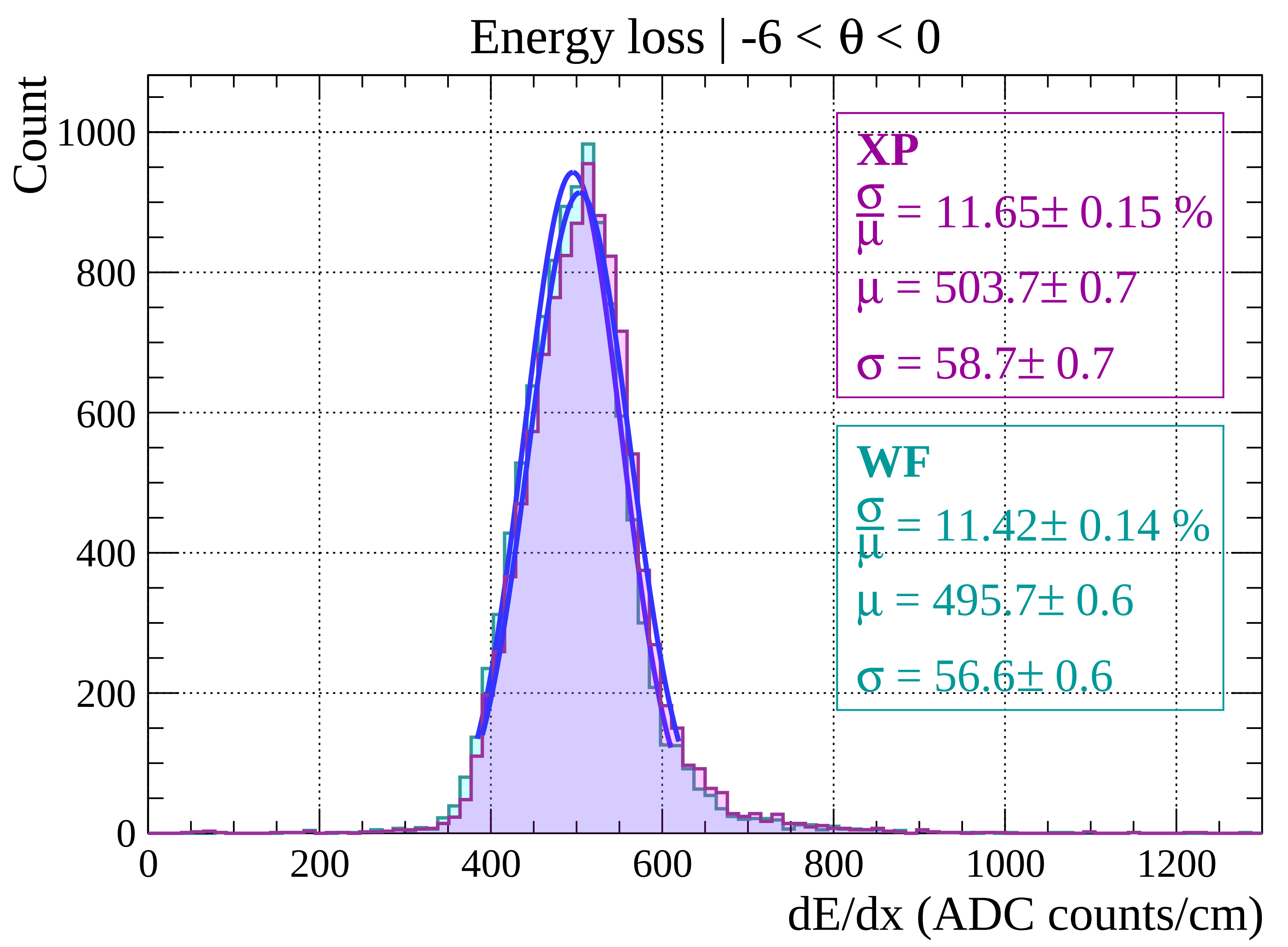
Energy loss | -24 < θ < -18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.81± 0.19 %
μ = 505.4± 0.9
σ = 59.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 11.96± 0.18 %
μ = 497.7± 0.8
σ = 59.5± 0.8



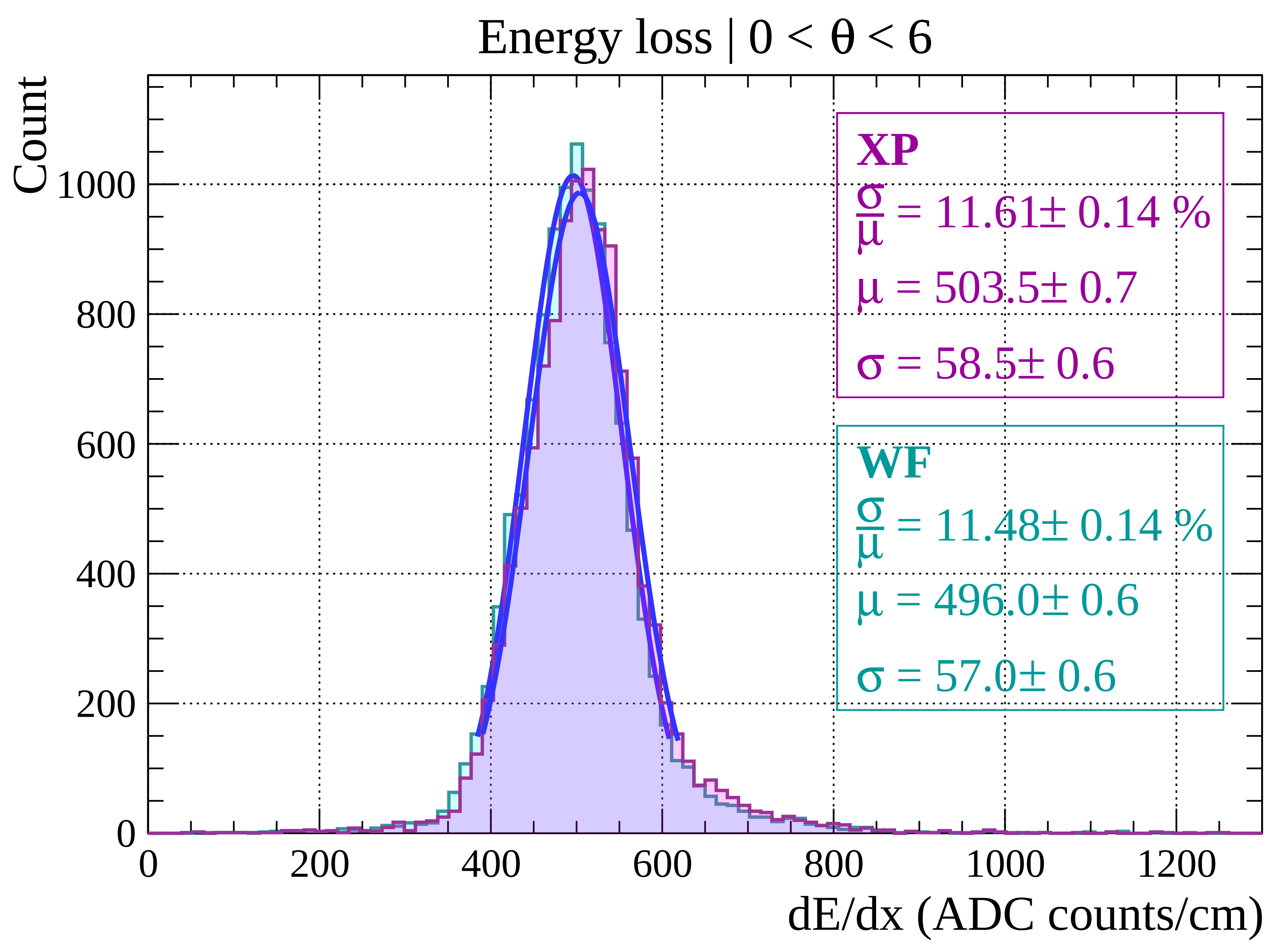
Energy loss | -18 < θ < -12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 11.49± 0.16 %
μ = 505.4± 0.8
σ = 58.1± 0.7
WF
σ/μ = 11.38± 0.17 %
μ = 497.7± 0.8
σ = 56.6± 0.8



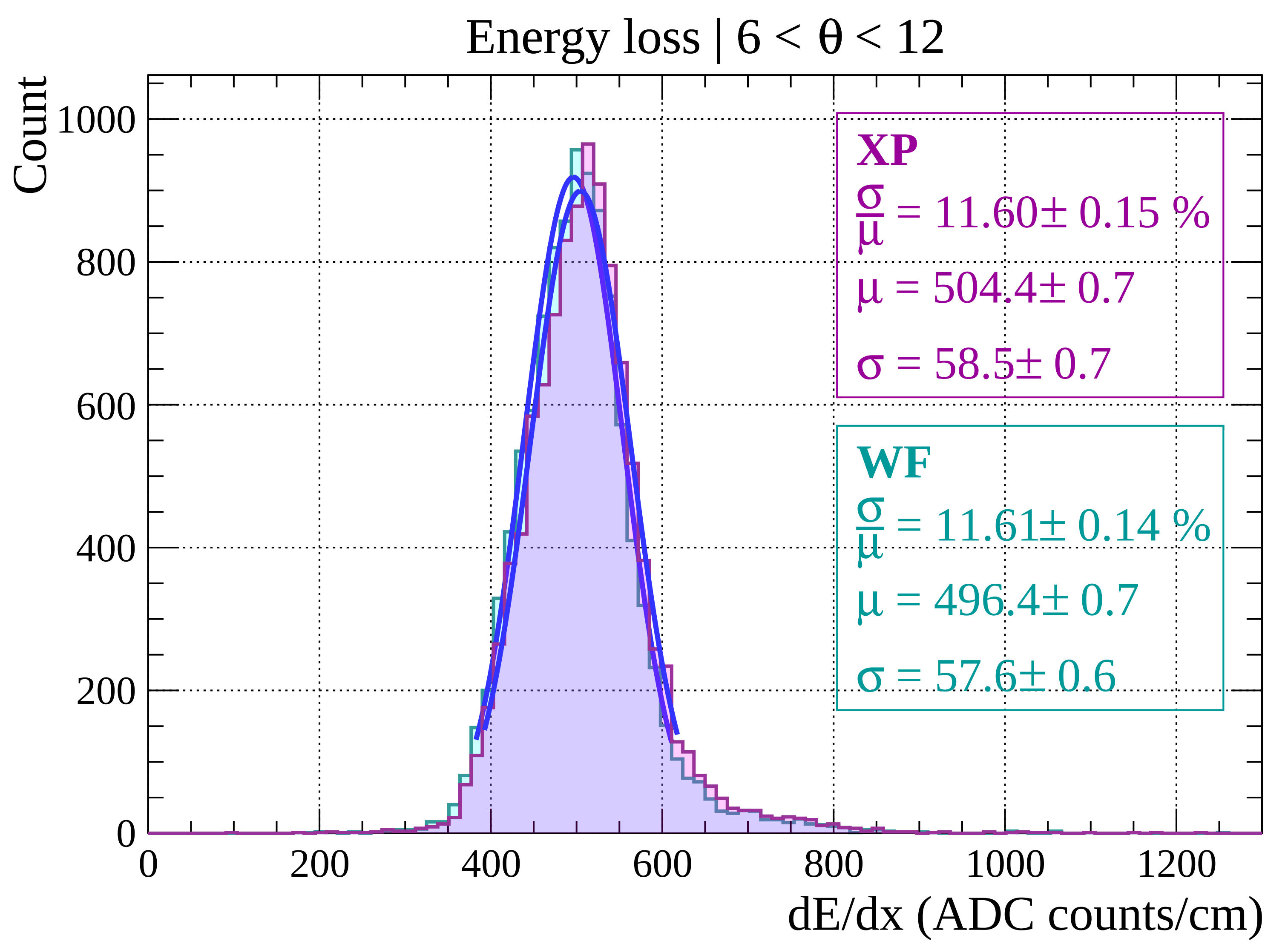
Energy loss | -12 < θ < -6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 11.69± 0.15 %
μ = 504.8± 0.7
σ = 59.0± 0.7
WF
σ/μ = 11.77± 0.15 %
μ = 496.3± 0.7
σ = 58.4± 0.7

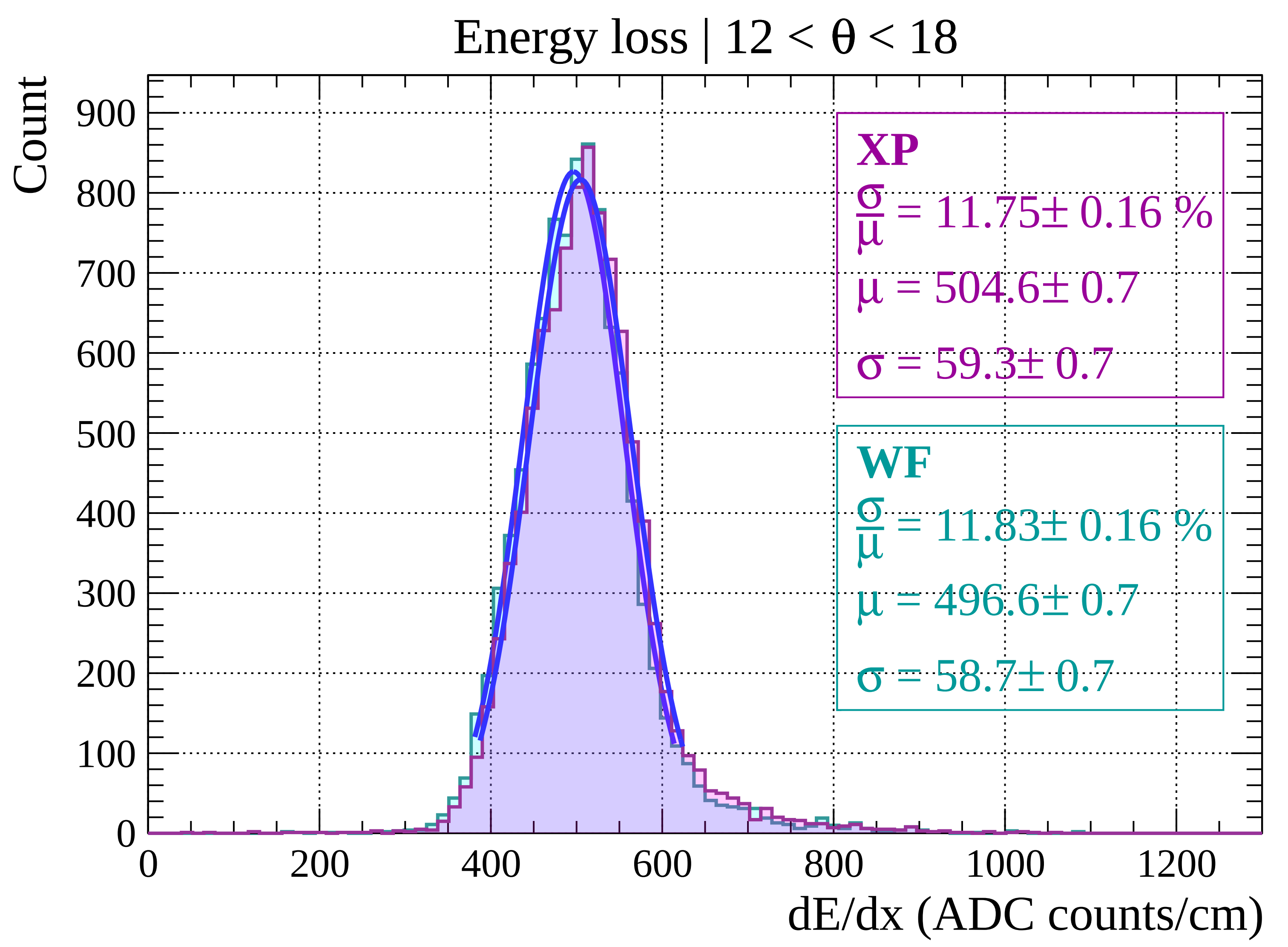


Energy loss | -6 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.65± 0.15 %
μ = 503.7± 0.7
σ = 58.7± 0.7
WF
σ/μ = 11.42± 0.14 %
μ = 495.7± 0.6
σ = 56.6± 0.6

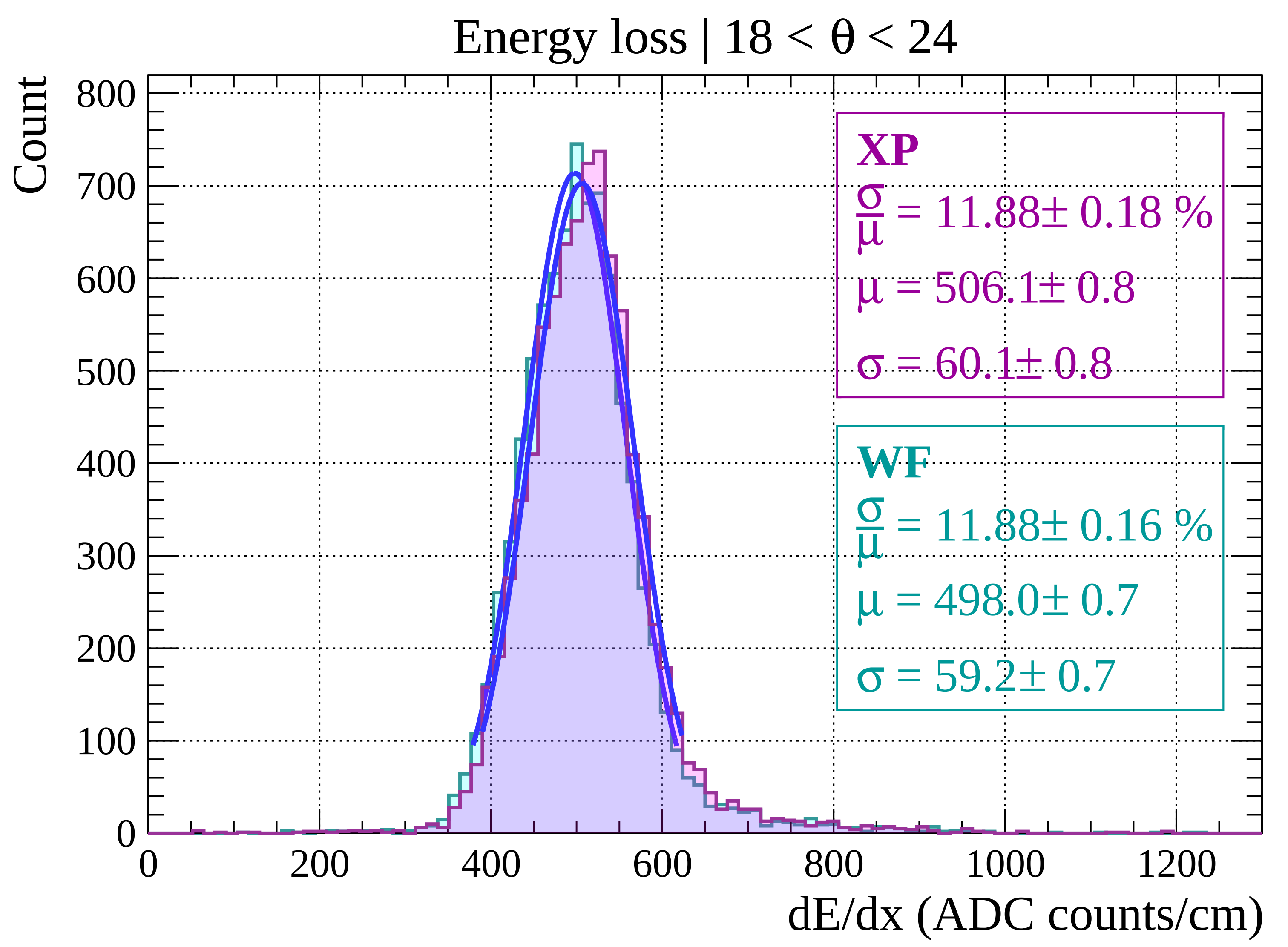


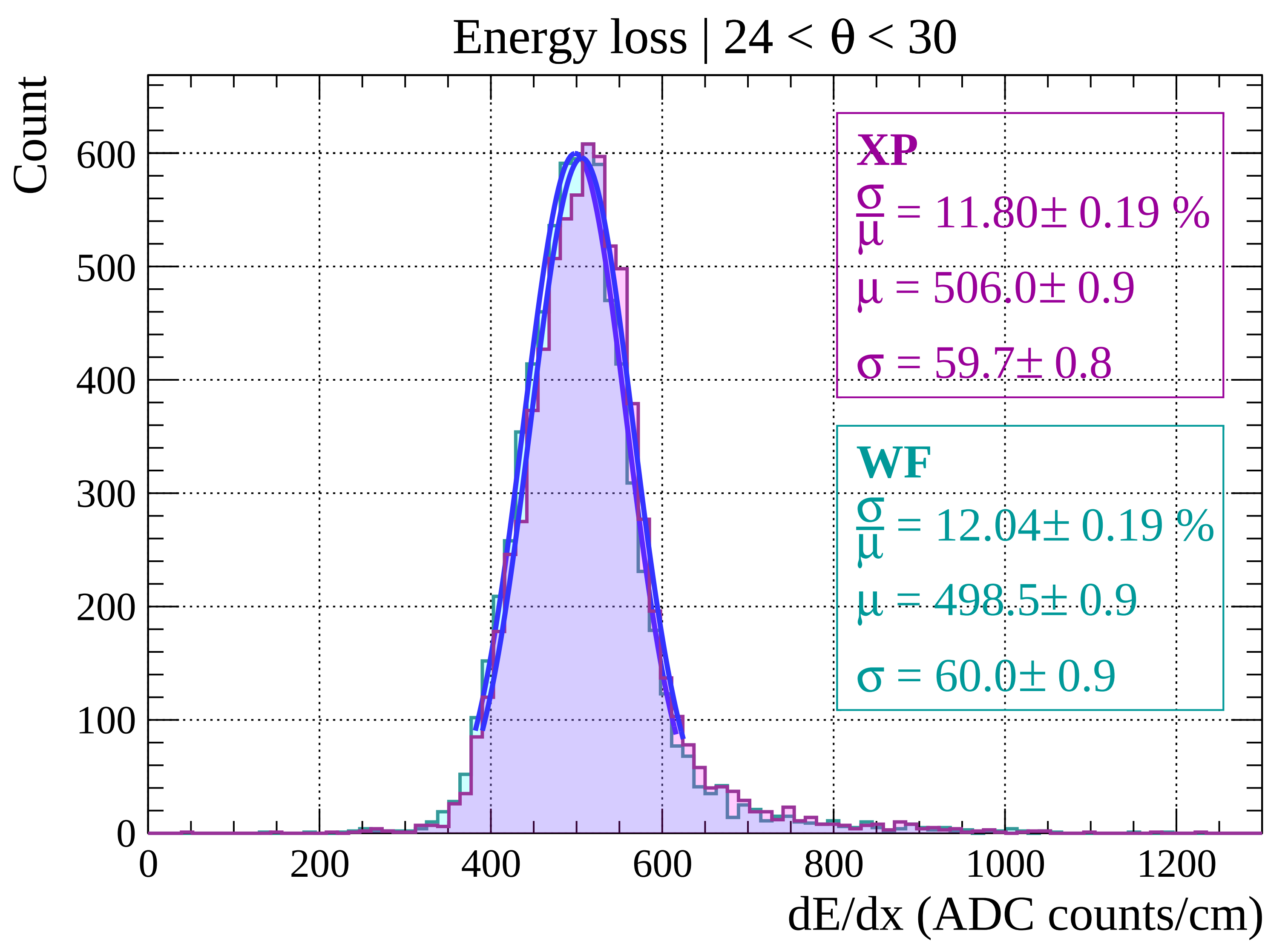
Energy loss | 0 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.61± 0.14 %
μ = 503.5± 0.7
σ = 58.5± 0.6
WF
σ/μ = 11.48± 0.14 %
μ = 496.0± 0.6
σ = 57.0± 0.6

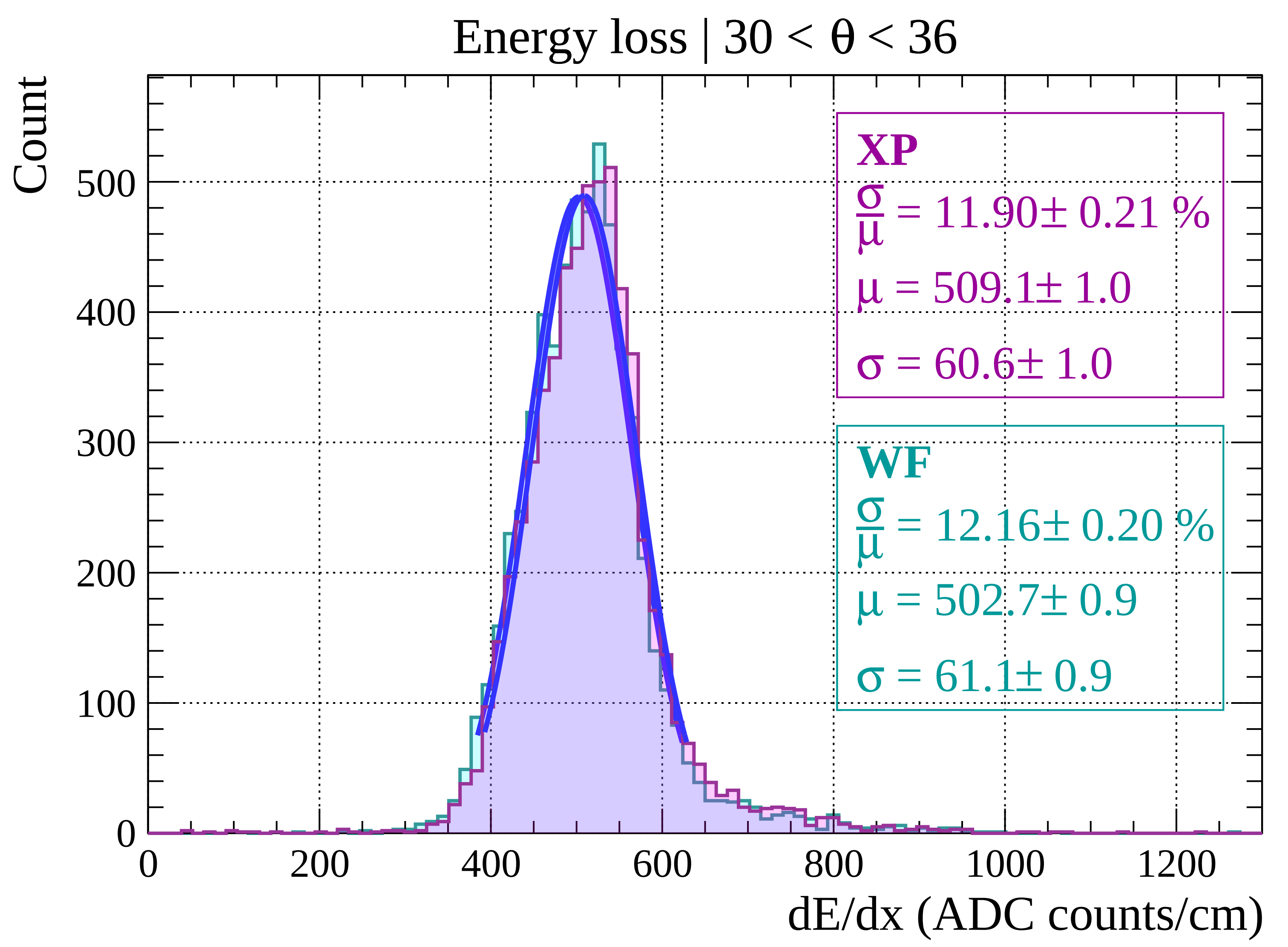

Energy loss | 6 < θ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.60± 0.15 %
μ = 504.4± 0.7
σ = 58.5± 0.7
WF
σ/μ = 11.61± 0.14 %
μ = 496.4± 0.7
σ = 57.6± 0.6
0
200
400
600
800
1000
1200

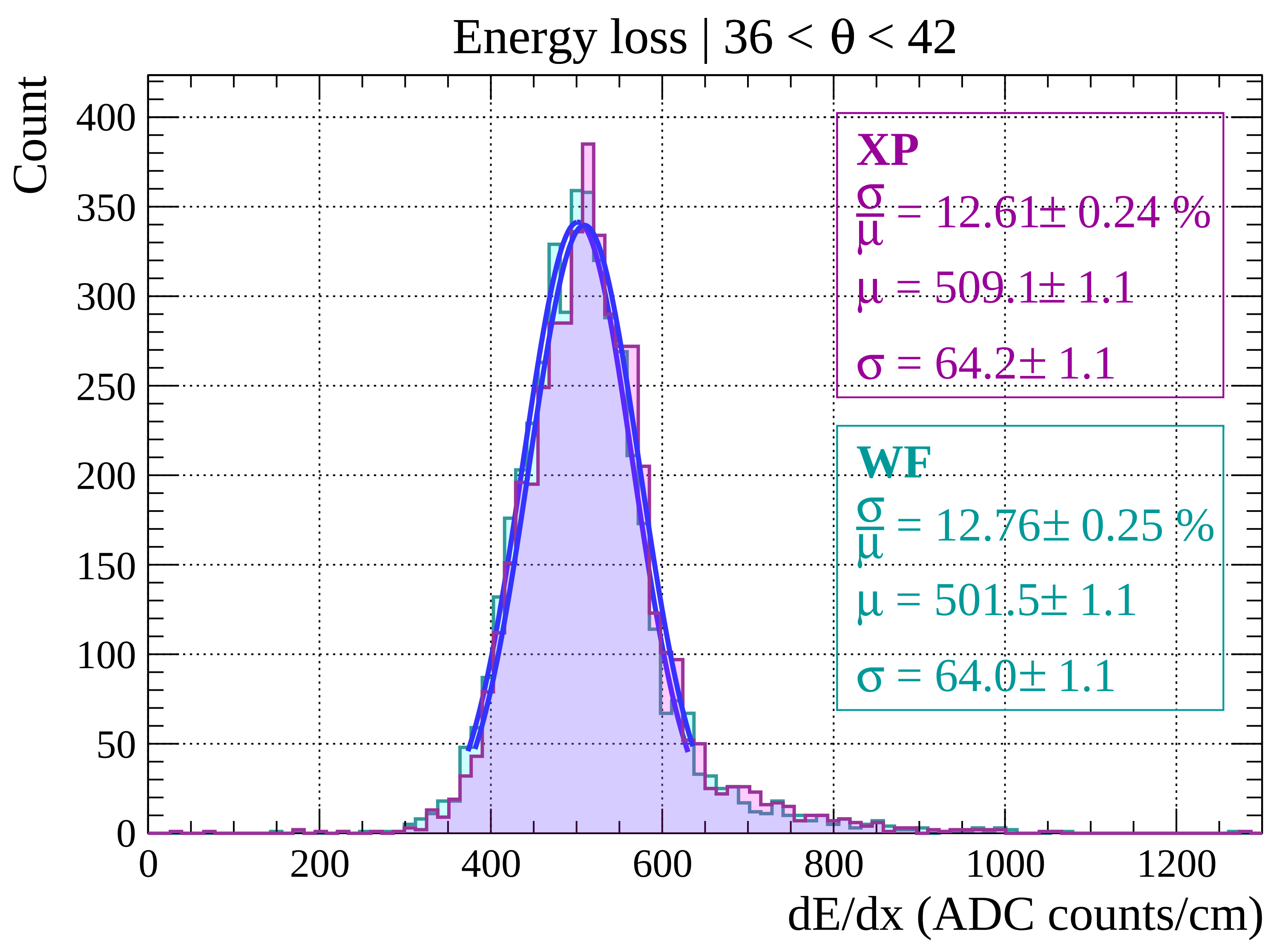


Energy loss | 12 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.75± 0.16 %
μ = 504.6± 0.7
σ = 59.3± 0.7
WF
σ/μ = 11.83± 0.16 %
μ = 496.6± 0.7
σ = 58.7± 0.7

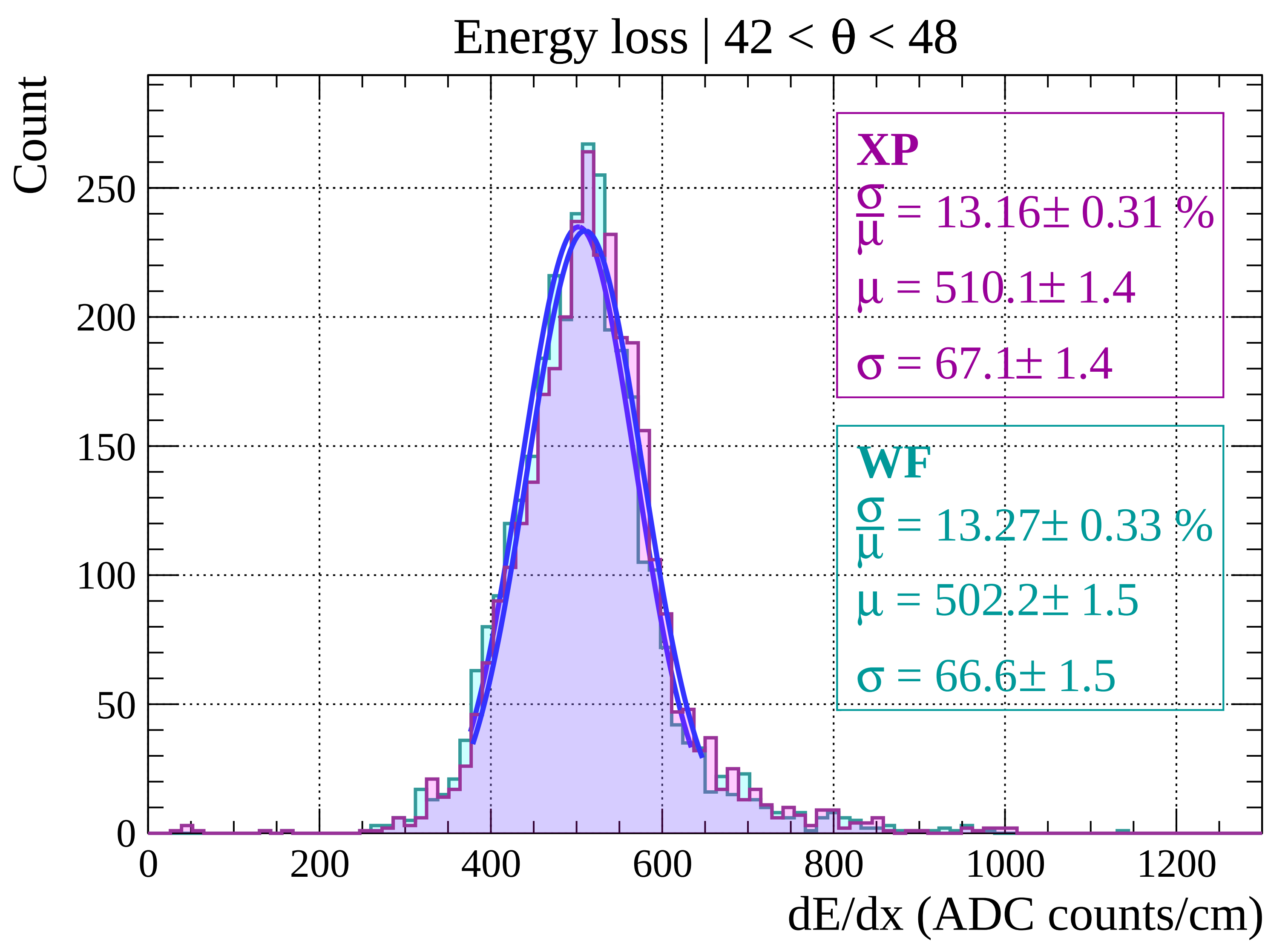

Energy loss | 18 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.88± 0.18 %
μ = 506.1± 0.8
σ = 60.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 11.88± 0.16 %
μ = 498.0± 0.7
σ = 59.2± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200


Energy loss | 24 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.80± 0.19 %
μ = 506.0± 0.9
σ = 59.7± 0.8
WF
σ/μ = 12.04± 0.19 %
μ = 498.5± 0.9
σ = 60.0± 0.9
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

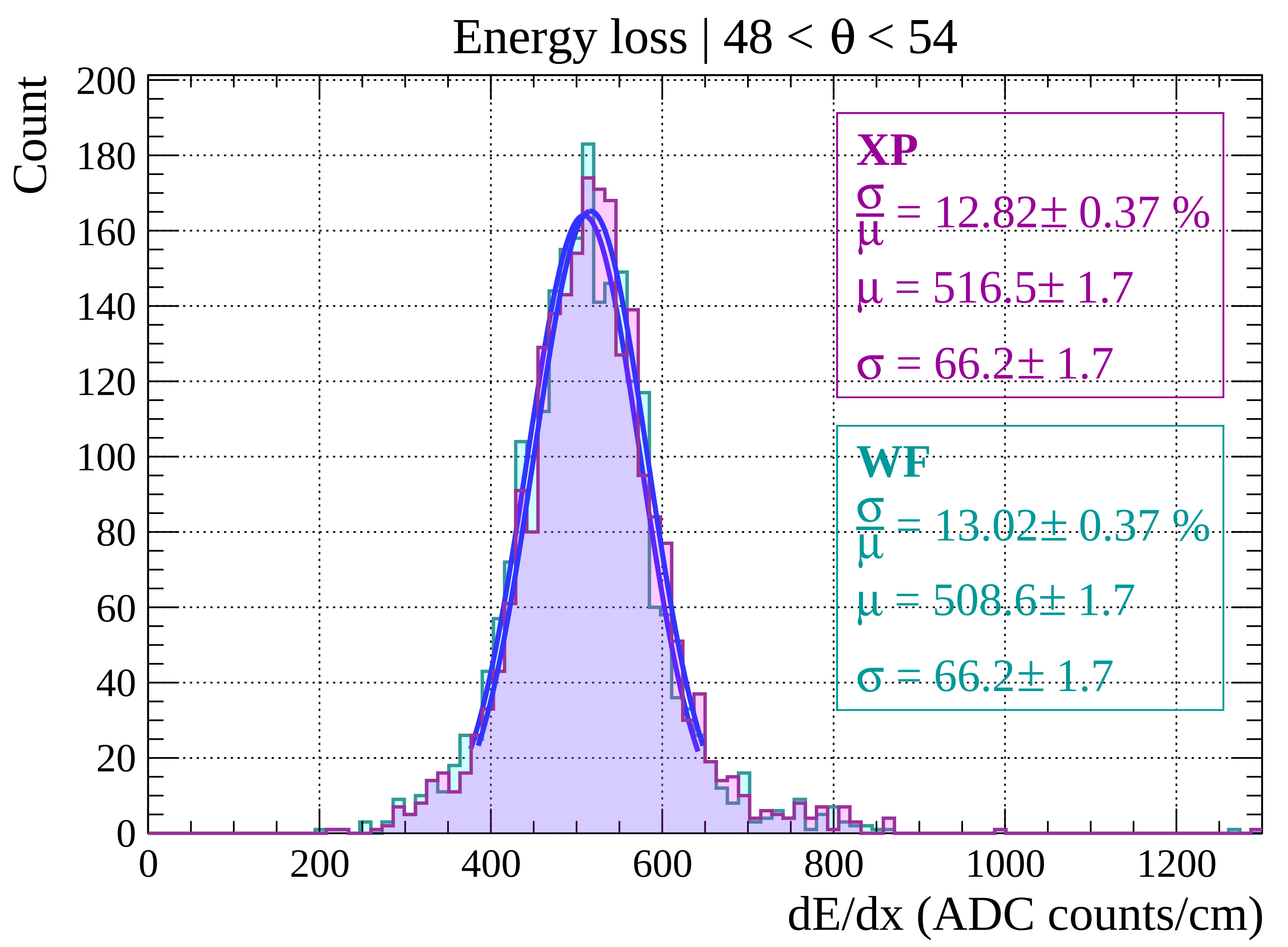

Energy loss | 30 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.90± 0.21 %
μ = 509.1± 1.0
σ = 60.6± 1.0
WF
σ/μ = 12.16± 0.20 %
μ = 502.7± 0.9
σ = 61.1± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200

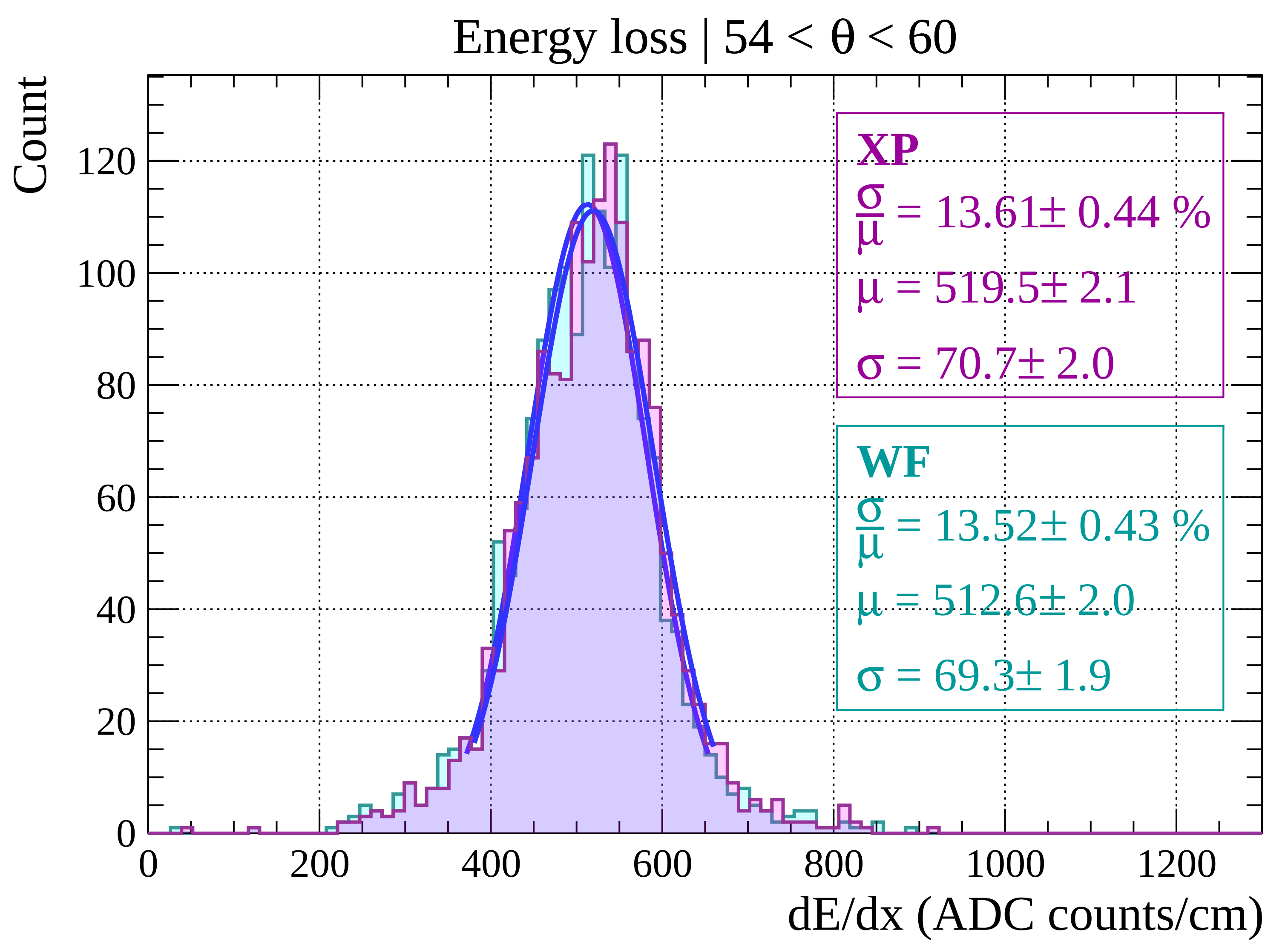


Energy loss | 36 < θ < 42
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 12.61± 0.24 %
μ = 509.1± 1.1
σ = 64.2± 1.1
WF
σ/μ = 12.76± 0.25 %
μ = 501.5± 1.1
σ = 64.0± 1.1

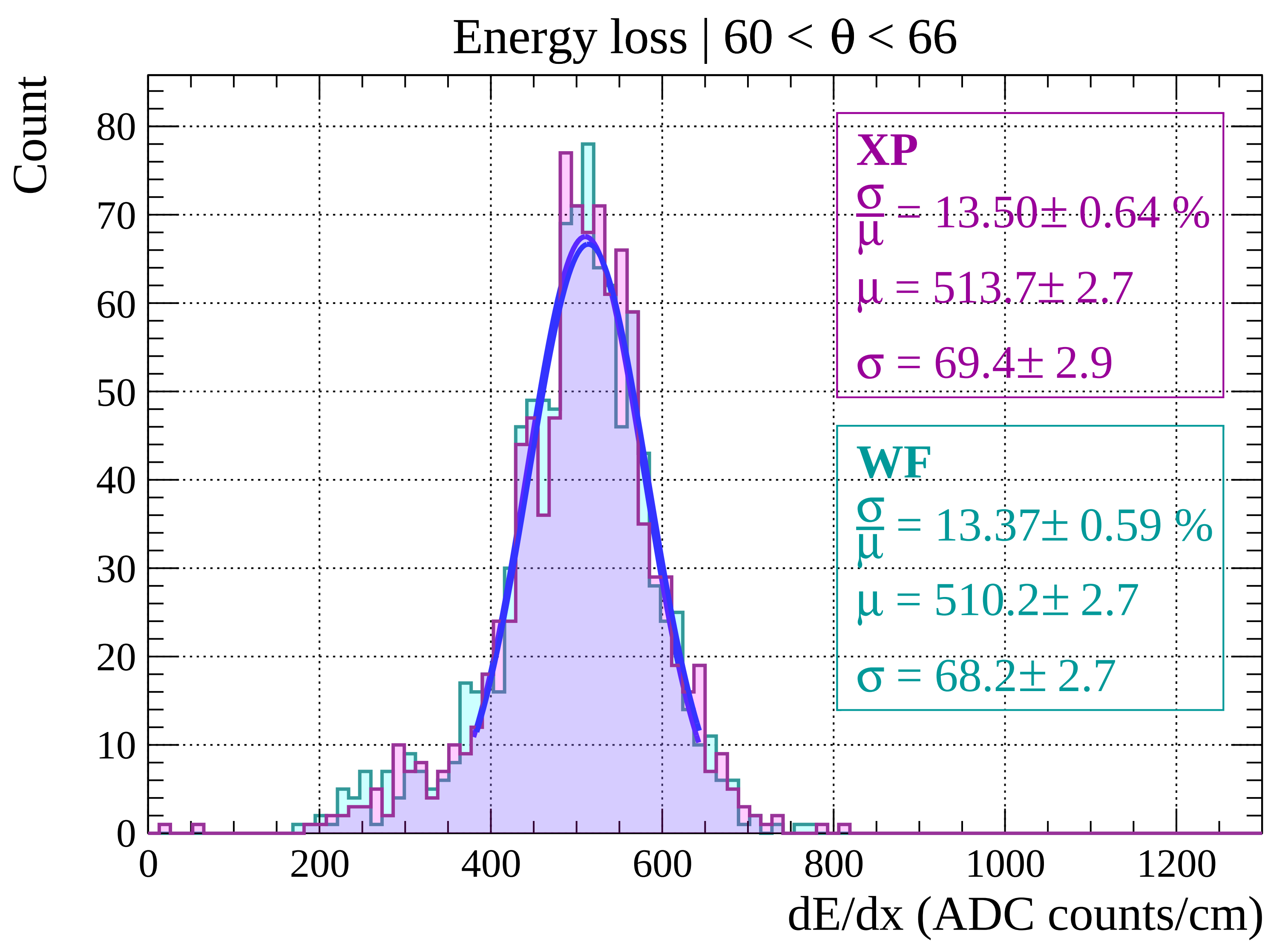


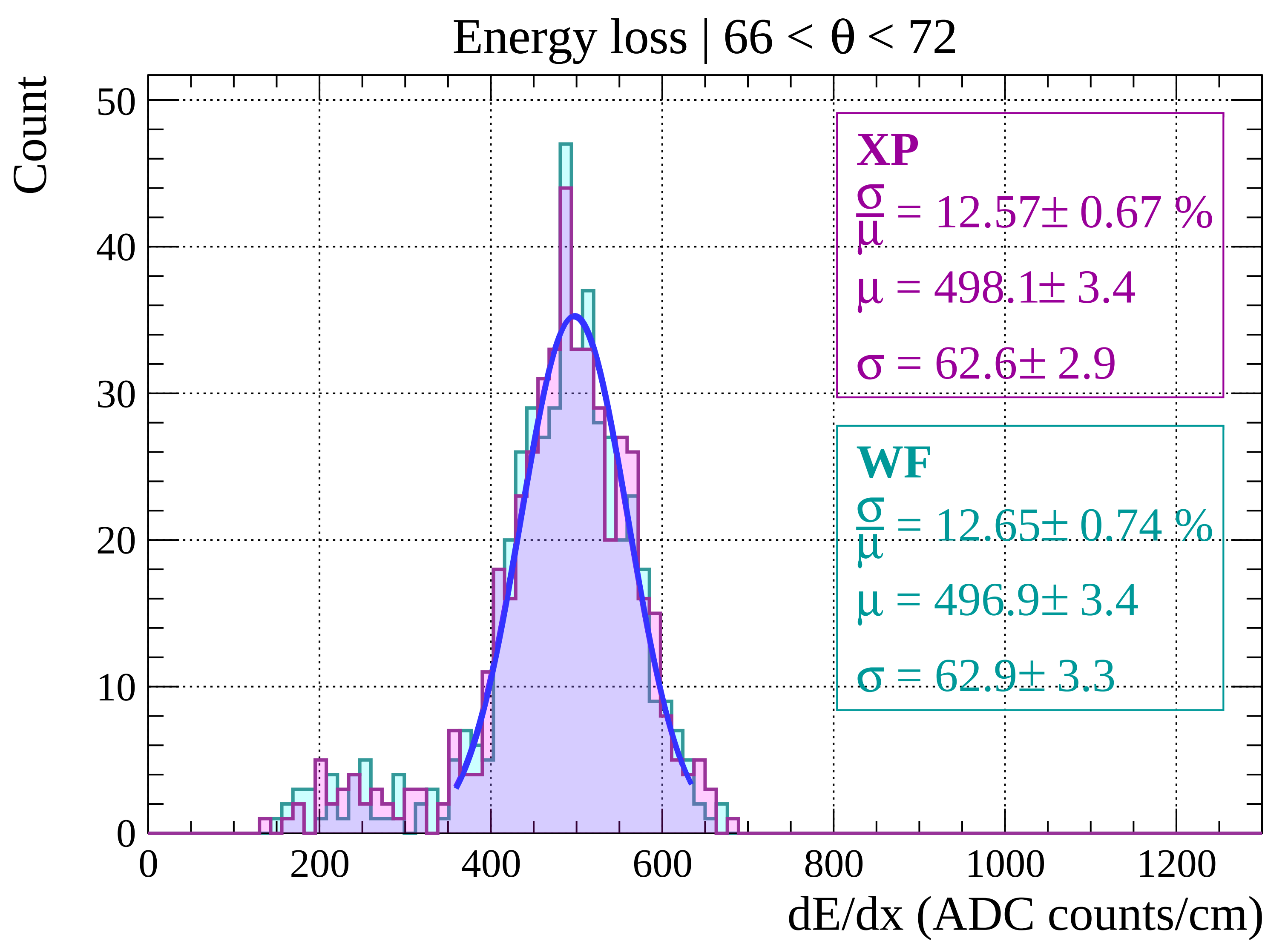
Energy loss | 42 < θ < 48
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 13.16± 0.31 %
μ = 510.1± 1.4
σ = 67.1± 1.4
WF
σ/μ = 13.27± 0.33 %
μ = 502.2± 1.5
σ = 66.6± 1.5


Energy loss | 48 < θ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.82 ± 0.37 %
μ = 516.5 ± 1.7
σ = 66.2 ± 1.7
WF
σ/μ = 13.02 ± 0.37 %
μ = 508.6 ± 1.7
σ = 66.2 ± 1.7
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200

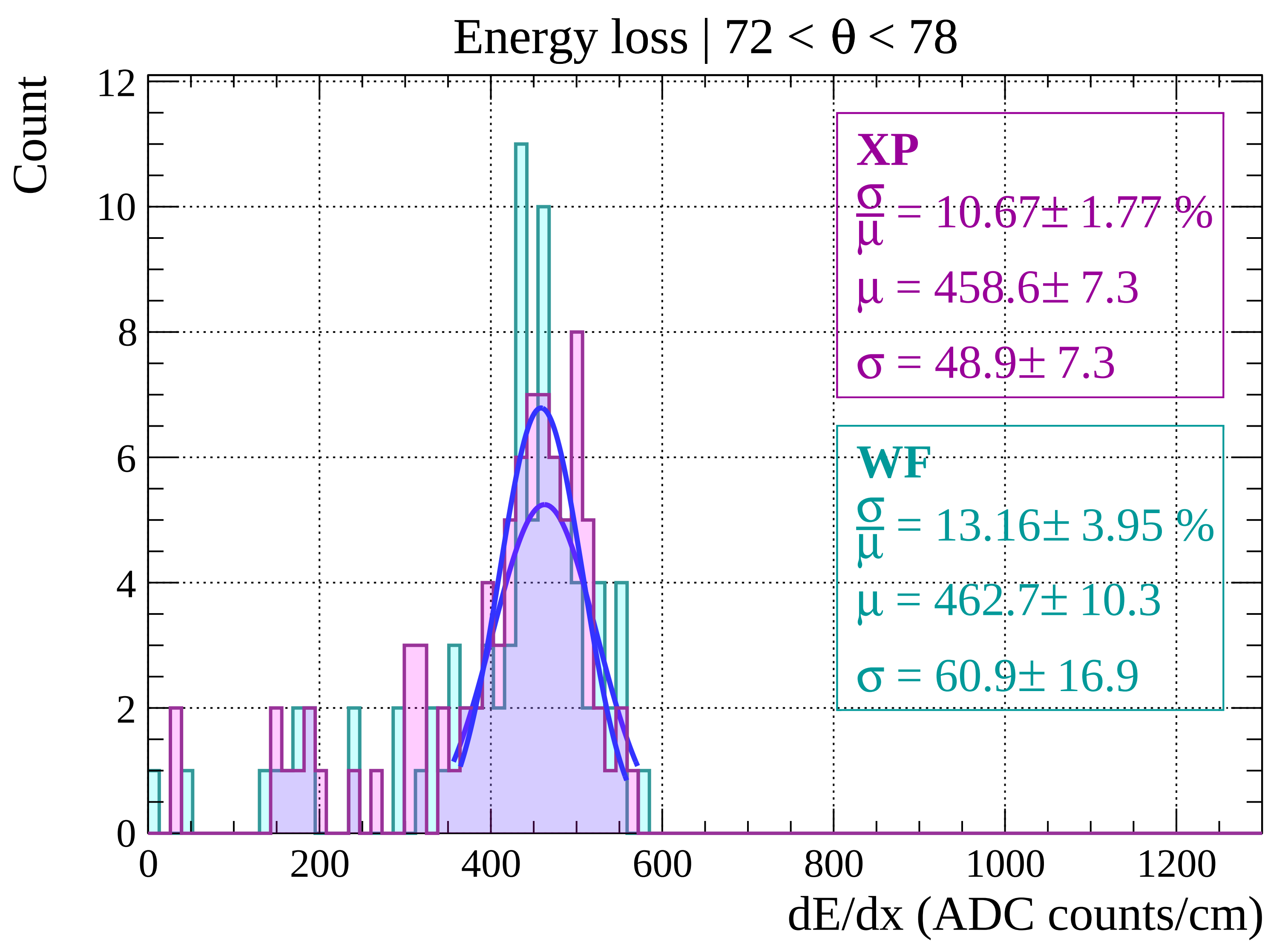


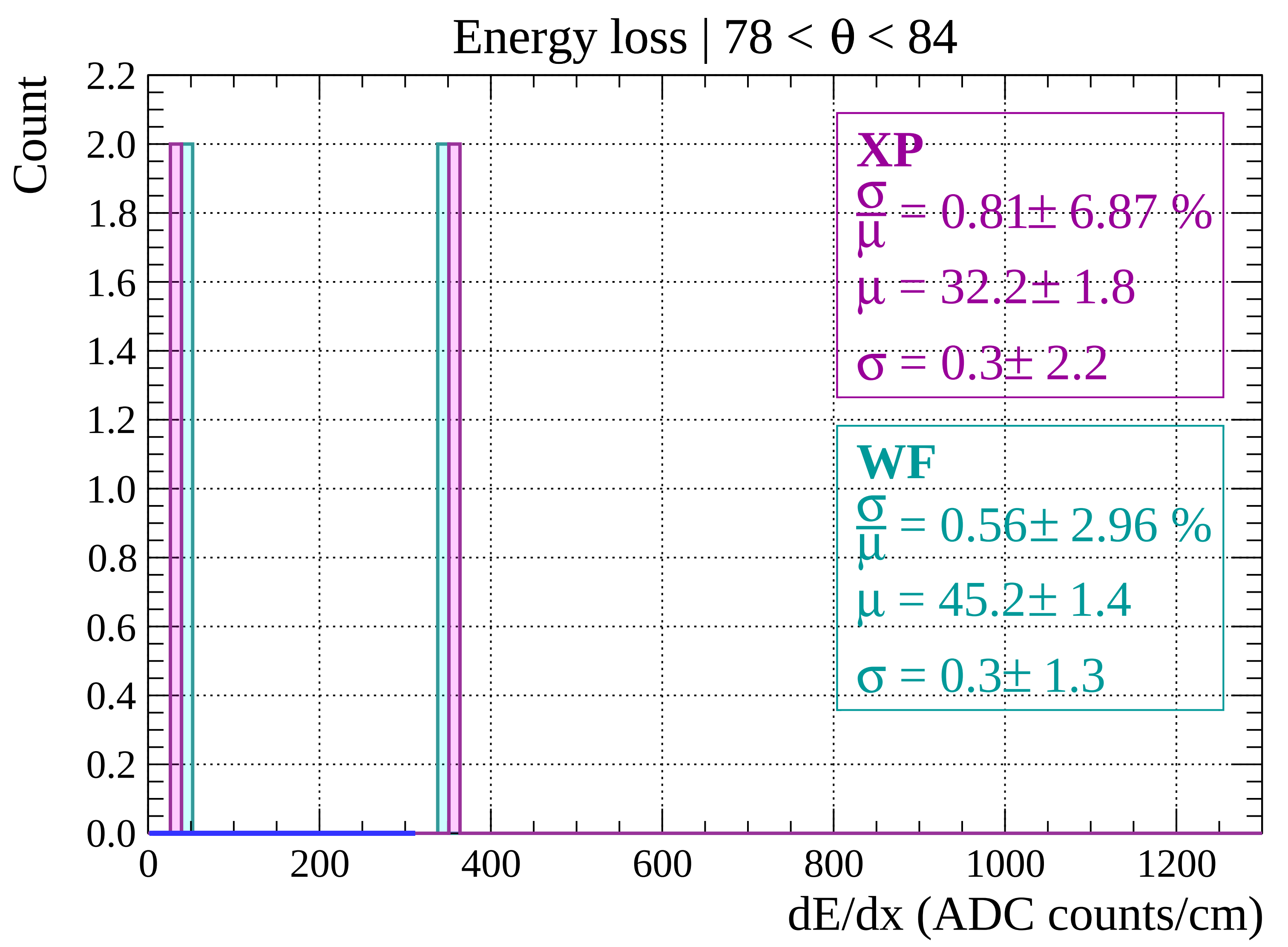
Energy loss | 54 < θ < 60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.61± 0.44 %
μ = 519.5± 2.1
σ = 70.7± 2.0
WF
$\frac{\sigma}{\mu}$ = 13.52± 0.43 %
μ = 512.6± 2.0
σ = 69.3± 1.9


Energy loss | 60 < θ < 66
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.50± 0.64 %
μ = 513.7± 2.7
σ = 69.4± 2.9
WF
$\frac{\sigma}{\mu}$ = 13.37± 0.59 %
μ = 510.2± 2.7
σ = 68.2± 2.7

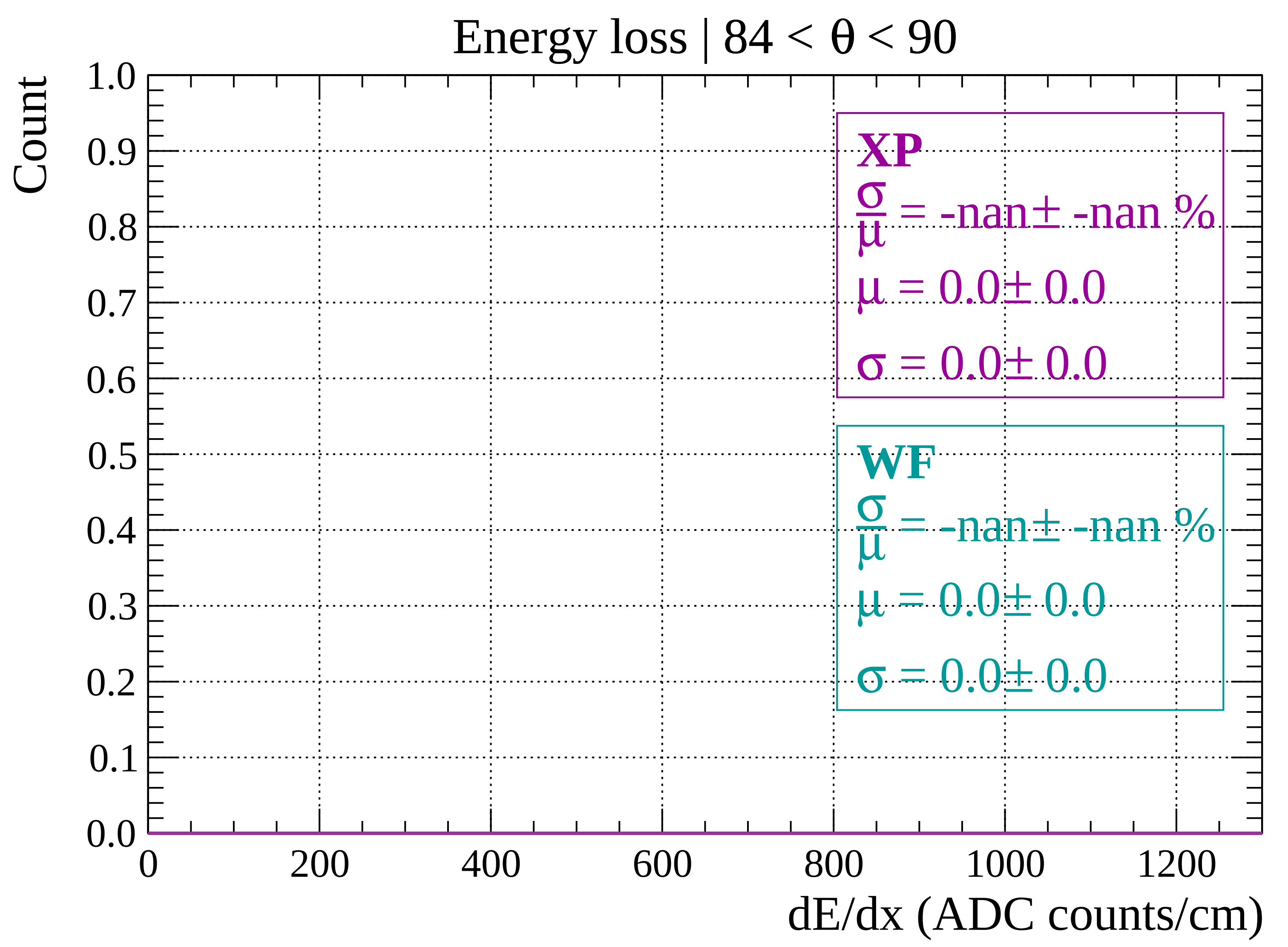


Energy loss | 66 < θ < 72
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.57± 0.67 %
μ = 498.1± 3.4
σ = 62.6± 2.9
WF
σ/μ = 12.65± 0.74 %
μ = 496.9± 3.4
σ = 62.9± 3.3

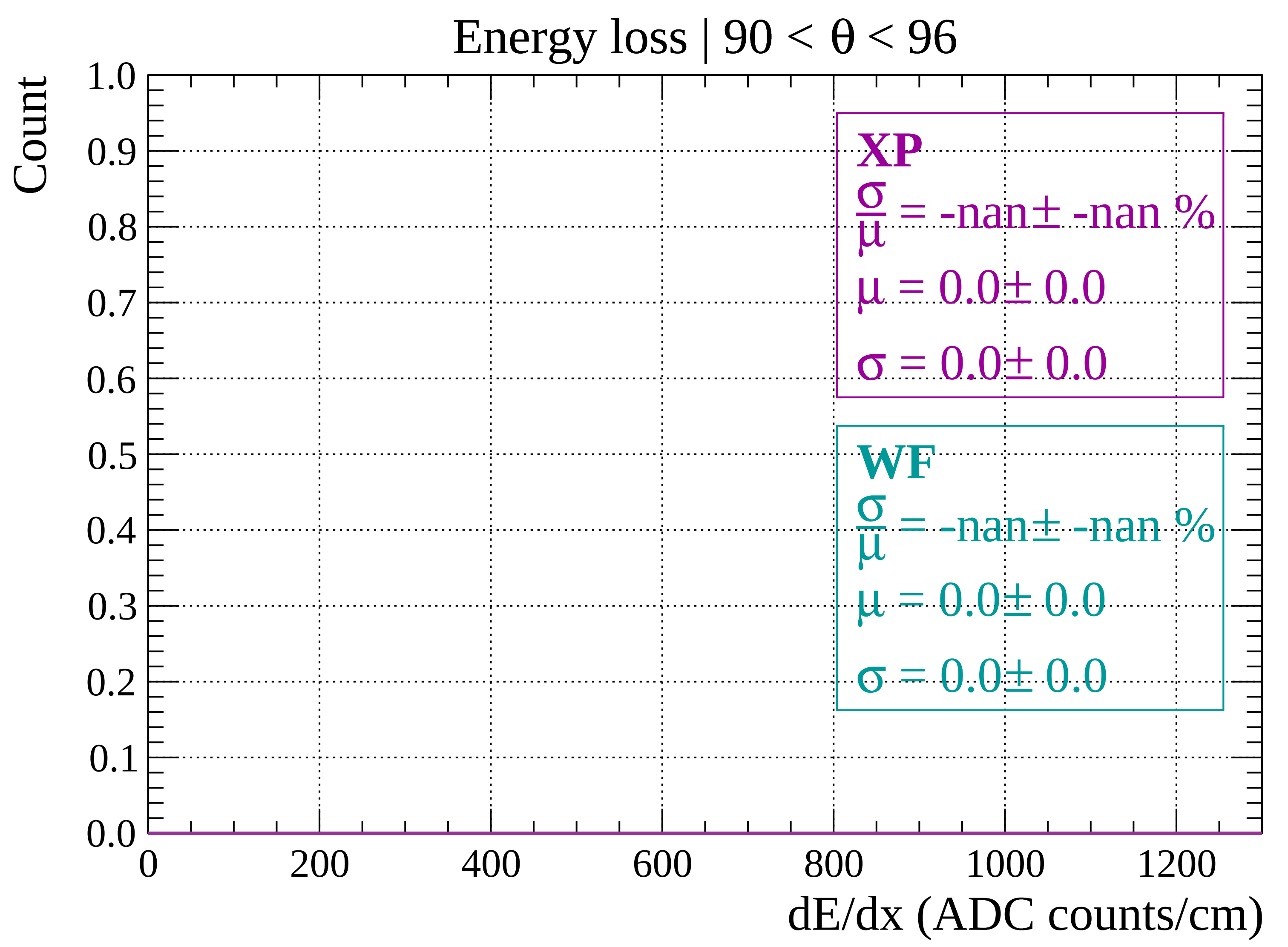

Energy loss | 72 < θ < 78
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.67± 1.77 %
μ = 458.6± 7.3
σ = 48.9± 7.3
WF
σ/μ = 13.16± 3.95 %
μ = 462.7± 10.3
σ = 60.9± 16.9



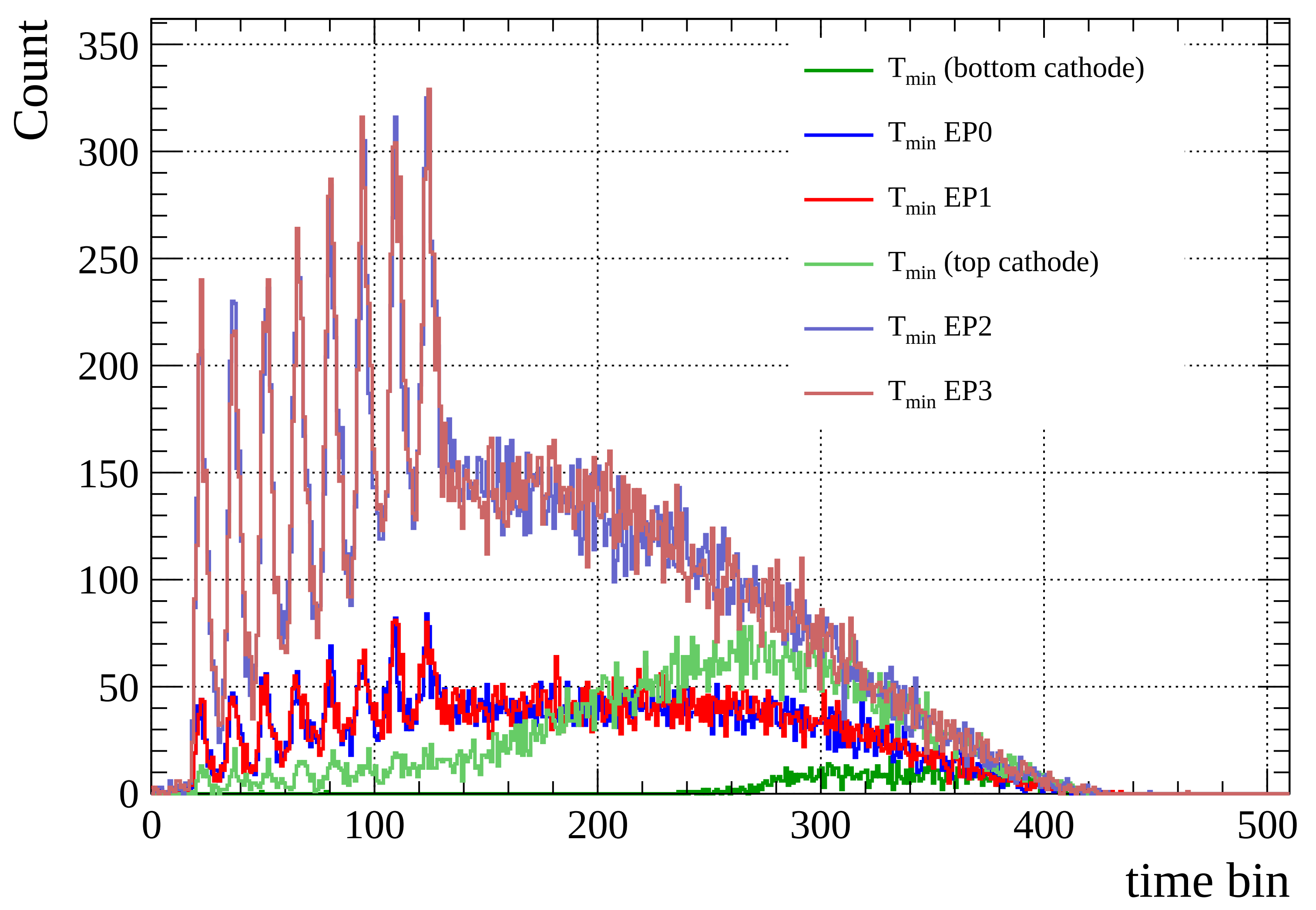
Energy loss | 78 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.81± 6.87 %
μ = 32.2± 1.8
σ = 0.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 0.56± 2.96 %
μ = 45.2± 1.4
σ = 0.3± 1.3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



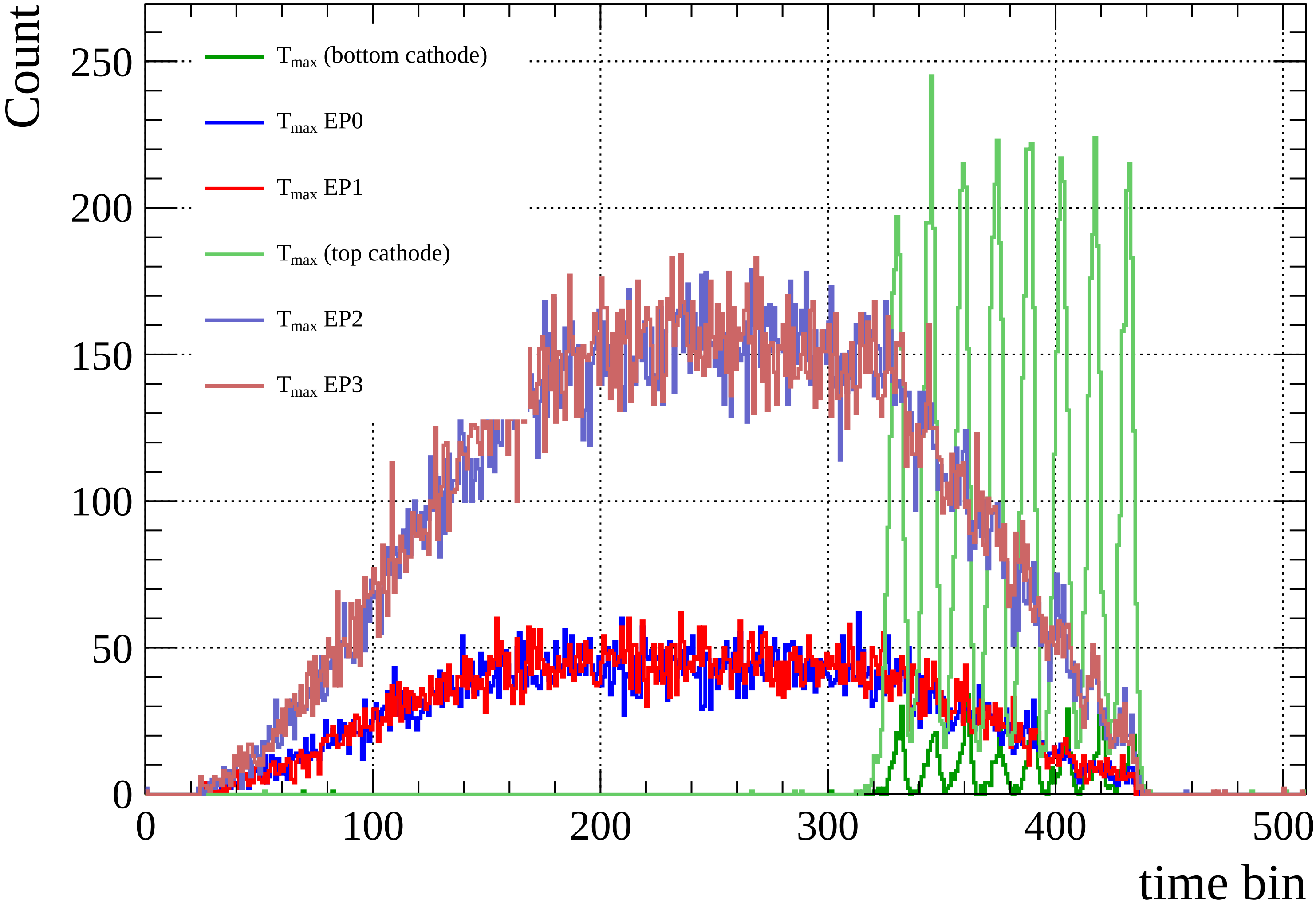
Energy loss | 84 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 96
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
50
100
150
200
250
300
350
0
100
200
300
400
500
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
250
200
150
100
50
0
0
100
200
300
400
500
time bin
Tmax (bottom cathode)
Tmax EP0
Tmax EP1
Tmax (top cathode)
Tmax EP2
Tmax EP3